



Lawful Interception Guide

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Affirmed Networks Inc.

A Microsoft Company

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Preface

This guide describes Lawful Interception in UnityCloud.

Audience

This guide is intended for all users who work with Affirmed Networks 5G or CN1 products.

Release Information

See the UnityCloud™ Software Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. You can access Affirmed Networks Guides and Release Notes on the [Manuals](#) site. From here, you can also download a PDF version of an individual guide, or the full manual set for a product. For access and instructions, please contact Support.

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Lawful Intercept

The UnityCloud Lawful Intercept solution is compliant with the ETSI TS 103.221 series standards along with the accompanying 3GPP TS 33.126, 33.127, and 33.128 standards. In addition to these 5G standards, UnityCloud Lawful Intercept provides interception for 4G sessions. Interception events for 4G sessions are reported using the ETSI TS 103.221 series standards in conjunction with the 3GPP TS 33.108 standard.

This section describes the LI-X1 and LI-X2 microservices:

- [LI-X1](#)
- [LI-X2](#)

LI-X1

LI-X1 is one of the microservices in Lawful interception functionality. LI-X1 is a common microservice which will be included by different Network Functions in their federations if lawful interception functionality is required.

For example, control plane federations like SMF and user plane network functions like UPF etc includes this common microservice. In the Network function this microservice exposes the service end point on URI /X1/NE towards the ADMF/Trigger Function.

It exposes the LI_X1 interface towards ADMF when this microservice is included in control plane federations which does not require trigger functionality ex: AMF.

It exposes the LI_X1, LI_T2, and LI_T3 interfaces towards ADMF and Triggered POI when this microservice is included in control plane federation which requires a trigger functionality. Ex: SMF.

When this microservice is included in user plane federation it exposes the LI_T2 and LI_T3 interfaces towards the Trigger Function (which is working as an AMDF). Ex: UPF.

LI-X1 is responsible for processing of LI_X1, LI_T2 and LI_T3 protocol messages from/to ADMF or Trigger Function. Within the Network Function LI-X1 is responsible for handling create/modify/remove destinations, activate/deactivate/modify tasks from ADMF/Trigger Function.

LI-X1 publishes the provisioned Destinations/Tasks learnt from ADMF/TF in ETCD key value data store in encrypted format in predefined paths. Subscribers for these etcd tables are LI-X2, Call control(ex: SMFCC, AMFCC etc) and PPE.

The LI-X1 also manages the statistics and provisioning status document for the Destinations/Tasks in Mongo Db in encrypted format. These Mongo Db documents are internal to the LI-X1 and used for storing the statistic and faults and providing details for get queries(ex: GetTaskDetails, GetDestinationDetails) from ADMF/TriggerFunction.

Pod interactions

LI-X1 microservice interacts with following microservices within the federation for LI functionality.

LI-X2

- Exposes the REST API towards the LI-X2 for collecting the X2 PDU statistics. LI-X2 pushes the statistics to LI-X1 on this interface. LI-X1 stores the collected statistics for the task in Mongo Db.

Application Call control

- Exposes the REST API for Trigger Function(TF) functionality which include CC/IRI Activate/Modify/Deactivate Task functionality.
- Uses the REST API exposed by call control to notify when Task is created/modified/deleted.

PPE

- Uses the REST API exposed by PPE to notify when Task is created/modified/deleted.
- Exposes the REST API towards the PPE for collecting the X2/X3 PDU statistics. PPE pushes the statistics to LI-X1 on this interface. LI-X1 stores the collected statistics for the task in MongoDB.

3GPP Specification Features

Following call flows are supported in current release for LI functionality:

- Activate Task - CPF / Activate Task - UPF
- CNF Readiness and Virtual Restart
- Create Destination - CPF / Create Destination - UPF
- Deactivate Task - CPF / Deactivate Task - UPF
- DeactivateAllTasks - CPF / DeactivateAllTasks - UPF
- Destinations - Creating, Modifying, and Removing
- DID Configuration Limitations
- GetAllDetails - CPF / GetAllDetails - UPF
- GetAllDetails - UPF (Management Interface)
- GetDestinationDetails - CPF / GetDestinationDetails - UPF
- GetNEStatus - CPF / GetNEStatus - UPF

- GetTaskDetails - CPF / GetTaskDetails - UPF
- Getting Information from NE
- HTTPS (TLS 1.3) Support for X1 Interface
- Internal Interfaces to Other Microservice
- IRI or/and CCTF Activate Task Procedure - CPF
- IRI or/and CCTF Deactivate Task Procedure - CPF
- Keep Alive from LIPF - CPF or UPF
- KeepAlive-P2
- KeepAlives
- ListAllDetails - CPF / ListAllDetails - UPF
- ListAllDetails - UPF (Management Interface)
- Management Interface Support
- Modify Task - CPF
- Modify Task - UPF
- Modify Task Support
- Multi-Instance Support
- Multiple UPF Per PDU Session Support
- Ping from LIPF - CPF or UPF
- Pings
- Provisioning Local DID
- Publication
- Remove All Destinations - CPF / Remove All Destinations - UPF
- Remove Destination - CPF / Remove Destination - UPF
- Report Destination Issue - CPF / Report Destination Issue – UPF
- Report NE Issue - CPF / Report NE Issue - UPF
- Report Task Issue - CPF / Report Task Issue - UPF
- Reporting Issues From NE

- Security Context for the Containers
- Support PDSR TaskDetailsExtensions
- Task and Destination Delivery Type Enhancements
- Tasks - Starting, Modifying, and Stopping
- Warnings and Faults
- Whitelisting
- X2 Task Stats Push Procedure - CPF

Non 3GPP Specification Features

Following call flows are supported in current release for LI functionality:

- ELF Core Files Generation
- Enabling/Disabling Task Sensitive Logs
- Encryption
- Node Failure Resiliency Through POD Spread Policy
- Private Destination Details Extension Support for Secondary Destination
- Publication
- Whitelisting
- Redundant ADMF Support
- LI TF Audit and Reprogram

Configuration

This section contains information for LI-X1 yang configuration. Following configurations are supported by the LI-X1 microservice.

LI related configurations at the NE is supposed to happen over LI_X0 interface. But as 3GPP is not yet defined the specification for the LI_X0, so LI-X1 supports the yang-based configuration for local NE/NF and Peer NFs/NES.

LI configurations are accessible to the LI admin user privileges and not available for regular NF admin.

On X1/T2/T3 interfaces when messages are exchanged between two peers, following attributes are used to identify the unique peer. These are mandatory attributes and present in all X1 related messages.

- ADMF Identifier
- NE Identifier

Timer-P1 is applicable only on fed-smf. SMF uses this configured value to send KeepAlive periodically to UPF. Timer-P2 is applicable to all application feds, and all tasks are deactivated if no message is received over X1/T2/T3 interface.

- Time-P1
- Time-P2
- Timer-P1-X2X3
- Timer-P2-X2X3

The configurations are modelled around these attributes.

Local Lawful Interception (LI) Profile

Following profile allows to configure the local NF related LI configurations:

```
li-x1 local-profile local-ne-identifier <local identifier name>

li-x1 local-profile operator-identifier opeid li-x1 local-profile time-p1
60

li-x1 local-profile time-p2 60

li-x1 local-profile time-p1-x2x3 60 li-x1 local-profile time-p2-x2x3 180
```

Peer LI Profile

Following profiles are to configure the Peer NF related configurations:

```
li-x1 peer-profile attpeer1

host      li-tools-sim.fed-li-test.svc.cluster.local https-port 9009

http-port 80

keep-alive-t2t3      [enabled | disabled] time-p2 60

!
```

Lawful Intercept

```
li-x1 peer-interface peerintf
peer-type      t2
ne-identifier  dummy.com
associated-peer-profile attpeer1 n4      dummy.com
!

li-x1 peer-interface peerintf1
peer-type      t3
ne-identifier  dummy.com associated-peer-profile attpeer1 n4      dummy.com
!
```

Note: As part of the LI-X1 configuration, configure Peer Profile with unique IP address or FQDN. This is due to the limitation on egress gateway that limits the creation of external host resources having the same host but a different port.

DID Profile

Following profile is used for the configuration of Destination (MDF2 or MDF3) server related details at the NF. This configuration is owned by the LI-X1 microservice:

```
li-x1 did-config f0054db4-f7ca-4420-9303-fbd2da6b2e74
delivery-type    x2andx3
delivery-address-type ipAddressAndPort primary-ipv4-address 10.11.11.23
primary-port     443
secondary-ipv4-address 10.11.11.24
secondary-port   443
switch-back-timer 60
time-p1-x2x3     60
time-p2-x2x3     180
!
```

LI Logging

The following configuration allows LI Admin user to enable log messages for LI Task related events. LI Task related log messages will potentially reveal Target Identifier values (IMSI/IMEI/MSISDN) of subscribers who are targeted for Lawful Intercept. Hence, this configuration (disabled by default) should only be enabled for use during lab trials and other debugging scenarios.

```
li-x1 li-logging task-logging disabled
```

Dashboard Description

Lawful-interception functionality is highly secured, sensitive, and not supposed to generate any information on NF dashboard. Hence, Dashboard is not applicable to lawful interception functionality.

Helm Customization

For LI-X1 Helm parameters, refer to the UnityCloud User Guide.

Statistics

Statistics for the LI functionality is obtained using the X1 or T2 or T3 interface of the GetTaskDetails/GetDestinationDetails/GetAllDetails messages defined in ETSI TS 103221-01. Hence, LI does not spill any statistics on NF.

Events

Events are available for LI-X1.

Related Topics

- [LI Architecture with SMF or UPF](#)
- [External Interfaces](#)
- [LI Egress Gateway Setup from fed-smf to External fed-upf](#)
- [LI Egress Gateway Setup from fed-smf to MDF2](#)
 - [LI Egress Gateway Setup from fed-smf to MDF2 Primary \(TLS Protocol\)](#)
 - [LI Egress Gateway Setup from fed-smf to MDF2 Secondary \(TLS Protocol\)](#)
- [LI Ingress Gateway Setup for fed-smf](#)

- [LI Ingress Gateway Setup for fed-upf](#)
- [Enabling or Disabling TLS Support](#)

Activate Task - CPF

This procedure is used by ADMF to add a new Task (XID) on NF.

1. ADMF sends the ActivateTask message to NF with the Task details. Message content includes following details:
 - XID - Uniquely identifies the Task
 - Target Identifiers - List of criteria which are used to identify the traffic to be intercepted.
 - Delivery Type - Statement of whether to deliver X2 and/or X3
 - List of DIDs - Details of where to send intercepted traffic.
 - ImplicitActivationAllowed - Indication that a Task may implicitly deactivate itself once the NE has determined that it has completed.
2. LI-X1 microservice provides the "/X1/NE" end point service for the LI_X1 interface.
3. ActivateTask message reaches to LI-X1 microservice through Istio Ingress LI gateway.
4. LI-X1 microservice parses the XML message content details in ActivateTask. LI-X1 generates the obfuscated XID from XID using one-way hash sha256 and generates the obfuscated Admflid from Admflid.
5. LI-X1 creates the XID related statistics and status document in Mongo Db to collect the statistics and provisioning status and acquires the soft lock on document. Primary key for the Mongo document is Obfuscated XID and secondary key for the Mongo Db document is Obfuscated ADMF id.
6. LI-X1 publishes the XID related details in ECTD and subscribes to same. Key for the ETCD entry is obfuscated XID. This is to protect the sensitive information being exposed. ETCD prefix for the XID entries is /<cpf-fixed-name>.<fed-unique-name>/<upcount>/li_x1/v1/xid_table/. ETCD typical DID entry complete key is /<cpf-fixed-name>.<fed-unique-name>/<upcount>/li_x1/v1/xid_table/<ofuscatedAdmflid>/<OfuscatedDID>.

When an encryption is disabled, content looks like below:

```
/smf.-10-163-64-206-07-21-07-56/1/li_x1/v1/xid_
table/8bb94fff066c8eaf4f4da81eee226befafa6a41c8c250fd46f68f2
7414b8991/82ac7b62e25295a8ddd2e978cb3366dbd5fee91838fc366c7a b1c78a7a89d

{"deliveryType":"X2andX3","listOfCcTFTasks":
[{"correlationID":"101","n4Address":"n4.testserver.com","tar
getIdentifiers":{"n1_targetIdentifier":[{"n1_
targetIdentifierExtension":{"n1_ UPFLIT3TargetIdentifierExtensions":{"ns1_
```

```
UPFLIT3TargetIdentifier": [{"ns1_FSEID": {"ns1_
IPv4Address": "10.25.21.234", "ns1_
IPv6Address": "fe80::20c:29ff:fee5:f188", "ns1_
SEID": "123456"}, "uPFLIT3TargetIdentifierType": "FSEID"}]}, {"t
argetIdentifierType": "targetIdentifierExtension"}]}, {"xid": "7 27487c1-c161-
4295-9810-79046c86e0da"}]}, {"listOfDIDs": {"n1_
dId": ["21ce308c-de30-42bd-a27b-287bcb5b3469", "21ce308c-de31- 42bd-a27b-
287bcb5b3469", "21ce308c-de32-42bd-a27b- 287bcb5b3469"}]}, {"listOfIRITasks":
[{"correlationID": "101", "n4Address": "n4.testserver.com", "tar
getIdentifiers": {"n1_targetIdentifier": [{"n1_
targetIdentifierExtension": {"n1_ UPFLIT3TargetIdentifierExtensions": {"ns1_
UPFLIT3TargetIdentifier": [{"ns1_FSEID": {"ns1_
IPv4Address": "10.25.21.234", "ns1_
IPv6Address": "fe80::20c:29ff:fee5:f188", "ns1_
SEID": "123456"}, "uPFLIT3TargetIdentifierType": "FSEID"}]}, {"t
argetIdentifierType": "targetIdentifierExtension"}]}, {"xid": "3 a809c30-4116-
4821-8ea3-af964ef00fcd"}]}, {"targetIdentifiers":
{"n1_targetIdentifier": [{"n1_
gpsiMsisdn": "919000000030", "targetIdentifierType": "gpsiMsisd n"}, {"n1_
supiimsi": "225097743010830", "targetIdentifierType": "supiims i"}, {"n1_
peiImei": "987654000000030", "targetIdentifierType": "peiIme i"}, {"n1_
gpsiNai": "929000000030", "targetIdentifierType": "gpsiNai"},
{"n1_ supinai": "235097743010830", "targetIdentifierType": "supina i"}, {"n1_
peiImeiCheckDigit": "937654000000030", "targetIdentifierType":
"peiImeiCheckDigit"}, {"n1_
peiImeiSv": "947654000000030", "targetIdentifierType": "peiImei
sv"}]}]}, {"version": "1.0.0", "xid": "29f28e1c-de30-486a-a860- f5a784ab9173"}
```

7. ETCD notifies the interested microservices about the XID addition. Typical subscribers for the XID entries are LI-X2 microservice and call control application.

8. LI-X1 releases the lock on Mongo XID document.

9. LI-X1 sends ActivateTaskResponse with OkAndAcknowledged to ADMF.

Activate Task - UPF

This procedure is used by Trigger Function (ex: SMF) working as ADMF to add a new Task (XID) on NF.

1. ADMF sends the ActivateTask message to NF with the Task details. Message includes following details:

- XID - Uniquely identifies the Task
- Target Identifiers - List of criteria which are used to identify the traffic to be intercepted.

- Delivery Type - Statement of whether to deliver X2 and/or X3
 - List of DIDs - Details of where to send intercepted traffic.
 - ImplicitActivationAllowed - Indication that a Task may implicitly deactivate itself once the NE has determined that it has completed.
 - Correlation ID - Correlation identifier to assign to intercepted material for this Task.
 - Product ID - When provided shall be used to populate x2/x3 header
2. LI-X1 microservice provides the "/X1/NE" end point service for the LI_T2 or LI_T3 interface.
 3. ActivateTask message reaches to LI-X1 microservice through Istio Ingress LI gateway.
 4. LI-X1 microservice parses the XML message content details in ActivateTask. LI-X1 generates the obfuscated XID from XID using one-way hash sha256 and generates the obfuscated Admflid from Admflid.
 5. LI-X1 creates the XID related statistics and status document in Mongo Db to collect the statistics and provisioning status and acquires the soft lock on document. Primary key for the Mongo document is Obfuscated XID and secondary key is obfuscated DID.
 6. LI-X1 publishes the XID related details in ECTD and subscribes to same. Key for the ETCD entry is obfuscated XID. This is to protect the sensitive information being exposed. ETCD prefix for the XID entries is /<cpf-fixed-name>.<fed-unique-name>/<upcount>/li_x1/v1/xid_table/. ETCD typical DID entry complete path is /<upf-fixed-name>.<fed-unique-name>/<upcount>/li_x1/v1/xid_table/<ofuscatedAdmflID>/<OfuscatedXID>.

When an encryption is disabled, content looks like below:

```
/upf.-10-163-65-1-07-08-04-39/1/li_x1/v1/xid_
table/qatestbedMaster206/4244e194-de33-4615-8130- a8cf8875b4f7

{"correlationID":"131","deliveryType":"X3Only","listOfDIDs":

{"n1_dId":["21ce308c-de33-42bd-a27b-
287bcb5b3469"]},"productID":"29f28e1c-de33-486a-a860-
f5a784ab9173","targetIdentifiers":{"n1_targetIden tifier":
[{"n1_targetIdentifierExtension":{"n1_
UPFLIT3TargetIdentifierExtensions":{"ns1_
UPFLIT3TargetIdentifier":[{"ns1_FSEID":{"ns1_

IPv4Address":"10.15.15.36","ns1_ IPv6Address":"fe80::20c:29ff:fee5:
f188","ns1_ SEID":"345326"},"uPFLIT3TargetIdentifierType":"FSEID"}]}}, "t
argetIdentifierType":"targetIdentifierExtension"}]},"version
":"1.0.0","xid":"4244e194-de33-4615-8130-a8cf8875b4f7"}

/upf.madan-maddela-10-163-65-1-07-08-04-39/1/li_x1/v1/xid_
table/qatestbedMaster206x/4244e194-de32-4615-8130- a8cf8875b4f7
```

```
{
  "correlationID": "131",
  "deliveryType": "X2Only",
  "listOfDIDs": [
    {
      "n1_dId": ["21ce308c-de32-42bd-a27b-287bcb5b3469"],
      "productID": "29f28e1c-de32-486a-a860-f5a784ab9173",
      "targetIdentifiers": {
        "n1_targetIdentifier": {
          "n1_targetIdentifierExtension": {
            "n1_UPFLIT3TargetIdentifierExtensions": {
              "ns1_UPFLIT3TargetIdentifier": [
                {
                  "ns1_FSEID": {
                    "ns1_IPv4Address": "10.15.15.36",
                    "ns1_IPv6Address": "fe80::20c:29ff:fee5:f188",
                    "ns1_SEID": "345156"
                  },
                  "uPFLIT3TargetIdentifierType": "FSEID"
                }
              ],
              "targetIdentifierType": "targetIdentifierExtension"
            }
          },
          "version": "1.0.0",
          "xid": "4244e194-de32-4615-8130-a8cf8875b4f7"
        }
      ]
    }
  ]
}
```

7. ETCD notifies the interested microservices about the XID addition. Typical subscribers for the XID entries are PPE microservice.
8. LI-X1 releases the lock on Mongo XID document.
9. LI-X1 sends ActivateTaskResponse with OkAndAcknowledged to ADMF.

CNF Readiness and Virtual Restart

With this feature, LI-X1 microservice participates in the CNF Readiness and Virtual functionality of federation. If it is part of the federation, it will publish present when it comes up. After receiving Service Available indication with status up from nf-monitor and after LI related etcd tables are loaded it publishes itself with ready to the consul.

Create Destination - CPF

This procedure is used by ADMF to add a new Destination(DID) on NF.

1. ADMF sends the CreateDestination message to NF with the details about MDF2 and/or MDF3 server details. Message includes following details:
 - DID - Destination identifier which uniquely identifies the destination
 - Delivery Type - Statement of whether to deliver X2 and/or X3 to this destination
 - Friendly Name - A human readable name associated with delivery destination
 - Delivery Address - Address of the MDF server
2. LI-X1 microservice provides the "/X1/NE" end point service for the LI_X1 interface.
3. CreateDestination message reaches to LI-X1 microservice through Istio Ingress LI gateway.

4. LI-X1 microservice parses the XML message content details in CreateDestination. LI_X1 generates the obfuscated DID from DID using one-way hash sha256 and generates the obfuscated Admflid from Admflid.
5. LI-X1 creates the DID related statistics and status document in Mongo Db to collect the statistics and provisioning status and acquires the soft lock on document. Primary key for Mongo document is obfuscated DID.
6. LI-X1 publishes the DID related details in ECTD and subscribes to same. Key for the ETCD entry is obfuscated DID. This is to protect the sensitive information being exposed. If encryption is enabled then the value part is encrypted and non-human-readable format. ETCD prefix for the DID entries is /<cpf-fixed-name>.<fed-unique- name>/<upcount>/li_x1/v1/did_table/. ETCD typical DID entry complete key is /<cpf-fixed-name>.<fed-unique-name>/<upcount>/li_x1/v1/did_table/<ofuscatedDID>.

Example samples are given below:

```
/smf.xyz-10-163-64-206-07-21-07-56/1/li_x1/v1/did_table/
```

```
/smf.xyz-10-163-64-206-07-21-07-56/1/li_x1/v1/did_table/b5742dfbf993ee1b4d25b2c13ff49fca215d2a28e0fd49951ca5d7 b2dc1
```

```
KeyVersion": "1", "Value": "
```

```
{\"cipheredData\": \"4NIH0E1ZdewSvUaOuy0c0yTbWL3QthxlVdOyPqnU/w9/knC0BOoGRY
auYlGz9vmpgQrQDHA8aWQK5iTKpdNpZ+6KwUnKNJyzRfOwoglqI1T0djr3KV+e7d95qU02ZA27
HG0mSSueTcb6LRlZyKjg0ZorRNWx7LjcUSiLoYFM0GEDVZ4q9L8G8XklQwpuLtlGHpxtfEd/Te
OK6x1M7IZyqTgm8Gpzgx50MiF9R7K6vFu/tQGmmsxE+YU/vPEeVoIrc2Lt8hiD1MWIZOXfWLwP
8EKhutsnA6GwhpJN7LhcmC1f3zoKJapfTbsSJU921xOMVFiKLlUnXBvtLn03111w4xghejVX2b
seWpNls6YtjKU0eJrkuUSO9RoBXao+Un5oXOEx2wrvBjuG2ndlwiZBIyobTZUCYZQcXXyYVxm
NoC3Z0NKJkG0mTgR0JKL5IK2uAicV08y1vkM56+xlGY0vEIEOlwWpU2NX8S02gzX51+Vekx3Vh
4tE0f4PV2r0309ipPpfkoNmLw7xqXGLFNqtOrCtPm+YuPUPQNR0DUQmswG8D8bMJS71AjvAEdq
diqOKMaZiDyD2kVhBwsBvcQbk+k/r9nhe8qVgm3G4jBQm3fAEvnLW6rUjSbQ1Dsvbv+DhliS/O
RXJ6Xa2NDAZwY/B+6RgiiKnQtABtHGFuCM4CzAWAXhG07sSsymPy9QcG27KGJsPgS/TLC22ur0
JJ6HgTWcSyvXmtiUIIlukeJugDv/t7gwW7RhsFolxhmZ+7pgo3HwtJWdRyaLhoPozKoZWn0Gu5M
YK6w11DNwZTUKcl7GrpPKs/alFzLX1jqXoX4FxOpL0FE1YNpns7XKd8Lgztg==\", \"keyVersio
n\": \" bd45862a9c4f475466be2e56bf849c847dd1fdb71d9fd139dbb89e779afb\"}
```

7. ETCD notifies the interested microservices about the DID addition. Typical subscribers for the DID ETCD entries are LI-X2 microservice which is responsible for creating and managing the actual TLS/TCP connections to MDF server.
8. Similarly, LI-X1 publishes the DID to LIPF mapping in ETCD. This is LI-X1 internal usage and required to know which ADMF provisioned the DID. Key for the ETCD entry is obfuscated DID. If encryption is enabled then the value part is encrypted and non-human-readable format.
9. ETCD prefix for the LIPF to DID mapping entry is /<cpf-fixed-name>.<fed-unique-name>/<upcount>/li_x1/v1/did_lipf_table/. ETCD typical LIPF to DID mapping entry complete key is /<cpf-fixed-name>.<fed-unique- name>/<upcount>/li_x1/v1/did_lipf_table/<ofuscatedDID>. Example samples are given below:

```
/smf.xyz-10-163-64-206-07-21-07-56/1/li_x1/v1/did_lipf_table/

/smf.xyz-10-163-64-206-07-21-07-56/1/li_x1/v1/did_lipf_table/a2bd2a6ac140697a137c469bedfffe9b112eae76aldeaffc7d9de8 786244b5

KeyVersion": "1", "Value": "

{ \"cipheredData\": \"4NIH0E1ZdewSvUaOuy0c0yTbWL3Qthx1VdOyPqnU/w9/knC0BooGRY
auYlGz9vmpgQrQDHA8aWQK5iTKpdNpZ+6KwUnKNJyzRfOwoglqI1TOdjr3KV+e7d95qU02ZA27
HG0mSSueTcb6LRlZyKjg0ZorRNWx7LjcUSiLoYFM0GEDVZ4q9L8G8XklQwpuLtlGHpxtfEd/Te
OK6x1M7IZyqTGM8Gpzgx50MiF9R7K6vFu/tQGmmsxE+YU/vPEeVoIrc2Lt8hiD1MWIZOXfWLwP
8EKhutsnA6GwhpJN7LhcmC1f3zoKJapfTbsSJU921xOMVFiKLlUnXBvtLn03111w4xghejVX2b
seWpPnls6YtjKU0eJrkuUSO9RoBXao+Un5oXOEx2wrvBjuG2nd1wiZBIyobTZUCYZQcXXyYVxm
NoC3Z0NKJkG0mTgr0JKL5IK2uAicV08y1vkM56+xlGY0vEIEOlwwpU2NX8S02gzX51+Vekx3Vh
4tE0f4PV2r0309ipPpfkoNmLw7xqXG1FNqtOrCtPm+YuPUPQNR0DUQmswG8D8bMJS71AjvAEdq
diqOKMaZiDyD2kVhBwsBvcQbk+k/r9nhe8qVgm3G4jBQm3fAEvnLW6rUjSbQ1Dsvbv+DhliS/O
RXJ6Xa2NDAZwY/B+6RgiiKnQtABtHGFuCM4CzAWAXhGO7sSsymPy9QcG27KGJsPgS/TLC22ur0
JJ6HgTWcSyvXmtiUIIlukeJugDv/t7gwW7RhsFolxhmZ+7pgo3HwtJWdRyaLhoPozKoZWn0Gu5M
YK6w11DNwZTUKcl7GrpPKs/alFzLX1jqXoX4FxOpL0FE1YNpns7XKd8Lgztg=\"\", \"keyVersio
n\": \" bd45862a9c4f475466be2e56bf849c847dd1fdb71d9fd139dbb89e779afb\" }
```

10. LI-X1 releases the lock on Mongo DID document.

11. LI-X1 sends CreateDestinationResponse with OkAndAcknowledged to ADMF.

Create Destination - UPF

UPF support to create destination with same DID from multiple Trigger Function (ex: SMF). All request contains same delivery address. UPF creates single connection towards MDF with the same DID configured by multiple Trigger Function.

UPF also supports to configure same DID with delivery type as X2 followed by X3 or X3 followed by X2. In such case, UPF updates its existing DID entry to X2AndX3.

This procedure is used by Trigger Function (ex: SMF) working as ADMF to add a new Destination(DID) on NF.

1. SMF working as ADMF sends the CreateDestination message to NF with the details about MDF2 and/or MDF3 server details. Message includes following details:

- DID - Destination identifier which uniquely identifies the destination
- Friendly Name - A human readable name associated with delivery destination
- Delivery Type - Statement of whether to deliver X2 or X3 to this destination
- Delivery Address - Address of the MDF server

2. LI-X1 microservice provides the "/X1/NE" end point service for the LI_T2 and LI_T3 interface.

3. CreateDestination message reaches to LI-X1 microservice through Istio Ingress LI gateway.
4. LI-X1 microservice parses the XML message content details in CreateDestination. LI-X1 generates the obfuscated DID from DID using one-way hash sha256 and generates the obfuscated Admfld from Admfld.
5. LI-X1 creates the DID related statistics and status document in Mongo Db to collect the statistics and provisioning status and acquires the soft lock on document. Primary key for the Mongo Db document is Obfuscated DID.
6. LI-X1 publishes the DID related details in ECTD and subscribes to same. Key for the ETCD entry is obfuscated DID. This is to protect the sensitive information being exposed. If encryption is enabled then the value part is encrypted and non-human-readable format ETCD prefix for the DID entries is /<upf-fixed-name>.<fed-unique- name>/<upcount>/li_x1/v1/did_table/. ETCD typical DID entry complete key is /ucpf-fixed-name>.<fed-unique-name>/<upcount>/li_x1/v1/did_table/<ofuscatedDID>.

Example samples are given below:

```
/upf.xyz-10-163-64-206-07-21-07-56/1/li_x1/v1/did_table/

/upf.xyz-10-163-64-206-07-21-07-56/1/li_x1/v1/did_
table/b5742dfbf993ee1b4d25b2c13ff49fca215d2a28e0fd49951ca5d7 b2dc1
```

```
KeyVersion": "1", "Value": "
```

```
{\"cipheredData\": \"uuqoSOFSoJceXFvBWK9sIxGtNw41x5a7RUAFEwvO
rY9Ihoz8ywMF1T1T1lwUJvKqph79nPQQYpGDQTGIH4Eiax/tWqUnZKOWIbj

D0FSTVvKUS6H8k1N37FcF5CYOLR4sRNmoWMapYeDv9axEkP8hN/BfA6TX3vj
DVv28/LnqOvXNOkmk95Szyjfw3Qdd6p0zzc30NZ5DGiafk3ZLr1XmB80tPCG
Ck/chWEK9mEv06rBoht41VF5CxUAMuGUMrMWIoVRLsGJ//2JOjW8CPJtehow
GRIG1pN6FgbQF79bOywlwNYY1/gUgMxsMPxHIZCkC9uE6CxsemcTNCpk5H0c
hceTexlXkKXoHS385pljsVicbWFooFSN3IIKABzJKWarh9MQakzPqHOBXVQh
bL8SNoJrvvhuy9sU2Kig1Lch1BdeEjgvgzG4nOr6gkK1RMiVCHiCE5H/mT+t
GFjvmTcaZslgpgli5A1PVz4cV14cWj4=\", \"keyVersion\": 1}
```

7. ETCD notifies the interested microservices about the DID addition. Typical subscribers for the DID entries are PPE microservice which is response the actual TLS/TCP connections to MDF server.
8. Similarly, LI-X1 publishes the DID to LIPF mapping in ETCD. This is LI-X1 internal usage and required to know which ADMF provisioned the DID. Key for the ETCD entry is obfuscated DID. If encryption is enabled then the value part is encrypted and non-human- readable format.
9. ETCD prefix for the LIPF to DID mapping entry is /<upf-fixed-name>.<fed- unique- name>/<upcount>/li_x1/v1/did_lipf_table/. ETCD typical LIPF to DID mapping entry is /<upf-fixed- name>.<fed-unique- name>/<upcount>/li_x1/v1/did_lipf_table/<ofuscatedDID>. Examples are given below:

```
/upf.xyz-10-163-64-206-07-21-07-56/1/li_x1/v1/did_lipf_ table/
```

```
/upf.xyz-10-163-64-206-07-21-07-56/1/li_x1/v1/did_lipf_
table/a2bd2a6ac140697a137c469bedfffe9b112eae76a1deaffc7d9de8 786244b5
```

```
KeyVersion": "1", "Value": "
```

```
{\"cipheredData\": \"uuqoSOFSoJceXFvBWK9sIxGtNw41x5a7RUAFEwvO
rY9Ihoz8ywMF1T1T1lwWUjvKqph79nPQQYpGDQTGIH4Eiax/tWqUnZKOWIbj
D0FSTVvKUs6H8k1N37FcF5CYOLR4sRNmoWMapYeDv9axEkP8hN/BfA6TX3vj
DVv28/LnqOvXNOkmk95Szyjfw3Qdd6p0zzc30NZ5DGiafk3ZLr1XmB80tPCG
Ck/chWEK9mEv06rBoht41VF5CxUAMuGUMrMWIoVRLsGJ//2JOjW8CPJtehow
GRIG1pN6FgbQF79bOywlwNYY1/gUgMxsMPxHIZCkC9uE6CxsemcTNCpk5HOc
hceTexlXkKXoHS385pljsVicbWFOoFSN3IIKABzJKWarh9MQakzPqHOBXVQh
bL8SNoJrvhhuy9sU2Kig1LchlBdeEjgvvgzG4nOr6gkK1RMiVCHiCE5H/mT+t
GFjvmTcaZslgpgli5A1PVz4cV14cWj4=\", \"keyVersion\": 1}
```

10. LI-X1 releases the lock on Mongo DID document.

11. LI-X1 sends CreateDestinationResponse with OkAndAcknowledged to ADMF.

Deactivate Task – CPF

This procedure is used by ADMF to remove the Task (XID) from NF.

1. ADMF sends the DeactivateTask message to NF with the XID
 - XID - uniquely identifies the Task
2. LI-X1 microservice provides the "/X1/NE" end point service for the LI_X1 interface.
3. DeactivateTask message reaches to LI-X1 microservice through Istio Ingress LI gateway.
4. LI-X1 microservice parses the XML message content details in DeactivateTask. LI-X1 generates the obfuscated XID from XID using one-way hash sha256.
5. LI_X1 read and lock XID related Mongo Db document.
6. If LI-X1 is working as a trigger function, it triggers a Deactivate Task towards the UPF for each IRI and/or CC task on LI_T2 and/or LI_T3 interfaces based on earlier provisioning.
7. Receives the DeactivateTask response from triggered POI.
8. LI_X1 deletes the XID related details in ECTD.
9. ETCD notifies the interested microservices about the XID deletion. Typical subscribers for the XID entries are Call control and LI-X2 microservice.
10. LI-X1 deletes the XID document from Mongo Db.
11. LI-X1 sends DeactivateTaskResponse with OkAndAcknowledged to ADMF.

Deactivate Task – UPF

This procedure is used by Trigger Function (ex: SMF) working as ADMF to remove the Task (XID) from NF.

1. ADMF sends the DeactivateTask message to NF with the XID
 - XID - uniquely identifies the Task
2. LI-X1 microservice provides the "/X1/NE" end point service for the LI_T2 and LI_T2 interfaces.
3. DeactivateTask message reaches to LI-X1 microservice through Istio Ingress LI gateway.
4. LI-X1 microservice parses the XML message content details in DeactivateTask. LI-X1 generates the obfuscated XID from XID using one-way hash sha256.
5. LI-X1 read and lock XID related Mongo Db document.
6. LI-X1 deletes the XID related details in ECTD.
7. ETCD notifies the interested microservices about the XID deletion. Typical subscribers for the XID entries are PPE microservice.
8. LI-X1 deletes the XID document from Mongo Db.

LI-X1 sends DeactivateTaskResponse with OkAndAcknowledged to ADMF.

DeactivateAllTask - CPF

This command is used by the ADMF to Deactivate all the Tasks on NE.

1. ADMF sends a DeactivateAllTasks request to NE.
2. DeactivateAllTasks request message reaches to LI-X1 pod through LI Ingress GW.
3. LI-X1 starts a DeactivateAllTasks progress timer with the timer microservice.
4. LI-X1 adds a DeactivateAllTasks progress in etcd table so that other Pod instances of LI-X1 knows this NE level procedure is ongoing in one of the pod instances.
5. If Trigger Function related tasks are provisioned on any of the UPFs, LI-X1 initiates a DeactivateAllTasks message to each relevant UPF. For a given UPF based on earlier provisioning, NE initiates the separate messages for LI_T2 and LI_T3.
6. LI-X1 receives the DeactivateAllTasks response from all the UPFs. Step 8 and 9 are only applicable to NE which also works as the Trigger Function.
7. LI-X1 removes all the XIDs provisioned by ADMF from ETCD tables.
8. LI-X1 also removes all the XIDs related records from mongoDB which are provisioned by ADMF.

9. LI-X1 removes the DeactivateAllTasks progress flag from the ETCD tables.
10. LI-X1 removes the DeactivateAllTasks progress timer from the timer microservice.
11. Once all DIDs are removed, NE sends DeactivateAllTasks Response with OK:AcknowledgedAndCompleted to ADMF through LI Ingress GW.

DeactivateAllTasks - UPF

This command is used by the ADMF to all the Tasks on NE.

1. ADMF sends a DeactivateAllTasks request to NE.
2. DeactivateAllTasks request message reaches to LI-X1 pod through LI Ingress GW.
3. LI-X1 starts a DeactivateAllTasks progress timer with the timer microservice.
4. LI-X1 adds a DeactivateAllTasks progress in etcd table so that other Pod instances of LI-X1 knows this NE level procedure is ongoing in one of the pod instances.
5. LI-X1 removes all the XIDs provisioned by ADMF from ETCD tables.
6. LI-X1 also removes all the XIDs related records from mongoDB which are provisioned by ADMF.
7. LI-X1 removes the DeactivateAllTasks progress flag from the ETCD tables.
8. LI-X1 removes the DeactivateAllTasks progress timer from the timer microservice.
9. Once all DIDs are removed, NE sends DeactivateAllTasks Response with OK:AcknowledgedAndCompleted to ADMF through LI Ingress GW.

Destinations – Create, Modify and Remove

Create Destination

Create Destination is used by the ADMF to add a new Destination to an NE. On successful processing, LI-X1 publishes the destination related information in ETCD. Consumer microservice interested in destinations subscribe to ETCD destinations table.

Modify Destination

Used by the ADMF to modify an existing Destination on the NE. All details for the Destination is given (i.e. the modified details and the information that is unchanged) to totally replace the previous Destination details.

LI-X1 updates the destination related information in ETCD. Consumer microservices (LI-X2 or PPE) handles creating connection to new DID Ip and close the connection to the old one.

Remove Destination

Remove Destination is used by the ADMF to remove a Destination from an NE. On successful processing, LI-X1 removes the destination entry from ETCD and also notifies the consumers about destination deletion.

Remove All Destinations

Remove All Destinations completely and permanently removes all Destinations on an NE. On successful processing, LI-X1 removes all destinations present in the NE for that ADMF.

DID Configuration Limitations

On control plane federations (SMF, AMF, etc), the Istio egress gateway is used for communication with the MDF2 server for DID configuration. Following are the limitations:

Due to a limitation in Istio, multiple X2 Destinations (DID) using the same IP address (the only difference is the port number) are unsupported.

- Example of unsupported configuration of multiple MDF2 Destinations:

- DID1: X2Only, IP address = 10.1.1.1 port=1001

- DID2: X2Only, IP address = 10.1.1.1 port=2002

- Example of supported configuration of multiple MDF2 Destinations:

- DID1: X2Only, IP address = 10.1.1.1 port=1001

- DID2: X2Only, IP address = 10.2.2.2 port=2002

For DID, the Primary and secondary address with the same IP and different port is unsupported due to Istio egress gateway limitations for TLS.

DSCP Marking

The LI-X1 microservice marks the configured DSCP on ingress/egress LI HTTP/HTTPS interfaces.

Supported interfaces are LI_X1, LI_T2, and LI_T3. DSCP can be configured per interfaces type. Different interface types are LIPF, LIPF-MGMT, T2, and T3.

When the peer is using the same IP for the T2 and T3 interfaces, lowest configured DSCP value among the T2/T3 interface is used for both the interfaces.

Sample Configuration:

```
li-x1 dscp-marking lipf df
```

```
li-x1 dscp-marking t2 df
li-x1 dscp-marking t3 df
li-x1 dscp-marking x2 df
li-x1 dscp-marking lipf-mgmt df
```

Sample Show Command:

```
show running-config li-x1 dscp-marking

li-x1 dscp-marking lipf df li-x1 dscp-marking t2 df li-x1 dscp-marking t3
df li-x1 dscp-marking x2 df

li-x1 dscp-marking lipf-mgmt df
```

ELF Core Files Generation

When a process inside a container fails in CNA environment, it results in the generation of mini- dump back-trace files that are useful in identifying the back-trace process at the time of failure. However, it does not allow to debug the possible root cause of the failure. ELF core files equip the user with a snapshot of the CPU and memory state of the process at the time of the failure.

ELF core files generation is enabled for LI-X1 pods and core files are written to the Host VM file system (the K8s worker node). This addresses the issue in generating ELF core files along with mini-dump files in CNA environment. The management of these core files generated on the K8s worker nodes improves the ability to debug the application failure.

Enabling/Disabling Task Sensitive Logs

Logs contain sensitive information (such as Task level info example: XID, Target info, Correlation Id etc) are categorized separately to ensure it is not logged into regular CNA observability by default.

To generate the sensitive logs using li-admin level, refer LI Logging section in LI-X1.

Enabling or Disabling TLS Support

By default, when NF interacts with the L1_X1/LI_T2/LI_T3 interface peers, HTTP/TLS is used. To enable/disable the TLS for the LI egress HTTP interfaces, follow the below configuration.

Sample Configuration:

```
li-x1 http-over-tls-egress enabled li-x1 x2-x3-over-tls-egress enabled
```

Sample Show Command:

```
show running-config li-x1 http-over-tls-egress

li-x1 http-over-tls-egress enabled
```

Encryption

LI-X1 supports encryption of etcd and Mongo DB data. Encryption is enabled at fed-level, that is all PODs contain the same value of encryption key and enable/disable flag. When encryption is enabled, etcd data for Task (xid_table) and Destination (did_table/did_lipf_table) tables and Mongo DB data for Task and Destination subscriber are stored in an encrypted format and its not human-readable format. The etcd and Mongo DB both also stores a version of the key at the time of encrypting and decrypting is the same.

Note: When the encryption key is retrieved from an external vault(such as Azure Key vault), rotation of the key at runtime is not supported.

It is recommended to clean up all LI-X1/LI-X2 etcd or Mongo DB data, change the encryption key, or enable/disable encryption, and then delete LI-X1/LI-X2 and application PODs (all application pod which is using LI data).

External Interfaces

LI_X1

ADMF provisions the tasks and destinations over the LI_X1 interface. LI-X1 microservice works as the both HTTP client and server for the LI-X1 interface.

For messages where the ADMF is the requester LI-X1 works as the HTTP server. For messages where NE is the requester, LI-X1 will work as the client.

Transport layer for the LI-X1 interface is HTTPS. For the HTTPS transport in the federation, there is a dedicated LI istio ingress gateway deployment. LI istio ingress exposes the IP/FQDN for the LI_X1 interface.

LI-X1 microservice exposes the "/X1/NE" uri for the ingress messaged from the ADMF. Security related certificates are provisioned at the istio LI ingress gateway.

LI_X1 interface uses XML over HTTPS interface for the protocol messages . XML stack and protocol messages encoding/decoding is done by the LI-X1 microservice.

LI-X1 microservice supports the HTTP/2 profile. This interface is applicable for all the control plane federations.

LI-X1 microservice implements the LI_X1 interface functionality compliant to ETSI TS 103 221- 1 V1.6.1 (2019-12).

LI_T2 and LI_T3

LI_T2 and LI_T3 interfaces are for the communication between Trigger Function and Triggered POI. For example, LI at SMF/UPF, SMF works as the trigger function (i.e nothing but ADMF) and UPF works as the triggered POI (nothing but NE).

Same common LI-X1 common microservice works as the Trigger Function or Triggered POI based on federation type in which it is included.

Transport layer for the LI_T2/LI_T3 interface is HTTPS. For the HTTPS ingress transport in the federation, there is a dedicated LI istio ingress gateway deployment. LI istio ingress exposes the IP/FQDN for the LI_T2/LI_T3 interfaces.

Similarly, for HTTPS egress transport in the federation we have dedicated LI istio egress gateway.

LI_T2/LI_T3 interfaces uses XML over HTTPS for the protocol messages . XML stack and protocol messages encoding/decoding is done by the LI-X1 microservice.

LI-X1 microservice supports the HTTP/2 profile for the LI_T3 and LI_T2 interfaces.

LI_T2 interface is used for the provisioning of the IRI (Interception related information) related functionality where as LI_T3 is used for the provisioning of content of communication (CC) related functionality.

LI-X1 microservice implements the LI_T2/LI_T3 interface functionality compliant to ETSI TS 103 221- 1 V1.6.1 (2019-12)

LI_X1 (Management)

LI-X1 management interface is supported on UPF and this interface is not applicable to control plane federations. Following messages are supported on management interface.

- GetTaskDetails
- GetDestinationDetails
- ListAllDetails
- GetAllDetails
- GetNeStatus

GetAllDetails - CPF

This procedure is used by the ADMF to retrieve the details of all XIDs, DIDs, and NE.

1. ADMF initiates a GetAllDetailsRequest message to the NE.
2. Message reaches to LI Ingress GW on control plane federation.
3. LI IngressGW forwards the messages to LI-X1 pod instance.
4. For each UPF in all did_lipf_peed etcd table, LI-X1 sends GetAllDetailsRequest message to LI Egress GW and waits for response from UPF.
5. LI Egress GW sends message to UPF.
6. LI-X1 receives GetAllDetailsResponse from UPF.
7. LI-X1 reads all tasks details from MongoDB using multi-read DB event.
8. LI-X1 reads all Destination details from Mongo DB using multi-read DB event.
9. LI-X1 reads NE status from DB.
10. LI-X1 populates the details from inmemory XID, DID, and details received in step 6-9.
11. LI-X1 sends the populated GetAllDetailsResponse to LIPF through LI Ingress gateway.

GetAllDetails - UPF (Management Interface)

This procedure is used by the ADMF to retrieve the details of all XIDs, DIDs, and NE over management interface.

1. ADMF initiates a GetAllDetailsRequest message to the NE.
2. Message reaches to LI Ingress GW on control plane federation.
3. LI IngressGW forwards the messages to LI-X1 pod instance.
4. LI-X1 reads all tasks details from Mongo DB using multi-read DB event (matchAll to read complete TaskCollection).
5. LI-X1 reads all Destination details from Mongo DB using multi-read DB event (matchAll to read complete DestinationCollection).
6. LI-X1 reads NE status from DB.
7. LI-X1 populates the details from in-memory XID, DID, and details received in step 4-6.
8. LI-X1 sends the populated GetAllDetailsResponse to LIPF through LI Ingress gateway.

GetAllDetails - UPF

This procedure is used by the ADMF to retrieve the details of all XIDs, DIDs, and NE.

1. ADMF initiates a GetAllDetailsRequest message to the NE.
2. Message reaches to LI Ingress GW on control plane federation.
3. LI IngressGW forwards the messages to LI-X1 pod instance.
4. LI-X1 reads all tasks details from Mongo DB using multi-read DB event (secondary key as Obfuscated Admf ID from which GetAllDetailsRequest is received).
5. LI-X1 reads all Destination details from Mongo DB using multi-read DB event (secondary key as Obfuscated Admf ID from which GetAllDetailsRequest is received).
6. LI-X1 reads NE status from DB.
7. LI-X1 populates the details from inmemory XID, DID, and details received in step 4-6.
8. LI-X1 sends the populated GetAllDetailsResponse to LIPF through LI Ingress gateway.

GetDestinationDetails - CPF

This procedure is used by ADMF to retrieve the details of Destination (DID) on NF.

1. ADMF sends the GetDestinationDetails message to NF with the DID.
 - DID - Destination identifier which uniquely identifies the destination.
2. LI-X1 microservice provides the "/X1/NE" end point service for the LI_X1 interface.
3. GetDestination Details message reaches to LI-X1 microservice through Istio Ingress LI gateway.
4. LI-X1 microservice parses the XML message content details in GetDestination. LI_X1 generates the obfuscated DID from DID using one-way hash sha256.
5. LI-X1 read and lock DID related Mongo Db document.
6. LI-X1 sends GetDestinationRequest to all UPFs where DID is provisioned.
7. LI-X1 aggregates GetDestinationResponse from UPFs.
8. LI-X1 fills the DestinationDetails and DestinationStatus using Mongo DID record, UPF Responses, and inmemory ETCD DID table details.
9. LI-X1 releases the lock Mongo Db document.
10. LI-X1 sends GetDestination Details Response with Destination Status and Destination Details to ADMF.

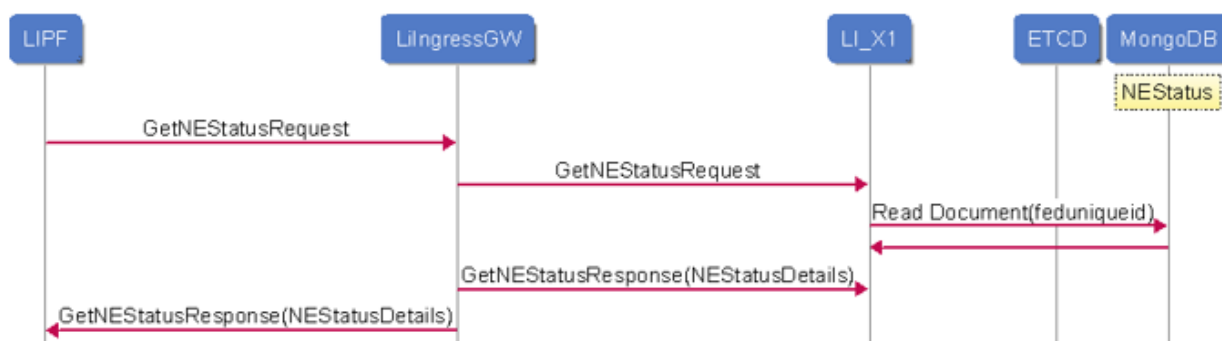
GetDestinationDetails - UPF

This procedure is used by Trigger Function (ex: SMF) working as ADMF to retrieve the details of particular Destination(DID) on NF.

1. ADMF sends the GetDestinationDetails message to NF with the DID.
 - DID - Destination identifier which uniquely identifies the destination.
2. LI-X1 microservice provides the "/X1/NE" end point service for the LI_T2 and LI_T3 interfaces.
3. GetDestination Details message reaches to LI-X1 microservice through Istio Ingress LI gateway.
4. LI-X1 microservice parses the XML message content details in GetDestination. LI-X1 generates the obfuscated DID from DID using one-way hash sha256.
5. LI-X1 read and lock DID related to Mongo Db document.
6. LI-X1 fills the DestinationDetails and DestinationStatus using Mongo DID record and inmemory ETCD DID table details.
7. LI-X1 releases the lock Mongo Db document.
8. LI-X1 sends GetDestination Details Response with Destination Status and Destination Details to ADMF.

GetNEStatus - CPF

Figure 1. GetNEStatus Call Flow



ADMF uses this procedure is to determine the status of NE.

1. ADMF initiates a GetNEStatus Request message to the NE.
2. Message reaches to LI Ingress GW on control plane federation.
3. LI IngressGw forwards the messages to LI-X1 pod instance.

4. If the Control plane federation is SMF, it will also fetch the GetNEStatus from configured peer UPFs.
5. After all the UPF responses, it includes the fetched NEStatus details in NEStatus Response.
6. LI-X1 reads the NE Status documents from the MongoDB.
7. LI-X1 sends the populated GetNEStatus Response to LIPF through LI Ingress gateway.

GetNEStatus - UPF

This procedure is used by the ADMF to determine the status of NE.

1. ADMF initiates a GetNEStatus Request message to the NE.
2. Message reaches to LI Ingress GW on user plane federation.
3. LI IngressGw forwards the messages to LI-X1 pod instance.
4. LI-X1 reads the NE Status documents from the MongoDB.
5. LI-X1 sends the populated GetNEStatus Response to LIPF through LI Ingress gateway.

GetTaskDetails - CPF

This procedure is used by ADMF to retrieve the details of Task (XID) on NF.

1. ADMF sends the GetTaskDetails message to NF with the XID
 - XID - Uniquely identifies the Task
2. LI-X1 microservice provides the "/X1/NE" end point service for the LI_X1 interface.
3. GetTaskDetails message reaches to LI-X1 microservice through Istio Ingress LI gateway.
4. LI-X1 microservice parses the XML message content details in GetTaskDetails. LI-X1 generates the obfuscated XID from XID using one-way hash sha256.
5. LI-X1 read and lock XID related Mongo Db document.
6. LI-X1 sends GetTaskDetailsRequest to all UPFs where child xid is provisioned for the xid.
7. LI-X1 aggregates GetTaskDetailsResponse from UPFs.
8. LI-X1 fills the TaskDetails and TaskStatus using Mongo DID record, UPF Responses, and inmemory ETCD DID table details.
9. LI-X1 releases the lock Mongo Db document.
10. LI-X1 sends GetTask Details Response with Task Status and Task Details to ADMF.

GetTaskDetails - UPF

This procedure is used by Trigger Function (ex: SMF) working as ADMF to retrieve the details of Task (XID) on NF.

1. ADMF sends the GetTaskDetails message to NF with the XID
 - XID - Uniquely identifies the Task
2. LI-X1 microservice provides the "/X1/NE" end point service for the LI_T2 and LI_T3 interface.
3. GetTaskDetails message reaches to LI-X1 microservice through Istio Ingress LI gateway.
4. LI-X1 microservice parses the XML message content details in GetTaskDetails. LI-X1 generates the obfuscated XID from XID using one-way hash sha256.
5. LI-X1 read and lock XID related Mongo Db document.
6. LI-X1 fills the TaskDetails and TaskStatus using Mongo DID record and inmemory ETCD DID table details.
7. LI-X1 releases the lock Mongo Db document.
8. LI-X1 sends GetTask Details Response with Task Status and Task Details to ADMF.

Get Information from NE

ADMF uses the following set of messages to request status information from the NE. This is distinct from Reporting Issues where the NE pushes information to the ADMF.

Message Details

GetTaskDetails

ADMF uses GetTaskDetails to retrieve the details of a Task. LI-X1 processes the message and responds with the task details.

GetDestinationDetails

ADMF uses GetDestinationDetails to retrieve the details of a Destination. LI-X1 processes the message and responds with the destination details.

GetNEStatus

ADMF uses GetNEStatus to determine the status of the NE.

GetAllDetails

ADMF uses GetAllDetails to determine the details of all Tasks, Destinations and the status of the NE. LI-X1 processes the message and collects the data from ETCD and DB responds with the all XIDs, DIDs, and NE status present in the NE.

ListAllDetails

ADMF uses ListAllDetails to retrieve the list of all XIDs, and DIDs (i.e. a list of identifiers) but not the details of them. LI-X1 processes the message and responds with the all XIDs, and DIDs present in the NE.

ADMF/LIPF Request for Info

TIME1

NFs must respond to LIPF requests within TIME1 seconds (default value 5 seconds). See ETSI TS 103.221-1

For GetAllDetails, the SMF and UPF must complete the transaction within TIME1 with OK – Acknowledged and Completed.

GetAllDetails/GetTaskDetails/GetDestinationDetails – Trigger Function (SMF)

When SMF works as a Trigger Function, upon Get Info Request (GetAllDetails / GetTaskDetails / GetDestinationDetails) from LIPF, SMF will always fetch the details from UPF. SMF will not pull the details upfront from the UPF and cache it. As SMF is pulling the details from the UPF on Get Request, if large number of UPF instances are deployed, it may not always be possible to complete the transaction in TIME1 default (5 secs) value.

Note: Time required to complete the transaction will most likely vary based on the number of UPF instances deployed.

If SMF is not able to receive the response from UPF (or on negative response) within the TIME1 value, SMF skips that particular UPF related details in GetAllDetails response.

HTTP Retry on Failure on ETCD/TF/CC Interface

ETCD Interface

Towards the etcd server transactions, LI-X1 retries the message for the HTTP status response codes 500, 502, 503, 504, 429, and 460.

LI-X1 retries three times for the negative response codes. If the message is not successful, the LI- X1 transaction fails.

Call-control (CC) Interface

Towards the call-control transactions, LI-X1 retries the message for the HTTP status response codes 500, 503, and 429.

LI-X1 retries three times (including the first message) for the negative response codes.

Trigger-Function (TF) Interface

Towards the LI-T2/LI-T3 interface transactions, LI-X1 retries the message for the HTTP status response codes 429, 460, 500, 502, 503, and 504.

LI-X1 retries three times (including the first message) for the negative response codes. If the message is not successful, the LI-X1 transaction fails.

HTTPS (TLS 1.3) Support for X1 Interface

Affirmed Networks NF supports HTTPS according to IETF RFC 2818. NF supports both as a client and a server.

NF supports TLS 1.3 version and is even compatible with version 1.2.

Internal Interfaces to other Microservice

Trigger Function

LI-X1 exposes the REST API towards the application call control for the Trigger function (TF) functionality. SMF federation uses this functionality.

LI-X1 exports the REST API swagger file named LixTriggerFunction.yaml in cna-li_x1/interface/swagger directory. Call control application which requires this functionality can import the swagger file for the REST API. Trigger functionality is applicable to SMF.

Fan-out Functionality

LI-X1 also supports the fan-out functionality when it works as a trigger function. For example, multiple tasks (XIDs) from ADMF are targeting the same session at the call control. When PDU session is established at the call control, do not send separate CC/IRI TF activate task messages for each XID. Application can send single activate task request with the all XIDs interested in particular target. Call control is supposed to provide target identifiers to be sent to triggered POI (UPF) and peer NRF registered FQDN.

LI-X1 will look in to the configuration using N4 Address provided by the call control for finding the peer related Neldentifier and T2/T3 host FQDN or IP address and port.

Notification - Task Add/Modify/Deactivate

After adding the successful task to the ETCD table which is provisioned by the ADMF, LI-X1 notifies the call control/PPE about the new target addition/deactivation. In notify request, the same data which is added to ETCD is included.

This notification is to help the call control to resolve the contention for the action on newly added target, for example, when interacting with DB. Otherwise, on ETCD notification every pod instance of the call control may attempt to take action on the task which is not desirable.

After successful task modification and update to the ETCD, LI-X1 notifies the call control/PPE about target modification. In notify request, the same data which is added to ETCD is included.

On deactivate task/Deactivate All Task before removing entry from ETCD, LI-X1 notifies applications/consumers about the task deactivation.

This notification functionality can be subscribed by microservices which are interested in task table addition/update/deletion.

For applications which are subscribing to notifications, the LI-X1 notification and actual ETCD server notifications are asynchronous and they can come in any order.

Subscribing to Notification

Unless application is interested in notification, LI-X1 will not send the notification message. For subscribing to the notifications, the applications must publish the notification interest "key":value" pair in consul.

To construct, the key-value pair call control applications can import and use the

LixServiceRegApi.yaml exported by LI-X1 in interface directory.

Report Statistics

LI-X1 is responsible for storing the statistics of the Task in mongo DB on behalf of federation.

LI-X1 exposes the following URI for the reporting of the statistics for the given Obfuscated XID and schema is defined in cna-li_x1/interface/swagger/LixReportStats.yaml file. Applications can import LixReportStats.yaml and use it for reporting the statistics. Application can choose to send single request for bunch of obfuscated XIDs. It supports up to 500 obfuscated in one go.

LI-X2 uses API to report the amount of X2 data bytes. PPE uses to report the amount of X2 and X3 Data bytes.

Report Task/Destination/NE Issues

LI-X1 is responsible for storing the faults of the Task/Destination/NE in mongo DB on behalf of federation and report faults to LIPF.

LI-X1 exposes the following URI for the reporting of the faults of Task/Destination/NE and schema is defined in `cna-li_x1/interface/swagger/LixReportIssues.yaml` file. Applications can import `LixReportIssues.yaml` and use it for reporting the faults.

IRI or/and CCTF Activate Task Procedure - CPF

This procedure is used by NF working as Trigger Function (ex: SMF) to program the Tasks on Triggered Function (UPF).

1. It is assumed that Create Destination and Activate Task are already completed. Otherwise, CC/IRI Task activation is not relevant.
2. In the step 1, Call control microservice already learnt the Task details through ETCD notification.
3. It is also assumed that UPF related FQDN Address and IP Port details are configured at the LI_X1 microservice.
4. When PDU session establishments starts at the call control, it will check if the Lawful interception is activated for the Session using target identifiers.
5. If the Lawful interception is activated for the session, call control initiates the TF activatetask message towards the LI_X1 microservices via REST interface. Message contains details about the Obfuscated XID, Correlation Id, T2 and T3 target identifiers and UPF N4 FQDN. Call control microservice can also send list of Obfuscated XIDs if multiple XIDs are interested in interception on same PDU session. This Obfuscated XID is the XID received from the LIPF in step 1.
6. LI-X1 microservice read and lock the XID related to Mongo Db document.
7. LI-X1 microservice on receiving the message it will find out whether it should activate the CC TF Task or IRI TF Task or both based on delivery type of the Task and target identifiers received from the call control.
8. LI-X1 generates the XID for the IRI and/or CC TF tasks.
9. LI-X1 will find the details about UPF FQDN/IP Address from the configuration using N4 FQDN look up.
10. LI-X1 microservice configures the Destination rules (whitelisting) for the UPF destination IP/FQDN at the LI egress gateway.
11. LI-X1 uses the matching DID delivery type related DIDs provisioned in the step 1 to program the UPF.
12. If the current IRI and/or CC task is the first one for the DID towards UPF, before initiating the activate task LI-X1 sends the CreateDestination request message to UPF for each DID supposed to be used by the CC or/and IRI Task.
13. On CreateDestination response from the UPF, LI-X1 updates the LIPF→DID mapping in ETCD for the downstream functionality to identify DID is programmed on which and all UPFs.

14. After successful CreateDestinationResponse, LI-X1 initiates CC and/or IRI task activation.
15. For the IRI tasks LI-X1 sends the ActivateTask message on LI_T2 interface to UPF.
16. For the CC TF tasks LI-X1 sends the ActivateTask message on LI_T3 interface to UPF.
17. After successful IRI and/or CC TF tasks activation LI-X1 updates the ETCD with these IRI and/or CC task details. Sample updated ETCD entry looks like below:

```
/smf.-10-163-64-206-07-21-07-56/1/li_x1/v1/xid_
table/8bb94fff066c8eaf4f4da81eee226befafa6a41c8c250fd46f68f2
7414b8991/82ac7b62e25295a8ddd2e978cb3366dbd5fee91838fc366c7a b1c78a7a89d

{"deliveryType":"X2andX3","listOfCcTFTasks":
[{"correlationID":"101","n4Address":"n4.testserver.com","tar
getIdentifiers":{"n1_targetIdentifier":[{"n1_
targetIdentifierExtension":{"n1_ UPFLIT3TargetIdentifierExtensions":{"ns1_
UPFLIT3TargetIdentifier":[{"ns1_FSEID":{"ns1_
IPv4Address":"10.25.21.234","ns1_
IPv6Address":"fe80::20c:29ff:fee5:f188","ns1_
SEID":"123456"},"uPFLIT3TargetIdentifierType":"FSEID"}]}},{"t
argetIdentifierType":"targetIdentifierExtension"}]},"xid":"7 27487c1-c161-
4295-9810-79046c86e0da"}],"listOfDIDs":{"n1_dId":["21ce308c-de30-42bd-
a27b-287bcb5b3469","21ce308c-de31- 42bd-a27b-287bcb5b3469","21ce308c-de32-
42bd-a27b- 287bcb5b3469"]},"listOfIRITasks":
[{"correlationID":"101","n4Address":"n4.testserver.com","tar
getIdentifiers":{"n1_targetIdentifier":[{"n1_
targetIdentifierExtension":{"n1_ UPFLIT3TargetIdentifierExtensions":{"ns1_
UPFLIT3TargetIdentifier":[{"ns1_FSEID":{"ns1_
IPv4Address":"10.25.21.234","ns1_
IPv6Address":"fe80::20c:29ff:fee5:f188","ns1_
SEID":"123456"},"uPFLIT3TargetIdentifierType":"FSEID"}]}},{"t
argetIdentifierType":"targetIdentifierExtension"}]},"xid":"3 a809c30-4116-
4821-8ea3-af964ef00fcd"}],"targetIdentifiers":

{"n1_targetIdentifier":[{"n1_
gpsiMsisdn":"919000000030","targetIdentifierType":"gpsiMsisd n"},"{"n1_
supiimsi":"225097743010830","targetIdentifierType":"supiims i"},"{"n1_
peiImei":"987654000000030","targetIdentifierType":"peiIme i"},"{"n1_
gpsiNai":"929000000030","targetIdentifierType":"gpsiNai"},

{"n1_ supinai":"235097743010830","targetIdentifierType":"supina i"},"{"n1_
peiImeiCheckDigit":"937654000000030","targetIdentifierType":
"peiImeiCheckDigit"},"{"n1_
peiImeiSv":"947654000000030","targetIdentifierType":"peiImei
sv"}]},"version":"1.0.0","xid":"29f28e1c-de30-486a-a860- f5a784ab9173"}
```

18. LI-X1 releases the lock on Mongo XID document.

19. LI-X1 sends CC/IRI TF activate task response to call control microservice.

IRI or/and CCTF Deactivate Task Procedure - CPF

This procedure is used by NF working as Trigger Function (ex: SMF) to deactivate the Tasks on Triggered Function (UPF)

1. It is assumed that Create Destination and Activate Task are already completed.
2. In the step 1 Call control microservice already learnt the Task details through ETCD notification.
3. It is also assumed that UPF related FQDN Address and IP Port details are configured at the LI-X1 microservice.
4. It is also assumed that CC and/or IRI task are already activated towards the UPF.
5. When PDU session establishments release happens at the call control, call control application will check if the Lawful interception is activated for the Session.
6. If the Lawful interception is activated for the session, call control initiates the TF Deactivate task message towards the LI-X1 microservices via REST interface. Message contains details about the Obfuscated XIDs which are to be deactivated. Call control microservice can send list of Obfuscated XIDs also if multiple XIDs are interested in interception on same PDU session. This Obfuscated XID is the XID received from the LIPF in step 1.
7. LI-X1 microservice read and lock the XID related Mongo Db document.
8. LI-X1 will find the details about UPF FQDN/IP Address from the configuration using the N4 FQDN stored in the IRI/CC TF task ETCD entry.
9. LI-X1 microservice configures the Destination rules (whitelisting) for the UPF destination at the LI egress gateway if not done so.
10. For the IRI tasks LI_X1 will send the Deactivate Task message on LI_T2 interface to UPF.
11. For the CC TF tasks LI_X1 will send the Deactivate Task message on LI_T3 interface to UPF.
12. After successful IRI and/or CC TF tasks deactivation LI-X1 updates the ETCD with these IRI and/or CC TF task removal details. Sample updated ETCD entry after deactivation looks like below:

```
/smf.-10-163-64-206-07-21-07-56/1/li_x1/v1/xid_
table/8bb94fff066c8eaf4f4da81eee226befafa6a41c8c250fd46f68f2
7414b8991/82ac7b62e25295a8ddd2e978cb3366dbd5fee91838fc366c7a b1c78a7a89d

{"deliveryType":"X2andX3","targetIdentifiers":{"n1_
targetIdentifier":[{"n1_
gpsiMsisdn":"919000000030","targetIdentifierType":"gpsiMsisd n"}, {"n1_
supiimsi":"225097743010830","targetIdentifierType":"supiims i"}, {"n1_
peiImei":"987654000000030","targetIdentifierType":"peiIme i"}, {"n1_
gpsiNai":"9290000000030","targetIdentifierType":"gpsiNai"},
```

```
{ "n1_supinai": "235097743010830", "targetIdentifierType": "supina i"}, {"n1_
peiImeiCheckDigit": "937654000000030", "targetIdentifierType":
"peiImeiCheckDigit"}, {"n1_
peiImeiSv": "947654000000030", "targetIdentifierType": "peiImei
sv"} ] }, {"version": "1.0.0", "xid": "29f28e1c-de30-486a-a860- f5a784ab9173" }
```

13. LI-X1 releases the lock on Mongo XID document.

14. LI-X1 sends CC/IRI TF Deactivate task response to call control microservice.

Keep Alive from LIPF - CPF or UPF

This message goes from ADMF to NF. It is intended as a means for the NE application to assert that the ADMF application is still operational, and remove all tasking information as a security measure if it is not. Removal of task on if ADMF is not operation is not supported. Partial functionality Responding to KeepAlive with OK is supported.

1. ADMF sends the KeepAlive message to NF
2. LI-X1 microservice provides the "/X1/NE" end point service for the LI_X1 or LI_T2 or LI_T3 interface.
3. KeepAlive message reaches to LI-X1 microservice through Istio Ingress LI gateway.
4. LI-X1 microservice parses the XML message and identifies the message.
5. LI-X1 sends Keep Alive Response with OkAndAcknowledged to ADMF.

KeepAlive-P2

Timer Update

This procedure is used by all the federation to update time stamp and start global timer.

1. ADMF initiates any message to the NE.
2. Message reaches to LI Ingress GW on the federation.
3. LI IngressGw forwards the messages to LI-X1 pod instance.
4. LI-X1 starts one-minute local timer.
5. On expiry of timer, LI-X1 automatically sets to update KeepAlive Subscriber record with time stamp in Mongo DB.
6. If Mongo DB subscriber does not exist, then LI-X1 create KeepAlive Subscriber record in Mongo DB with current time stamp.

7. If LI-X1 in-memory subscriber does not exist, then LI-X1 sends KeepAlive timer start request with configured time-P2 value.

Timer Expiry

This procedure is used by the all federation to handle timer-p2 expiry.

1. Timer microservice sends timer expiry notification to LI-X1 microservice.
2. LI-X1 performs multiRead of keepAlive subscriber to get last message received time stamp.
3. LI-X1 compares current time stamp with last message time stamp and if no message is received in time-P2 duration then LI-X1 starts DeactiveAllTask procedure.
4. If any message is received from LIPF in time-P2 duration, then LI-X1 sends update global timer request to timer microservice with remaining duration of time-P2.

KeepAlives

LI-X1 supports keepalive over X1 interface. It also supports Time-P1 for egress KeepAlive over T2/T3 interface (only in fed-smf) and Time-P2 to deactivate all tasks when the message is not received over X1 interface.

KeepAlive (Time-P1)

If LI-X1 is part of Fed-SMF, then it sends KeepAlive to all the configured UPF after every Time- P1 interval. Time-P1 can be defined at Local configuration.

At the peer-level egress, KeepAlive can be enabled/disabled per peer. If the values are configured at both places, then the value in peer-interface takes precedence. KeepAlive is sent based on the duration configured in Local configuration.

If KeepAlive is disabled (time-p1=0) in local configuration and enabled in peer-profile, then KeepAlive will be sent every 60 seconds. If config is modified, then the new timer value is applicable after current timer expires.

KeepAlive (Time-P2)

Time-P2 is supported on fed-smf, fed-amf, and fed-upf. LI-X1 deactivates all tasks if no message is received over LI-X1 interface in Time-P2 duration. Time-P2 can be defined at two places:

- Local configuration
- Peer profile associated with peer-interface

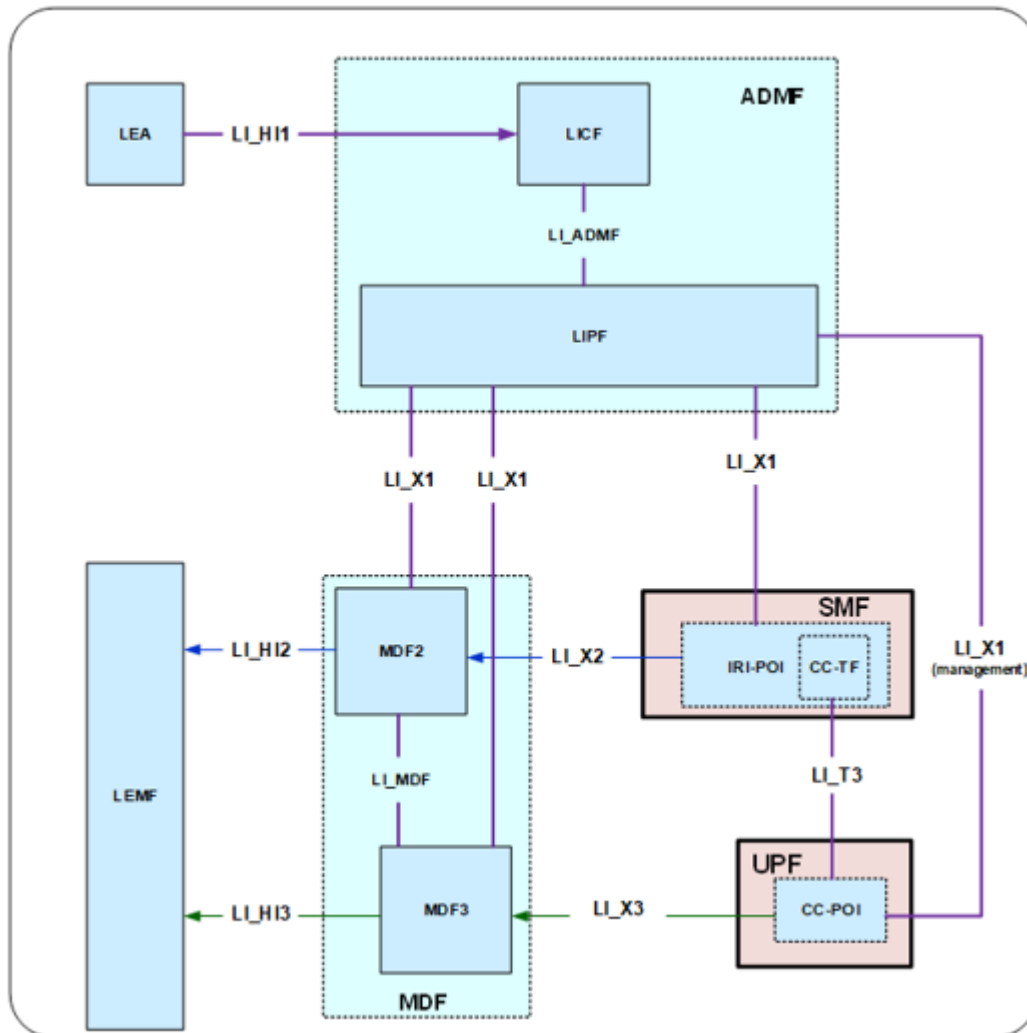
If the values are configured at both places, then the value in peer-interface takes precedence. If Timer-P2 is configured in peer-profile associated with peer-interface, then peer-profile value is used but If Time-P2 is not configured in peer-profile associated with peer-interface, then local- profile value is used. If config is modified, then the new timer value is applicable after current timer expires.

LI Architecture with SMF or UPF

For example, when LI-X1 is included as part of SMF federation it supports the LI_T3 and LI_T2 TF functionality in addition to LI_X1 interface.

Similarly, when LI-X1 is included as part of UPF federation it works as Triggered POI and supports LI_T2 and LI_T3 interface functionality.

Figure 2. LI Architecture



LI TF Audit and Reprogram

The feature:

- Periodically audits and synchronizes the difference between SMF (TF/Trigger Function) and UPF (triggered POI).

- Auto-Synchronizes out of sync Tasks (XIDs) between SMF and UPF.
- Auto-Synchronizes out of sync Destinations (DIDs) between SMF and UPF.

Configuration

Execute the following command to enable/disable TF audit re-program:

```
li-x1 local-profile tf-audit-reprogram <Enabled/Disabled, default = disabled>
```

Execute the following command to configure the audit and re-program in minutes:

```
li-x1 local-profile tf-audit-reprogram-interval <>
```

Here,

- Default value is 15 minutes.
- Min value 15 minutes.
- Max value 1440 minutes (24 Hours).

Execute the following command to re-program rate per second and per pod:

```
li-x1 local-profile tf-audit-reprogram-rate < >
```

Here,

- Default value is 20.
- Min value is 5.
- Max value is 150.

Troubleshooting

Following commands are used for debugging:

- li_x1_tf_list_all_details_request_attempts
- li_x1_tf_list_all_details_request_success
- li_x1_tf_error_response_list_all_details_request

A tag is added for existing LIX1 debug stats to keep track of the number of times reprogram is succussed and failed:

- li_x1_tf_activate_task_request_attempts
- li_x1_tf_activate_task_request_success

- li_x1_tf_error_response_activate_task_request

Figure 3. Troubleshooting Example

```
li_x1_tf_activate_task_request_success
{
  admfid="ga-smf-li", appName="TF_T2T3_interface", canaryVersion="4.1.0-35-main-int", container="li-
x1", dtype="X3Only", fed_unique_name="noble-thattil-10-163-66-211-03-06-07-53", job="li-x1", namespace="fed-
smf", neid="neID", pod="li-x1-4-1-0-35-main-int-ff85d44cf-hlt7q", status="AcknowledgedAndCompleted", tfreprogram="true"
}
```

- li_x1_tf_modify_task_request_attempts
- li_x1_tf_modify_task_request_success
- li_x1_tf_error_response_modify_task_request
- li_x1_tf_deactivate_task_request_attempts
- li_x1_tf_deactivate_task_request_success
- li_x1_tf_error_response_deactivate_task_request
- li_x1_tf_create_destination_request_attempts
- li_x1_tf_create_destination_request_success
- li_x1_tf_error_response_create_destination_request
- li_x1_tf_modify_destination_request_attempts
- li_x1_tf_modify_destination_request_success
- li_x1_tf_error_response_modify_destination_request
- li_x1_tf_remove_destination_request_attempts
- li_x1_tf_remove_destination_request_success
- li_x1_tf_error_response_remove_destination_request

ListAllDetails - CPF

This procedure is used by the ADMF to retrieve the list of all XIDs and DIDs (i.e a list identifiers) but no details.

1. ADMF initiates a ListAllDetailsRequest message to the NE.
2. Message reaches to LI Ingress GW on control plane federation.
3. LI IngressGw forwards the messages to LI-X1 pod instance.
4. LI-X1 fills the list of XIDs and list of DIDs from inmemory tables.

5. LI-X1 sends the populated ListAllDetailsResponse to LIPF through LI Ingress gateway.

ListAllDetails - UPF (Management Interface)

This procedure is used by the ADMF to retrieve the list of all XIDs and DIDs (i.e a list identifiers) but no details.

1. ADMF initiates a ListAllDetailsRequest message to the NE.
2. Message reaches to LI Ingress GW on user plane federation.
3. LI IngressGw forwards the messages to LI-X1 pod instance.
4. LI-X1 fills the list of all XIDs and list of all DIDs from inmemory tables. ADMF ID in ListAllDetailsRequest.
5. LI-X1 sends the populated ListAllDetailsResponse to LIPF through LI Ingress gateway.

ListAllDetails - UPF

This procedure is used by the ADMF to retrieve the list of all XIDs and DIDs provisioned by one SMF (i.e a list identifiers) but no details.

1. ADMF initiates a ListAllDetailsRequest message to the NE.
2. Message reaches to LI Ingress GW on user plane federation.
3. LI IngressGw forwards the messages to LI-X1 pod instance.
4. LI-X1 fills the list of XIDs and list of DIDs from in memory tables with ADMF ID matches with ADMF id in ListAllDetailsRequest.
5. LI-X1 sends the populated ListAllDetailsResponse to LIPF through LI Ingress gateway.

Management Interface Support

This topic describes the management interface support for GetAllDetails/ListAllDetails/GetNEStatus/GetDestinationDetails/GetTaskDetails.

Management interface is supported when LI-X1 is part of fed-upf. LI-X1 provides information of task and destination provisioned by all SMF, if GetAllDetails/ListAllDetails/GetNEStatus/GetDestinationDetails/GetTaskDetails are received on management interface from LIPF/ADMF.

LI-X1 accepts the request from management interface if ADMF identifier in GetAllDetails/ListAllDetails/GetNEStatus/GetDestinationDetails/GetTaskDetails matches with ADMF identifier present in peer-interface configuration and interface type is LIPF_MGMT.

Modify Task - CPF

ADMF uses modify procedure to modify an existing task on the NE. All the modified details and the unchanged information are provided to replace the task details. The NF support for modification of different attributes is captured in LIPFModifyTask.

1. ADMF sends the ModifyTask message to NF with the XID - uniquely identifies the Task.
2. LI-X1 microservice provides the /X1/NE endpoint service for the LI_X1 interface.
3. ModifyTask Request message reaches the LI-X1 microservice through Istio Ingress LI gateway.
4. LI-X1 microservice parses the XML message content details in ModifyTask. LI-X1 generates the obfuscated XID from XID using one-way hash sha256.
5. LI-X1 reads and locks XID related MongoDB document and validates the new data received in ModifyTaskRequest with existing task details.
6. If modification is not supported for any of the received attributes, NE rejects the Modify Task Request.
7. If modification is accepted and if LI-X1 is working as a trigger function, it triggers a ActivateTask/ModifyTask/Deactivate request to UPF on a need basis.

For example, if:

- PDSR attribute T2 task is present, it leads to T2 modify task.
 - Task Delivery type is upgraded from X2Only to X2andX3, it leads to T3 activate task.
 - Task Delivery type is downgraded from T2andT3 to T2Only, it leads to T3 deactivate task.
8. If signaling sent to UPF at step 7, it receives the ActivateTask/DeactivateTask/ModifyTask response from triggered POI.
 9. LI-X1 updates the ETCD with the replaced task details.
 10. LI-X1 updates MongoDB record.
 11. LI-X1 deletes the XID related details in ECTD.
 12. LI-X1 sends ModifyTaskResponse with OK to ADMF.
 13. LI-X1 sends ModifyTaskNotify to Call control and LI-X2 microservices.

Modify Task - UPF

ADMF uses modify procedure to modify an existing task on the NE. All the modified details and the unchanged information are provided to replace the task details. The NF support for modification of different attributes is captured in LIPFModifyTask.

1. Trigger Function (TF) sends the ModifyTask message to NF with the XID - uniquely identifies the Task.
2. LI-X1 microservice provides the /X1/NE endpoint service for the LI_X1 interface.
3. ModifyTask Request message reaches the LI-X1 microservice through Istio Ingress LI gateway.
4. LI-X1 microservice parses the XML message content details in ModifyTask. LI-X1 generates the obfuscated XID from XID using one-way hash sha256.
5. LI-X1 reads and locks XID related MongoDB document and validates the new data received in ModifyTaskRequest with existing task details.
6. If modification is not supported for any of the received attributes, NE rejects the Modify Task Request.
7. If modification is accepted, LI-X1 updates the ETCD with the replaced task details.
8. LI-X1 updates MongoDB record.
9. LI-X1 deletes the XID related details in ECTD.
10. LI-X1 sends ModifyTaskResponse with OK to ADMF.
11. LI-X1 sends ModifyTaskNotify to PPE if it requires notification.

Modify Task Support

LIPF Modify Task

When there is a change in the warrant(s) on the targeted subscriber, LIPF issues a ModifyTask request to modify the data collection type.

For example, LIPF uses the X1 ModifyTask request to modify the following:

- DeliveryType (X2Only or X2andX3)
- PDSR (enabled/disabled)
- ListOfDIDs (one or more DIDs as required to implement the Task DeliveryType)

Following table describes the overall modify task support for the various attributes present in the TaskDetails.

Table 1. Modify Task Support

NF Type	Parameter/Attribute	Modify Supported?	Comments
CPF,UPF	Target Identifiers	No	Reject the ModifyTask Request
CPF,UPF	Delivery Type	Yes	
CPF,UPF	ListofDIDs	Yes	
UPF	CorrelationId	No	Reject the ModifyTask Request
CPF	CorrelationId	NA	
CPF	ImplicitDeactivationAllowed	Yes	
UPF	ImplicitDeactivationAllowed	NA	
UPF	ProductID	No	Reject the ModifyTask Request
CPF	ProductID	NA	
CPF,UPF	TaskDetailsExtensions/HeaderReporting	Yes	

In case of SMF, when LIPF initiates the Modify Task, as a trigger function, it initiates the ActivateTask/ModifyTask/DeactivateTask Request to respective UPF's on need basis if IRI/CC TF tasks are present.

TF Modify Task

TF Modify Task is only applicable to SMF. As a trigger function on its own, the only probable parameter that can modify is TargetIdentifiers. In SMF, there is no scenario where it needs to trigger the TF task modification to UPF.

Multi-Instance Support

LI-X1 supports the call processing with multiple instances. Support is added for all the messages. It also supports dynamic scaling of PODs using horizontal pod auto scaling (HPA). Currently, HPA is supported based on the CPU.

Multiple UPF Per PDU Session Support

The SMF can decide to add one or more UPFs in the datapath of a PDU Session during or after the PDU Session Establishment procedure. This enables selective routing of traffic to a local DN or

connectivity when a UE moves out of the service area of the existing UPF. The SMF can also relocate or remove an added UPF due to events such as UE's location change.

The LI-X1 supports the IRI/CC TF task activation/modification/deactivation on UPF addition/removal and also performs fan-out of T2//T3 task for the UPF. Call control initiates the trigger for each UPF separately.

Node Failure Resiliency Through POD Spread Policy

LI-X1 supports node failure resiliency through POD spread policy used in Fed-level default constraints.

To handle the worker node failure, there must be multiple application pods running across multiple nodes to manage the traffic. It is recommended to enable HPA and to have at least two LI-X1 pod instances running for an application.

Note: Without multiple LI-X1 pod instances, spreading the pods across multiple worker nodes do not work.

Operational CLI

Ingress Keep-Alive Status

Following command is used to query the incoming keep-alive request status on LI-X1/LI-T2/LI- T3 interfaces.

Table 2. show ingress keep alive Parameters

Parameter	Description
LocalNEID	Configured Local NE identifier.
PeerADMfid	ADMF identifier of the peer.
TIME_P2	Configured time at the NF. If no message is received from ADMF within the configured time, it leads to DeactivateAllTasks on NF.
KeepAliveReqRcvdTime	Timestamp when last keep alive/any other message received from the ADMF.

Sample Output

Note: Status can be queried for particular ADMF ID or all peers.

```
show li-x1 keep-alive ingress-x1 status admfid smf136r-li
```

```
LocalNEID      PeerADMFID  TIME_P2  KeepAliveReqRcvdTime
```

```
-----
```

```
ne_id_1  smf136r-li  60Tue Nov 16 10:07:15 2021 GMT
```

```
-----
```

```
Current Time: Tue Nov 16 10:07:19 2021 GMT
```

```
-----
```

```
show li-x1 keep-alive ingress-x1 status all
```

```
LocalNEID      PeerADMFID  TIME_P2  KeepAliveReqRcvdTime
```

```
-----
```

```
ne_id_1  smfanimesh-li      60Tue Nov   16
```

```
10:06:43 2021 GMT
```

```
ne_id_1  smf201s-li  60Tue Nov   16
```

```
10:06:39 2021 GMT
```

```
ne_id_1  smf136r-li  60Tue Nov   16
```

```
10:07:15 2021 GMT
```

```
-----
```

```
Current Time: Tue Nov 16 10:07:29 2021 GMT
```

Trigger-Function Keep-Alive status

When NF is working as a Trigger Function, and Keep-Alive is enabled, as a Trigger Function (SMF), NF continuously sends Keep-Alive requests to Triggered-POI (UPF) in configured TIME_P1 interval.

Note: This is only applicable to SMF.

Table 3. show li-x1 keep-alive trigger-functional t2/t3 Parameters

Parameter	Description
LocalADMfid	Configured Local NE identifier. It is used as ADMFID when sending a Keep-Alive Request to a peer.
PeerNEID	Peer NE-Identifier
PeerAddress:Port	Peer IP address and port details.
TIME_P1	Configured time in which Keep-Alive Requests are sent to a peer.
KeepAliveStatus	OK - If successful keep-alive response is received. ERROR - If successful keep-alive response is not received.
KeepAliveReqSent	Timestamp when the last keep-alive request is sent to the peer.
KeepAliveRspRcvd	Timestamp when the last keep-alive response is received.

Sample Output

Note: Status can be queried for particular peer NEID or all trigger-function peers.

```
show li-x1 keep-alive trigger-function t2 status all
```

```
-----
LocalADMfid    PeerNEID    PeerAddress:Port  TIME_P1  KeepAliveStatus
KeepAliveReqSent  KeepAliveRspRcvd
-----

smf136r-li          ne_id_1          10.163.206.60:443
60 OK              Tue   Nov   16   09:16:38 2021 GMT Tue Nov
16 09:16:38      2021   GMT
smf136r-li          neID             10.163.209.139:443
```

Lawful Intercept

60 OK Tue Nov 16 09:16:38 2021 GMT Tue Nov

16 09:16:38 2021 GMT

Current Time: Tue Nov 16 09:16:40 2021 GMT

show li-x1 keep-alive trigger-function t2 status neid ne_id_1

LocalADMfid PeerNEID PeerAddress:Port TIME_P1 KeepAliveStatus
KeepAliveReqSent KeepAliveRspRcvd

smf136r-li ne_id_1 10.163.206.60:443

60 OK Tue Nov 16 09:16:38 2021 GMT Tue Nov

16 09:16:38 2021 GMT

Current Time: Tue Nov 16 09:16:44 2021 GMT

show li-x1 keep-alive trigger-function t3 status all

LocalADMfid PeerNEID PeerAddress:Port TIME_P1 KeepAliveStatus
KeepAliveReqSent KeepAliveRspRcvd

smf136r-li ne_id_1 10.163.206.60:443

60 OK Tue Nov 16 09:55:38 2021 GMT Tue Nov

16 09:55:38 2021 GMT

smf136r-li neID 10.163.209.139:443

60 OK Tue Nov 16 09:55:38 2021 GMT Tue Nov

16 09:55:39 2021 GMT

Lawful Intercept

Current Time: Tue Nov 16 09:55:55 2021 GMT

```
show li-x1 keep-alive trigger-function t3 status neid ne_id_1
```

```
-----  
LocalADMFID      PeerNEID      PeerAddress:Port TIME_P1 KeepAliveStatus  
KeepAliveReqSent KeepAliveRspRcvd
```

```
smf136r-li      ne_id_1      10.163.206.60:443
```

```
60 OK      Tue Nov 16 09:55:38 2021 GMT Tue Nov
```

```
16 09:55:38 2021 GMT  
-----
```

Current Time: Tue Nov 16 09:56:06 2021 GMT

Task Details

This CLI is used to query the task details using XID. Details contain provisioned Task details, provisioning status, and faults if any.

Sample Output

```
show li-x1 task-details xid 29f28e1c-f222-486a-a860- f5a784ab9172
```

```
-----  
Task Details  
-----
```

```
{  
  "task Details": { "obfuscatedAdmfId":  
    "ba6b2ee76a2880cece6279a5d85b5c3123df41ad71fe241ed661c8aadfe a25",  
    "obfuscatedXid":  
    "212e7560512463bf99bda6adb6bb7595dc86ad96e5ff8386914bb5db9ec aa2",  
    "taskEntryValue": { "admfiIdentifier": "admfiID", "deliveryType": "X2andX3",  
    "listOfCcTFTasks": [  
      {  
        "correlationID": "2923", "n4Address": "upf.test.com", "targetIdentifiers":  
        {  
          "n1_targetIdentifier": [  
            {  
              "n1_targetIdentifierExtension": { "n1_  
UPFLIT3TargetIdentifierExtensions": {  
                "ns1_UPFLIT3TargetIdentifier":  
  
            ]  
            "10.15.15.36",  
  
            "fe80::20c:29ff:fee5:f188",
```

Lawful Intercept

```
{
  "ns1_FSEID": {
    "ns1_IPv4Address": "ns1_IPv6Address":

    "ns1_SEID": "123456"
  },

  "uPFLIT3TargetIdentifierType": "FSEID"
}
]
}
},
"targetIdentifierType": "targetIdentifierExtension"
}
]
},
"xid": "e8307a14-55e5-4802-93b6-5155618d77ab"
}
],
"listOfDIDs": { "n1_dId": [
  "b0ce308c-aa22-42bd-a27b-287bcb5b3468"
]
},
"listOfIRITasks": [
{
  "correlationID": "2923", "n4Address": "upf.test.com", "targetIdentifiers":
  {
    "n1_targetIdentifier": [
      {
        "n1_targetIdentifierExtension": { "n1_
UPFLIT3TargetIdentifierExtensions": {
          "ns1_UPFLIT3TargetIdentifier":

          [
            "10.15.15.36",

            "fe80::20c:29ff:fee5:f188",

            {
              "ns1_FSEID": {
                "ns1_IPv4Address": "ns1_IPv6Address":
                "ns1_SEID": "123456"
              },

              "uPFLIT3TargetIdentifierType": "FSEID"
            }
          ]
        }
      }
    ],
    "targetIdentifierType": "targetIdentifierExtension"
  }
}
]
```

Lawful Intercept

```
}
]
},
"xid": "e95068b1-f395-44d6-bb6b-f7fdfa1eeef7"
},
],
"targetIdentifiers": { "n1_targetIdentifier": [
{
"n1_supiimsi": "225097743010822", "targetIdentifierType": "supiimsi"
}
]
},
"taskDetailsExtensions": [
{
"n1_extension": { "n1_X1Extensions": [
{
"ns1_HeaderReporting": { "ns1_PDHType": {
"ns1_PDSR": {
"ns1_PDSRTriggerType": [
{

"20000",

"pDSRTriggerEnumType": "ByteCount"
},
{

"50",

"pDSRTriggerEnumType": "PacketCount"
},
{

300,

"pDSRTriggerEnumType": "TimerExpiry"
}
]
}
}
}
}

"ns1_ByteCount":
"ns1_PacketCount":
"ns1_TimerExpiry":

}
]
}
}
],
"version": "1.0.0",
```

Lawful Intercept

```
"xid": "29f28e1c-f222-486a-a860-f5a784ab9172"
},
"taskStatus": "
{"claimant\":"unassigned\","version\":"1.0.0\","secretKeyVersion\":"1\","obfuscatedxid\":"212e7560512463bf99bda6adb6bb7595dc86ad96e5ff8386914bb5db9ecaa2\","obfuscatedadmfid\":"ba6b2ee76a2880cece6279a5d85b5c3123df41ad71fe241ed661c8aadfea25\","data\":"
{"numberOfModifications\":"0\","provisionstatus\":"eComplete"}, {"timeOfLastIntercept\":"2021-11-16T08:57:33.000000\","amountOfX2Data\":"344}"
}
}
```

Destination Details

This CLI is used to query the destination details using DID. Details contain provisioned DID details, provisioning status, and faults if any.

Sample Output

```
show li-x1 destination-details did b0ce308c-aa21-42bd-a27b- 287bcb5b3468
```

```
-----
Destination Details
-----
```

```
{
"destinationDetails": { "DIDPeerEntryValue": {
"listOfDIDPeers": { "value": [
{
"admfIdentifier": "admfID"
}
]
},
"version": "1.0.0"
},
"admfIdPair": [
{
"admfId": "admfID", "obfuscatedAdmfId":
"ba6b2ee76a2880cece6279a5d85b5c3123df41ad71fe241ed661c8aadfe a25",
"taskCountValue": 0
}
],
"destinationEntryValue": { "destinationDetails": {
"n1_dId": "b0ce308c-aa21-42bd-a27b- 287bcb5b3468",
"n1_deliveryAddress": { "deliveryAddressType": "ipAddressAndPort",
"n1_ipAddressAndPort": {
"etsi103280_address": { "etsi103280_IPv4Address":
"10.163.209.140"
```

```
,
"etsi103280_port": { "etsi103280_TCPPort": 7117
}
},
"n1_deliveryType": "X3Only"
},
"destinationProvisioningSource": "provSourceLipf", "switchBackTimer": 0,
"time_p1": 60,
"time_p2": 180,
"version": "1.0.0"
},
"destinationStatus": "
{"destinationstatus\":"eActiveAndWorking\"}", "obfuscatedDid":
"1398d22050682cb16f71662b9c75c0945db1b4aa4125829d7fc1268e08e d583",
"taskCountValue": 0
}
}
```

NE Status

This CLI is used to query the status of the NE. Details contain faults if any.

Sample Output

```
show li-x1 ne-status
```

NE Status Details

```
{
"neDetails": {
"neStatus": "
{"claimant\":"unassigned\", \"version\":"1.0.0\", \"obfusca
tedneid\":"40e420ad28c887d7d32c48242a567f8255e04bafc1a48175
bd539aa78d3e7a37\", \"data\":
{"nestatus\":"eOk\", \"listoffaults\":
[{"reporttype\":"eNrtWarning\", \"errorcode\":"9010\", \"issueDe
tails\":"CONFIG-
MODIFICATION\", \"faultCount\":"4\", \"faultOrigin\":"eLiX1\", \"
timeInSecs\":"1636647668\"},
{"reporttype\":"eNrtWarning\", \"errorcode\":"9010\", \"issueDe
tails\":"PING-HANDLING-
FAILURE\", \"faultCount\":"1\", \"faultOrigin\":"eLiX1\", \"timeI
nSecs\":"1636543643\"},
{"reporttype\":"eNrtWarning\", \"errorcode\":"9010\", \"issueDe
tails\":"PING-HANDLING-
SUCCESS\", \"faultCount\":"5\", \"faultOrigin\":"eLiX1\", \"timeI
nSecs\":"1636581247\"},
{"reporttype\":"eNrtWarning\", \"errorcode\":"9010\", \"issueDe
tails\":"CONFIG-
MODIFICATION\", \"faultCount\":"1\", \"faultOrigin\":"eLiX2\", \"
timeInSecs\":"1636647670\"}]]}"
```

```
}  
}
```

Debug Clear Commands

LI-X1 contains the following manual debug clear CLI commands to clean up the Destination (DID) locally in any unrecoverable error scenarios.

```
debug li-x1 clear destination did <b0ce308c-aa21-42bd-a27b- 287bcb5b3468>
```

Ping from LIPF - CPF or UPF

Ping can go from ADMF→NF or NF→ADMF direction.

From ADMF→NF direction:

1. ADMF sends the Ping message to NF.
2. LI-X1 microservice provides the /X1/NE end point service for the LI_X1 or LI_T2 or LI_T3 interface.
3. Ping message reaches to LI-X1 microservice through Istio Ingress LI gateway.
4. LI-X1 micro service parses the XML message and identifies the Ping message.
5. LI-X1 sends Ping Response with OK to ADMF.

From NF→ADMF direction (li-admin can check the peer connectivity using operational CLI):

1. LI-admin issue show li-x1 ping v1 peer interfaceType \[lipf] peerId <admfCustId1> with ADMF identifier.
2. LI-X1 microservice present in NF reads the respective IP address from config and send the ping request on /X1/ADMF URI to ADMF.
3. Ping message reaches to ADMF through Istio egress LI gateway.
4. ADMF sends with Ping response OK.
5. LI-X1 microservice parses the XML message and relay back the response to operational CLI.

From UPF(NF)→SMF(ADMF) direction (li-admin can check the peer connectivity using operational CLI):

1. LI-admin issue show li-x1 ping v1 peer interfaceType \[lipf] peerId <neid> with SMF NE identifier.
2. LI-X1 microservice present in UPF reads the IP address from config and send the ping request on /X1/ADMF URI to SMF through Istio egress gateway.
3. Ping message reaches to LI-X1 microservice present in SMF through Istio ingress LI gateway on /X1/ADMF URI.

4. SMF sends with Ping response OK.
5. LI-X1 microservice present in UPF parses the XML message and relay back the response to operational CLI.

From SMF(ADMF)→UPF(NF) direction (li-admin can check the peer connectivity using operational CLI):

1. LI-admin issue show li-x1 ping v1 peer interfaceType \[t2|t3] peerId <neid> with UPF NE identifier.
2. LI-X1 microservice present in SMF reads the IP address from config and send the ping request on /X1/NE URI to UPF through Istio egress gateway.
3. Ping message reaches to LI-X1 microservice present in UPF through Istio ingress LI gateway on /X1/NE URI.
4. UPF sends with Ping response OK.
5. LI-X1 microservice present in SMF parses the XML message and relay back the response to operational CLI.

Pings

Message Details

- At any time from the ADMF or NE, to get a response over the X1 interface (does not test X2 or X3 or onward delivery) .
- Ping can be issued from either direction: ADMF→NE or NE→ADMF or SMF→UPF or UPF→SMF.
- LI-X1 microservice support ingress ping from both LIPF/ADMF, and UPF.

Ping Operational CLI

NF provides the operational CLI to check the connectivity to peer Nodes (ADMF/NE) using PING.

For example, to check the PING to t2 interface on peer NE node with NE identifier ne_id246:

```
show li-x1 ping v1 peer interfaceType t2 peerId ne_id246

{
  "result": "Ping http://10.165.5.254:80/X1/NE Success"
}
```

Similarly, to check the PING to lipf interface on peer ADMF node with ADMF identifier admfCustId1:

```
show li-x1 ping v1 peer interfaceType lipf peerId admfCustId1

{

"result":"Ping http://10.165.5.246:80/X1/ADMF Success"

}
```

Example for negative scenarios where peer node is not reachable:

```
show li-x1 ping v1 peer interfaceType lipf peerId admfCustId1

{ "cause":"MESSAGE_HANDLING_FAILED",

"detail":"", "status":500,

"title":"Ping procedure failed http://10.165.5.246:80/X1/ADMF"

}

show li-x1 ping v1 peer interfaceType t2 peerId ne_id246

{

"cause":"MESSAGE_HANDLING_FAILED",

"detail":"", "status":500,

"title":"Ping procedure failed for http://10.165.5.254:80/X1/NE"

}
```

Private Destination Details Extension Support for Secondary Destination

In Affirmed SMF and UPF, the concept of Secondary Destination is implemented through a proprietary CLI that LI Admin users can configure.

To support a specific DestinationDetailsExtension to allow the LIPF to send CreateDestinationRequests to the NF, the CreateDestinationRequest includes the vendor-specific DestinationDetailsExtension containing the Secondary Destination information. This eliminates the usage of Affirmed proprietary CLI when provisioning Destinations on the Affirmed NF (SMF). Affirmed SMF propagates the Secondary Destination information to the Affirmed UPF through the DestinationDetailsExtension when programming Destinations on the UPF.

When both primary and secondary destination address details are provided in CreateDestinationRequest, Affirmed NF(SMF/UPF) establishes the connections to both the primary and secondary MDF servers. IRI/CC data is delivered to Secondary MDF when the primary MDF connection is down.

DID operator must choose either CLI configuration model or LIPF provisioning, mix and match are not allowed for the given DID.

- When using LIPF created Destinations, the Secondary Extension provided by LIPF to SMF is sent to the UPF.
- When using CLI-created Destinations, the SMF does not send the Secondary address to UPF. Instead, configure the UPF directly through Affirmed CLI with a Destination containing the desired Secondary IP address and port.

Note: The 5G Lawful Intercept standards do not currently support the concept of a Secondary Destination.

ETSI 103221-01 LI-X1 specification doesn't have a provision to include the secondary address for the DID in CreateDestinationRequest message.

Provisioning Local DID

According to Lawful Interception specifications, destinations are provisioned by LIPF using the LI_X1 interface on Network Functions.

The NF supports custom feature Local Destination (DID) Provisioning using LI-Admin CLI.

NF also allows LIPF creation of Destinations using the X1 Create Destination command as per LI specifications.

NF Support for Create Destination

- LI Admin CLI created Destinations must be managed by the LI Admin CLI:
 - Destinations created by LI Admin CLI must be deleted by LI Admin CLI
 - LIPF cannot delete a Destination created by LI Admin CLI
 - LIPF cannot modify a Destination created by the LI Admin CLI
- LIPF created Destinations must be managed by the LIPF:
 - Destinations created by LIPF must be deleted by LIPF using X1 Remove Destination
 - LI Admin CLI cannot delete a Destination created by LIPF

- LI Admin CLI cannot modify a Destination created by LIPF (the only exception being LI Admin CLI should be able to provide additional details for the Destination as specified in the Partial DID configuration config section):

LI Admin CLI - Local Configuration of Primary and Secondary Destinations(DIDs)

LI-Admin can configure the local DID using following configurations.

If the LI admin expects to establish the TLS/TCP connections towards the MDF2/MDF3 server, it is mandatory to configure all the primary server-related details under did-config. If any of the primary address-related details are missing NF does not establish the TLS/TCP connections towards the MDF2/MDF3 server. NF simply treats the configuration as a supplement configuration for LIPF created destinations. If primary and secondary server-related complete details are configured NF establishes the TLS/TCP connections to both primary and secondary MDFs on configuration commit.

LI admin CLI - Full Configuration of DID

Note: CLI allows configuration of all info for the specified DID.

```
li-x1 did-config f0054db4-f7ca-4420-9303-fbd2da6b2e74
delivery-type x2andX3
primary-ipv4-address 10.1.1.1
primary-port 111
secondary-ipv4-address 10.2.2.2
secondary-port 222
switch-back-timer 60 <== this functionality not yet supported
```

Note: After switching over to secondary, the switch-back-timer specifies the time in seconds (the connection to primary must remain active before it switch-backs to primary). This is an unsupported functionality.

LI Admin CLI – Partial DID Configuration to Supplement LIPF Create Destination with Secondary

Use CLI to specify all fields that cannot be provided using standard-based X1 Create Destination.

In lawful interception specifications, there is no secondary server address related details are specified in LI_X1 interfaces. LI admin can use the following secondary related configuration to add the secondary server for the DID. TLS/TCP connection for secondary from NF→MDF2/MDF3 will happen only after receiving LIPF initiated Create Destination for the DID.

Partial DID Config

```
li-x1 did-config f0054db4-f7ca-4420-9303-fbd2da6b2e74 secondary-ipv4-  
address 10.2.2.2
```

```
secondary-port 222
```

```
switch-back-timer 60
```

Note: Supports modification of DID related config attributes like IP and port.

Publication

LI-X1 sends multiple publication messages when it receives configuration. These publication messages are sent at the boot-up time after configuration is received. Publication messages are also sent if new T2/T3/LIPF interfaces are added and configuration is received at LI-X1 microservice. LI-X1 sends a publication message with a unique publication service id for each external-Host it needs to connect.

Redundant ADMF Support

The purpose of this feature is to maintain the LIPF peer's state (active/standby) and send reports to only active LIPFs.

Currently, there is no method to detect LIPF peers state, and this feature provides mechanism to detect peer states. This is a non-3gpp compliant feature.

Note: This feature is currently NOT supported for UPF.

- The feature uses the existing LIPF peer interface configuration. LI-X1 peer interfaces are configured with same ADMF identifiers with different peer-profiles having distinct IP addresses.
- Ping request is triggered to all the peer IP addresses associated with all the distinct ADMF identifier.
- Peers sending a successful ping response is marked as “active” and the ones not sending response OR sending negative response is marked as “standby”.

Note: It is assumed that only the active peers send successful ping response.

- Reports (NEIssue/TaskIssue/DID issue) are sent only to peers marked as active.
- On change of state of a LIPF Peer from active to standby , GetAllDetails message is received from the respective ADMF. On receiving GetAllDetails ping request is sent to all the peers to fetch the latest peer status.
- On configuration modification, ping procedure is initiated based on the new configuration to fetch the latest peer status.

Single LIPF Peer Configuration

If a single LIPF peer is configured , its state (active/standby) is also determined based on the ping response. If a positive response is not received from the peer, it is marked as standby and reports are not sent, since no peer is active.

However, peer returns to active state if it sends a successful response to the subsequent ping request.

Sample Configuration

The following is the sample configuration:

```
li-x1 peer-interface lipf-interface
peer-type          lipf
admf-identifier     admfID associated-peer-profile lipf
!
li-x1 peer-interface lipf-interface1
peer-type          lipf
```

Lawful Intercept

```
admf-identifier      admfID associated-peer-profile lipf1 li-x1 peer-  
profile lipf
```

```
host      10.10.10.10
```

```
https-port xxxx http-port  yyyy
```

```
!
```

```
li-x1 peer-profile lipf1
```

```
host      10.10.10.11
```

```
https-port xxxx http-port  yyyy
```

```
!
```

Change in li-x1 ping Operation CLI

With this feature, ping is sent to all the LIPF peers associated with the given ADMF-ID and the response is displayed in their corresponding status.

```
liadmin@pod-cfgmgr-2-4-0-1005-main-int-dbg-56df7758b7-zq7dn 05:52:48# show  
li-x1 ping v1 peer interfaceType lipf peerId admfID
```

```
{
```

```
"lipfPeerPingResult":[
```

```
{
```

```
"errorDetails":{ "status":500,
```

```
"title":"Ping procedure failed "
```

```
},
```

```
"result":"Ping http://10.10.10.10:80/X1/ADMF Failed - Response data is  
null"
```

```
},
```

```
{
  "result": "Ping http://10.10.10.11:80/X1/ADMF
  Success"
}

],
  "result": "Success"
}
```

Remove All Destination - CPF

This command is used by the ADMF to remove all the Destinations on NE permanently.

1. ADMF sends a RemoveAllDestinations request to NE.
2. RemoveAllDestinations message reaches to LI-X1 pod through LI Ingress GW.
3. LI-X1 validates if any outstanding tasks are using any of the DIDs provisioned by the ADMF.
4. If any tasks are using any of the DID provisioned by the ADMF, LI_X1 responds with the "Destinations In Use" (8010) error code.
5. If no outstanding tasks are present for the any of the DIDs, LI-X1 continues with RemoveAllDestinations procedure.
6. LI-X1 starts a "RemoveAllDestinations" progress timer with the timer microservice.
7. LI-X1 adds a "RemoveAllDestinations" progress in etcd table so that other Pod instances of LI-X1 knows this NE level procedure is ongoing in one of the pod instances.
8. If Trigger Function related Destinations are provisioned on any of the UPFs, NE initiates a RemoveAllDestinations message to each relevant UPF. For a given UPF based on earlier provisioning, NE initiates the separate messages for LI_T2 and LI_T3.
9. LI-X1 receives the RemoveAllDestinations response from all the UPFs. Step 8 and 9 are only applicable to NE which also works as the Trigger Function.
10. LI-X1 removes all the DIDs provisioned by ADMF from ETCD tables.
11. LI-X1 also remove all the DIDs related records from mongoDB which are provisioned by ADMF.
12. LI-X1 removes the RemoveAllDestination progress flag from the ETCD tables.
13. LI-X1 removes the RemoveAllDestinations progress timer from the timer microservice.

14. Once all DIDs are removed, LI-X1 sends RemoveAllDestinations Response with OK:AcknowledgedAndCompleted to ADMF through LI Ingress GW.

Remove All Destinations - UPF

This command is used by the ADMF to remove all the Destinations on NE permanently.

1. ADMF sends a RemoveAllDestinations request to NE.
2. RemoveAllDestinations message reaches to LI-X1 pod through LI Ingress GW.
3. LI-X1 validates if any outstanding tasks are using any of the DIDs provisioned by the ADMF.
4. If any tasks are using any of the DID provisioned by the ADMF, LI-X1 responds with the Destinations In Use (8010) error code.
5. If no outstanding tasks are present for the any of the DIDs, LI-X1 continues with RemoveAllDestinations procedure.
6. LI-X1 starts a RemoveAllDestinations progress timer with the timer microservice.
7. LI-X1 adds a RemoveAllDestinations progress in etcd table so that other Pod instances of LI- X1 knows this NE level procedure is ongoing in one of the pod instances.
8. LI-X1 removes all the DIDs provisioned by ADMF from ETCD tables.
9. LI-X1 also remove all the DIDs related records from mongoDB which are provisioned by ADMF.
10. LI-X1 removes the RemoveAllDestination progress flag from the ETCD tables.
11. LI-X1 removes the RemoveAllDestinations progress timer from the timer microservice.
12. Once all DIDs are removed, LI-X1 sends RemoveAllDestinations Response with OK:AcknowledgedAndCompleted to ADMF through LI Ingress GW.

Remove Destination - CPF

This procedure is used by ADMF to remove the Destination(DID) from NF.

1. ADMF sends the RemoveDestination message to NF with the DID
 - DID - Destination identifier which uniquely identifies the destination
2. LI-X1 microservice provides the "/X1/NE" end point service for the LI_X1 interface.
3. RemoveDestination message reaches to LI-X1 microservice through Istio Ingress LI gateway.
4. LI-X1 microservice parses the XML message content details in RemoveDestination. LI-X1 generates the obfuscated DID from DID using one-way hash sha256.

5. LI-X1 read and lock DID related Mongo Db document.
6. If LI-X1 is working as a trigger function it will trigger a RemoveDestination towards the UPF for each IRI/CC interface based on earlier provisioning.
7. Receives the RemoveDestination response from triggered POI.
8. LI-X1 deletes the DID related details in ECTD.
9. ETCD notifies the interested microservices about the DID deletion. Typical subscribers for the DID entries are LI-X2 microservice which manages the actual TLS/TCP connections to MDF server.
10. Similarly, LI-X1 deletes the DID to LIPF mapping in ETCD.
11. LI-X1 deletes the DID document from Mongo Db.
12. LI-X1 sends RemoveDestinationResponse with OkAndAcknowledged to ADMF.

Remove Destination - UPF

If UPF is configured for both X2 and X3 with same DID, first destination removal causes to remove entry and send removal results into error response.

This procedure is used by ADMF which is working as SMF to remove the Destination(DID) on NF.

1. ADMF sends the RemoveDestination message to NF with the DID.
 - DID - Destination identifier which uniquely identifies the destination.
2. LI-X1 microservice provides the "/X1/NE" end point service for the LI_T2 and LI_T3 interfaces.
3. RemoveDestination message reaches to LI_X1 microservice through Istio Ingress LI gateway.
4. LI-X1 microservice parses the XML message content details in RemoveDestination. LI-X1 generates the obfuscated DID from DID using one-way hash sha256.
5. LI-X1 read and lock DID related Mongo Db document.
6. If the destination entry is configured by last SMF, then it deletes did_table and did_lipf_table entry. Otherwise, it updates did_lipf_table entry.
7. ETCD notifies the interested microservices about the DID deletion. Typical subscribers for the DID entries are PPE microservice which manages the actual TLS/TCP connections to MDF server.
8. Similarly, LI-X1 deletes the DID to LIPF mapping in ETCD.
9. UPF deletes Mongo Db entry if no more SMF has DID configured.
10. LI-X1 sends RemoveDestinationResponse with OkAndAcknowledged to ADMF.

Report Destination Issue - CPF

This procedure is used by the control plane federation to report Destination issue to LIPF.

1. LI-X1 detects Destination issue or LI-X2/Application microservice reports Destination issue to LI-X1 microservice.
2. LI-X1 reads and locks DID subscriber.
3. LI-X1 updates Destination record, writes, and unlocks DB record.
4. LI-X1 sends ReportDestinationIssue message to LI Egress GW.
5. LI Egress GW sends ReportDestinationIssue message to LIPF.
6. LIPF sends ReportDestinationIssueResponse message with OKAndAcknowledge to LI Egress GW.
7. LI Egress GW sends ReportDestinationIssueResponse message to LI-X1.

Report Destination Issue - UPF

This procedure is used by the user plane federation to report Destination issue to LIPF.

1. LI-X1 detects Destination issue or PPE microservice reports Destination issue to LI-X1 microservice.
2. LI-X1 reads and locks DID subscriber.
3. LI-X1 updates Destination record, writes, and unlocks DB record.
4. LI-X1 sends ReportDestinationIssue message to LI Egress GW for all LIPF (configured LIPF or SMFs which provisioned this DID).
5. LI Egress GW sends ReportDestinationIssue message SMF.
6. SMF sends ReportDestinationIssue message LIPF.
7. LIPF sends ReportDestinationIssueResponse message with OKAndAcknowledge to SMF.
8. SMF sends ReportDestinationIssueResponse with OKAndAcknowledge to UPF LI Egress GW.
9. LI Egress GW sends ReportDestinationIssueResponse message to LI-X1.

Report NE Issue - CPF

This procedure is used by the control plane federation to report NE issue to LIPF.

1. LI-X1 detects NE issue or LI-X2/Application microservice reports NE issue to LI-X1 microservice.

2. LI-X1 reads and locks NE subscriber.
3. LI-X1 updates NE record, writes, and unlocks DB record.
4. LI-X1 sends ReportNEIssue message to LI Egress GW.
5. LI Egress GW sends ReportNEIssue message to LIPF.
6. LIPF sends ReportNEIssueResponse message with OKAndAcknowledge to LI Egress GW.
7. LI Egress GW sends ReportNEIssueResponse message to LI-X1.

Report NE Issue - UPF

This procedure is used by the user plane federation to report NE issue to LIPF.

1. LI-X1 detects NE issue or PPE microservice reports NE issue to LI-X1 microservice.
2. LI-X1 reads and locks NE subscriber.
3. LI-X1 updates NE record, writes, and unlocks DB record.
4. LI-X1 sends ReportNEIssue message to LI Egress GW for all configured LIPF (SMFs).
5. LI Egress GW sends ReportNEIssue message SMF.
6. SMF sends ReportNEIssue message LIPF.
7. LIPF sends ReportNEIssueResponse message with OKAndAcknowledge to SMF.
8. SMF sends ReportNEIssueResponse with OKAndAcknowledge to UPF LI Egress GW.
9. LI Egress GW sends ReportNEIssueResponse message to LI-X1.

Report Task Issue - CPF

This procedure is used by the control plane federation to report Task issue to LIPF.

1. LI-X1 detects Task issue or LI-X2/Application microservice reports Task issue to LI-X1 microservice.
2. LI-X1 reads and locks Task subscriber.
3. LI-X1 updates Task record, writes, and unlocks DB record.
4. LI-X1 sends ReportTaskIssue message to LI Egress GW.
5. LI Egress GW sends ReportTaskIssue message to LIPF.
6. LIPF sends ReportTaskIssueResponse message with OKAndAcknowledge to LI Egress GW.

7. LI Egress GW sends ReportTaskIssueResponse message to LI-X1.

Report Task Issue - UPF

This procedure is used by the user plane federation to report Task issue to LIPF.

1. LI-X1 detects Task issue or PPE microservice reports Task issue to LI-X1 microservice.
2. LI-X1 reads and locks Task subscriber.
3. LI-X1 updates Task record, writes, and unlocks DB record.
4. LI-X1 sends ReportTaskIssue message to LI Egress GW.
5. LI Egress GW sends ReportTaskIssue message SMF.
6. SMF sends ReportTaskIssue message LIPF.
7. LIPF sends ReportTaskIssueResponse message with OKAndAcknowledge to SMF.
8. SMF sends ReportTaskIssueResponse with OKAndAcknowledge to UPF LI Egress GW.
9. LI Egress GW sends ReportTaskIssueResponse message to LI-X1.

Reporting Issues From NE

Message Details

There are set of message defined for NE to report issues (warning and faults) to the ADMF. It is distinct from Getting Status, in which the ADMF retrieves information from the NE.

ReportTaskIssue on given XID

Direction is from NE to ADMF. The NE sends a ReportTaskIssue request when it becomes aware of an issue (warning or fault) relating specifically to a particular XID. It is also used to follow up on an OK - Acknowledged response, to signal that a request has been completed successfully or unsuccessfully.

ReportDestinationIssue on given DID

The NE sends a ReportDestinationIssue request when it becomes aware of an issue (warning or fault) relating specifically to a particular DID. It is also used to follow up on an OK - Acknowledged response, to signal that a request has been completed successfully or unsuccessfully.

ReportNEIssue

The NE sends a ReportNEIssue request when it becomes aware of an issue (warning or fault) relating to the whole NE.

Report Issues handling at NF

LI-X1 handles issues reported by LI-X2 and other application microservice, update them in MongoDB and report issues to LIPF. LI-X1 can also generate issues. Below are the different types of faults:

- Terminating (This fault type is not supported in current release)
- Non-terminating
- Warning

When LI-X1 report Non-terminating issues to LIPF, it saves same to the MongoDB and sends clear when issue is resolved. Warnings are one-time reporting and no clear message is sent for such issues. LI-X1 adds ReportType and FaultDetails in free text to provide details about issues to LIPF.

LI-X1 reports issues as below:

- Task Issues: xIRI event report failure CC/IRI TF Activate Task, Deactivate Task, and Get Task failure.
- Destination Issues: Whitelisting failure, TCP connection failure, TCP socket failure, CC/IRI TF Create Destination, Remove Destination, and Get Destination failure.
- NE Issues: Config modification and Ciphering key is unavailable (when Ciphering is enabled).

Note: If NF is not able to report all the issues due to the network-connectivity issues between NF and LIPF, all reports are not available later for Get-Status message from LIPF, for example, warning reports. Warning reports are not stored at NF whereas non-terminating type reports are stored, hence non-terminating type reports are available.

Report Issues Handling as a Trigger Function - SMF

While SMF working as a Trigger Function, it will work as ADMF towards the UPF. It exposes the /X1/ADMF URI as a server towards the UPF.

When UPF initiates the reports as a Trigger Function, SMF tries to handle the issues. If the report type is not possible to handle at the SMF, it will relay the UPF report as it is to LIPF.

Report NE Issue

- When multiple SMFs have provisioned a single UPF, the UPF sends ReportNEIssue to all SMFs, which have provisioned UPF.
- SMF will not relay the ReportNEissues from the UPF to LIPF.

- Upon Report NE Issue indicating UPF reboot, SMF will reprogram the targets on UPF. Feature not supported yet.

Report Destination Issue

- When multiple SMFs have provisioned the same DID at UPF, UPF sends ReportDestinationIssue to all SMFs when the issue occurs. In this scenario, it is possible that the same DID Report can be received from different paths at LIPF. Hence, LIPF should be able to handle duplicate ReportDestinationIssue reports for the same issue.
- The LI X1 now sends ReportDestinationIssue reports on a per-connection basis. Single ReportDestinationIssue now contain per connection info with primary and secondary connections status. ReportDestinationIssue also carries Connection-interrupts details for both primary and secondary connections. Bufferflow/Bufferloss details are included in ReportDestinationIssue, if any. Corresponding alarms now report issues on a per-connection basis.

Note: SMF/UPF populate ErrorDescription field to include the NE identifier + DID problem details as a free text. LIPF can use the free text to distinguish whether ReportDestinationIssue is from SMF or UPF.

SMF Report Destination Issue Configuration

Operators can use the following configuration to control the amount of frequency of the Report Destination Issue requests the LIPF receives, as well configuration to suppress the Report Destination Issue altogether. The intent of this configuration is to limit the total number of Report Destination Issues sent to the LIPF. Each Report Destination Issue message contains all statistics for the current interval, where current interval = the period of time based on configuration for 'frequency' (default 120 seconds).

li-x1 local-profile report-destination-issue state enabled/disabled (default enabled) li-x1 local-profile report-destination-issue frequency 60-300 (In secs. default 120)

li-x1 local-profile report-destination-issue connection-interrupt state disabled (default enabled)

li-x1 local-profile report-destination-issue connection-interrupt state enabled threshold 1-100 (default 10)

Setting the connection-interrupt state to disabled causes the system to never generate a Report Destination Issue based solely on the occurrence of connection interrupts. However, if Report Destination Issue is generated due to some other reason (e.g. connection loss or buffer overflow) then the message includes connection-interruption statistics for the current interval, i.e. the actual count of connection-interrupts that occurred during the previous 'frequency' interval (120 seconds).

Setting the connection-interrupt state to enabled causes the system to generate Report Destination Issue messages if the only changes over the previous interval were connection-interrupts.

When connection-interrupt state is enabled, the 'threshold' value is a count of how many connection-interrupts can occur before the Report Destination Issue message is sent.

For example, with 'connection-interrupt state enabled threshold 10' (the default settings) if only 4 connection-interrupts occur within the 'report-destination-issue frequency' time period (120 second default), then the Report Destination Issue message is not sent and the internal counter of connection-interrupts are reset to 0.

If during the next time period (120 second default) there are 10 or more connection-interruptions, then the Report Destination Issue message is sent and it reports the actual number of connection-interruptions which occurred in that time period (10 or more).

Report Task Issue

When UPF reports TF task related issues to SMF, SMF first tries to reconcile/reprogram the TF task on UPF. If this is unsuccessful, it reports the issue back to LIPF.

SMF/UPF populates ErrorDescription field with the NE identifier + TF task XID details. The PPE handling the session generates the Report Task issue to SMF, including a PPE Identifier in the ErrorDescription. The SMF will then act on or relay the Report Task issue to the LIPF.

LIPF can distinguish using the free text to identify the ReportTaskIssue is from the task programmed by LIPF on SMF versus TF task generated by the SMF.

Security Context for the Containers

With this feature, LI-X1 pod security context is auto injected. A security context defines privilege and access control settings for a Pod or Container.

Support PDSR TaskDetailsExtensions

Due to a typo in the 3GPP TS 33.128 Annex C: XSD Schema for LI_X1 extensions, the syntax for LI-X1 implementation for TaskDetailsExtension is invalid.

Due to the backward-compatibility issue, vendors have issues in changing the XSD that is already implemented:

```
<xs:element name="X1Extensions" type="X1Extension"></xs:element>
```

Note: The type="X1Extension" (singular) is an expected implementation standard.

NF strictly complies with the 3GPP TS 33.128 Annex C:XSD schema and also supports the multiple PDSRTriggerType as per 33.128 v 16.6.0.

Task and Destination Delivery Type Enhancements

In the UnityCloud Lawful Intercept deployment, LIPF provisions a Task with DeliveryType=X2andX3 associated with a ListOfDIDs, where one Destination has DeliveryType=X2Only and another Destination has DeliveryType=X3Only.

The task requires DeliveryType X2andX3 to associate with a Destination of DeliveryType X2andX3.

The current ETSI 103.221-1 5G Lawful Intercept specifications are unclear as to whether this is a valid configuration.

SMF allows LIPF to provision a single Task with DeliveryType=X2andX3 associated with a ListOfDIDs, where one DID has DeliveryType=X2Only and another DID has DeliveryType=X3Only.

The UnityCloud LI solution complies with the 5G Lawful Intercept standards for DeliveryType configurations.

Tasks - Starting, Modifying, and Stopping

Activate Task

ADMF uses this task to add a new Task to an NE. On successful processing, LI-X1 publishes the task related information in ETCD. Consumer microservices interested in task subscribe to ETCD tasks table.

Deactivate Task

ADMF uses this Task to deactivate (permanently stop and remove) a Task on an NE. On successful processing, LI-X1 removed the task entry from ETCD and also notifies the consumers about task deletion.

Modify Task

ADMF uses this task to modify an existing Task on the NE. All details for the task is provided (i.e. the modified details and the information that is unchanged) to completely replace the previous task details.

Deactivate All Tasks

LI-X1 supports the deactivate all tasks. On receiving, LI-X1 deletes all the Tasks present on an NE for that ADMF.

Warnings and Faults

The errors shown in this section are not X1 protocol errors. The issues associated with the XID/DID/NE which occurs on NE are reported as warnings and faults using Reporting Issues mechanism.

Warnings are one-off problems i.e. sent by the NE and then not referred to again over X1. Warnings should not be used for issues that are affecting traffic (i.e. losing content or intercept- related information). For example, warnings may include resources being nearly exhausted but not yet traffic-affecting. Warnings should include that keys/certificates that are about to expire.

Faults are problems, which the NE continues to be aware of and which the NE is trying to manage and/or rectify. Any issue which loses traffic is categorized as a fault.

Warnings are reported using issue-reporting messages but then are not included in any future Status-Getting messages.

Note: Logging for warnings on NF is not yet supported.

NF supports issuing a report when warning or fault occurs on XID/DID/NE. Faults are stored at the NF in database and are available for Status-Getting messages. NF also supports issuing another report when Fault is cleared.

The following table describes the different warning and faults supported by the NF.

Table 4. Reports NE Issue Table

SL	Event Type	Event Trigger	Reported From	Report Type	Are Reports Cleared
1	CONFIG-MODIFICATION	Config addition / modification / deletion	x1	Warning	NO
2	CONFIG-MODIFICATION-IMPACTING	Config addition / modification / deletion any important config impacted, its triggered when any of the config is missing	x1	Non Terminating Fault, Clear will also be sent when issue is cleared	YES

Lawful Intercept

SL	Event Type	Event Trigger	Reported From	Report Type	Are Reports Cleared
3	CIPHERING-FAILURE	Encryption / Decryption Failure	x1	Future work Item	NA
4	CIPHERING-KEY- VERSION-UPDATED	Encryption / Decryption key updated	x1	Future work Item	NA
5	PKI-CERT-EXPIRING	Certificates Expiring	Future work Item	Future work Item	NA
6	PKI-CERT-REVOKED	Certificates Revoked	Future work Item	Future work Item	NA
7	NODE-FAILURE	Node Failure	Future work Item	Future work Item	NA
8	CLUSTER-RESTARTED	Virtual Service Restarted	x1	Warning	NO
9	WHITELISTING-REGISTER-FAILURE	Whitelisting register failure	x1	Non Terminating Fault, Clear will also be sent when issue is cleared	YES
10	WHITELISTING-DEREGISTER-FAILURE	Whitelisting deregister failure	x1	Future work Item	NA
11	SERVICE-UNAVAILABLE	When x2 or ppe pod goes down	x1	Any pending nonterminating fault is cleared if set by x2 or ppe	YES, if Terminating or Non Terminating faults were reported by x2 or ppe

Table 5. Reports Destination Issue Table

SL	Event Type	Event Trigger	Reported From	Report Type	Are Reports Cleared
1	Create Destinations Failure Towards UPF	Create Destination Failure Towards UPF as part CCTF/IRITF Activate Task Request.	x1	Warning	NO

Lawful Intercept

SL	Event Type	Event Trigger	Reported From	Report Type	Are Reports Cleared
		Note: If the destination is already provisioned on UPF, no report destination issue is reported to LIPF.			
2	Service Unavailability	When x2 or ppe pod goes down	x1	Any pending non terminating fault is cleared if set by x2 or ppe	YES, if Terminating or Non Terminating faults were reported by x2 or ppe
3	WHITELISTING-REGISTER-FAILURE	Whitelisting register failure	x2	Non Terminating Fault, Clear will also be sent when issue is cleared	YES
4	WHITELISTING-DEREGISTER-FAILURE	Whitelisting deregister failure	x2	Future work Item	NA

Table 6. Reports Tasks Issue Table

SL	Event Type	Event Trigger	Reported From	Report Type	Are Reports Cleared
1	ActivateTask Failure Towards UPF	ActivateTask Failure Towards UPF as part CCTF/IRITF Activate Task Request	x1	Warning	NO
2	DeactivateTask Failure Towards UPF	DeactivateTask Failure Towards UPF as part CCTF/IRITF Deactivate Task Request	x1	Warning	NO
3	Service Unavailability	When x2 or ppe pod goes down	x1	[Future work item]: Any pending non	YES, if Terminating or

Lawful Intercept

SL	Event Type	Event Trigger	Reported From	Report Type	Are Reports Cleared
				terminating fault is cleared if set by x2 or ppe	Non Terminating faults were reported by x2 or ppe
4	GetTaskDetails failure towards UPF	GetTaskDetails failure towards UPF as part of GetTaskDetails request	x1	Warning	NO
5	WHITELISTING-FAILURE	Whitelisting failure of the peer (UPF) for sending either ActivateTask / DeactivateTask	x1	Warning	NO
6	DESTINATION-NOT-CONNECTED	For an XID, no connection for corresponding DID is available	x2	Non Terminating Fault, Clear will also be sent when issue is cleared	YES
7	DESTINATION-SOCKET-ERROR	For an XID, for a corresponding socket for a DID, some generic error, such as write failed, etc.	x2	Non Terminating Fault, Clear will also be sent when issue is cleared	YES

Whitelisting

Whitelisting is required for LI-X1 initiated egress messages. LI-X1 microservice sends a register message to istio-config-agent for each configured UPF/LIPF. Register message is sent before microservice initiates first message towards UPF/LIPF.

In order to select between secure and non-secure transport, namely https / http and tcp-tls / tcp, application would initiate this register API towards istio-cfg-agent accordingly depending on the **CNA_LIX1_TO_LIPEER_TLS_DISABLED** environmental flag set. This feature is currently only during startup, which means during fed deployment, you need to decide if it needs to be secure/non-secure mode of transport.

LI-X1 pod environment variable **CNA_LIX1_TO_LIPEER_TLS_DISABLED** can be set during the deployment of the federation using the **liX1ToLipeerTLSDisabled** helm variable defined in the federation values.yaml.

X2 Task Stats Push Procedure - CPF

This procedure is used by LI-X1 and LI-X2 microservices to collect the Amount of X2 PDU data collected.

1. It is assumed that LIPF initiated Activate Task is done
2. LI-X2 and Call control microservice learnt the Task details via ETCD notification.
3. LI-X2 starts the Task stats timer for the group of XIDs instead of starting timer for each XID. It will make sure all XIDs related statistics are reported to LI-X1 within fixed interval, by default which is 60 secs.
4. When Call control posts the IRI event to LI-X2 microservice to dispatch the event to MDF server, LI-X2 calculates the amount of x2 data and collects the statistics in inmemory.
5. Once group relates Tasks stats timer is fired all the statistics related to XIDs belongs that group are reported to LI-X1 microservice over REST interface. In the single message up to 512 XIDs related statistics can be reported.
6. On receiving the stats report, LI-X1 atomically updates the statistics in Mongo Db atomically.
7. One successful stats reporting to LI-X1, LI-X2 will reset the stats and start collecting next interval delta statistics.
8. And this cycle continues.

LI-X2

The LI-X2 microservice is responsible to send the IRI event to a Mediation and Delivery Function (MDF). It is one of the microservice included in Lawful interception functionality and is a common microservice that is added to different Network Functions to enable lawful interception functionality. LI_X2 is deployed along with Control Plane Function (CPF) federation like SMF/AMF.

LI-X2 communicates with LI-X1 to receive the Task details and Destination Details.

The LI-X2 microservice gets IRI events based on correlation id from call control microservice. It receives a list of target-ids along with json format xIRI event data. It can fan-out xIRI event for the correlation id, XID, and Domain ID. LI-X2 generates a unique sequence number and encodes the xIRI events in the required formats as expected by MDF. LI-X2 maintains target-ids based subscriber. Currently, LI-X2 supports ASN.1 format in BER encoding.

LI-X2 is responsible to create and maintain connections with MDF. It also increments x2 event statistics and periodically sends them to LI_X1.

LI-X2 microservice interacts with the following microservices within the federation for LI functionality:

LI-X1

- LI-X2 microservice uses the REST API exposed by LI-X1 microservice to periodically send statistics for set of XIDs.
- LI-X2 microservice uses the REST API exposed by LI-X1 to receive REST based notifications for activating/modifying/deactivating tasks for corresponding XIDs.
- Subscribes for Task details and Destination Details ETCD events published by LI-X1.

Call control

- Exposes the REST API to handle IRI event for a Correlation ID.

Features

3GPP Specification Features

Supports the following call flows in Phase 0 LI functionality:

- [Creating and Maintaining the Destination Details at X2](#)
- [Creating and Maintaining the Task Details at X2](#)
- [IRI Event](#)
- [KeepAlives](#)
- [Modify Task](#)
- [Report Issues](#)

Non 3GPP Specification Features

- [Alerts](#)
- [ELF Core Files Generation](#)
- [Enabling or Disabling Task Sensitive Logs](#)
- [Encryption](#)
- [Node Failure Resiliency Through POD Spread Policy](#)
- [Publication](#)
- Provisioning

- [Switchback Timer](#)

Other Features

- [Target ID Based Subscriber Management](#)
- [Sequence Number Generation](#)
- [Fan-out Functionality](#)
- [Connection Management Towards MDF Server](#)
- [Statistics Collection at X2](#)
- [Local DID Config Related Connection Management Towards MDF Server](#)
- [Multi-Instance Support](#)
- [Support for Conditional Attributes Network Function ID and Interception Point ID in X2 PDU Header](#)
- [In Memory Buffering of X2 Events](#)

Security Considerations of PaaS Data

LI data is critical in nature and data stored in PaaS components must be in encrypted format and only the LI applications must understand the data in etcd and DB using the secret keys generated by common fed-level pass-phrase.

Configuration

LI-X2 microservice does not own any configuration.

LI-X2 microservice subscribes to the configuration of LI-X1 microservice for knowing the operator-identifier to be filled in the X2 PDUs towards MDF.

Dashboard Description

Lawful interception functionality is highly secured, sensitive, and does not generate any information on NF dashboard.

Related Topics

- [External Interfaces](#)

- [LI Egress Gateway Setup from fed-smf to External fed-upf](#)
- [LI Egress Gateway Setup from fed-smf to MDF2](#)
 - [LI Egress Gateway Setup from fed-smf to MDF2 Primary \(TLS Protocol\)](#)
 - [LI Egress Gateway Setup from fed-smf to MDF2 Secondary \(TLS Protocol\)](#)
- [LI Ingress Gateway Setup for fed-smf](#)
- [LI Ingress Gateway Setup for fed-upf](#)
- [Enabling or Disabling TLS Support](#)

Activate Task

1. LI-X2 subscribes for Etcd Task table(xid_table) at bootup.
2. LI-X1 adds etcd entry when a new Task is added.
3. LI-X1 sends notify to X2 when a new Task is added.
4. LI-X2 receives etcd notification for the Task.
5. All LI-X2 creates string from ordered list of target ids for the Task and one-way hash of the string gives obfuscated target ID and updates in-memory data structure.
6. When LI-X2 receives activate task notify, it creates a DB entry obfuscated target ID. Go to step 9, if target ID subscriber create is successful.
7. If DB entry already exist for obfuscated target ID, then create fails. LI-X2 reads and locks DB entry.
8. LI-X2 updates obfuscated target ID record with a new XID of the task.
9. LI-X2 save target ID subscriber and release the lock.

Alerts

Destination related alerts for DID related issues - Parallel to LIPF reports. Refer auto-generated format for more information.

- TCP-CONNECTION-FAILURE-PRIMARY-PEER(set/clear)
- TCP-CONNECTION-FAILURE-SECONDARY-PEER(set/clear)
- WHITELISTING-FAILURE-PRIMARY-PEER(clear/set)
- WHITELISTING-FAILURE-SECONDARY-PEER(set/clear)

CNF Readiness and Virtual Restart

With this feature, LI-X2 microservice participates in the CNF Readiness and Virtual functionality of federation. If it is part of the federation, it publishes **present** when it comes up. After receiving Service Available indication with status **up** from nf-monitor and after LI related etcd tables are loaded it publishes itself with **ready** to the consul.

Connection Management Towards MDF Server

LI-X2 is responsible to create/maintain/delete connection towards MDF server. Each LI-X2 pod instance provisions the MDF server IP and creates TCP connection for the DID, when CreateDestination notification from etcd is received. It deletes TCP connection, when RemoveDestination notification from etcd is received. LI-X2 opens the connections for DIDs that contains delivery type X2Only or X2andX3.

Create Destination

1. LI-X2 subscribers for Etcd Destination table(DID_table) at boot-up.
2. LI-X1 adds Etcd entry when a new Destination is added.
3. LI-X2 receives Etcd notification for the Destination.

LI-X2 creates a TCP socket and connect to destination IP address for the DID if the delivery type is X2Only or X2AndX3.

Remove Destination

1. LI-X2 subscribers for Etcd Destination table at boot up.
2. LI-X1 removes etcd entry when an existing Destination is removed.
3. LI-X2 receives etcd notification for the Destination removal.
4. LI-X2 close the TCP socket.

Creating and Maintaining the Destination Details at X2

The LI-X2 microservice subscribes to the Destination table in the path `/li_x1/did_table`, where the key is Obfuscated Destination ID.

X1 creates, modifies, and deletes the entries in the destination table based on the CreateDestination, ModifyDestination, and RemoveDestination messages received by X1, and the notifications is received by X2.

X2 maintains a map of <DID, DestinationSchema> in local memory.

Upon receipt of the notification to "Create" of a new key in Destination table, the receiving POD

(s) opens up a connection towards that destination. LI-X2 establishes a TCP connection towards MDF for DID if the delivery type is X2Only or X2AndX3. If destination connection is unsuccessful, then X2 issues an DestinationIssue towards X1.

- [Create Destination](#)
- [Remove Destination](#)

Creating and Maintaining the Task Details at X2

LI-X2 microservice subscribes to the Task table in the path /li_x1/xid_table, where the key is the obfuscated XID.

X1 creates, modifies, and deletes the entries in the task table based on the ActivateTask, ModifyTask, and DeactivateTask message received, the notifications of which is received at X2.

X2 maintains a map of <XID, TaskSchema> in local memory.

Upon receipt of the notification for Create of a new key in the task table, the receiving POD(s) creates an entry in the local map, to be referenced later, when an xIRI event is received from the CC.

- [Activate Task](#)
- [Deactivate Task](#)
- [Modify Task](#)

Deactivate Task

1. LI-X2 subscribes for Etcd Task table(xid_table) at bootstrap.
2. LI-X1 removes etcd entry when a Task is deactivated.
3. LI-X1 sends notify to X2 when a Task is removed.
4. LI-X2 receives etcd notification for the removal of Task.
5. LI-X2 creates string from ordered list of target ids for the Task and one-way hash of the string gives obfuscated target id and updates in-memory data structure.
6. When LI-X2 receives REST based deactivate task notify, it reads and locks DB entry for the obfuscated target id.
7. LI-X2 updates obfuscated target ID record with removing XID of the task.

8. If XID is the last XID in obfuscated target ID record, then LI-X2 deletes obfuscated target ID record, otherwise it writes updated record and releases lock.

DSCP Marking

The LI-X2 microservice marks the configured DSCP on X2 interface towards the MDF.

Sample DSCP Configuration

```
li-x1 dscp-marking x2 df
```

Sample Show Command

```
show running-config li-x1 dscp-marking
```

```
li-x1 dscp-marking x2 df
```

ELF Core Files Generation

When a process inside a container fails in CNA environment, it results in the generation of mini- dump back-trace files that are useful in identifying the back-trace process at the time of failure. However, it does not allow to debug the possible root cause of the failure. ELF core files equip the user with a snapshot of the CPU and memory state of the process at the time of the failure.

ELF core files generation is enabled for LI-X2 pods and core files are written to the Host VM file system (the K8s worker node). This addresses the issue in generating ELF core files along with mini-dump files in CNA environment. The management of these core files generated on the K8s worker nodes improves the ability to debug the application failure.

Enabling or Disabling Task Sensitive Logs

Logs contain sensitive information (such as Task level info example: XID, Target info, Correlation Id etc) are categorized separately to ensure it is not logged into regular CNA observability by default.

To generate the sensitive logs using li-admin level, refer LI Logging section in [LI-X1](#).

Enabling or Disabling TLS Support

By default, when NF interacts with the L1_X2 interface peer MDF2, TLS/TCP is used. To enable/disable the TLS/TCP, follow the below configuration.

Sample Configuration

```
li-x1 x2-x3-over-tls-egress enabled
```

Sample Show Command

```
show running-config li-x1 x2-x3-over-tls-egress
```

```
li-x1 x2-x3-over-tls-egress enabled
```

Encryption

LI-X2 decrypts etcd data for Task (xid_table) and Destination (did_table) tables. TargetID subscriber is stored in an encrypted format by LI-X2 and it's not a human-readable format. Both etcd and DB also stores the version of a key at the time of encrypting and decrypting is the same.

Note: The current release does not support the change of encryption key or enable/disable encryption at run time.

It is recommended to clean up all LI-X1/LI-X2 etcd/DB data, change encryption key or enable/disable encryption and then delete LI-X1/LI-X2 and application PODs (all application pod which is using LI data).

External Interfaces

Towards MDF

This is a mandatory interface. The interface shall comprise of connection(s) towards the Destination where the interception product (IRI) shall be sent as a stream of X2 PDUs.

But in the current implementation, only the type "IPAddressAndPort" shall be supported. The IP address could be an IPV4 or an IPV6 address, but not both.

The transport on this interface shall be TLS/TCP.

For each destination, each X2 POD shall create a TCP connection towards that destination via the Egress LI GW.

The Egress LI GW shall have to be programmed to allow a TLC/TCP connection to an external host whenever a new destination is received. This shall be achieved by using the services of istio- config-agent.

Before opening a TLS/TCP connection towards a destination, X2 shall call a "Register" API of istio- config-agent to configure rules to allow connection to the destination.

On success, X2 shall open a TLC/TCP connection to that destination.

Fan-out Functionality

Fan-out functionality in current context refers to the capability of LI-X2 microservice to generate and send multiple PDUs towards MDF2 server for a single IRI message received from call control, based on the number of tasks that are interested in intercepting a given target. A given target is identified by list of TargetIDs that is received in an IRI message. Fan-out is performed to DIDs that contains delivery type X2Only or X2andX3.

LI-X2 receives set of TargetIDs and correlation ID from call control microservice. It supports fan-out functionality when it receives IRI events.

LI-X2 generates fan-out request as follows:

1. Create a targetIDs string from the target IDs in list of target IDs received from call-control and fetch the record from the DB corresponding to the targetIDs key.
2. Iterate through each XID in target ID subscriber in step 1.
3. Iterate through each correlation ID for each XID in step 2.
4. Iterate through each DID of the XID, generate sequence, and send IRI event.

HTTP Retry on Failure on ETCD Interface

ETCD Interface

Towards the etcd server transactions, LI-X2 retries the message for the HTTP status response codes 500, 502, 503, 504, 429, and 460.

LI-X2 retries for three times for the negative response codes. If the message is not successful, LI- X2 transaction fails.

In Memory Buffering of X2 Events

This feature supports buffering of the X2 PDU events for a DID in-memory if the PDU delivery to MDF has failed due to a connection unavailability (initial connection or in between).

This feature:

- Replays buffered PDUs once the path to either primary or secondary MDF comes up.
- Provides configuration option to provide a queue size per DID per pod with min, max and default values (0 is disabled).
- Delivers the PDUs whenever either the primary or secondary path for that DID has come up, on FIFO basis.

Configuration

Execute the following command to configure the buffer size:

```
li-x1 local-profile intercept-buffer-size [cpx2, upx2, upx3] [size]
```

Where;

- cpx2: for control plane X2(IRI) events
- upx2: for user plane X2(PDSR|PDHR) events
- upx3: for user plane X3(CC) events
- Size: This has to be mentioned in terms of KiloBytes.
- "0" means that buffering is disabled.
- Default value : 102,400
- Max size for cpx2 : 1,048,576 KB
- Max size for upx2 : 1,048,576 KB
- Max size for upx3 : 10,485,76 KB

Example Configuration

```
li-x1 local-profile intercept-buffer-size cpx2 1048576
```

Statistics

Following are the statistics added to monitor the buffering of X2 events:

- inmemorybuffer_pdu_added
- inmemorybuffer_pdu_removed
- inmemory_buffer_pdu_dropped
- inmemory_buffer_pdu_remove_all

```
"inmemory_buffer_pdu_added": {
```

```
  "description": "Count of the PDUs added to the inmemorybuffer", "help" :  
  "Count of the PDUs added to the inmemorybuffer", "labels": ["DID",  
  "position"]
```

```
},
```

```
"inmemory_buffer_pdu_removed": {
```

```

"description": "Count of the PDUs removed from the inmemorybuffer", "help"
: "Count of the PDUs removed from the inmemorybuffer", "labels": ["DID",
"position"]
},

"inmemory_buffer_pdu_dropped": {

"description": "Count of the PDUs dropped from the inmemorybuffer",

"help": "Count of the PDUs dropped from the inmemorybuffer", "labels":
["DID", "position"]

},

"inmemory_buffer_pdu_remove_all":{

"description": "Count of times removeAllPdus is called for the inmemory-
buffer",

"help": "Count of times removeAllPdus is called for the inmemorybuffer",
"labels": ["did", "reason"]
}

```

IRI Event

1. Call control microservice sends IRI event to LI-X2 microservice.
2. LI-X2 creates string from ordered list of target IDs for the task and one-way hash of the string generates obfuscated target ID.
3. LI-X2 reads and locks DB entry for the obfuscated target ID.
4. LI-X2 generates sequence number:
 - Sequence number starts with 0, if the event type is start and entry is added in target ID subscriber.
 - Sequence number is incremented by 1 for continue events and same is updated in target ID subscriber.
 - Sequence number is incremented by 1 and entry is deleted from target ID subscriber.
5. LI-X2 encodes PDU as per format and sends PDU to MDF server.
6. LI-X2 writes updated record and releases lock.

SMF**Table 7. List of 5G IRI messages supported by LI-X2 microservice for the SMF**

Event	NF	Encoding	Spec
pDUSessionEstablishment	smf	ber	33.128
pDUSessionModification			
pDUSessionRelease			
startOfInterceptionWithEstablishedPDUSession			
unsuccessfulSMProcedure			

PGW/SMF+PG**Table 8. List of 4G EPS IRI messages supported by LI-X2 microservice for the PGW/SMF+PGW**

Event	NF	Encoding	Spec
bearerActivation	pgw/ pgw+smf	ber	33.108
startOfInterceptionWithActiveBearer			
bearerModification	pgw+smf	ber	33.108
bearerDeactivation			
uERequestedBearerResourceModification			
uERequestedPDNConnectivity			
uERequestedPDNDisconnection			
pDPContextActivation	pgw+smf	ber	33.108
startOfInterceptionWithPDPCContextActive			
pDPContextDeactivation			
pDPContextModification			
packetDataHeaderInformation			

AMF**Table 9. List of 5G IRI messages supported by LI-X2 microservice for the AMF**

Event	NF	Encoding	Spec
AMFRegistration	amf	ber	33.128
AMFDeregistration			

Event	NF	Encoding	Spec
AMFLocationUpdate			
AMFStartOfInterceptionWithRegisteredUE			
AMFUnsuccessfulProcedure			

KeepAlives

LI-X2 sends keepalive periodically (after every timer-p1-x2x3) over the X2 interface and handles keepAlive-ack response PDU. It resets TCP connection towards MDF if no keepAlive-ack is received till time-p2-x2x3.

KeepAlive (Time-P1-X2X3)

If LI-X2 is part of fed-smf/fed-amf, then it sends KeepAlive to all primary and secondary destinations after every Time-P1-X2X3 interval. Time-P1-X2X3 is defined in two places:

- Local configuration
- did-config

If values are configured at both places then the value in did-config takes precedence. If Timer- P1-X2X3 is configured in did-config, then the did-config value is used but If Time-P1-X2X3 is not configured in did-config, then the local-profile value is used. If config is modified then new the timer value is applicable after the current timer expires.

KeepAlive (Time-P2-X2X3)

Time-P2-X2X3 is supported on fed-smf/fed-amf. LI-X2 resets TCP connection towards MDF if no keepAlive-ack received till time-p2-x2x3 duration. Time-P2-X2X3 is defined in two places:

- Local configuration
- did-config

If values are configured at both places then the value in did-config takes precedence. If Timer- P2-X2X3 is configured in did-config, then the did-config value is used but If Time-P2-X2X3 is not configured in did-config, then the local-profile value is used. If config is modified then the new timer value is applicable after the current timer expires.

Local DID Config Related Connection Management Towards MDF Server

LI-X2 is responsible to create/maintain/delete connections towards the MDF server for the locally configured destinations. LI-X2 provisions MDF server IP and creates TCP connection for the DID, when LI-Admin commits the local DID configuration. It deletes TCP connection when the admin

removes the local DID configuration. For the locally provisioned DID, if any primary server related details are not configured, LI-X2 does not establish the TLS/TCP connection towards the MDF server. LI-X2 opens the connections for DIDs that contains delivery type X2Only or X2andX3.

Modify Task

1. LI-X2 subscribes for Etcd Task table(xid_table) at boot-up.
2. LI-X1 updates Etcd entry when a Task is updated.
3. LI-X2 receives Etcd notification for the ModifyTask.
4. All LI-X2 creates string from ordered list of target IDs for the task and one-way hash of the string generates obfuscated target ID and updates in-memory data structure.
5. LI-X1 sends notify to X2 when an existing task is modified.
6. When LI-X2 receives REST based modify task notification, LI-X2 reads and locks DB entry.
7. LI-X2 updates the DBentry corresponding to the obfuscated target ID. Typical parameters updated can be Delivery type and DID list of the task. On delivery type change, LI-X2 invalidates the relevant session data parameters. For example, X2andX3/X2Only→X3Only delivery type change, LI-X2 invalidates the sequence numbers and updates the DID list addition/removal so that IRI replication can be started/stopped for the DIDs.
8. LI-X2 saves target ID subscriber and releases the lock

Multi-Instance Support

LI-X2 supports the call processing with multiple instances. Support is added for all the messages. It also supports dynamic scaling of PODs using horizontal pod auto scaling (HPA). Currently, HPA is supported based on the CPU.

Node Failure Resiliency Through POD Spread Policy

LI-X2 supports node failure resiliency through POD spread policy used in Fed-level default constraints.

To handle the worker node failure, there must be multiple application pods running across multiple nodes to manage the traffic. It is recommended to enable HPA and to have at least two LI-X2 pod instances running for an application.

Note: Without multiple LI-X2 pod instances, spreading the pods across multiple worker nodes do not work.

Operational CLI

Peer Status

Following command is used to query the peer connection status.

```
show li-x2 peer-status summary
```

Table 10. Peer Status Parameters

Parameter	Description
DID	Destination identifier
MDFAddress	Peer MDF IP address and port.
MDF2Address(P/S)	Peer MDF IP address and port details. Indicates Primary or Secondary MDF2 address.
PodLinkStatus(up/total)	Aggregated link status.
LinkStatus	UP/Down
PodName	Pod name from which the connection is opened.
Reason	If LinkStatus is Down, describes the reason.

Sample Output

Note: Status can be queried for particular destination or all destinations.

```
show li-x2 peer-status summary
```

```
DID      MDF2Address  PodsLinkStatus (up/total)
b0ce308c-aa22-42bd-a27b-287bcb5b3468  10.163.209.139:7117
1/1
TLS Status: True
```

```
show li-x2 peer-status verbose
```

Lawful Intercept

DID MDF2Address (P/S) LinkStatus

PodName Reason

b0ce308c-aa22-42bd-a27b-287bcb5b3468 10.163.209.139:7117 (P)

Up li-x2-2-4-0-219-main-int-dbg-7d9b4c58dc-wrfms

N/A

TLS Status: True

Keep-Alive Status

Following command is used to query the Keep alive status to MDF2.

show li-x2 keep-alive x2 summary

Table 11. Keep Alive Status Parameters

Parameter	Description
DID	Destination identifier
MDF2Address	Peer MDF IP address and port details.
MDF2Address (P/S)	Peer MDF IP address and port details. Indicates Primary or Secondary MDF2 address.
LinkStatus	UP/Down
PodName	Pod name from which the connection is opened.
Reason	If LinkStatus is Down, describes the reason.
PodID	Pod name
Time_P1	Time interval by which the KeepAlive PDU is sent.
Time_P2	Time interval by which if no KeepAlivePDU_ack is received connection will be reset.
KeepAlivePDU_status	KeepAlive status OK or ERROR.
KeepAlivePDU_sent	Timestamp at which the KeepAlive PDU is sent.
KeepAlivePDU_ack	Timestamp at which the last KeepAlive successful PDU response is received.

Sample Output

Note: Status can be queried for particular destination or all destinations.

show li-x2 keep-alive x2 summary

```
Time_P1 Time_P2 KeepAlivePDU_status(OK/FAILED) TotalPods b0ce308c-aa22-42bd-a27b-287bcb5b3468 10.163.209.139:7117
```

```
60 180 1/0 1
```

```
TLS Status: True
```

show li-x2 keep-alive x2 did b0ce308c-aa22-42bd-a27b- 287bcb5b3468

```
DID MDF2Address(P/S) PodID Time_P1 Time_P2
```

```
KeepAlivePDU_status KeepAlivePDU_sent KeepAlivePDU_ack
```

```
b0ce308c-aa22-42bd-a27b-287bcb5b3468 10.163.209.139:7117(P)
```

```
li-x2-2-4-0-219-main-int-dbg-7d9b4c58dc-wrfms 60
```

```
180 OK Tue Nov 16 12:37:45 2021:1 Tue Nov
```

```
16 12:37:45 2021:1
```

```
TLS Status: True
```

Publication

LI-X2 sends multiple publication messages when it receives configuration. These publication messages are sent at the boot-up time after configuration is received. Publication messages are also sent if new did-configs are added and configuration is received at LI-X2 microservice. LI-X2 sends a publication message with a unique publication service id for each external-Host it needs to connect.

LI-X2 sends Task and Destination related issues to LI-X1:

- Destination Issue: Provisioning failure, TCP connection failure, and TCP socket failure.
- Task Issue: xIRI event failure is reported by LI-X2 to LI-X1.

ReportTaskIssue

LI-X2 uses two triggers to send task issues towards LI-X1, **SET** and **CLEAR**. SET is used when a fault is reported to LI-X1 and CLEAR is used to clear a previously SET fault from LI-X1.

For task issues, for the fault-type non-terminating the **SET** is supported in two situations:

- For an incoming IRI dispatch procedure, while trying to send LI-X2 PDU towards MDF, LI- X2 identifies that a DID connection does not exist, LI-X2 sends a **SET** trigger to LI-X1- Reported as Non-Terminating fault.
 - Upon subsequent IRI message from call control, whenever for this corresponding domain object, a DID connection is identified to be up, LI-X2 sends a **CLEAR** trigger to LI-X1.
- For an incoming IRI dispatch procedure, while trying to send LI-X2 PDU towards MDF, LI- X2 identifies that a DID connection exists, but some error occurred on network while sending the packet, LI-X2 sends a **SET** trigger to LI-X1 - Reported as Non-Terminating fault.
 - Upon subsequent IRI message from call control, whenever for this corresponding domain object, packet is successfully sent on the corresponding connection, LI-X2 sends a **CLEAR** trigger to LI-X1.

Table 12. Report Task Issue

SL	Event Type	Event Trigger	Reported From	Report Type	Fault Clear Sent?
1	DESTINATION-NOT-CONNECTED	For an XID, no connection for corresponding DID is available.	X2	Non Terminating Fault, Clear will also be sent when issue is cleared	YES
2	DESTINATION-SOCKET-ERROR	For an XID, for a corresponding TCP socket for a DID, some generic error occurred, such as write failed, etc.	X2	Non Terminating Fault, Clear will also be sent when issue is cleared	YES

LI-X2 also supports ReportTaskIssue of the type “warning”.

- When in-memory buffering is enabled for a DID and the in-memory buffer is full, if an IRI event is received during that time, Li-X2 will generate a ReportTaskIssue of the type warning.
- Since this report is of the type “warning”, no CLEAR is reported to LIPF.

ReportDestinitionIssue

LI-X2 uses two triggers to send destination issues towards LI-X1, **SET** and **CLEAR**. SET is used when a fault is reported to LI-X1 and CLEAR is used to clear a previously SET fault from LI-X1.

For destination issues, the **SET** is supported in 4 situations:

- Whenever LI-X2 identifies that a TCP link for the primary connection is down, LI-X2 sends a **SET** trigger to LI-X1- Reported as Non-Terminating fault.

- Whenever the TCP link comes up for the primary connection, LI-X2 sends a **CLEAR** trigger to LI-X1.
- Whenever LI-X2 identifies that a TCP link for the secondary connection is down, LI-X2 sends a **SET** trigger to LI-X1- Reported as Non-Terminating fault.
 - Whenever the TCP link comes up for the secondary connection, LI-X2 sends a **CLEAR** trigger to LI-X1.
- Whenever LI-X2 identifies that whitelisting fails for a primary MDF for a Destination, LI-X2 sends a **SET** trigger to LI-X1- Reported as Non-Terminating fault.
 - At a later stage, whenever for this MDF, if whitelisting is allowed, LI-X2 sends a **CLEAR** trigger to LI-X1.
- ▪ Whenever LI-X2 identifies that whitelisting fails for a secondary MDF for a Destination, LI-X2 sends a **SET** trigger to LI-X1- Reported as Non-Terminating fault.
 - At a later stage, whenever for this MDF, if whitelisting is allowed, LI-X2 sends a **CLEAR** trigger to LI-X1.

Table 13. Report Destination Issue Table

SL	Event Type	Event Trigger	Reported From	Report Type	Fault Clear Sent?
1	TCP-CONNECTION-FAILURE-PRIMARY-PEER	Tcp link failure towards MDF on primary configured address	X2	Non Terminating Fault, Clear will also be sent when issue is cleared	YES
2	TCP-CONNECTION-FAILURE-SECONDARY-PEER	Tcp link failure towards MDF on secondary configured address	X2	Non Terminating Fault, Clear will also be sent when issue is cleared	YES
3	PROVISIONING-FAILURE-PRIMARY-PEER	REGISTER request towards Istio config agent either times-out, or negative response from Istio config agent for the request is received, resulting into provisioning	X2	Non Terminating Fault, Clear will also be sent when issue is cleared	YES

Lawful Intercept

SL	Event Type	Event Trigger	Reported From	Report Type	Fault Clear Sent?
		denied for the external host for primary MDF address			
4	PROVISIONING-FAILURE-SECONDARY-PEER	REGISTER request towards Istio config agent either times out, or negative response from Istio config agent for the request is received, resulting into provisioning denied for the external host for secondary MDF address	X2	Non Terminating Fault, Clear will also be sent when issue is cleared	YES

Security Context for the Containers

With this feature, LI-X2 pod security context is auto injected. A security context defines privilege and access control settings for a Pod or Container

Sequence Number Generation

LI-X2 is responsible to generate unique incremental sequence number for each IRI event. It generates unique sequence number for each unique combination of correlation ID, XID, and Domain ID. LI-X2 maintain last sequence number in Db. When call control sends IRI event, LI-X2 fetches subscriber from dB, and generates a new sequence number for each fan-out request, update, sends IRI event to MDF and update subscriber to Db.

LI-X2 follows below rule to generate sequence number for each unique combination of correlation ID, XID, and Domain ID:

1. Sequence number is started with 0 if event type is start and entry is added in target ID subscriber.
2. Sequence number is incremented by 1 for continue events and same is updated in target ID subscriber.
3. Sequence number is incremented by 1 and entry is deleted from target ID subscriber.

Statistics Collection at X2

Task statistics collection

LI-X2 is responsible to increment statistics for the corresponding task and provide it to LI-X1. LI-X1 aggregates the statistics for each task XID from all LI-X2 pods and maintain it in DB. For this release, LI-X2 maintains following statistics per XID:

- amount of X2 data in bytes

LI-X2 increments x2 stats on successful TCP write when it sends IRI event to MDF server. The task statistics is collected at the LI-X2 microservice instance level for every xIRI event, that is processed by that particular instance.

LI-X2 pushes the statistics to X1 for all the tasks in the X2 instance periodically.

LI-X2 byte count for the task in differential manner, as in it will reset the statistics to 0 after a successful push towards X1 and start counting from 0. In this way, each X2 instance stores the delta statistics from the last successful push to X1.

Stats Entry Creation for a Task

Whenever a notification is received for the "Create" of a new key in the task table, the receiving POD (s) creates an entry in the statistics table in the local memory.

Stats Entry Update

Whenever an xIRI event is received from the CC on the X2 instance, the byte count sent towards the destination as inferred from the task table for the received XID is added to the existing entry in the stats table.

If multiple PDUs are sent towards MDF2 server due to the fan-out functionality described above, then the stats are added for all the PDUs that are successfully sent.

Stats Entry Resetting to 0

At X2 instance, a timer is maintained which gets fired every 500 milliseconds. The tasks are divided into 128 buckets and every 500 milliseconds, stats of one bucket is pushed to LI_X1 on REST based interface.

Once a successful response is received, the stats entry is reset to 0 for all tasks of corresponding bucket.

Similarly, statistics for all tasks are synchronized every 64 seconds.

Stats Entry Deletion for the Task

Whenever a notification is received for the "Delete" of the key in the task table, the receiving POD pushes the pending statistics to X1 and upon success response, remove the entry from the stats table, and then delete the entry from the task table in the local memory.

* Whenever a task is modified, no change to the stats table is observed, since the XID remains the same.

Support for Conditional Attributes Network Function ID and Interception Point ID in X2 PDU Header

LI-X2 supports the conditional attributes Network Function ID (NFID) and Interception Point ID (IPID) in X2 PDU towards MDF.

NFID is populated with the NF address, which is registered with the NRF. IPID is populated with the POD IP, which is intercepting the IRI events. This is call control POD IP. LI-X2 receives the NFID and IPID on internal REST interface from call control along with IRI event. LI-X2 encodes these conditional attributes in X2 PDU header.

Switchback Timer

LI-X2 supports redundancy towards MDF for a DID in the form of primary and secondary destination TCP connections. Configure the secondary destination details as below:

- The CLI configuration under `li-x1 did-config <did config> path`.
- Programming through LIPF in the X1SecondaryDestination extension in the X1 CreateDestinationRequest message.

Once LI-X2 identifies the primary TCP link to the MDF is down, it switches to the secondary TCP link to send the subsequent X2 PDUs. When the primary TCP link comes up, the switchback timer enables the operator to allow the primary TCP link to stabilize before shifting to it to send subsequent X2 PDUs.

Configure the switchback timer in the following sections:

- DID specific: `li-x1 did-config <did config> switch-back-timer` (Default value is 90)
- Local-Profile: `li-x1 local-profile switch-back-time` (Default value is 90)

Value configured in `li-x1 local-profile switch-back-time` is a global value, applicable to all the destinations. However, if configured under both the above config paths, the DID specific value takes precedence.

LI-X2 waits for the configured seconds after the primary TCP link has come up to send the traffic. This is to have a stabilized (unbroken) primary TCP connection before shifting to it.

Note: To disable this feature, configure the switchback timer value as **0**. The Switchback time must be greater than keep-alive-time-p2.

Target ID Based Subscriber Management

Target ID is domain object for LI-X2 and LI-X2 maintains target ID subscriber. Obfuscated target id is the key to Target ID subscriber. LI_X2 maintains target ID subscriber in Db. All task with same target IDs is part of same target ID subscriber, so the target ID subscriber holds one or more XID task.

Unique obfuscated target id is generated as follows:

1. Iterate for each target ID in the list of target ID given in ETSI 103221-1 and check if target id is present in target IDs of XID. If present, then add target id name and value to string.
2. Do one-way hash of string generated above.

Target ID subscriber is added/removed/modified at following event:

1. LI-X2 adds a new subscriber when Task ID notification is received and obfuscated target ID for this XID task is not present at X2.
2. LI-X2 modify target subscriber when:
 - a. New XID notification is received and target ID subscriber for the obfuscated target ID is already present.
 - b. Delete XID notification is received and this XID is not last XID in target ID subscriber.
 - c. IRI event is received.
3. LI-X2 deletes target subscriber when last XID task delete notification is received for the obfuscated target ID.

Provisioning

Provisioning is required for LI-X2 initiated egress messages. LI-X2 microservice sends a register message to istio-config-agent for each of the configured primary and secondary destinations.

Register message is sent before microservice initiates first message towards MDF.

To select between secure and non-secure transport tcp-tls/tcp, the application would initiate this register API towards istio-cfg-agent depending on the **CNA_LIX2_MDF2_TLS_DISABLED** environmental flag set. This means during fed deployment one needs to decide if it needs to be a secure/non-secure mode of transport.

LI-X2 pod environment variable **CNA_LIX2_MDF2_TLS_DISABLED** can be set during the deployment of the federation using the **liX2Mdf2TlsDisabled** helm variable defined in federation values.yaml.

Certificate Configuration to Support TLS and mTLS for Istio Gateways for LI

Certificates can be configured by two ways:

1. via values.yaml of the fed-smf and
2. via fetching from a keyVault (AKV or hashicorp)

Following section provides details on Configuring the certificates to be used for TLS and MTLS traffic in ingress and egress at the time of deployment in respective fed's **values.yaml** file.

Note: Refer to respective fed's *interface/values.yaml* for certificate configuration.

```
globals:
  mesh:
    istio:
      ingressGwTLSLi:
        enabled: true
        # Set the tls mode
        # SIMPLE --> to enable only TLS
        # MUTUAL --> to enable mTLS (default)
        mode: MUTUAL
        userProvided:
          enabled: true
          cacert: |-
```

Lawful Intercept

<Copy & Paste the CA Certificate, please refer to below "Example for certificate and key format">

tlscrt: |-

<Copy & Paste the Server Certificate, please refer to below "Example for certificate and key format">

tlskey: |-

<Copy & Paste the Server Key, please refer to below "Example for certificate and key format">

egressGwTLSLi:

enabled: true

Set the tls and mtls mode

SIMPLE --> to enable only TLS

MUTUAL --> to enable mTLS
(default)

mode: MUTUAL

userProvided:

enabled: true

cacert: |-

<Copy & Paste the CA Certificate, please refer to below "Example for certificate and key format">

tlscrt: |-

<Copy & Paste the Server Certificate, please refer to below "Example for certificate and key format">

tlskey: |-

<Copy & Paste the Server Key, please refer to below "Example for certificate and key format">

Following section provides details on Configuring the certificates to be used for TLS and MTLS traffic in ingress and egress using the certManager configuration using the fed's values.yaml.

Note: Refer to respective fed's *interface/values.yaml* for certificate configuration.

```
globals:
  mesh:
    istio:
      ingressGwTLSli:
        enabled: true <- Please set it to true when TLS has
        to be enabled for ingress LI traffic
        mode: MUTUAL
        certManager:
          enabled: false # Please set it to false when AKV is
          used
          certDuration: 2160h #90 days
          renewBefore: 648h #27 days
          organization: affirmednetworks
          issuerName: affirmed-clusterissuer-li
          issuerType: vault
        userProvided:
          enabled: false
          cacert: |-
          tlscrt: |-
          tlskey: |-
      egressGwTLSli:
        enabled: false # Please set it to true when TLS has
        to be enabled for egress LI traffic
```

Lawful Intercept

```
# Set the tls and mtls mode

# SIMPLE --> to enable only TLS

# MUTUAL --> to enable mTLS (default)

mode: MUTUAL

certManager:

  enabled: false

  certDuration: 2160h #90 days

  renewBefore: 648h #27 days

  organization: affirmednetworks

  issuerName: affirmed-clusterissuer-li

  issuerType: vault

userProvided:

  enabled: false

  cacert: |-

  tlscrt: |-

  tlskey: |-
```

Configuration Examples

This section provides configuration examples for:

- [TLS/HTTPS Without Mutual TLS](#)
- [TLS/HTTPS With Mutual TLS](#)
- [TCP/HTTP \(non-TLS\)](#)

TLS/HTTPS Without Mutual TLS

Configure the target LI server as TLS or HTTPS with the below configuration:

- pod-istio_gw_li: ingressGateway.enabled: true (enables ingressGateway)
- pod-istio_gw_li: egressGateway.enabled: true (enables egressGateway if required (optional))
- global.mesh.istio.ingressGwTLSLi.enabled: true (enables ingressGateway configuration for tls)
- global.mesh.istio.ingressGwTLSLi.mode: SIMPLE
- global.mesh.istio.ingressGwTLSLi.userProvided.enabled: true
- global.mesh.istio.ingressGwTLSLi.userProvided.cacert: <configure CA certificate here>
- global.mesh.istio.ingressGwTLSLi.userProvided.tlsCRT: <configure Server certificate here>
- global.mesh.istio.ingressGwTLSLi.userProvided.tlskey: <configure Server Key here>
- global.mesh.istio.egressGwTLSLi.enabled: true (enables egressGateway configuration for tls)
- global.mesh.istio.egressGwTLSLi.mode: SIMPLE
- global.mesh.istio.egressGwTLSLi.userProvided.enabled: true
- global.mesh.istio.egressGwTLSLi.userProvided.cacert: <configure CA certificate here>
- global.mesh.istio.egressGwTLSLi.userProvided.tlsCRT: <configure Server certificate here>
- global.mesh.istio.egressGwTLSLi.userProvided.tlskey: <configure Server Key here>

TLS/HTTPS With Mutual TLS

Configure the target LI server as TLS or HTTPS with mutual TLS as below:

- pod-istio_gw_li: ingressGateway.enabled: true (enables ingressGateway)
- pod-istio_gw_li: egressGateway.enabled: true (enables egressGateway if required (optional))
- global.mesh.istio.ingressGwTLSLi.enabled: true (enables ingressGateway configuration for tls)
- global.mesh.istio.ingressGwTLSLi.mode: MUTUAL
- global.mesh.istio.ingressGwTLSLi.userProvided.enabled: true
- global.mesh.istio.ingressGwTLSLi.userProvided.cacert: <configure CA certificate here>
- global.mesh.istio.ingressGwTLSLi.userProvided.tlsCRT: <configure Server certificate here>
- global.mesh.istio.ingressGwTLSLi.userProvided.tlskey: <configure Server Key here>
- global.mesh.istio.egressGwTLSLi.enabled: true (enables egressGateway configuration for tls)
- global.mesh.istio.egressGwTLSLi.mode: MUTUAL
- global.mesh.istio.egressGwTLSLi.userProvided.enabled: true

- global.mesh.istio.egressGwTLSLi.userProvided.cacert: <configure CA certificate here>
- global.mesh.istio.egressGwTLSLi.userProvided.tlsrct: <configure Server certificate here>
- global.mesh.istio.egressGwTLSLi.userProvided.tlskey: <configure Server Key here>

TCP/HTTP (non-TLS)

Configure the target LI server as TLS or HTTP with non-TLS as below:

- pod-istio_gw_li: ingressGateway.enabled: true (enables ingressGateway)
- pod-istio_gw_li: egressGateway.enabled: true (enables egressGateway if required (optional))

Example Configuration

The following section provides an example configuration:

```
cacert: |-
-----BEGIN CERTIFICATE-----
MIIFezCCA2OgAwIBAgIDEAISMA0GCSqGSIb3DQEBCwUAMFcxZzA1BjBGNVBAYTAlVT
MQ8wDQYDVQQIDAZEZW5pYWwxFDASBgNVBACMC1NwcmluZ2ZpZWxkMQwwCgYDVQK
DANEaXMxEzARBGNVBAMMCmxpdGVzdC5jb20wHhcNMjEwNDE1MTMzMNDMxWhcN MzEw
NDEzMjMzMNDMxWjBBMQswCQYDVQQGEwJVUzEPMA0GA1UECAwGRGVuaWFsMQww CgYD
VQKQDANEaXMxEzARBGNVBAMMCmxpdGVzdC5jb20wggiMA0GCSqGSIb3DQEB AQUA
A4ICDwAwggIKAoICAQDIY8JMwxElMod0iA8WtSwbWAD51o+3G9A0yYkZzRCMmmFo
Jt+AWAkIny/mY9wB5P6pLqJD3fpMfi9eirUS/u8azWtGLJUmPnHlH8n/BkRogXem
TlNzohaQVaH+cyLJ9mJ/Y71812b8kjPCYFvJyjf0QZv8I1jjk+WKVcS7h9FxAGgS
EJAKi2tjCFyjc973SP5aSCqfhvxnGUd+9JlVqlfEGEVjvsTAbcRT5ysbk2AkQeik
zviA8V17vHRRDs7p+O235ecpQU+abjg0i6HapOqDMRyBxqBLoX8sk5ZJbM5tERki
RCU6d2LZXZJkd91xMT+pz3GLQCJBVAGgzW46uxZC/o9ps0HzsJpsnYVeSQ1aIVp
d+xo8pfm1lXV0h6/k7LpddD7GgcWdAWuwz+5Hflm5wp96MWAj0YC6lVRSkaZFE0b
2PsKrpUgOYg5BietkJMULwIeZ3B8lsKdHB8dmyN5RTepN//SPuiW+G/PS5ISWpki
6B0JrquWEIX/N2xxQ39F7gP7twjgAZlIiGxXQ9gOkokj+fc6UhXq0P+IF0s3 S/jC
oMLAvA6u+xcHoKkXT4z/FcTEGFd91hklSMmrfzQfcbEufloJnmXKpErrfHsG 6Ety
jBpvJA5yKKyKkTUjlX3AYijIvmSZXZoZ6y3Yca3iDm4wpfiVtvMZH7QkQudKhQID
```

Lawful Intercept

AQABo2YwZDAdBgNVHQ4EFgQUYyg5CGktgLoddPZ1QHWJ1/x2Fzb4wHwYDVR0jBBgw
FoAUP8d/e2Ex8yl2RYKgVlgsgJrNKK0wEgYDVR0TAQH/BAgwBgEB/wIBADA0BgNV
HQ8BAf8EBAMCAYYwDQYJKoZIhvcNAQELBQADggIBACPP4o9dF02piT/zUxdC8y1A
b9CpzeqvFtElzRec8yiUX352Lc6K/G9oNofYjCWt4gxcb+iTlXzUDjMMzzEs+DsB
DvtAmmE0HE94YCBTCSjcn0SGJ6TDQhNn1btCT0YECE8IzGoKzIVUk5JXOrbXUT6j
zEeIIlvdh5DjRFIEFucp8Nqnd80tazGNdaW/dVmxanp2SL+02bDtxSGSfXr4gtD0
zL/kD8jdSBbhaqhMTokUC/5wkj8SUwFdQy7CsBBJb8j6MtU5NSQEWi4axTEcl65b
Pxp6VKc8LofMChrV3uoVj+rdLVcXYaUypK78gIqaa0Rl2S2D3oRi8ynRD2M+4VqZ
Q+iG36kFeKITbK9JSQ0EcYN6Wp6FBsanaXZQEQQggCLsO+DzQYgn4GM/MYEmHi7uu
6YlZgyo9BjlpZ2EAbsQdreYNX2pbfftAE5zLhTDnZNMDH5FMfE+L6UBatVoqj/6G
hO/GcZEJhcprRnvk77hPpYNsr4MwQrsOWU20y4aT+e8EsiNGtqbUbmMQ54bIGoZ5
iIkSXR7ifkAowqsPipB8lyRwHH0nybE3ZxDWj2ejJV0o/0L3AUH/cEgjjwxNO7KN
n05LnPjICTysxv3BI6Cy3eb64UaBAQbFhk3XqZl6T3nwcgw5TRkA+ZYH/b4NCK/5
pDMNTboqXrEph02poYfs

-----END CERTIFICATE-----

tlscrt: |-

-----BEGIN CERTIFICATE KEY-----

MIIEpAIBAACAQEAAnsXwQzLKMkn5XmFAwETIf5z3ZPscDbJ5jderH3Zu02yvRWBy
HP9/2ywqksWe3U+uiaDqYCfzrCqRBQBYLzk1D6E8hThVLnOL18Q6BbUOS1ecRo1n
kbleqMYJqCNOMrS9TM4CkDQmKNYgep5+fSnhms8e1W1vAH8fU1G1MroZD7dG7mEJ
QD/ZFBCfE0aHKuzB8eOWPcEBX2wdk6yY3FZzoQmaxNJ9t2PXumyvNcPWFd0zm2jV
f5q93q0Ibw17S4WPpOJ/42F+QGtUWBKPPEG0hVmWiU4H640I514k1hGrsWbPzxaq
GiiOcUQmD/xilQJqu6M42UnVTS0m6eFEsl2VZQIDAQABAOIBAAbOam1m6uW25/pU
nsre590YZD4uySgJI06iFNKXZ+6l9Ok7VgjKJrfFL9n0oHeJ58g8Xm9apwYVGPwm
5KC5if2ugoKPRI9mInNQFNCTzxYJnkgj3XGQtVL0kgtj4AWfg5GcLt3xbkvH0EKR
Moi419OjlwV82EokKLqQgPy4n0ReHGJdz5/ABFp0LQBwoTWohpeLFbpyYqMxm2mj
oPaDPufqNN0H0xlUigIffSYSaB9mzo6iKI48tJQ6lYl2DvvKq/I0zfBCP5s/QQvm

Lawful Intercept

YyL6ideSX05AvdBkBIi4w86ws/UUjlJYvaR9wztk45DXnK+IhqFJr3xzewIthqE2
yduM1+0CgYEAyxKuheB0eq6TmTL4qS3FjzHqluvXwuNaxzI7Sq62bmt0fvul0z20
00JeXYGJfHuU6lCR1WFBPgCYzAH0tUPmrNF+2xRzWe6qWcK85APRlUcSL+xdWhc
ANJwDL4u1ZRITHS03gWQgZPXJVGk5LYFomOIRy0r3cq36OIln1h/ajsCgYEAyC+0
4jH8Y+mtjoYFuDuDlkOCxQksrUC3VM8TMtMb5HpBGghcIsKgDLUbDaptAPH2FfFe
3dDE9xQy4QNTpfYpPStLVLgqoRYyLfV5LGWO6rknvgImfzfeNYc471u21Njpf5g
T2LAW4Au2iBjMze7+GO96kckTfJS6TTl7SKTZN8CgYAtaPULzOBfGrC4vlQqiKi3
16KssvphgzOmFY4SqjTzRiYbyyW0q6rGVwhhAnl7E6t+JhwmSgQr2DhSz1Ep
Q5DOrl4eoVkWEC+ZuNlCZ7QzHFxtYQBWWDLSoDwcLV8NtYXhSukEFNx2jVe76aDk3+ss
n/M2b1xy0UUE16UwF7sORwKBgQC+lTSXOFF56YmzK4Mcf4RqNUG1RmPDWd5wHmi/
tV5Tqdr67USST0pCbtUfsIXwJWtMPqw7bztlo1RBBoJkegeUm+CXStTbYf6IsQQw
oqVNjwfzFF5UA/vZNbbxVnLH5iXzbXbRfU12k1WHktf+dew9sO+osCwvDSnEHflc
RtOebwKBgQDJVl0Xz+ORLaHtIGPquZ7vBtsOKW9Fc6SS1xbiizEgZE13QD6N+gP/
SgsdXuYkptdXofQ9VKb6axjxaP2RaE0OpR/oOvJ51O5X5yfILuZVAabsEVXfOEYt
utoBOKDJsmzJHF+qZqviGE5OTTdryhit/Y8HKPcnq7uDhzS0HUO7qg==
-----END CERTIFICATE KEY-----
tlskey: |-
-----BEGIN CERTIFICATE KEY-----
MIIEpgIBAAKCAQEAx3UlzZFfAKIp4tLEFZRkErcR+9r0MGzah+aEF17u/Rj98IET
ajp8c9JI81Fc5dXBKtB31hNa0J0HudIVCXEEiZNtMYifISKGm8A6Xs43WV9Ct0+R
6llnnSqCIHlGCl9Pk9xakeO91OA63wySaRkhoTy00wcbnQQndNKyQ+bIFyFJJcbI
wM18tTMvMBVRBVAudCGXK1hJ3p+fhkvxdz+Ll/0elvBoRy2Wd4oOY2vXPobzi/Hc
aio+v1Xpl7YLoA+pm5lLOgX+VmcxKilMj01Q5Wu7lLwylGvzwdp/ZrLJsFB6/mBR
neaoYwlmUBA9zslvdspFYyCJwwRM7IxBzV6uQIDAQABaoIBAQCSE2kVvIDVrxGr
2mHirEjZzI77VZ6t2HuAG2OCNijAujr6rqDtWJ/d3GGLaXpX3t3T7h401mXW7GKp
8ZCsoKbT0hi2T0PZ9ktjO6gf+pDNzd1lAITGRXOap/2zvmyqriedTNHk/Exdx0WD
APmfKktVK2ZlWRB9mgPgSZX59rRPvVqnLTd65+iGfS2OPBY/CI3+b4FY7ZfU
bDRt214QRKARSEuoBeaSUP95JDoc5qjx+3/GREoXTyejOkU96fQb2Av5Fmt+NQuh N0nY

W937RU/qR/NdxtDStOAZ8uSzz658NtxiFcyErJGN4OgeqtkaMvSx11CukGrZfvm
fSpphgWxAoGBAPTjVa2GnjJ4800ESUQM1miSn0rxKug2JOeSoBOZfxJs3fqzjWU3
xjjJepWPTd4sE5KAXUYuP4Dt8+qp8RoB4YEOkxDHKFFdm7bmsb157469zjbUkXkv
jYXh1QFdf7rJ2mEFUxB5d9xj98K7rVXU73RhiiyjkWO0uU7oVEDWCCVPAoGBANCC
FbAQQqc+NQTiyK4JTMkQ9ASmfHy5nL7TIG4M1BYJtH95d1bEwHt/ybrItB+aadsk
X8qc81ARdyEagd6Okv8m9yU4POnZ3bOa07vSW8UXMG2YzTkbPVrL+sVPL5wyjD66
PlmsAPOeh17nCO5Rq9UtsL9S14/Mg907uZ2+fu13AoGBAPCLyiIshIKBh+7ZcfpV
Pdr0g2IC6Sm13aRWw8TclxRLOfTDoFVXZu9rAW2dmVhuJT+XtYd1cE1970WHbtwk
EtVJDIZriq/56jXL/7V17CljSgdT4B2Q10wcksIHjKr3EfqRV1lsZ4915v4yif4w
rpG8KhTVCsBMymW3EuOpGdFDAoGBAJS69fVs496gRPmDO1rJUKEpcqaOi2X8kI6d
0xgtFmvR6Gq1nibKmfJ+KMuPtsyEPNYQkDPa2MRYcP2wBstWLuQMy2ZFQ04cFkSz
QC1N1CA2dlj3EgyWHaHStsJfxbpJZDiqaSdVxigBANb1Pi2hHE0oFlei9nGglL8O
h0y6QZfpAoGBAN2D8199HVf1PWYIhZSj9h3nAegDjffBWRIOj/b1/z4LVtXLFeRo
KpF46LzdlEKVZWct/lrp7SBFimRAinwe7d/4uHVsBT8qITG82v5gJADq+veYjTtO
XIxahI6XZQMZRjyXygm1HVQsDOEf5jDS4t9LSzm5EabjVa4CkFhuJhGy
-----END CERTIFICATE KEY-----

Add includeIPRanges and excludeIPRanges in fed-istio Namespace

To add includeIPRanges and excludeIPRanges in fed-istio namespace, follow the below procedure:

1. On OPM, execute the below command to get cluster metallb IPRanges: `curl -s -X GET http://<opm- ip>:90/vnfmgr/api/v1/stack/1000/cluster`

Table 14. Metallb Addresses

SMF	UPF
metallb_peer_ip: 10.170.4.2	metallb_peer_ip: 10.170.4.2
metallb_peer_asn: 65102	metallb_peer_asn: 65102
metallb_asn: 65522	metallb_asn: 65522
metallb_pool_len: 27	metallb_pool_len: 27
metallb_pool_net: 10.170.65.0/27	metallb_pool_net: 10.170.65.64/27

2. From SMF/UPF CLI, execute the below command to verify installed fed-istio version:

```
helm3 ls -A
```

3. From SMF/UPF CLI, execute the below command to backup config:

```
helm3 get values -n fed-istio fed-istio -o yaml > fed-istio-values_DATE.yaml
```

4. From OPM, edit the config, refer to preceding table, and upgrade fed-istio.

```
istioproxy:
```

```
  accessLogFile:
```

```
    accessLogEncoding: TEXT
```

```
      includeIPRanges: *
```

```
      excludeIPRanges: 127.0.0.1/24
```

```
      excludeOutboundPorts: ''
```

```
      enableTracing: false
```

Where, `excludeIPRanges: 127.0.0.1/24` is used to exclude local host and `includeIPRanges: *` is used to include all other IPs.

5. To restart SMF/UPF namespace, execute the below command:

```
kubectl -n fed-smf rollout restart deploy
```

LI Egress Gateway Setup from fed-smf to External fed-upf

Lawful Interception (LI) requires communication to happen over secure channels. The messages from application traverse through its local istio-proxy container then to egress gateway in its namespace before hitting ingress of external network function. As a result, Ingress and Egress gateways must be setup with proper certificates, keys, and Kubernetes resources for the request and a response to work seamlessly.

This guide illustrates the steps required to setup the egress gateway for LI.

Prerequisite

The user/operator must have generated CA certificates and keys.

Procedure

As fed-upf cluster is in external IP range and message must intercepted by the local istio-proxy before hitting the egress gateway, the specific IP range CIDR must be updated as part of pod-istio helm variable 'includeIPRanges'. The range is extended by adding comma separated CIDR.

1. Login to the K8s master node as **kuberadmin** user.

2. Verify that the egress gateway pod is running:

```
kubectl get pods -n fed-smf
```

3. Provision the CA certificates and keys:

```
kubectl create -n fed-smf secret generic istio- egressgateway-ca-certs --
from-file=<ca-chain .pem file name> kubectl create -n fed-smf secret tls
istio-egressgateway-certs --key <client key file name> --cert <client cert
file name>
```

Example:

```
kubectl create -n fed-smf secret generic istio- egressgateway-ca-certs --
from-file=<ca-chain.pem file name>
```

```
kubectl create -n fed-smf secret tls istio-egressgateway- certs --key
clientlittest.com.key --cert clientlittest.com.crt
```

Verify the output

```
kubectl create -n fed-smf secret generic istio- egressgateway-ca-certs --
from-file=ca-chain.cert.pem
```

```
secret/istio-egressgateway-ca-certs created
```

```
kubectl create -n fed-smf secret tls istio-egressgateway- certs --key
clientlittest.com.key --cert clientlittest.com.crt
```

```
secret/istio-egressgateway-certs created
```

Verify the secret created above

```
kubectl get secret -n fed-smf
```

NAME	DATA	AGE	TYPE
istio-egressgateway-ca-certs	1	2m1s	Opaque
istio-egressgateway-certs	kubernetes.io/tls	2	2s

Note: 1. The Resource is generated automatically when the application initiates register API as described in whitelisting section.

2. The virtual service for application is also generated as per values specified in values.yaml at startup.

4. CLI Configuration for LI-X1: The user must configure the UPF server configuration in the fed-smf as per LI-X1. Here, host is example of LB IP of Ingress GW of fed-upf.

```
show full-configuration li-x1 peer-profile

li-x1 peer-profile <peer profile name>

host      <UPF Host IP>

https-port <UPF HTTPS PORT>

!

li-x1 peer-interface <peer interface name 1>

peer-type      t2

ne-identifier  <network peer ID>

associated-peer-profile <peer profile name>

n4 <n4.upfserver.com>

!

li-x1 peer-interface <peer interface name 2>

peer-type      t3

ne-identifier  <network peer ID>

associated-peer-profile <peer profile name>

n4 <n4.upfserver.com>

!

!
```

Note: If anytime the fed-smf and fed-upf cluster is redeployed, then this configuration is removed. So the user/operator has to once again repeat the same steps to configure the LI Ingress and LI Egress GW communications.

LI Egress Gateway Setup from fed-smf to MDF2 Primary (TLS Protocol)

As MDF2 is in external IP range and message must be intercepted by the local istio-proxy before hitting the egress gateway, the specific IP range CIDR must be updated as part of pod-istio helm variable 'includeIPRanges'. The range is extended by adding comma separated CIDR.

Procedure

1. The Resource is generated automatically when the application initiates register API as described in whitelisting section.
2. The virtual service for application is also generated as per values specified in values.yaml at startup.

LI Egress Gateway Setup from fed-smf to MDF2 Secondary (TLS Protocol)

As MDF2 is in external IP range and message must be intercepted by the local istio-proxy before hitting the egress gateway, the specific IP range CIDR must be updated as part of pod-istio helm variable 'includeIPRanges'. The range is extended by adding comma separated CIDR.

Procedure

1. The Resource is generated automatically when the application initiates register API as described in whitelisting section.
2. The virtual service for application is also generated as per values specified in values.yaml at startup.

LI Egress Gateway Setup from fed-smf to MDF2

As MDF2 is in external IP range and message must be intercepted by the local istio-proxy before hitting the egress gateway, the specific IP range CIDR must be updated as part of pod-istio helm variable 'includeIPRanges'. The range is extended by adding comma separated CIDR.

Procedure

1. The Resource is generated automatically when the application initiates register API as described in whitelisting section.
2. The virtual service for application is also generated as per values specified in values.yaml at startup

LI Ingress Gateway Setup for fed-smf

Lawful Interception (LI) requires communication to happen over secure channels. As a result, Ingress must be setup with proper certificates, keys, and Kubernetes resources for the request and response to work seamlessly.

This guide illustrates the steps required to setup the ingress gateway for LI in SMF cluster (fed- smf).

Prerequisite

The user/operator must have generated CA certificates and keys.

Procedure

1. Login to the K8s master node as **kuberadmin** user.
2. Create the secret using the CA certificate: (A default ca.crt PEM file is present as part of **fed-pass-helpers-cert-ctrl-ca**. The user must verify the available config and can delete it before applying the new secret).

First verify the already available secret:

```
kubect1 describe secret fed-paas-helpers-cert-ctrl-ca -n fed-paas-helpers
```

Now delete the available secret ca.crt:

```
kubect1 delete secret fed-paas-helpers-cert-ctrl-ca -n fed- paas-helpers
```

Create new secret.

```
kubect1 create -n fed-paas-helpers secret generic fed-paas- helpers-cert-ctrl-ca --from-file=<ca-chain pem file name>
```

For example, the user uses a new **ca-chain.cert.pem** CA file.

```
kubect1 create -n fed-paas-helpers secret generic fed-paas- helpers-cert-ctrl-ca --from-file=ca-chain.cert.pem
```

Note: It is always recommended to rename the CA file to `ca-chain.cert.pem` to generate correct end path name.

3. Verify the created secret:

```
kubectl describe secret fed-paas-helpers-cert-ctrl-ca -n fed-paas-helpers
```

4. Share the Application (Server) certificate and keys:

- a. To apply, the user must find the **credentialName** of **ingress-gateway-li** gateway.

```
kubectl get gateway -n fed-smf ingress-gateway-li
-o yaml
```

or

Example:

```
$ kubectl get gateway -n fed-smf ingress-gateway-li -o yaml | grep
credentialName
credentialName: smfli-com-ingress-cert-manager-cert-controller
```

- b. From the o/p get the credentialName, and now use only **smfli-com-ingress-cert-manager**.

```
kubectl create -n fed-smf secret tls smfli-com-ingress-cert-manager --
key=<server key file name>
--cert=<server cert file name>
```

Example

```
kubectl create -n fed-smf secret smfli-com-ingress-cert-manager --
key=applitest.com.key -- cert=applitest.com.cert
```

Note: Remove `-cert-controller` part and apply the `tls` secret on `fed-smf`.

5. Verify the Application (Server side) certificate configuration applied on `fed-smf` for `smfli-com-ingress-cert-manager-cert-controller`:

```
kubectl get secret -n fed-smf
```

And verify:

```
smfli-com-ingress-cert-manager
```

```
kubernetes.io/tls 2 2d7h
```

```
smfli-com-ingress-cert-manager-cert-controller kubernetes.io/generic 3
2d7h
```

- a. The Resource is generated automatically when the application initiates register API as described in whitelisting section.

- b. The virtual service for application is also generated as per values specified in values.yaml at startup.

Note: Secret smfli-com-ingress-cert-manager-cert-controller is auto generated.

6. Verify that the virtual service is created as part of SMF deployment:

```
kubectl get virtualservice -n fed-smf
# Verify the created test-external-li Virtual Service Name:
li-x1-external-http
[ingress-gateway-li] [10.163.204.229
smfli.com] 2d23h
li-x1-external-https
[ingress-gateway-li] [smf206.litest.com] 2d23h
```

7. Find **LB IP Address** that is reachable from the external Client to send requests towards Ingress GW.

For example, here 10.163.204.240 is LB EXTERNAL-IP and port 443 to reach from external ADMF LI PF Clients for sending LI_X1 messages.

```
kubectl get service istio-ingressgateway-li -n fed-smf
```

NAME	TYPE	CLUSTER-IP
EXTERNAL-IP	PORT(S)	AGE
istio-ingressgateway-li	LoadBalancer	10.98.67.88
10.163.204.240	443:32214/TCP	22h

Note: If anytime the fed-smf and fed-upf cluster is redeployed, then this configuration is removed. So the user/operator has to once again repeat the same steps to configure the LI Ingress and LI Egress GW communications.

LI Ingress Gateway Setup for fed-upf

Lawful Interception (LI) requires communication to happen over secure channels. As a result, Ingress must be setup with proper certificates, keys, and Kubernetes resources for the request and response to work seamlessly.

This guide illustrates the steps required to setup the ingress gateway for LI in UPF cluster (fed- upf).

Prerequisite

The user/operator must have generated CA certificates and keys.

Procedure

1. Login to the K8s master node as **kuberadmin** user.
2. Create the secret using the CA certificate: (A default ca.crt PEM file is present as part of fed-pass-helpers-cert-ctrl-ca. The user must verify the available config and can delete it before applying the new secret).

First verify the already available secret:

```
kubectl describe secret fed-paas-helpers-cert-ctrl-ca -n fed-paas-helpers
```

Now delete the available secret ca.crt:

```
kubectl delete secret fed-paas-helpers-cert-ctrl-ca -n fed- paas-helpers
```

Create new secret.

```
kubectl create -n fed-paas-helpers secret generic fed-paas- helpers-cert-ctrl-ca --from-file=<ca-chain pem file name>
```

For example, the user uses a new ca-chain.cert.pem CA file.

```
kubectl create -n fed-paas-helpers secret generic fed-paas- helpers-cert-ctrl-ca --from-file=ca-chain.cert.pem
```

3. Verify the created secret:

```
kubectl describe secret fed-paas-helpers-cert-ctrl-ca -n fed-paas-helpers
```

4. Share the Application (Server) certificate and keys:

- a. To apply, the user must find the credentialName of ingress-gateway-li gateway.

```
kubectl get gateway -n fed-upf ingress-gateway-li -o yaml
```

or

Example:

```
$ kubectl get gateway -n fed-upf ingress-gateway- li -o yaml | grep credentialName credentialName: upfli-com-ingress-cert-manager- cert-controller
```

- b. From the o/p get the credentialName, and now use only upfli-com-ingress-cert- manager.

```
kubectl create -n fed-upf secret tls upfli-com- ingress-cert-manager --
```

```
key=<server key file name>
--cert=<server cert file name>
Example:
kubectl create -n fed-upf secret upfli-com- ingress-cert-manager --
key=applitest.com.key -- cert=applitest.com.cert
```

Note: Remove -cert-controller part and apply the tls secret on fed-upf.

5. Verify the Application (Server side) certificate configuration applied on fed-upf for upfli- com- ingress-cert-manager-cert-controller:

```
kubectl get secret -n fed-upf
# And verify:
upfli-com-ingress-cert-manager
kubernetes.io/tls      2      13h
upfli-com-ingress-cert-manager-cert-controller kubernetes.io/generic 3
13h
```

- a. The Resource is generated automatically when the application initiates register API as described in whitelisting section.
- b. The virtual service for application is also generated as per values specified in values.yaml at startup.

Note: Secret upfli-com-ingress-cert-manager-cert-controller is auto generated.

6. Verify that the virtual service is created as part of UPF deployment:

```
kubectl get virtualservice -n fed-upf

# Verify the created test-external-li Virtual Service Name:
li-x1-external-http
[ingress-gateway-li] [10.163.206.34 upfli.com] 4d12h

li-x1-external-https
[ingress-gateway-li] [upfreal.litest.com] 4d12h
```

7. Find **LB IP** Address that is reachable from the external Client which can send requests towards Ingress GW of UPF.

Note: Use this IP address and port in Create a Service Entry section in [LI Egress Gateway Setup from fed-smf to External fed-upf](#). The same is used in the LI-X1 CLI configuration of fed-smf.

For example, here 10.163.206.38 is LB EXTERNAL-IP and port 443 to reach from external clients to sent LI_T2 and LI_T3 messages.

```
kubectl get service istio-ingressgateway-li -n fed-upf
```

NAME	EXTERNAL-IP	PORT(S)	TYPE	AGE	CLUSTER-IP
istio-ingressgateway-li	10.163.206.38	443:31411/TCP	LoadBalancer	22h	10.106.112.179

This section describes information related to SMF Lawful Interception.

- Interception events for 5G sessions are reported via ETSI TS 103.221-2 X2 PDU header encapsulating 3GPP TS 33.128 ASN.1 encoded X2 events (ETSI TS 103.221-2 payload format 2)
- Interception events for 4G sessions are reported via ETSI TS 103.221-2 X2 PDU header encapsulating 3GPP TS 33.108 ASN.1 encoded X2 events (ETSI TS 103.221-2 payload format 3)

SMF-LI ePDG/2Sb/EPS/5G Interworking Message Flows with Lawful Intercept

Message Flows document the interactions between the SMF/PGW-C and other network elements such as the UPF, PCF, LI MDF2, and LI MDF3.

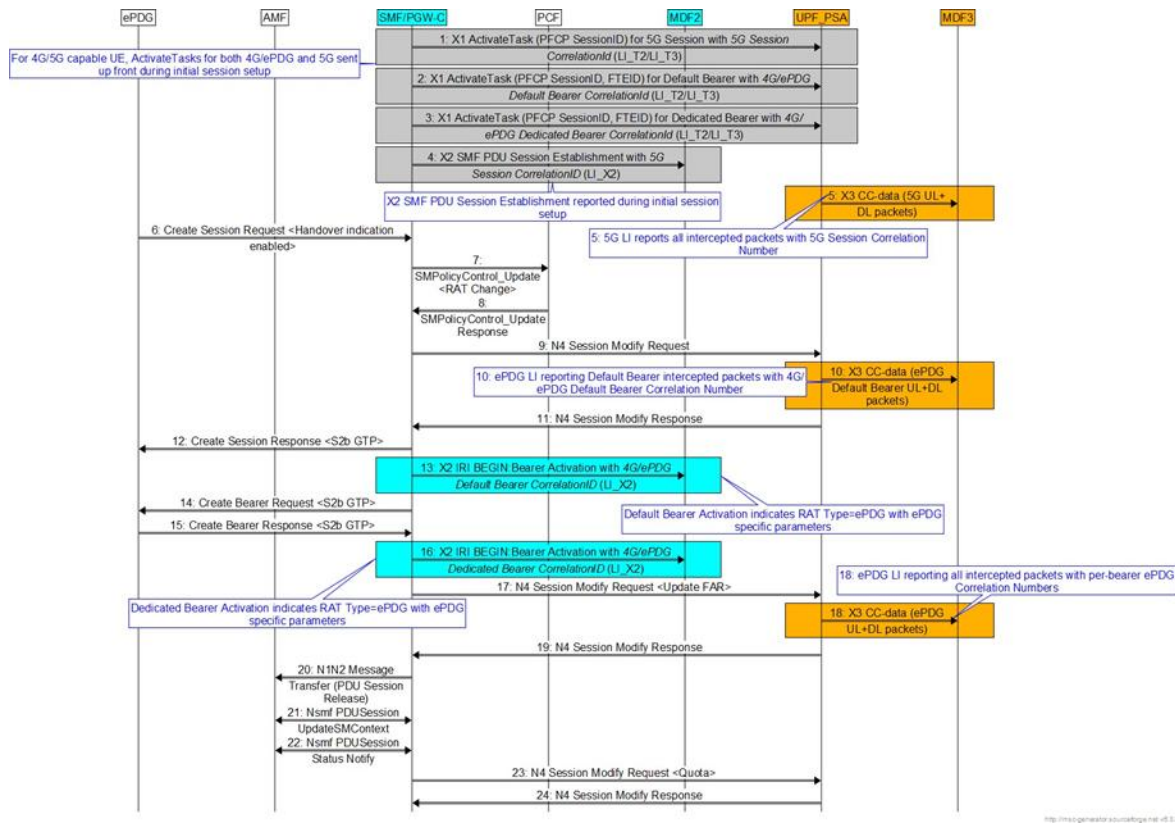
Handover Between ePDG and 4G/5G:

- [5GS to ePDG-S2b Handover with Lawful Intercept](#)
- [ePDG-S2b to 5GS Handover with Lawful Intercept](#)
- [ePDG-S2b to EPS Handover with Lawful Intercept](#)
- [EPS to ePDG-S2b Handover with Lawful Intercept](#)

5GS to ePDG-S2b Handover with Lawful Intercept

Callflow

Figure 4. 5GS to ePDG-S2b Handover Call Flow



This procedure is used to handover a UE that is currently served by the 5GC to ePDG. The Initial condition is to establish PDU Session with 5GC. This procedure consists of four transactions:

- Creating Default Bearer in ePDG based on the established 5G PDU session.
- Creating N/W initiated dedicated bearer (existing 5G dedicated QoS flows).
- Releasing the 5G PDU Session from AMF and NG-RAN.
- Delete/Modify/Create related to old/new PCC rules as given by PCF during the HO procedure.

Initial PDU Session

UE established PDU session in 5GC. SMF+PGW-C has the active 5G session and the PDU session is stored in the DB with a secondary key (DNN, SUPI <IMSI>, PduSessionId). This step is not shown in the call flow diagram.

Step 1: SMF+PGW-C sends X1 ActivateTask with 5G Session Correlation Number to UPF-PSA (LI_T2/LI_T3 interface).

Step 2: Since UE is 4G/5G capable, SMF+PGW-C sends X1 ActivateTask with 4G/ePDG Default Bearer Correlation Number to UPF-PSA (LI_T2/LI_T3 interface).

Step 3: If the session includes SDFs that correspond to 4G/ePDG dedicated bearers, for each dedicated bearer SMF+PGW-C sends X1 ActivateTask with 4G/ePDG Dedicated Bearer Correlation Number to UPF-PSA (LI_T2/LI_T3 interface).

Step 4: SMF+PGW-C sends X2 SMF PDU Session Establishment to MDF2 with 5G Session Correlation Number.

Step 5: UPF-PSA is intercepting traffic on all flows in session. UPF-PSA sends X3 CC-data to MDF3 with 5G Session Correlation Number.

ePDG-S2b Session Creation

Step 6: ePDG sends the Create Session Request message with Handover Indication Flag enabled. ePDG sends PduSessionId and enables the 5G-Interworking Flag in the CSR message.

Step 7: SMF+PGW-C informs PCF about the event that occurred (RAT change) for the UE by sending SM Policy Update message.

Step 8: SMF+PGW-C receives the Policy Update Response message from PCF. If there are any new PCC rules to be installed or any old PCC rules to be deleted, SMF+PGW-C stores them to handle in a separate transaction procedure.

Step 9: SMF+PGW-C sends N4 Session Modify message towards UPF-PSA through PFCP to update the existing default flows DL FAR.

- DL FAR is updated with a new Outer Header Creation field with ePDG's GTP-U TEID.
- At this point, SMF+PGW-C also updates DL FARs of all the existing dedicated QoS flows (created in 5G).
- The DL FARs are updated with Outer Header Creation field with ePDG's GTP-U TEID and transport level marking with Default Flows Transport level marking.
- For dedicated bearers, QERs are updated for Maximum Bitrate with APNAMBR values and unset the Guaranteed BitRate.
- All the dedicated Bearers are updated till dedicated bearers are created in ePDG so that DL- data can flow through the default bearer.
- Gets the usage reports for all the bearers from UPF.
- UPF sends the response towards SMF+PGW-C with all usage reports.

Step 10: UPF-PSA is now intercepting ePDG traffic on ePDG Default Bearer. UPF-PSA sends X3 CC-data to MDF3 (LI_X3 interface) with 4G/ePDG Default Bearer Correlation Number.

Step 11: UPF-PSA sends N4 Session Modify Response to SMF+PGW-C.

Step 12: SMF+PGW-C sends the Create Session Response with Default Bearer Information (GTP-U TEID) to ePDG.

Step 13: SMF+PGW-C sends X2 IRI BEGIN: BearerActivation with 4G/ePDG Default Bearer Correlation Number to MDF2 (LI_X2 interface).

Dedicated Bearer Creation

SMF+PGW-C starts creating legacy dedicated Bearers (dedicated bearers related to QoS Flows existed in 5G). SMF+PGW-C identifies a list of existing Bearers to create an ePDG.

Step 14-15: Send the Create Bearer Request for all the legacy bearers. In the Create Bearer Response, it receives the eps-bearerId for all the legacy bearers. Updates all the dedicated Bearer Context with new eps-bearerId. Receives the remote GTP-U TEID for all the dedicated bearers.

Step 16: For each dedicated bearer, send X2 IRI BEGIN: BearerActivation with 4G/ePDG Dedicated Bearer Correlation Number to MDF2 (LI_X2 interface).

Step 17: SMF+PGW-C sends N4 Session Modify Request to UPF-PSA to update UPF for all the dedicated bearers with FAR, QER, and URR (if needed).

Step 18: At this point, UPF-PSA is intercepting all ePDG traffic and sends X3 CC-data to MDF3 (LI_X2 interface) with 4G/ePDG per-bearer correlation numbers.

Step 19: UPF-PSA sends N4 Session Modify Response to SMF+PGW-C.

Step 20: SMF+PGW-C send N1N2Message Transfer for N2Release. It does not contain any N1 container.

Step 21: Get N2 release confirmation in UpdateSMContxt message.

Step 22: Sends the StatusNotify to inform PDU Session is moved to non-3GPP access.

Step 23-24: SMF+PGW-C updates the new Quota to the UPF by sending the N4 Session Modify message.

New PCF Rule Processing

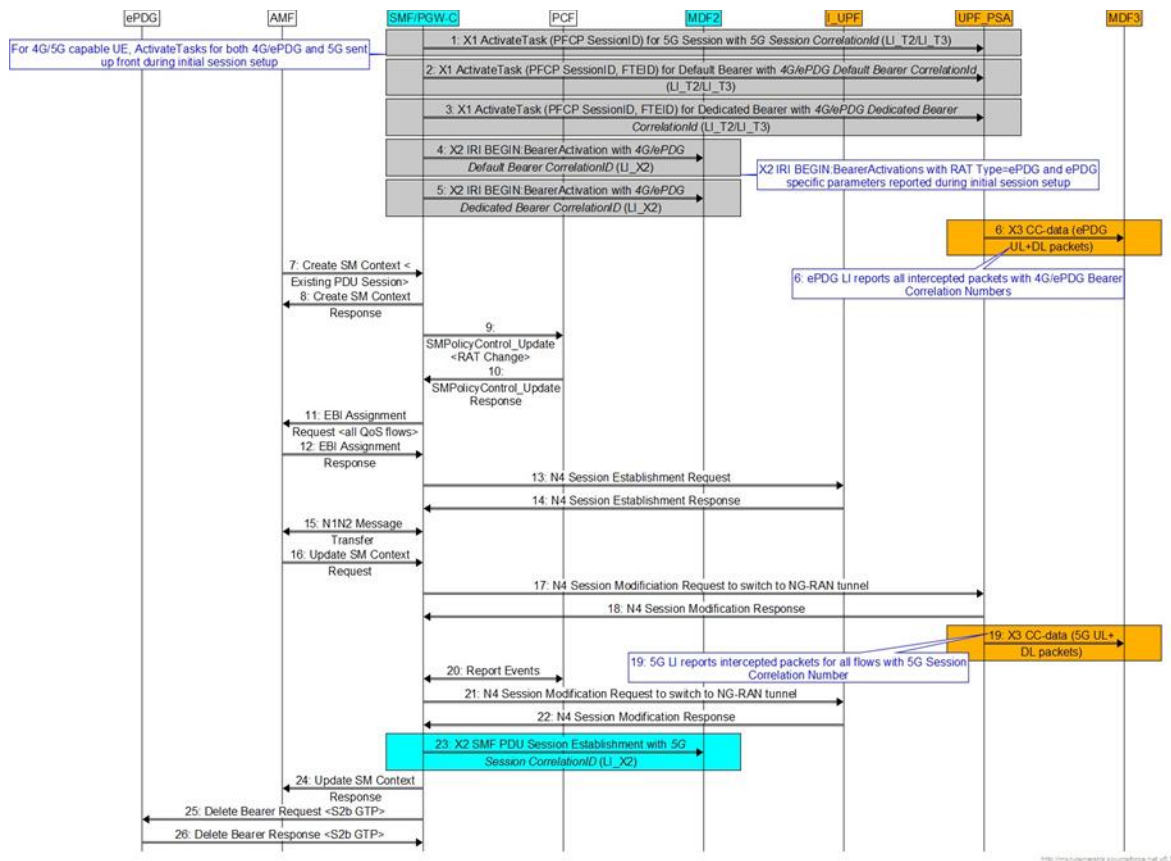
After Step 24, SMF+PGW-CC initiates the processing of any rules added/modified/removed by PCF.

Note: This subsequent rule processing is not shown in the call flow diagram.

ePDG-S2b to 5GS Handover with Lawful Intercept

Call Flow

Figure 5. ePDG-S2b to 5GS Handover Call Flow



This procedure is used to handover a UE that is currently served by the ePDG to 5GC. This procedure consists of four transactions.

Prerequisite

Create an ePDG-S2b Session.

Initial Setup

ePDG-S2b session is created in SMF+PGW-C. The session is created with PduSessionId and Interworking flag is enabled by ePDG. This step is not shown in the call flow diagram.

SMF

Step 1: If the UE is 4G/5G capable, SMF+PGW-C sends X1 ActivateTask with 5G Session Correlation Number to UPF-PSA (LI_T2/LI_T3 interface).

Step 2: SMF+PGW-C sends X1 ActivateTask with 4G/ePDG Default Bearer Correlation Number to UPF-PSA (LI_T2/LI_T3 interface).

Step 3: If the ePDG session includes a dedicated bearer, SMF+PGW-C sends X1 ActivateTask with 4G/ePDG Dedicated Bearer Correlation Number to UPF-PSA (LI_T2/LI_T3 interface).

Step 4: SMF+PGW-C sends X2 IRI BEGIN: Bearer Activation to MDF2 with 4G/ePDG Default Bearer Correlation Number.

Note: The same Default Bearer Correlation Number is used for 4G Default Bearer and ePDG Default Bearer.

Step 5: If the ePDG session includes a dedicated bearer, SMF+PGW-C sends X2 IRI BEGIN: Bearer Activation to MDF2 with 4G/ePDG Dedicated Bearer Correlation Number.

Note: The same Dedicated Bearer Correlation Number is used for 4G Dedicated Bearer and ePDG Dedicated Bearer.

Step 6: UPF-PSA is reporting all intercepted ePDG packets to MDF3 through X3 CC-data messages with 4G/ePDG per-bearer correlation numbers.

Create SM Context Transaction

Step 7: AMF sends the CreateSMContext message with Request Type as Existing PDU Session. The NSSAI, PLMNid, DNN, SUPI, and PDU-SessionId is the same as the Session that existed in Step 1.

Step 8: SMF+PGW-C sends the CreateSmContext Response towards AMF.

Step 9-10. SMF+PGW-C updates PCF about the RAT change event using the existing PCF session for the PDU Session.

Update SM Context Transaction

Step 11-12: Identifies the list of existing QoS flows (if PCF installs new PCC rules or updates/deletes existing rules those will be taken at end of the HO procedure) to be created. Get the corresponding EPS-Bearer Id for all the QoS Flows (existing). Update the Bearer Context with a new eps-BearerId.

Step 13-14: If an I-UPF is selected, SMF+PGW-C initiates the N4 Session Establishment procedure with I-UPF including or requesting the Tunnel IDs for N3 and N9 interfaces. The N9 tunnel information for the original IP Anchor UPF is also provided. The Uplink and Downlink PDRs for all QoS Flows are installed on the I-UPF with only catch-all packet filters. The QERs are installed with only QFIs set.

Step 15: SMF+PGW-C sends N1N2 message transfer with N2 container (Resource setup Request) and N1 NAS for all the existing QoS flows.

Step 16: Receives UpdateSMContext Request with NGRAN TEIDs.

Step 17-18: N4 Session Modification to switch to NG-RAN tunnel DL FARs for new outer header creation with a new GTP-U TEID. Gets the usage report for all the Rating Groups.

Step 19: At this point, UPF-PSA is intercepting 5G traffic for all flows. UPF-PSA sends X3 CC- data to MDF3 for all flows with a 5G Session Correlation Number.

Step 20: Reports Events to PCF.

Step 21-22: SMF+PGW-C initiates the N4 Session Modification procedure with the I-UPF to set the target NG-RAN Tunnel information at the I-UPF.

Step 23: SMF+PGW-C sends 5G SMF PDU Session Establishment to MDF2 with 5G Session Correlation Number (LI_X2 interface).

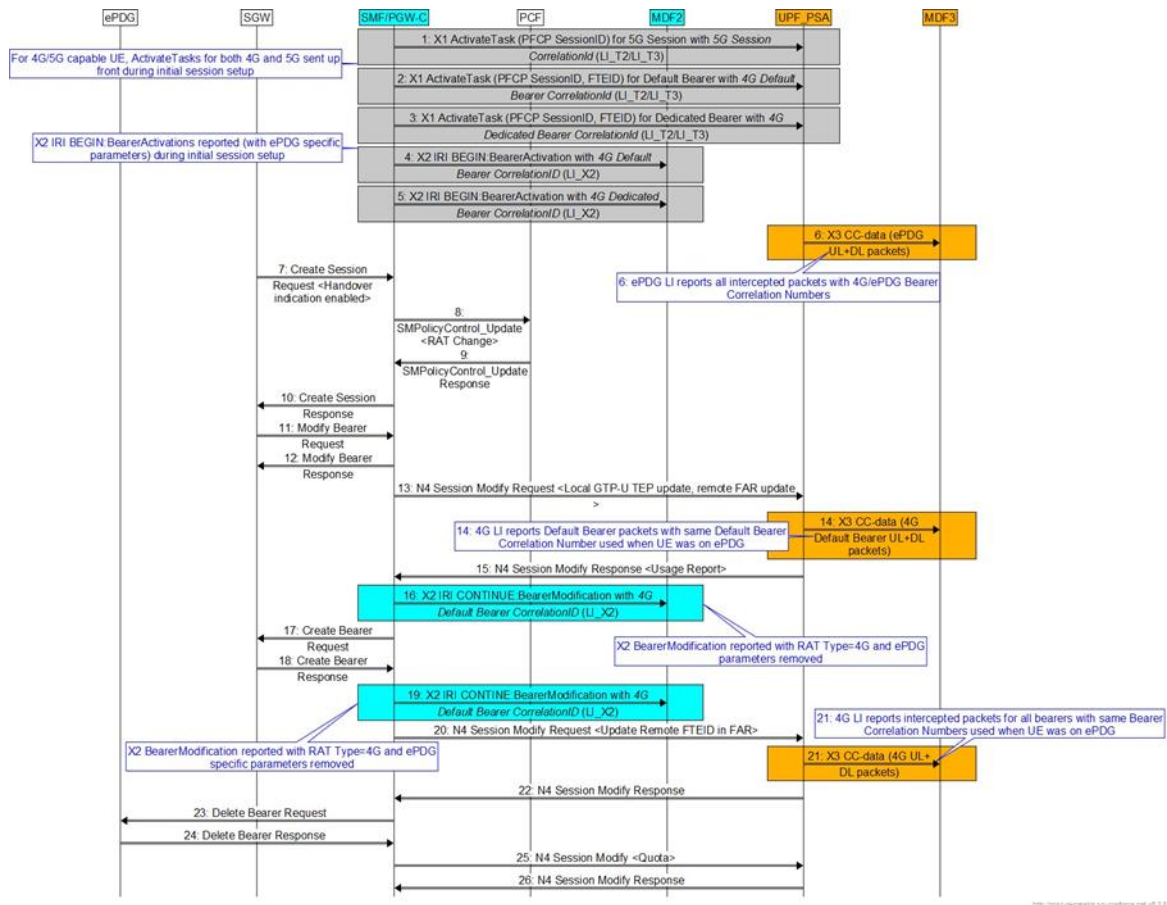
Delete Bearer for ePDG

Step 25-26: SMF+PGW-C triggers the Delete Bearer Request towards ePDG for all ePDG Bearers existed when the handover is started. Process the response from ePDG.

ePDG-S2b to EPS Handover with Lawful Intercept

Call Flow

Figure 6. ePDG-S2b to EPS Handover Call Flow



This procedure is used to handover a UE that is currently served by the ePDG to EPC. This procedure consists of four transactions.

Prerequisite

Create an ePDG-S2b Session.

Initial Setup

ePDG-S2b session is created in SMF+PGW-C (this step is not shown in the call flow diagram). The session can be created in two ways:

- ePDG can include the PDU Session ID in the Additional Protocol Config Options (APCO) parameter in the Create Session Request. In that case, during handover to 3GPP EPC the SGW

includes the same PDU Session ID in the Create Session Request sent to SMF/PGW-C, see Create Session Transaction. The Unity Cloud product currently expects 5G capable UEs, hence this should always be the case in Unity Cloud deployments.

- When ePDG does not include PDU Session ID in the Create Session Request, the SGW will not have the PDU Session ID. Hence, during handover to 3GPP EPC, the SGW will not provide a PDU Session ID in the Create Session Request sent to SMF/PGW-C in Step 7. This case is not currently supported by the Unity Cloud products - the handover to EPS will fail.

Step 1: If the UE is 4G/5G capable, SMF+PGW-C sends X1 ActivateTask with 5G Session Correlation Number to UPF-PSA (LI_T2/LI_T3 interface).

Step 2: SMF+PGW-C sends X1 ActivateTask with 4G/ePDG Default Bearer Correlation Number to UPF-PSA (LI_T2/LI_T3 interface).

Step 3: If the ePDG session includes a dedicated bearer, SMF+PGW-C sends X1 ActivateTask with 4G/ePDG Dedicated Bearer Correlation Number to UPF-PSA (LI_T2/LI_T3 interface).

Step 4: SMF+PGW-C sends X2 IRI BEGIN: Bearer Activation to MDF2 with 4G/ePDG Default Bearer Correlation Number.

Note: The same Default Bearer Correlation Number is used for 4G Default Bearer and ePDG Default Bearer.

Step 5: If the ePDG session includes a dedicated bearer, SMF+PGW-C sends X2 IRI BEGIN: Bearer Activation to MDF2 with 4G/ePDG Dedicated Bearer Correlation Number.

Note: The same Dedicated Bearer Correlation Number is used for 4G Dedicated Bearer and ePDG Dedicated Bearer.

Step 6: UPF is intercepting packets for all ePDG bearers and reporting X3 CC-data to MDF3 with the appropriate 4G/ePDG bearer correlation numbers (LI_X3 interface).

Create Session Transaction

Step 7: SGW sends the Create Session Request message with the Handover indication flag enabled. The NSSAI, PLMNid, DNN, SUPI.

- When the PDU Session ID is received from ePDG, SGW includes the same PDU Session ID that is created during Initial Setup.

- When the PDU Session ID is not received from ePDG, SGW sends Create Session Request without PDU Session ID. This case is not currently supported by the Unity Cloud products - the handover will fail.

Step 8-9: SMF+PGW-C updates PCF about the RAT change event using the existing PCF session for the PDU Session. PCF can update the PCC rules in the Response message. Any rules-related updates happen in Step 15. For now, EPC creates default bearer and dedicated bearer as per Step 1 (whatever established in ePDG).

Note: As per spec 23.402, section 8.2.1.1 - Since Handover Indication is included, the PDN GW defers any modification to the PCC Rules (due to changes received from the PCRF, if there is PCRF interaction) and still applies the existing PCC Rules for charging and policy until the creation of the legacy (existing) dedicated bearers.

Step 10: SMF+PGW-C sends Create Session Response to SGW with Default Bearer details.

Step 11-12: SMF+PGW-C receives Modify Bearer Request Message with Handover Indication. It sends the MBResp message to SGW.

Step 13: SMF+PGW-C sends N4 Session Modify Message to UPF updating outer header creation for DL FAR in default QoS flow and query the usage reports.

Step 14: UPF-PSA starts intercepting Default Bearer traffic and sending X3 CC-data to MDF3 with 4G/ePDG Default Bearer Correlation Number.

Step 15: UPF-PSA sends N4 Session Modify Response to SMF+PGW-C.

Step 16: SMF+PGW-C sends IRI CONTINUE: BearerModification with 4G/ePDG Default Bearer Correlation Number to MDF2.

Note: The same Default Bearer Correlation Number is used for 4G Default Bearer and ePDG Default Bearer.

Create Legacy Dedicated Bearer Transaction

As per spec 23.402, 8.2.1.1:

Step 17-18: SMF+PGW-C sends Create Bearer Request for each existing Dedicated Bearer and SGW sends back Create Bearer Response.

Step 19: SMF+PGW-C sends X1 IRI CONTINUE: BearerModification with 4G/ePDG Dedicated Bearer Correlation Number to MDF2 for each Dedicated Bearer.

Note: The same Dedicated Bearer Correlation Number is used for a given 4G Dedicated Bearer and its corresponding ePDG Dedicated Bearer.

Step 20: SMF+PGW-C sends N4 Session Modify Request to UPF-PSA to update existing FARs with Remote GTPU TEID received in Create Bearer Response.

Step 21: UPF-PSA starts intercepting traffic for the Dedicated Bearer(s) and reports all intercepted traffic to MDF3 with appropriate per-bearer Correlation Numbers.

Step 22: UPF-PSA sends N4 Session Modify Response to SMF+PGW-C.

Delete Bearer for ePDG

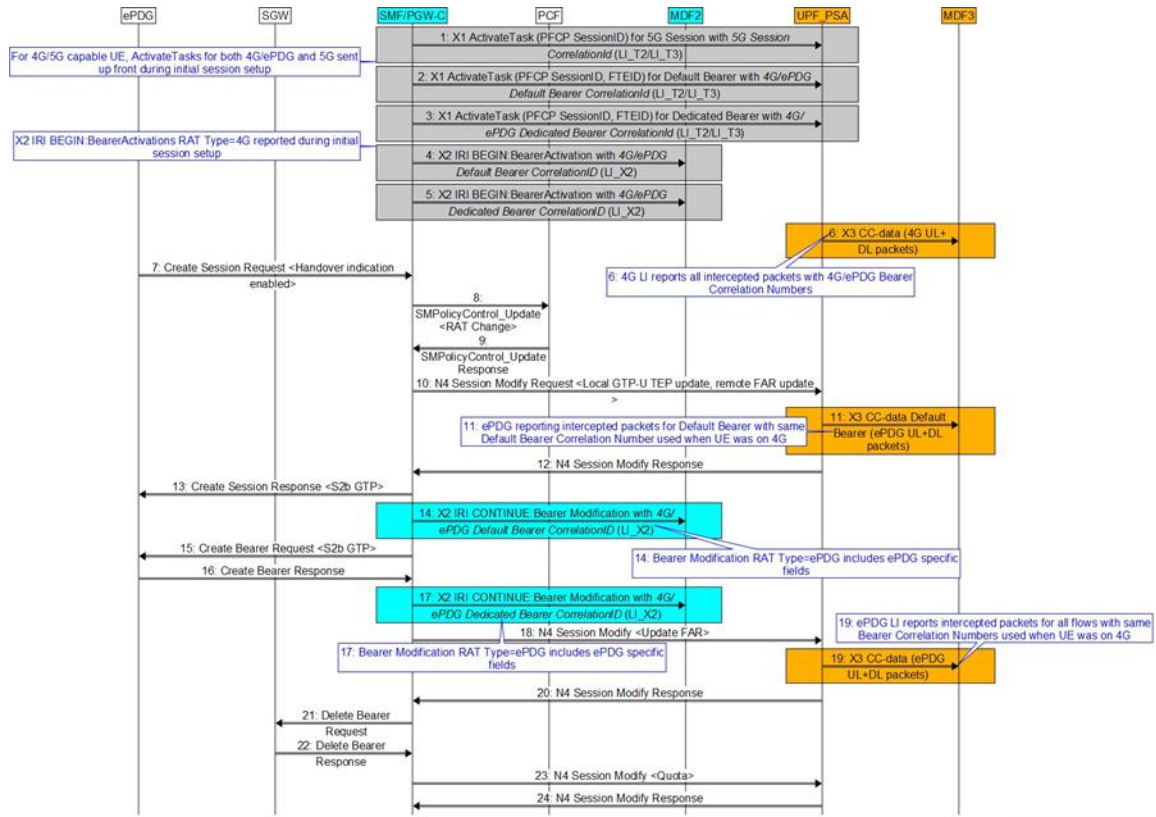
Step 23-24: SMF+PGW-C triggers the Delete Bearer Request towards ePDG for all the Bearers existed in Step 1 and processes the response from ePDG.

Step 25-26: SMF+PGW-C sends N4 Modify Session message to UPF to update the new Quota. UPF acknowledges by sending N4 Modify Session Response.

EPS to ePDG-S2b Handover with Lawful Intercept

Call Flow

Figure 7. EPS to ePDG-S2b Handover Call Flow



This procedure is used to handover a UE that is currently served by the EPC to ePDG. This procedure consists of three transactions. The initial condition is to establish PDU Session with EPC.

- Creating Default Bearer in ePDG based on the established EPC PDN session.
- Creating N/W initiated dedicated bearer (existing 4G dedicated Bearers). Creating new dedicated bearers as provided by PCF.
- Releasing the 4G PDN Session from SGW.

Initial PDU Session

UE established PDU Session in 4G EPC. SMF+PGW-C has the active 4G session and the PDU session is stored in the DB. This step is not shown in the call flow diagram. The session can be created in two ways:

- When the 4G Session is created with PDU Session ID provided in the Create Session Request, during handover to ePDG, the same PDU Session ID is present in the Create Session Request sent by ePDG. The Unity Cloud product currently expects 5G capable UEs, hence this should always be the case in Unity Cloud deployments.
- When the 4G Session did not include PDU Session ID in the Create Session Request. In this case, during handover to ePDG, the PDU Session ID will not be present in the Create Session Request sent to SMF+PGW-C. This case is not currently supported by the Unity Cloud products - the handover to ePDG will fail.

Step 1: If the UE is 4G/5G capable, SMF+PGW-C sends X1 ActivateTask with 5G Session Correlation Number to UPF-PSA (LI_T2/LI_T3 interface).

Step 2: SMF+PGW-C sends X1 ActivateTask with 4G/ePDG Default Bearer Correlation Number to UPF-PSA (LI_T2/LI_T3 interface).

Step 3: If the ePDG session includes a dedicated bearer, SMF+PGW-C sends X1 ActivateTask with 4G/ePDG Dedicated Bearer Correlation Number to UPF-PSA (LI_T2/LI_T3 interface).

Step 4: SMF+PGW-C sends X2 IRI BEGIN: Bearer Activation to MDF2 with 4G/ePDG Default Bearer Correlation Number.

Note: The same Default Bearer Correlation Number is used for 4G Default Bearer and ePDG Default Bearer.

Step 5: If the ePDG session includes a dedicated bearer, SMF+PGW-C sends X2 IRI BEGIN: Bearer Activation to MDF2 with 4G/ePDG Dedicated Bearer Correlation Number.

Note: The same Dedicated Bearer Correlation Number is used for 4G Dedicated Bearer and ePDG Dedicated Bearer.

Step 7: ePDG sends the Create Session Request message with Handover Indication Flag enabled.

- If the PduSessionId is received in the CSR message then the same is used to create a composite key to query the session from DB.
- If the PduSessionId is not received in the CSR message then the handover will fail. This case is not currently supported by the Unity Cloud products.

Step 8: SMF+PGW-C informs PCF about the event that occurred (RAT change) for the UE by sending SM Policy Update message.

Step 9: SMF+PGW-C receives the Policy Update Response message from PCF. If there are any new PCC rules to be installed or any old PCC rules to be deleted, SMG+PGW-C stores them to handle in a separate transaction procedure.

Step 10: SMF+PGW-C sends the N4 Session Modify message towards UPF (via PFCP) to update the existing default flows DL FAR.

- DL FAR is updated with a new Outer Header Creation field with ePDG's GTP-U TEID.
- At this point, SMG+PGW-C also updates DL FARs of all the existing dedicated QoS bearers (created in 4G).
- The DL FARs are updated with Outer Header Creation field with ePDG's GTP-U TEID and transport level marking with Default Flows Transport level marking.
- For dedicated bearers, QERs are updated for Maximum Bitrate with APNAMBR values and unset the Guaranteed Bitrate.
- All the dedicated Bearers are updated till dedicated bearers are created in ePDG, so that DL-data can flow through the default bearer.
- Gets the usage reports for all the bearers from UPF.
- UPF sends the response towards SMF+PGW-C with all usage reports.

Step 11: UPF-PSA starts intercepting traffic for ePDG Default Bearer. UPF-PSA sends X3 CC- data to MDF3 with 4G/ePDG Default Bearer Correlation Number.

Note: The same Default Bearer Correlation Number is used for 4G Default Bearer and ePDG Default Bearer.

Step 12: UPF-PSA sends N4 Session Modify Response to SMF+PGW-C.

Step 13: SMF+PGW-C sends the Create Session Response with Default Bearer Information (GTP-U TEID).

Step 14: SMF+PGW-C sends X2 IRI CONTINUE: BearerModification with 4G/ePDG Default Bearer Correlation number to MDF2 (LI_X2 interface). BearerModification indicates RAT Type=ePDG and includes ePDG specific parameters.

SMF+PGW-C Starts Creating Existing Dedicated Bearers

Step 15-16: SMF+PGW-C sends Create Bearer Request for all dedicated bearers. In Create Bearer Response, if it receives the eps-bearerId for the bearers, updates the dedicated Bearer Context with eps-bearerId. Receives the remote GTP-U TEID for all the existing dedicated bearers.

Step 17: SMF+PGW-C sends X2 IRI CONTINUE: BearerModification for each Dedicated Bearer with the corresponding 4G/ePDG Dedicated Bearer Correlation Number.

Note: The same Dedicated Bearer Correlation Number is used for a given 4G Dedicated Bearer and the corresponding ePDG Dedicated Bearer.

Step 18: SMF+PGW-C sends N4 Session Modification to UPF-PSA to update UPF for all the existing dedicated bearers with FAR, QER, and URR (if needed).

Step 19: UPF-PSA starts intercepting traffic for all bearers and reports to MDF3 with the same per-bearer correlation number as was used for the given bearer when UE was on 4G.

Step 20: UPF-PSA sends N4 Session Modification Response to SMF+PGW-C.

SMF+PGW-C Starts Releasing 4G Session Towards SGW

Step 21-22: SMF+PGW-C triggers the Delete Bearer Request towards SGW for all the Bearers existed in step 1. Process the response from SGW.

Step 23-24: SMG+PGW-C updates the new Quota to the UPF-PSA by sending the N4 Session Modify message.

After Step 24: SMF+PGW-CC initiates the processing of any rules added/modified/removed provisioned from PCF.

Note: This subsequent rule processing is not shown in the call flow diagram.

SMF-LI 4G and 5G Interworking Message Flows with Lawful intercept

This section documents Message Flows for Lawful Intercept 4G/5G interworking sessions handled by the SMF/PGW-C.

UE starts on 4G, handover to 5G:

- [4G to 5G \(Single UPF\) with Lawful Intercept](#)

- [4G to 5G \(Insertion of I-UPF\) with Lawful Intercept](#)

UE starts on 5G, handover to 4G:

- [5G \(Single UPF\) to 4G with Lawful Intercept](#)

Multiple 4G/5G handovers: (LI Task in place before session setup):

- [4G to 5G \(Single UPF\) - Followed by Return to 4G - with Lawful Intercept](#)
- [5G \(Single UPF\) to 4G Followed by Return to 5G - with Lawful Intercept](#)

Multiple 4G/5G handovers: StartOfIntercept (mid-call addition of LI Task):

- [4G to 5G \(Single UPF\) - 5G StartOfIntercept - Followed by Return to 4G with Lawful Intercept](#)
- [5G \(Single UPF\) to 4G - 4G StartOfIntercept - Followed by Return to 5G - with Lawful Intercept](#)

Release on 5G after 4G/5G handovers:

- [Access Side Release on 5G after 4G/5G Handovers with Lawful Intercept](#)

Release on 4G after 4G/5G handovers:

- [Access Side Release on 4G after 5G/4G Handovers with Lawful Intercept](#)

4G to 5G (Insertion of I-UPF) with Lawful Intercept

This procedure is used to handover a UE that is currently served by the EPS to 5GC using the N26 interface. This procedure consists of three transactions:

Lawful Intercept Tasks are activated on the UPF-PSA as part of the initial session setup:

1. SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Default Bearer FTEID) for Default Bearer with 4G Default Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface).
2. When a dedicated bearer is present, SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Dedicated Bearer FTEID) with 4G Dedicated Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface).
3. For 4G/5G capable UE, SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID) with 5G Session Correlation Number to UPF-PSA during initial session setup. (LI_T2/LI_T3 interface).
4. SMF/PGW-C sends X2 IRI BEGIN: Bearer Activation with 4G Default Bearer Correlation Number to MDF2 at time of initial session setup. (LI_X2 interface).

5. When a dedicated bearer is present, SMF/PGW-C sends X2 IRI BEGIN: Bearer Activation with 4G Dedicated Bearer Correlation Number to MDF2. (LI_X2 interface).
6. UPF-PSA is intercepting 4G UE traffic and reporting X3 CC-data to MDF3 with the appropriate 4G bearer correlation number(s). (LI_X3 interface).

Create SM Context transaction

7. The first transaction of the procedure is triggered when the AMF sends a Create SM Context message including the EPS PDN Connection Container IE with PGW Control F-TEID and bearer information and hoState set to PREPARING. SMF finds the PDN Connection based on the PGW S5-S8 Control TEID.
8. If the PCF has subscribed for event triggers and SMF received trigger information, SMF reports the events to PCF.
9. If an I-UPF has been selected, SMF initiates the N4 Session Establishment procedure with I-UPF including or requesting the Tunnel IDs for N3 and N9 interfaces. The N9 tunnel information for the original IP Anchor UPF is also provided. The Uplink and Downlink PDRs for all QoS Flows are installed on the I-UPF with only catch-all packet filters. The QERs are installed with only QFIs set.
10. SMF responds with a Create SM Context response message including N2 PDU Session Resource Setup Request message with N3 Tunnel ID information of the UPF in it and hoState set to PREPARING. SMF also returns EBI and QFI mapping information and allocated EBI List
11. Handover has begun. Uplink is now 5G, Downlink is still 4G. UPF reports 5G Uplink traffic with 5G Session Correlation Number, 4G Downlink traffic with 4G bearer correlation numbers to the MDF3. (LI_X3 interface).

Update SM Context transaction with N2 Handover Request Ack

12. The second transaction is triggered by receipt of an Update SM Context request with N2 Handover Request Ack and hoState set to PREPARED. SMF receives the Target NG-RAN N3 Tunnel IDs as well as data forwarding Tunnel information.
13. SMF ends the transaction after responding with an Update SM Context response with hoState PREPARED and an N2 Handover Command message including the forwarding Tunnel Info to be sent to the source NG-RAN.

Update SM Context transaction with hoState COMPLETED

14. The third transaction is triggered when AMF sends an Update SM Context message with hoState set to COMPLETED.
15. SMF initiates the N4 Session Modification procedure with the I-UPF to set the target NG-RAN Tunnel information at the I-UPF.

16. Handover to 5G complete. Both Uplink and Downlink are now 5G. UPF reports all intercepted traffic with 5G Session Correlation Number to MDF3.
17. If the PCF has subscribed for event triggers and SMF received trigger information, SMF reports the events to PCF.
18. SMF initiates the N4 Session Modification procedure with the original UPF to update it with the N9 tunnel information of the I-UPF.
19. SMF sends X2 SMF PDU Session Establishment with 5G Session Correlation Number to MDF2.
20. SMF responds to the AMF with an Update SM Context response with hoState set to COMPLETED.

Reference

3GPP TS 23.502, Section 4.11.1.2.2

4G to 5G (Single UPF) - 5G StartOfIntercept - Followed by Return to 4G with Lawful Intercept

Initial Handover from 4G to 5G

Create SM Context transaction initiates the handover to 5G

1. The first transaction of the procedure is triggered when the AMF sends a Create SM Context message including the EPS PDN Connection Container IE with PGW Control F-TEID and bearer information and hoState set to PREPARING. SMF finds the PDN Connection based on the PGW S5-S8 Control TEID.
2. If the PCF has subscribed for event triggers and SMF received trigger information, SMF reports the events to PCF.
3. SMF initiates an N4 Session Establishment procedure with the UPF including or requesting the Tunnel IDs for N3 and N9 interfaces. The N9 tunnel information for the original IP Anchor UPF is also provided. The Uplink and Downlink PDRs for all QoS Flows are installed on the I-UPF with only catch-all packet filters. The QERs are installed with only QFIs set.
4. SMF responds with a Create SM Context response message including N2 PDU Session Resource Setup Request message with N3 Tunnel ID information of the UPF in it and hoState set to PREPARING. SMF also returns EBI and QFI mapping information and allocated EBI List.

Update SM Context Transaction with N2 Handover Request Ack

5. The second transaction is triggered by receipt of an Update SM Context request with N2 Handover Request Ack and hoState set to PREPARED. SMF receives the Target NG-RAN N3 Tunnel IDs as well as data forwarding Tunnel information.

6. SMF ends the transaction after responding with an Update SM Context response with hoState PREPARED and an N2 Handover Command message including the forwarding Tunnel Info to be sent to the source NG-RAN.

Update SM Context Transaction with hoState COMPLETED

7. The third transaction is triggered when AMF sends an Update SM Context message with hoState set to COMPLETED.

8. SMF initiates an N4 Session Modification procedure with the UPF to set the target NG-RAN Tunnel information at the UPF.

9. If the PCF has subscribed for event triggers and SMF received trigger information, SMF reports the events to PCF.

10. SMF responds to the AMF with an Update SM Context response with hoState set to COMPLETED.

LIPF Activates Task on Existing Session

11. LIPF sends X1 ActivateTask for the subscriber with established sessions (LI_X1 interface).

12. SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID) with 5G Session Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)

13. For 4G/5G capable UE, SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Default Bearer FTEID) for Default Bearer with 4G Default Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)

14. For 4G/5G capable UE with dedicated bearer present, SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Dedicated Bearer FTEID) with 4G Dedicated Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)

15. UPF-PSA is intercepting 5G UE traffic and reporting X3 CC-data to MDF3 with 5G session correlation number. (LI_X3 interface)

16. SMF/PGW-C sends 5G X2 StartOfInterceptWithEstablishedPDUSession with 5G Session Correlation Number to MDF2. (LI_X2 interface)

Handover Return to 4G

Retrieve SM Context transaction initiates the handover to 4G (return to 4G)

17. The first transaction of the procedure is triggered when AMF requests SMF to provide SM Context information using the Retrieve SM Context Request message. SMF/PGW-C returns the SM context information including the PGW Control plane F-TEID and the Bearer Contexts created during PDU Session establishment inside the EPS PDN Connection Container IE. The SGW TEID information in the Bearer Context IE is set to 0xFFFFFFFF and the IPv4 address is set to 0.0.0.0, or the IPv6 Prefix Length and IPv6 prefix and Interface Identifier is set to zero.

Modify Bearer Request for S1 Handover

18. The second transaction is triggered by the receipt of a Modify Bearer Request message that starts the S1 handover procedure. The request includes the SGW-U Tunnel ID information. If the default EPS Bearer ID is not available at this point, SMF/PGW-C rejects the handover request and deletes the PDU Session.

19. If the PCF has subscribed for event triggers and SMF/PGW-C received trigger information, SMF/PGW-C reports the events to PCF.

20. SMF/PGW-C initiates an N4 Session Modification procedure with the UPF to switch the tunnel to point to the SGW-U at the UPF.

21. Handover in progress. Uplink is now 4G, Downlink is still 5G. UPF reports intercepted 4G packets with appropriate 4G bearer correlation number, and 5G packets with 5G Session Correlation Number. (LI_X3 interface)

22. Events reported to PCF.

23. SMF/PGW-C ends the transaction after responding with a Modify Bearer Response.

24. Handover (return to 4G) has been completed. Both Uplink and Downlink packets are now 4G. UPF reports intercepted packets to MDF3 with appropriate 4G bearer correlation numbers.

25. SMF/PGW-C sends 4G IRI BEGIN: Bearer Activation with 4G Default Bearer Correlation Number to MDF2. (LI_X2 interface)

26. SMF/PGW-C sends 4G IRI BEGIN: Bearer Activation with 4G Dedicated Bearer Correlation Number to MDF2. (LI_X2 interface)

27. After the handover, the UE initiates a Tracking Area Update procedure which may result in the SMF/PGW-C receiving a Modify Bearer Request message. A new procedure is triggered to report any subscribed event information to the PCF and/or the CHF.

Reference

3GPP TS 23.502, Section 4.11.1.2.1

3GPP TS 23.502, Section 4.11.1.2.2

4G to 5G (Single UPF) - Followed by Return to 4G - with Lawful Intercept

Lawful Intercept Tasks are activated on the UPF-PSA as part of the initial session setup:

1. SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Default Bearer FTEID) for Default Bearer with 4G Default Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)

2. When a dedicated bearer is present, SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Dedicated Bearer FTEID) with 4G Dedicated Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
3. For 4G/5G capable UE, SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID) with 5G Session Correlation Number to UPF-PSA during initial session setup to prepare for possible future handover to 5G. (LI_T2/LI_T3 interface)
4. SMF/PGW-C sends X2 IRI BEGIN: Bearer Activation with 4G Default Bearer Correlation Number to MDF2 at time of initial session setup. (LI_X2 interface)
5. When a dedicated bearer is present, SMF/PGW-C sends X2 IRI BEGIN: Bearer Activation with 4G Dedicated Bearer Correlation Number to MDF2. (LI_X2 interface)
6. UPF-PSA is intercepting 4G UE traffic and reporting X3 CC-data to MDF3 with the appropriate 4G bearer correlation number(s). (LI_X3 interface)

Initial Handover from 4G to 5G

Create SM Context Transaction Initiates the Handover to 5G

7. The first transaction of the procedure is triggered when the AMF sends a Create SM Context message including the EPS PDN Connection Container IE with PGW Control F-TEID and bearer information and hoState set to PREPARING. SMF finds the PDN Connection based on the PGW S5-S8 Control TEID.
8. If the PCF has subscribed for event triggers and SMF received trigger information, SMF reports the events to PCF.
9. SMF initiates an N4 Session Establishment procedure with the UPF including or requesting the Tunnel IDs for N3 and N9 interfaces. The N9 tunnel information for the original IP Anchor UPF is also provided. The Uplink and Downlink PDRs for all QoS Flows are installed on the I-UPF with only catch-all packet filters. The QERs are installed with only QFIs set.
10. SMF responds with a Create SM Context response message including N2 PDU Session Resource Setup Request message with N3 Tunnel ID information of the UPF in it and hoState set to PREPARING. SMF also returns EBI and QFI mapping information and allocated EBI List.
11. Handover has begun. Uplink is now 5G, Downlink is still 4G. UPF reports 5G Uplink traffic with 5G Session Correlation Number and 4G Downlink traffic with 4G bearer correlation numbers to the MDF3. (LI_X3 interface)

Update SM Context Transaction with N2 Handover Request Ack

12. The second transaction is triggered by receipt of an Update SM Context request with N2 Handover Request Ack and hoState set to PREPARED. SMF receives the Target NG-RAN N3 Tunnel IDs as well as data forwarding Tunnel information.

13. SMF ends the transaction after responding with an Update SM Context response with hoState PREPARED and an N2 Handover Command message including the forwarding Tunnel Info to be sent to the source NG-RAN.

Update SM Context Transaction with hoState COMPLETED

14. The third transaction is triggered when AMF sends an Update SM Context message with hoState set to COMPLETED.

15. SMF initiates an N4 Session Modification procedure with the UPF to set the target NG-RAN Tunnel information at the UPF.

16. Handover to 5G complete. Both Uplink and Downlink are now 5G. UPF reports all intercepted traffic with 5G Session Correlation Number to MDF3. (LI_X3 interface)

17. If the PCF has subscribed for event triggers and SMF received trigger information, SMF reports the events to PCF.

18. SMF sends X2 SMF PDU Session Establishment with 5G Session Correlation Number to MDF2. (LI_X2 interface)

19. SMF responds to the AMF with an Update SM Context response with hoState set to COMPLETED.

20. The session continues for sometime as 5G. UPF-PSA intercepts UE traffic from all flows and reports to MDF3 with 5G Session Correlation Number. (LI_X3 interface)

Handover return to 4G

Retrieve SM Context transaction initiates the handover to 4G (return to 4G)

21. The first transaction of the procedure is triggered when AMF requests SMF to provide SM Context information using the Retrieve SM Context Request message. SMF/PGW-C returns the SM context information including the PGW Control plane F-TEID and the Bearer Contexts created during PDU Session establishment inside the EPS PDN Connection Container IE. The SGW TEID information in the Bearer Context IE is set to 0xFFFFFFFF and the IPv4 address is set to 0.0.0.0, or the IPv6 Prefix Length and IPv6 prefix and Interface Identifier is set to zero.

Modify Bearer Request for S1 handover

22. The second transaction is triggered by the receipt of a Modify Bearer Request message that starts the S1 handover procedure. The request includes the SGW-U Tunnel ID information. If the default EPS Bearer ID is not available at this point, SMF/PGW-C rejects the handover request and deletes the PDU Session.

23. If the PCF has subscribed for event triggers and SMF/PGW-C received trigger information, SMF/PGW-C reports the events to PCF.

24. SMF/PGW-C initiates an N4 Session Modification procedure with the UPF to switch the tunnel to point to the SGW-U at the UPF.

25. Handover in progress. Uplink is now 4G, Downlink is still 5G. UPF reports intercepted 4G packets with appropriate 4G bearer correlation number, 5G packets with 5G Session Correlation Number. (LI_X3 interface)
26. Events reported to PCF.
27. SMF/PGW-C ends the transaction after responding with a Modify Bearer Response.
28. Handover (return to 4G) has been completed. Both Uplink and Downlink packets are now 4G. UPF reports intercepted packets to MDF3 with appropriate 4G bearer correlation numbers.
29. SMF/PGW-C sends 4G IRI BEGIN: Bearer Activation with 4G Default Bearer Correlation Number to MDF2. (LI_X2 interface)
30. SMF/PGW-C sends 4G IRI BEGIN: Bearer Activation with 4G Dedicated Bearer Correlation Number to MDF2. (LI_X2 interface)
31. After the handover, the UE initiates a Tracking Area Update procedure which may result in the SMF/PGW-C receiving a Modify Bearer Request message. A new procedure is triggered to report any subscribed event information to the PCF and/or the CHF.

Reference

3GPP TS 23.502, Section 4.11.1.2.1

3GPP TS 23.502, Section 4.11.1.2.2

4G to 5G (Single UPF) with Lawful Intercept

This procedure is used to handover a UE that is currently served by the EPS to 5GC using the N26 interface. This procedure consists of three transactions:

Lawful Intercept Tasks are Activated on the UPF-PSA as part of the Initial Session Setup

1. SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Default Bearer FTEID) for Default Bearer with 4G Default Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
2. When a dedicated bearer is present, SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Dedicated Bearer FTEID) with 4G Dedicated Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
3. For 4G/5G capable UE, SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID) with 5G Session Correlation Number to UPF-PSA during initial session setup. (LI_T2/LI_T3 interface)

4. SMF/PGW-C sends X2 IRI BEGIN: Bearer Activation with 4G Default Bearer Correlation Number to MDF2 at the time of initial session setup. (LI_X2 interface)
5. When a dedicated bearer is present, SMF/PGW-C sends X2 IRI BEGIN: Bearer Activation with 4G Dedicated Bearer Correlation Number to MDF2. (LI_X2 interface)
6. UPF-PSA is intercepting 4G UE traffic and reporting X3 CC-data to MDF3 with the appropriate 4G bearer correlation number(s). (LI_X3 interface)

Create SM Context Transaction

7. The first transaction of the handover procedure is triggered when the AMF sends a Create SM Context message including the EPS PDN Connection Container IE with PGW Control F- TEID and bearer information and hoState set to PREPARING. SMF finds the PDN Connection based on the PGW S5-S8 Control TEID.
8. If the PCF has subscribed for event triggers and SMF received trigger information, SMF reports the events to PCF.
9. SMF initiates an N4 Session Establishment procedure with the UPF including or requesting the Tunnel IDs for N3 and N9 interfaces. The N9 tunnel information for the original IP Anchor UPF is also provided. The Uplink and Downlink PDRs for all QoS Flows are installed on the I-UPF with only catch-all packet filters. The QERs are installed with only QFIs set.
10. SMF responds with a Create SM Context response message including N2 PDU Session Resource Setup Request message with N3 Tunnel ID information of the UPF in it and hoState set to PREPARING. SMF also returns EBI and QFI mapping information and allocated EBI List.
11. Handover has begun. Uplink is now 5G, Downlink is still 4G. UPF reports 5G Uplink traffic with 5G Session Correlation Number and 4G Downlink traffic with 4G bearer correlation numbers to the MDF3. (LI_X3 interface)

Update SM Context Transaction with N2 Handover Request Ack

12. The second transaction is triggered by receipt of an Update SM Context request with N2 Handover Request Ack and hoState set to PREPARED. SMF receives the Target NG-RAN N3 Tunnel IDs as well as data forwarding Tunnel information.
13. SMF ends the transaction after responding with an Update SM Context response with hoState PREPARED and an N2 Handover Command message including the forwarding Tunnel Info to be sent to the source NG-RAN.

Update SM Context Transaction with hoState COMPLETED

14. The third transaction is triggered when AMF sends an Update SM Context message with hoState set to COMPLETED.

15. SMF initiates an N4 Session Modification procedure with the UPF to set the target NG-RAN Tunnel information at the UPF.
16. Handover to 5G complete. Both Uplink and Downlink are now 5G. UPF reports all intercepted traffic with 5G Session Correlation Number to MDF3. (LI_X3 interface)
17. If the PCF has subscribed for event triggers and SMF received trigger information, SMF reports the events to PCF.
18. SMF sends X2 SMF PDU Session Establishment with 5G Session Correlation Number to MDF2. (LI_X2 interface)
19. SMF responds to the AMF with an Update SM Context response with hoState set to COMPLETED.

Reference

3GPP TS 23.502, Section 4.11.1.2.2

5G (Single UPF) to 4G - 4G StartOfIntercept - Followed by Return to 5G - with Lawful Intercept

Initial Handover from 5G to 4G

Retrieve SM Context Transaction Initiates the Handover to 4G

1. The first transaction of the procedure is triggered when AMF requests SMF to provide SM Context information using the Retrieve SM Context Request message. SMF/PGW-C returns the SM context information including the PGW Control plane F-TEID and the Bearer Contexts created during PDU Session establishment inside the EPS PDN Connection Container IE. The SGW TEID information in the Bearer Context IE is set to **0xFFFFFFFF** and the IPv4 address is set to **0.0.0.0**, or the IPv6 Prefix Length and IPv6 prefix and Interface Identifier is set to **zero**.

Modify Bearer Request for S1 Handover

2. The second transaction is triggered by the receipt of a Modify Bearer Request message that starts the S1 handover procedure. The request includes the SGW-U Tunnel ID information. If the default EPS Bearer ID is not available at this point, SMF/PGW-C rejects the handover request and deletes the PDU Session.
3. If the PCF has subscribed for event triggers and SMF/PGW-C received trigger information, SMF/PGW-C reports the events to PCF.
4. SMF/PGW-C initiates an N4 Session Modification procedure with the UPF to switch the tunnel to point to the SGW-U at the UPF
5. Events reported to PCF.

6. SMF/PGW-C ends the transaction after responding with a Modify Bearer Response
7. After the handover, the UE initiates a Tracking Area Update procedure which may result in the SMF/PGW-C receiving a Modify Bearer Request message. A new procedure is triggered to report any subscribed event information to the PCF and/or the CHF.

LIPF Activates Task to Intercept Established Session

8. LIPF sends X1 ActivateTask for the subscriber with an established session (LI_X1 interface)
9. SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Default Bearer FTEID) for Default Bearer with 4G Default Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
10. When a dedicated bearer is present, SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Dedicated Bearer FTEID) with 4G Dedicated Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
11. For 4G/5G capable UE, SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID) with 5G Session Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
12. UPF-PSA is intercepting 4G UE traffic and reporting X3 CC-data to MDF3 with the appropriate 4G bearer correlation number(s). (LI_X3 interface)
13. SMF/PGW-C sends X2 IRI BEGIN: StartOfInterceptWithActiveBearer with 4G Default Bearer Correlation Number to MDF2. (LI_X2 interface)
14. When a dedicated bearer is present, SMF/PGW-C sends X2 IRI BEGIN: StartOfInterceptWithActiveBearer with 4G Dedicated Bearer Correlation Number to MDF2. (LI_X2 interface)

Handover Return to 5G

Create SM Context Transaction Initiates the Handover to 5G

15. The first transaction of the procedure is triggered when the AMF sends a Create SM Context message including the EPS PDN Connection Container IE with PGW Control F-TEID and bearer information and hoState set to PREPARING. SMF finds the PDN Connection based on the PGW S5-S8 Control TEID.
16. If the PCF has subscribed for event triggers and SMF received trigger information, SMF reports the events to PCF.
17. SMF initiates an N4 Session Establishment procedure with the UPF including or requesting the Tunnel IDs for N3 and N9 interfaces. The N9 tunnel information for the original IP Anchor UPF is also provided. The Uplink and Downlink PDRs for all QoS Flows are installed on the I-UPF with only catch-all packet filters. The QERs are installed with only QFIs set.
18. SMF responds with a Create SM Context response message including N2 PDU Session Resource Setup Request message with N3 Tunnel ID information of the UPF in it and hoState set to PREPARING. SMF also returns EBI and QFI mapping information and allocated EBI List.

19. Handover has begun. Uplink is now 5G, Downlink is still 4G. UPF reports 5G Uplink traffic with 5G Session Correlation Number, and 4G Downlink traffic with 4G bearer correlation numbers to the MDF3. (LI_X3 interface)

Update SM Context Transaction with N2 Handover Request Ack

20. The second transaction is triggered by receipt of an Update SM Context request with N2 Handover Request Ack and hoState set to PREPARED. SMF receives the Target NG-RAN N3 Tunnel IDs as well as data forwarding Tunnel information.

21. SMF ends the transaction after responding with an Update SM Context response with hoState PREPARED and an N2 Handover Command message including the forwarding Tunnel Info to be sent to the source NG-RAN.

Update SM Context Transaction with hoState COMPLETED

22. The third transaction is triggered when AMF sends an Update SM Context message with hoState set to COMPLETED.

23. SMF initiates an N4 Session Modification procedure with the UPF to set the target NG-RAN Tunnel information at the UPF.

24. Handover to 5G complete. Both Uplink and Downlink are now 5G. UPF reports all intercepted traffic with 5G Session Correlation Number to MDF3. (LI_X3 interface)

25. If the PCF has subscribed for event triggers and SMF received trigger information, SMF reports the events to PCF.

26. SMF responds to the AMF with an Update SM Context response with hoState set to COMPLETED.

27. SMF sends X2 SMF PDU Session Establishment with 5G Session Correlation Number to MDF2. (LI_X2 interface)

Reference

3GPP TS 23.502, Section 4.11.1.2.1

3GPP TS 23.502, Section 4.11.1.2.2

5G (Single UPF) to 4G Followed by Return to 5G - with Lawful Intercept

Lawful Intercept Tasks are activated on the UPF-PSA as part of the initial session setup:

1. SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID) with 5G Session Correlation Number to UPF-PSA during initial session setup. (LI_T2/LI_T3 interface)

2. For 4G/5G capable UE, SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Default Bearer FTEID) for Default Bearer with 4G Default Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
3. For 4G/5G capable UE when a dedicated bearer is present, SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Dedicated Bearer FTEID) with 4G Dedicated Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
4. SMF/PGW-C sends X2 SMF PDU Session Establishment with 5G Session Correlation Number to MDF2 at time of initial session setup. (LI_X2 interface)
5. UPF-PSA is intercepting 5G UE traffic and reporting X3 CC-data to MDF with the 5G Session Correlation Number. (LI_X3 interface)

Initial Handover from 5G to 4G

Retrieve SM Context Transaction Initiates the Handover to 4G

6. The first transaction of the procedure is triggered when AMF requests SMF to provide SM Context information using the Retrieve SM Context Request message. SMF/PGW-C returns the SM context information including the PGW Control plane F-TEID and the Bearer Contexts created during PDU Session establishment inside the EPS PDN Connection Container IE. The SGW TEID information in the Bearer Context IE is set to 0xFFFFFFFF and the IPv4 address is set to 0.0.0.0, or the IPv6 Prefix Length and IPv6 prefix and Interface Identifier is set to zero.
7. Handover in progress. Uplink is now 4G, Downlink is still 5G. UPF reports intercepted 4G Uplink packets with 4G bearer correlation numbers, and 5G Downlink packets with 5G Session Correlation Number. (LI_X3 interface)

Modify Bearer Request for S1 Handover

8. The second transaction is triggered by the receipt of a Modify Bearer Request message that starts the S1 handover procedure. The request includes the SGW-U Tunnel ID information. If the default EPS Bearer ID is not available at this point, SMF/PGW-C rejects the handover request and deletes the PDU Session.
9. If the PCF has subscribed for event triggers and SMF/PGW-C received trigger information, SMF/PGW-C reports the events to PCF.
10. SMF/PGW-C initiates an N4 Session Modification procedure with the UPF to switch the tunnel to point to the SGW-U at the UPF.
11. Events reported to PCF.
12. SMF/PGW-C ends the transaction after responding with a Modify Bearer Response.
13. Handover complete. Both Uplink and Downlink packets are now 4G. UPF reports intercepted packets to MDF3 with appropriate 4G bearer correlation numbers.

14. SMF/PGW-C sends 4G IRI BEGIN: Bearer Activation with 4G Default Bearer Correlation Number to MDF2. (LI_X2 interface)
15. SMF/PGW-C sends 4G IRI BEGIN: Bearer Activation with 4G Dedicated Bearer Correlation Number to MDF2. (LI_X2 interface)
16. After the handover, the UE initiates a Tracking Area Update procedure which may result in the SMF/PGW-C receiving a Modify Bearer Request message. A new procedure will be triggered to report any subscribed event information to the PCF and/or the CHF.
17. The session continues for sometime as 4G. UPF-PSA intercepts UE traffic and reports to MDF3 with appropriate 4G bearer correlation numbers. (LI_X3 interface)

Handover Return to 5G

Create SM Context Transaction Initiates the Handover to 5G

18. The first transaction of the procedure is triggered when the AMF sends a Create SM Context message including the EPS PDN Connection Container IE with PGW Control F-TEID and bearer information and hoState set to PREPARING. SMF finds the PDN Connection based on the PGW S5-S8 Control TEID.
19. If the PCF has subscribed for event triggers and SMF received trigger information, SMF reports the events to PCF.
20. SMF initiates an N4 Session Establishment procedure with the UPF including or requesting the Tunnel IDs for N3 and N9 interfaces. The N9 tunnel information for the original IP Anchor UPF is also provided. The Uplink and Downlink PDRs for all QoS Flows are installed on the I-UPF with only catch-all packet filters. The QERs are installed with only QFIs set.
21. SMF responds with a Create SM Context response message including N2 PDU Session Resource Setup Request message with N3 Tunnel ID information of the UPF in it and hoState set to PREPARING. SMF also returns EBI and QFI mapping information and allocated EBI List.
22. Handover has begun. Uplink is now 5G, Downlink is still 4G. UPF reports 5G Uplink traffic with 5G Session Correlation Number, 4G Downlink traffic with 4G bearer correlation numbers to the MDF3. (LI_X3 interface)

Update SM Context Transaction with N2 Handover Request Ack

23. The second transaction is triggered by receipt of an Update SM Context request with N2 Handover Request Ack and hoState set to PREPARED. SMF receives the Target NG-RAN N3 Tunnel IDs as well as data forwarding Tunnel information.
24. SMF ends the transaction after responding with an Update SM Context response with hoState PREPARED and an N2 Handover Command message including the forwarding Tunnel Info to be sent to the source NG-RAN.

Update SM Context Transaction with hoState COMPLETED

25. The third transaction is triggered when AMF sends an Update SM Context message with hoState set to COMPLETED.

26. SMF initiates an N4 Session Modification procedure with the UPF to set the target NG-RAN Tunnel information at the UPF.

27. Handover to 5G complete. Both Uplink and Downlink are now 5G. UPF reports all intercepted traffic with 5G Session Correlation Number to MDF3. (LI_X3 interface)

28. If the PCF has subscribed for event triggers and SMF received trigger information, SMF reports the events to PCF.

29. SMF responds to the AMF with an Update SM Context response with hoState set to COMPLETED.

30. SMF sends X2 SMF PDU Session Establishment with 5G Session Correlation Number to MDF2. (LI_X2 interface)

Reference

3GPP TS 23.502, Section 4.11.1.2.1

3GPP TS 23.502, Section 4.11.1.2.2

5G (Single UPF) to 4G with Lawful Intercept

5G (Single UPF) to 4G with Lawful Intercept

This procedure is used to handover a UE that is currently served by the 5GC to EPS using the N26 interface. This procedure consists of two transactions:

Lawful Intercept Tasks are activated on the UPF-PSA as part of the initial session setup:

1. SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID) with 5G Session Correlation Number to UPF-PSA during initial session setup. (LI_T2/LI_T3 interface)
2. For 4G/5G capable UE, SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Default Bearer FTEID) for Default Bearer with 4G Default Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
3. For 4G/5G capable UE when a dedicated bearer is present, SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Dedicated Bearer FTEID) with 4G Dedicated Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
4. SMF/PGW-C sends X2 SMF PDU Session Establishment with 5G Session Correlation Number to MDF2 at the time of initial session setup. (LI_X2 interface)
5. UPF-PSA is intercepting 5G UE traffic and reporting X3 CC-data to MDF with the 5G Session Correlation Number. (LI_X3 interface)

Retrieve SM Context Transaction

6. The first transaction of the handover procedure is triggered when AMF requests SMF to provide SM Context information using the Retrieve SM Context Request message. SMF/PGW-C returns the SM context information including the PGW Control plane F-TEID and the Bearer Contexts created during PDU Session establishment inside the EPS PDN Connection Container IE. The SGW TEID information in the Bearer Context IE is set to **0xFFFFFFFF** and the IPv4 address is set to **0.0.0.0**, or the IPv6 Prefix Length and IPv6 prefix and Interface Identifier is set to **0**.
7. Handover has begun. Uplink packets are now 4G, Downlink packets are still 5G. UPF reports intercepted Uplink packets to MDF3 with appropriate 4G bearer correlation number(s) and 5G packets with 5G Session Correlation Number. (LI_X3 interface)

Modify Bearer Request for S1 Handover

8. The second transaction is triggered by the receipt of a Modify Bearer Request message that starts the S1 handover procedure. The request includes the SGW-U Tunnel ID information. If the default EPS Bearer ID is not available at this point, SMF/PGW-C rejects the handover request and deletes the PDU Session.
9. If the PCF has subscribed for event triggers and SMF/PGW-C received trigger information, SMF/PGW-C reports the events to PCF.
10. SMF/PGW-C initiates an N4 Session Modification procedure with the UPF to switch the tunnel to point to the SGW-U at the UPF.
11. SMF/PGW-C ends the transaction after responding with a Modify Bearer Response.
12. Handover has been completed. Both Uplink and Downlink packets are now 4G. UPF reports intercepted packets to MDF3 with appropriate 4G bearer correlation numbers. (LI_X3 interface)
13. MF/PGW-C sends 4G IRI BEGIN: Bearer Activation with 4G Default Bearer Correlation Number to MDF2. (LI_X2 interface)
14. SMF/PGW-C sends 4G IRI BEGIN: Bearer Activation with 4G Dedicated Bearer Correlation Number to MDF2. (LI_X2 interface)
15. After the handover, the UE initiates a Tracking Area Update procedure which may result in the SMF/PGW-C receiving a Modify Bearer Request message. A new procedure is triggered to report any subscribed event information to the PCF and/or the CHF.

Note: It is possible that intercepted 4G packets with 4G bearer correlation numbers are reported to MDF3 before the corresponding IRI BEGIN: Bearer Activation events have been reported to MDF2. Prepare MDFs to handle this.

Reference

3GPP TS 23.502, Section 4.11.1.2.1

Access Side Release on 4G after 5G/4G Handovers with Lawful Intercept

UE triggers the detach procedure by sending a Detach Request to MME. MME sends a Delete Session Request message to SGW to trigger the release of the PDN Session. SGW relays this message to SMF+PGW-C.

1. Delete Session Request sent by SGW is received by SMF/PGW-C.
2. SMF/PGW-C releases the PFCP session by sending the Sx Session Release Request message towards the UPF.
3. At this point, UE data is no longer flowing. X3 CC-data will no longer be reported to MDF3 for the session/bearers.
4. SMF/PGW-C sends X1 DeactivateTask (TargetIdentifier: PFCP SessionID, Default Bearer FTEID) for Default Bearer with 4G Default Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
5. When a dedicated bearer is present, SMF/PGW-C sends X1 DeactivateTask (TargetIdentifier: PFCP SessionID, Dedicated Bearer FTEID) with 4G Dedicated Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
6. SMF/PGW-C sends X1 DeactivateTask (TargetIdentifier: PFCP SessionID) with 5G Session Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
7. If Dynamic PCC is applied to this session the SMF/PGW-C invokes SM Policy Association Termination procedure towards the PCF to delete the PDU session.
8. SMF/PGW-C sends 4G IRI END: Bearer Deactivation with 4G Default Bearer Correlation Number to MDF2. (LI_X2 interface)
9. When a dedicated bearer was present, SMF/PGW-C sends 4G IRI END: Bearer Deactivation with 4G Dedicated Bearer Correlation Number to MDF2. (LI_X2 interface)
10. SMF/PGW-C sends 5G SMF PDU Session Release with 5G Session Correlation Number to MDF2. This ends the outstanding 5G Session Correlation used before handover to 4G. (LI_X2 interface)
11. SMF/PGW-C acknowledges by sending Delete Session Response to SGW.

Reference

3GPP TS 23.401, Section 5.3.8.2.1

Access Side Release on 5G after 4G/5G Handovers with Lawful Intercept

1. From previous message exchanges (not shown here), UPF-PSA has been programmed to intercept Uplink & Downlink packets from/to UE. UPF-PSA reports intercepted packets in X3 CC-data events to MDF3 (LI_X3 interface).
2. AMF sends Release SM Context Request.
3. SMF sends N4 Session Release Request (PFCP message) to UPF-PSA.
4. UPF-PSA releases session resources and sends N4 Session Release Response (PFCP message) to SMF.
5. At this point session no longer exists on UPF-PSA. No further X3 CC-data events are sent to MDF3 for the session which is releasing.
6. SMF sends X1 DeactivateTask (TargetIdentifier:PFCP SessionID) with 5G Session Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
7. SMF sends X1 DeactivateTask (TargetIdentifier:PFCP SessionID, Default Bearer FTEID) with 4G Default Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
8. SMF sends X1 DeactivateTask (TargetIdentifier:PFCP SessionID, Dedicated Bearer FTEID) with 4G Dedicated Bearer Correlation Number to UPF-PSA. (LI_T2/LI_T3 interface)
9. SMF sends Release SM Context Response to AMF to indicate the session has been released.
10. SMF sends 5G SMF PDU Session Release event with 5G Session Correlation Number to MDF2. This ends the outstanding 4G Default Bearer Correlation in use before the handover to 5G. (LI_X2 interface)
11. SMF sends 4G IRI END: Bearer Deactivation with 4G Default Bearer Correlation Number of MDF2. This ends the outstanding 4G Default Bearer Correlation in use before the handover to 5G. (LI_X2 interface)
12. SMF sends 4G IRI END: Bearer Deactivation with 4G Dedicated Bearer Correlation Number of MDF2. This ends the outstanding 4G Dedicated Bearer Correlation in use before the handover to 5G. (LI_X2 interface)
13. SMF performs SM Policy Control Delete procedure to delete the SM Policy Association at the PCF.

SMF-LI Message Flows with Lawful Intercept X1 ModifyTask

The following call flows with ModifyTask are expected scenarios:

ModifyTask is used in cases where a warrant change on the targeted subscriber requires a change in the Task DeliveryType.

LIPF uses the X1 ModifyTask request to modify the following fields in an existing Task:

- DeliveryType: X2Only or X2andX3
- PDSR: enabled/disabled
- ListOfDIDs: One or more DIDs as required to implement the Task DeliveryType

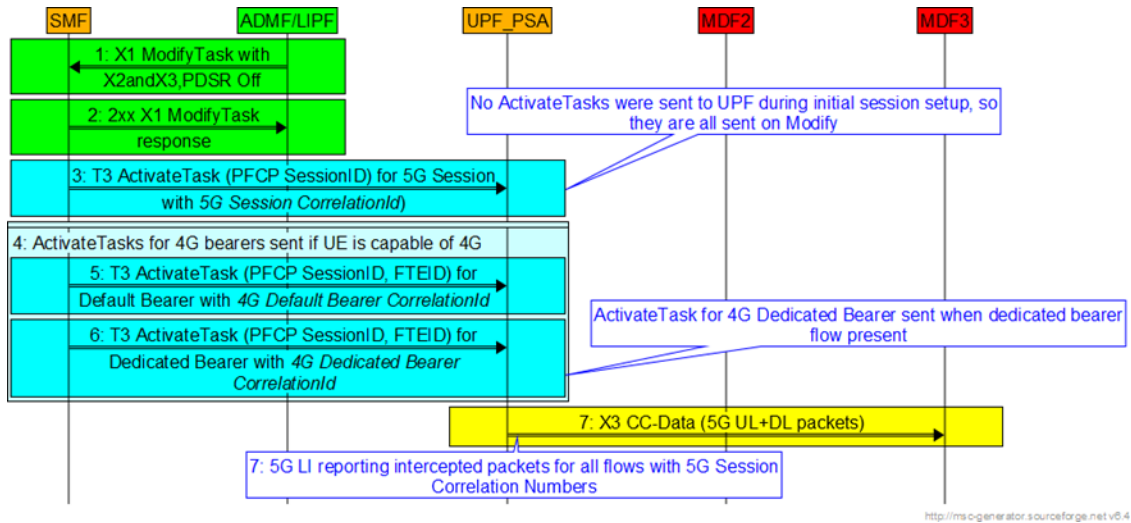
Note: LIPF provisioning an SMF Task with DeliveryType=X3Only is unexpected.

The Modify flows documented below include only the flows used in the current deployment:

1. [X1 ModifyTask: Previous \(DeliveryType=X2,PDSR=OFF\), New \(DeliveryType=X2andX3,PDSR=OFF\)](#)
2. [X1 ModifyTask: Previous \(DeliveryType=X2,PDSR=OFF\), New \(DeliveryType=X2andX3,PDSR=ON\)](#)
3. [X1 ModifyTask: Previous \(DeliveryType=X2, PDSR=ON\), New \(DeliveryType=X2andX3, PDSR=OFF\)](#)
4. [X1 ModifyTask: Previous \(DeliveryType=X2,PDSR=ON\), New \(DeliveryType=X2andX3,PDSR=ON\)](#)
5. [X1 ModifyTask: Previous \(DeliveryType=X2andX3,PDSR=OFF\), New \(DeliveryType=X2,PDSR=OFF\)](#)
6. [X1 ModifyTask: Previous \(DeliveryType=X2andX3,PDSR=OFF\), New \(DeliveryType=X2,PDSR=ON\)](#)
7. [X1 ModifyTask: Previous \(DeliveryType=X2andX3,PDSR=ON\), New \(DeliveryType=X2,PDSR=OFF\)](#)
8. [X1 ModifyTask: Previous \(DeliveryType=X2andX3,PDSR=ON\), New \(DeliveryType=X2,PDSR=ON\)](#)

X1 ModifyTask: Previous (DeliveryType=X2,PDSR=OFF), New (DeliveryType=X2andX3,PDSR=OFF)

Figure 8. X1 ModifyTask: Previous (DeliveryType=X2,PDSR=OFF), New (DeliveryType=X2andX3,PDSR=OFF)



This call flow starts with the initial state where an ActivateTask is X2Only with PDSR disabled. The task is then modified to X2andX3 with PDSR disabled.

Initial Setup - steps not shown in the diagram

1. LIPF sends ActivateTask with DeliveryType=X2Only, PDSR=Off.
2. ActivateTask establishes and intercepts the session.

Note: ActivateTask is not sent to UPF during the initial session setup since the LIPF ActivateTask is currently X2Only.

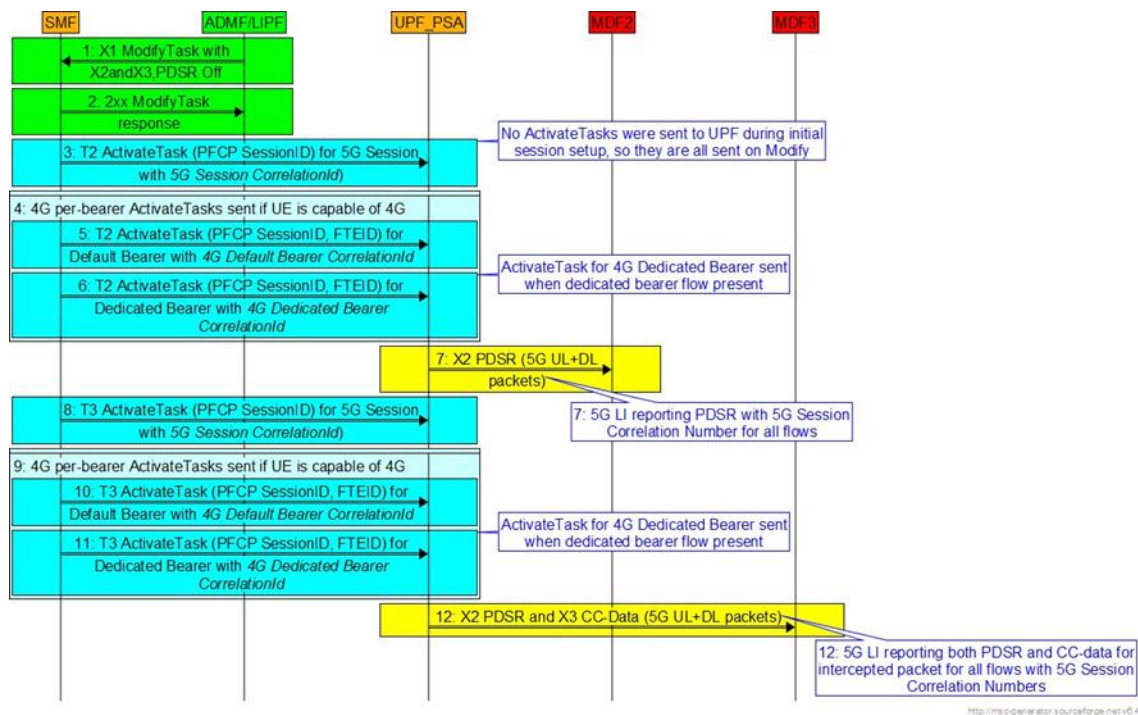
Modify Task

1. LIPF sends ModifyTask with DeliveryType=X2andX3, PDSR=Off.
2. SMF processes the ModifyTask and responds to LIPF with 200 OK.
3. SMF sends T3 ActivateTask to UPF for 5G session with 5G Session Correlation Number.
4. If UE is capable of 4G, T3 ActivateTask for 4G bearers are sent to UPF.

5. SMF sends T3 ActivateTask to UPF for 4G default bearer with 4G Default Bearer Correlation Number.
6. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T3 ActivateTask to UPF for 4G dedicated bearer with 4G Dedicated Bearer Correlation Number.
7. UPF starts sending X3 CC-data to MDF3 for intercepted 5G Uplink and Downlink packets using 5G Session Correlation Number.

X1 ModifyTask: Previous (DeliveryType=X2,PDSR=OFF), New (DeliveryType=X2andX3,PDSR=ON)

Figure 9. X1 ModifyTask: Previous (DeliveryType=X2,PDSR=OFF), New (DeliveryType=X2andX3,PDSR=ON)



This call flow starts with the initial state where an ActivateTask is X2Only with PDSR disabled. The task is then modified to X2andX3 with PDSR enabled.

Initial Setup - steps not shown in diagram

1. LIPF sends ActivateTask with DeliveryType=X2Only, PDSR=Off.
2. ActivateTask establishes and intercepts the session.

Note: *ActivateTask is not sent to UPF during the initial session setup since the LIPF ActivateTask is currently X2Only.*

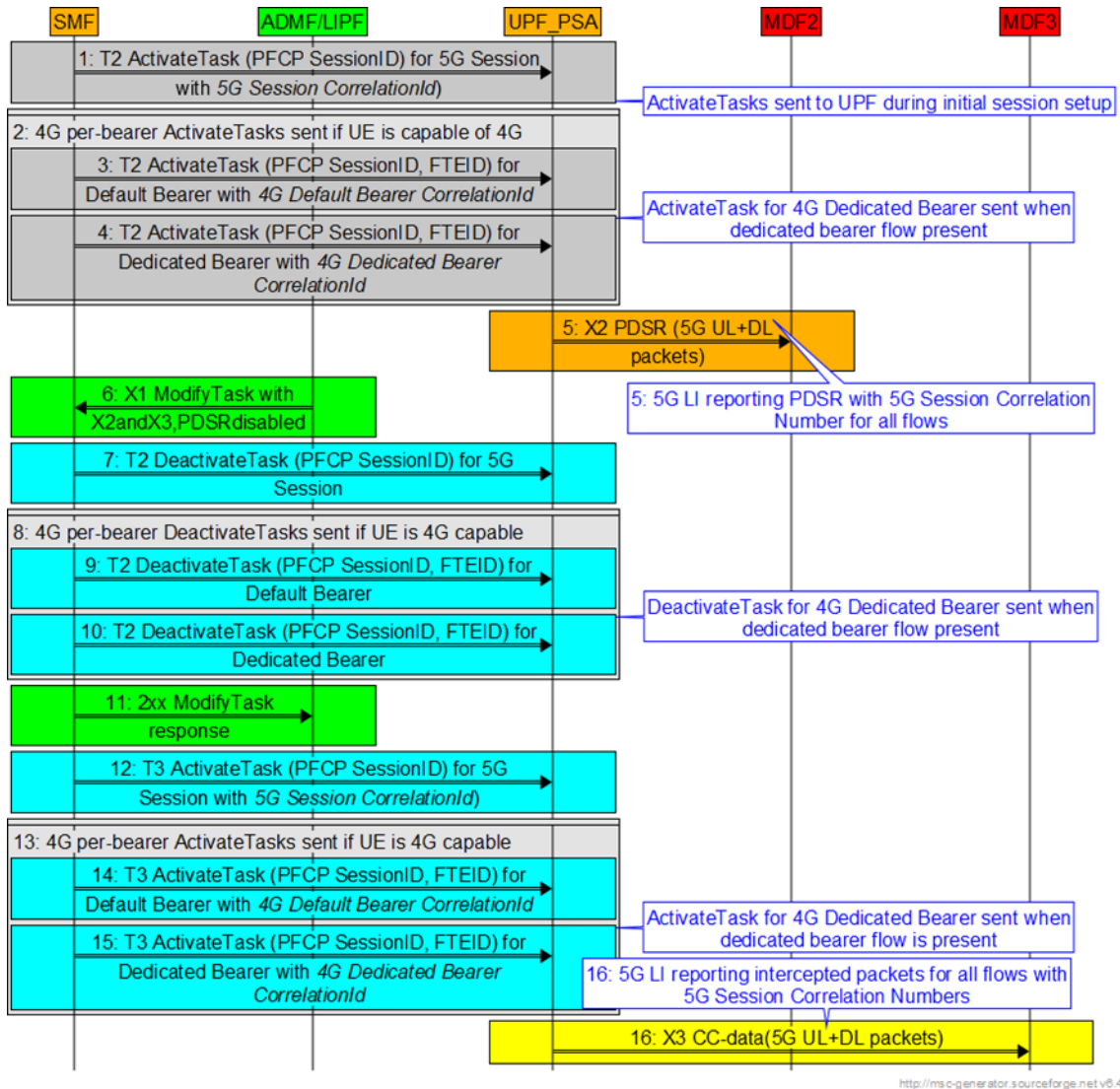
ModifyTask

1. LIPF sends ModifyTask with DeliveryType=X2andX3, PDSR=On.
2. SMF processes the ModifyTask and responds to LIPF with 200 OK.
3. SMF sends T2 ActivateTask to UPF for 5G session with 5G Session Correlation Number.
4. If UE is capable of 4G, T2 ActivateTask for 4G bearers are sent to UPF.
5. SMF sends T2 ActivateTask to UPF for 4G default bearer with 4G Default Bearer Correlation Number.
6. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T2 ActivateTask to UPF for 4G dedicated bearer with 4G Dedicated Bearer Correlation Number.
7. UPF starts sending X2 PDSR events to MDF2 for intercepted 5G Uplink and Downlink packets using 5G Session Correlation Number.
8. SMF sends T3 ActivateTask to UPF for 5G session with 5G Session Correlation Number.
9. If UE is capable of 4G, T3 ActivateTask for 4G bearers are sent to UPF.
10. SMF sends T3 ActivateTask to UPF for 4G default bearer with 4G Default Bearer Correlation Number.
11. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T3 ActivateTask to UPF for 4G dedicated bearer with 4G Dedicated Bearer Correlation Number.
12. UPF sends both X2 PDSR events to MDF2 and X3 CC-data to MDF3 for intercept 5G Uplink and Downlink packets using the 5G Session Correlation Number.

Note: *When MDF is capable of generating PDSR events from X3 CC-data, it is not expected that the LIPF would provision the SMF for both PDSR and X3 CC-data as shown in this call flow. Instead, when warrants require both X2 PDSR and X3 CC-data, the LIPF provisions the SMF for X3 CC-data, and then the MDF constructs PDSR events from the collected X3 CC-data.*

X1 ModifyTask: Previous (DeliveryType=X2, PDSR=ON), New (DeliveryType=X2andX3, PDSR=OFF)

Figure 10. X1 ModifyTask: Previous (DeliveryType=X2, PDSR=ON), New (DeliveryType=X2andX3, PDSR=OFF)



This call flow starts with the initial state where an ActivateTask is X2Only with PDSR enabled. The task is then modified to X2andX3 with PDSR disabled.

Initial Setup - not shown on the diagram

1. LIPF sends ActivateTask with DeliveryType=X2Only, PDSR=On.
2. ActivateTask establishes and intercepts the session.

Initial T2 ActivateTask Signaling

SMF

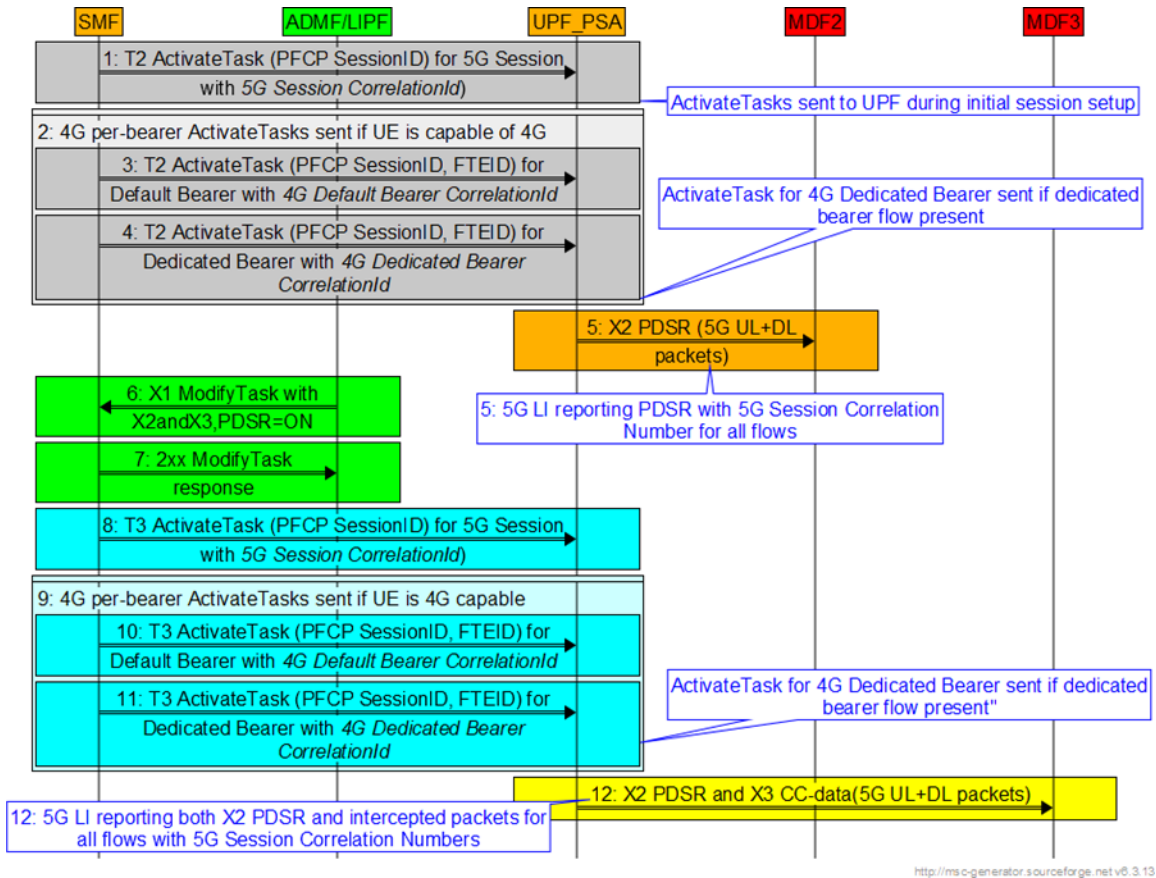
1. SMF sends T2 ActivateTask to UPF with 5G Session Correlation Number.
2. If UE is capable of 4G, SMF sends T2 ActivateTask for 4G bearers to UPF.
3. SMF sends T2 ActivateTask for 4G default bearer with 4G Default Bearer Correlation Number.
4. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T2 ActivateTask to UPF for 4G dedicated bearer with 4G Dedicated Bearer Correlation Number.
5. UPF reports X2 PDSR events for all 5G Uplink and Downlink packets.

ModifyTask

6. LIPF sends ModifyTask with DeliveryType=X2andX3, PDSR=Off.
7. SMF sends T2 DeactivateTask with 5G Session Correlation Number to UPF.
8. If UE is capable of 4G, SMF sends T2 ActivateTask for 4G bearers to UPF.
9. SMF sends T2 DeactivateTask with 4G Default Bearer Correlation Number to UPF.
10. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T2 DeactivateTask to UPF with 4G Dedicated Bearer Correlation Number.
11. SMF processes respond to LIPF with 200 OK for the ModifyTask.
12. SMF sends T3 ActivateTask to UPF for 5G session with 5G Session Correlation Number.
13. If UE is capable of 4G, T3 ActivateTask for 4G bearers are sent to UPF.
14. SMF sends T3 ActivateTask to UPF for 4G default bearer with 4G Default Bearer Correlation Number.
15. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T3 ActivateTask to UPF for 4G dedicated bearer with 4G Dedicated Bearer Correlation Number.
16. UPF starts sending X3 CC-data to MDF3 for intercepted 5G Uplink and Downlink packets using 5G Session Correlation Number.

X1 ModifyTask: Previous (DeliveryType=X2,PDSR=ON), New (DeliveryType=X2andX3,PDSR=ON)

Figure 11. X1 ModifyTask: Previous (DeliveryType=X2,PDSR=ON), New (DeliveryType=X2andX3,PDSR=ON)



This call flow starts with the initial state where an ActivateTask is X2Only with PDSR enabled. The task is then modified to X2andX3 with PDSR enabled.

Initial Setup - not shown on the diagram

1. LIPF sends ActivateTask with DeliveryType=X2Only, PDSR=On.
2. ActivateTask establishes and intercepts the session.

Initial T2 ActivateTask Signaling

1. SMF sends T2 ActivateTask to UPF with 5G Session Correlation Number.
2. If UE is capable of 4G, SMF sends T2 ActivateTask for 4G bearers to UPF.
3. SMF sends T2 ActivateTask for 4G default bearer with 4G Default Bearer Correlation Number.

SMF

4. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T2 ActivateTask to UPF for 4G dedicated bearer with 4G Dedicated Bearer Correlation Number.
5. UPF reports X2 PDSR events for all 5G Uplink and Downlink packets.

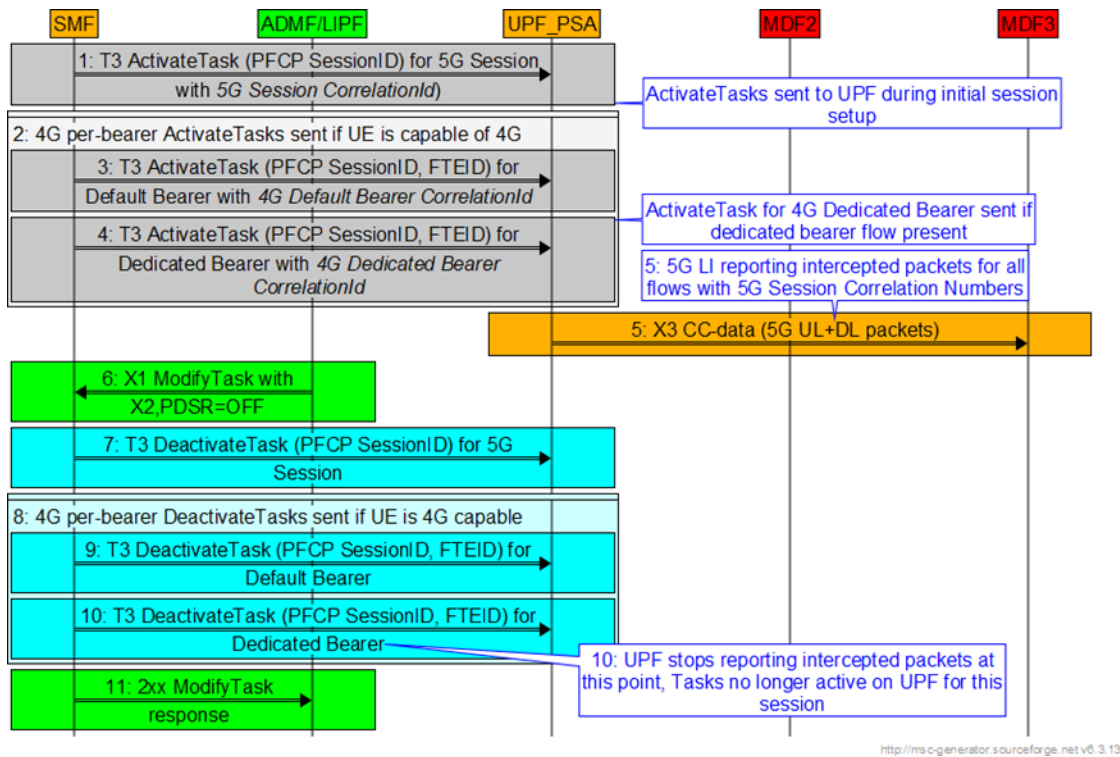
ModifyTask

6. LIPF sends ModifyTask with DeliveryType=X2andX3, PDSR=On.
7. SMF responds to LIPF with 200 OK for the ModifyTask request.
8. SMF sends T3 ActivateTask with 5G Session Correlation Number to UPF.
9. If UE is capable of 4G, SMF sends T3 ActivateTask for 4G bearers to UPF.
10. SMF sends T3 ActivateTask with 4G Default Bearer Correlation Number to UPF.
11. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T3 ActivateTask to UPF with 4G Dedicated Bearer Correlation Number.
12. UPF sends both X2 PDSR events and X3 CC-data to MDF3 for intercepted 5G Uplink and Downlink packets using 5G Session Correlation Number.

Note: When MDF is capable of generating PDSR events from X3 CC-data, it is not expected that the LIPF would provision the SMF for both PDSR and X3 CC-data as shown in this call flow. Instead, when warrants require both X2 PDSR and X3 CC-data, the LIPF provisions the SMF for X3 CC-data, and then the MDF constructs PDSR events from the collected X3 CC-data.

X1 ModifyTask: Previous (DeliveryType=X2andX3,PDSR-OFF), New (DeliveryType=X2,PDSR=OFF)

Figure 12. X1 ModifyTask: Previous (DeliveryType=X2andX3,PDSR-OFF), New (DeliveryType=X2,PDSR=OFF)



This call flow starts with an initial state where an ActivateTask is X2andX3 with PDSR disabled. The task is then modified to X2Only with PDSR disabled.

Initial Setup - not shown on the diagram

1. LIPF sends ActivateTask with DeliveryType=X2andX3, PDSR=Off.
2. ActivateTask establishes and intercepts the session.

Initial T2 ActivateTask Signaling

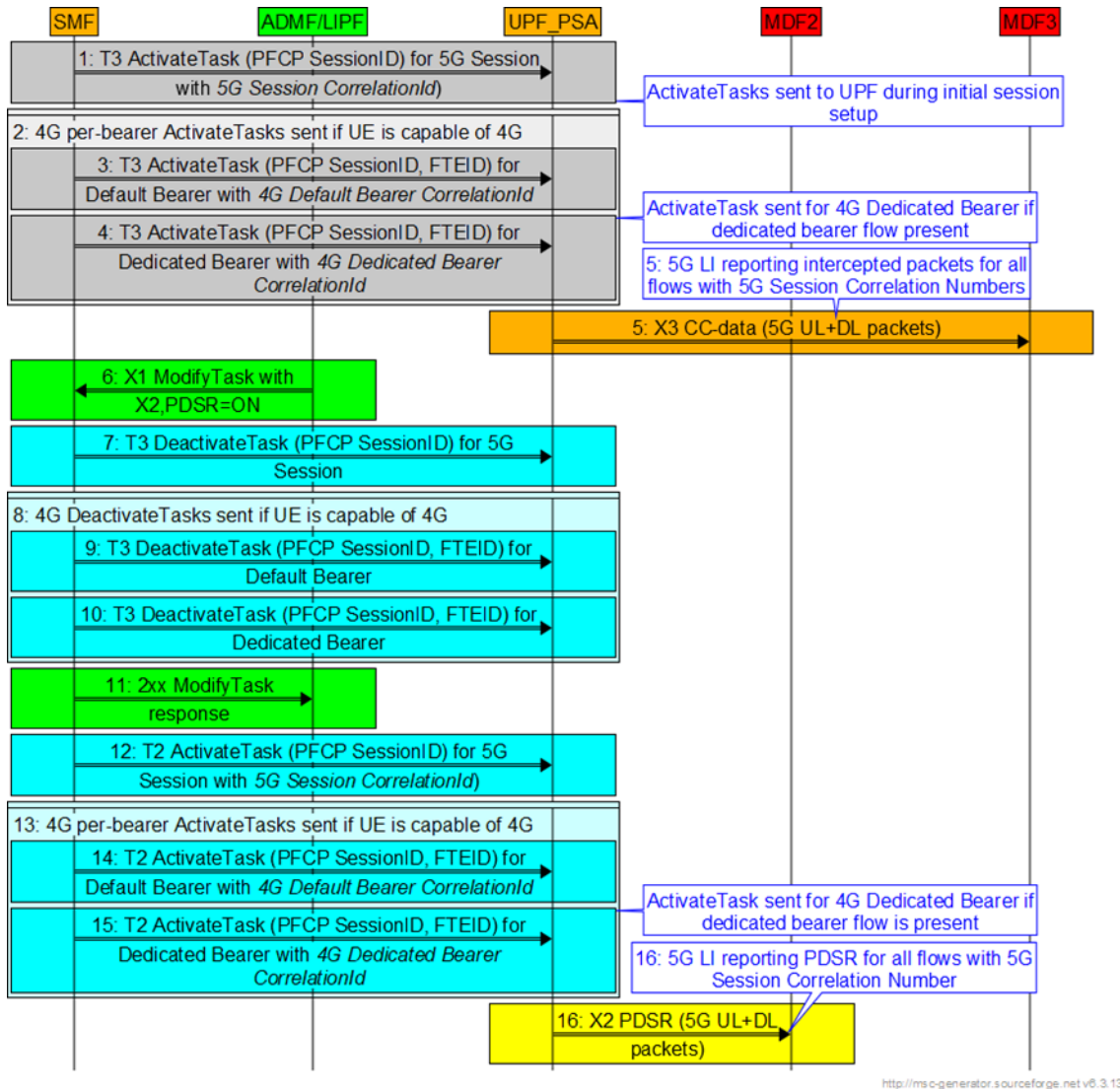
1. SMF sends T3 ActivateTask to UPF with 5G Session Correlation Number.
2. If UE is capable of 4G, SMF sends T3 ActivateTask for 4G bearers to UPF.
3. SMF sends T3 ActivateTask for 4G default bearer with 4G Default Bearer Correlation Number.
4. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T3 ActivateTask to UPF for 4G dedicated bearer with 4G Dedicated Bearer Correlation Number.
5. UPF reports X3 CC-data for all 5G Uplink and Downlink packets.

ModifyTask

6. LIPF sends ModifyTask with DeliveryType=X2Only,PDSR=Off.
7. SMF sends T3 DeactivateTask with 5G Session Correlation Number to UPF.
8. If UE is capable of 4G, SMF sends T3 DeactivateTask for 4G bearers to UPF.
9. SMF sends T3 ActivateTask with 4G Default Bearer Correlation Number to UPF.
10. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T3 DeactivateTask to UPF with 4G Dedicated Bearer Correlation Number. At this point, tasks are no longer active on the UPF for this session. UPF stops reporting intercepted packets for this session.
11. SMF responds to LIPF with 200 OK for ModifyTask.

X1 ModifyTask: Previous (DeliveryType=X2andX3,PDSR=OFF), New (DeliveryType=X2,PDSR=ON)

Figure 13. X1 ModifyTask: Previous (DeliveryType=X2andX3,PDSR=OFF), New (DeliveryType=X2,PDSR=ON)



This call flow starts with the initial state where an ActivateTask is X2andX3 with PDSR disabled. The task is then modified to X2Only with PDSR enabled.

Initial Setup - not shown on the diagram

1. LIPF sends ActivateTask with DeliveryType=X2andX3, PDSR=Off.
2. ActivateTask establishes and intercepts the session.

Initial T2 ActivateTask Signaling

SMF

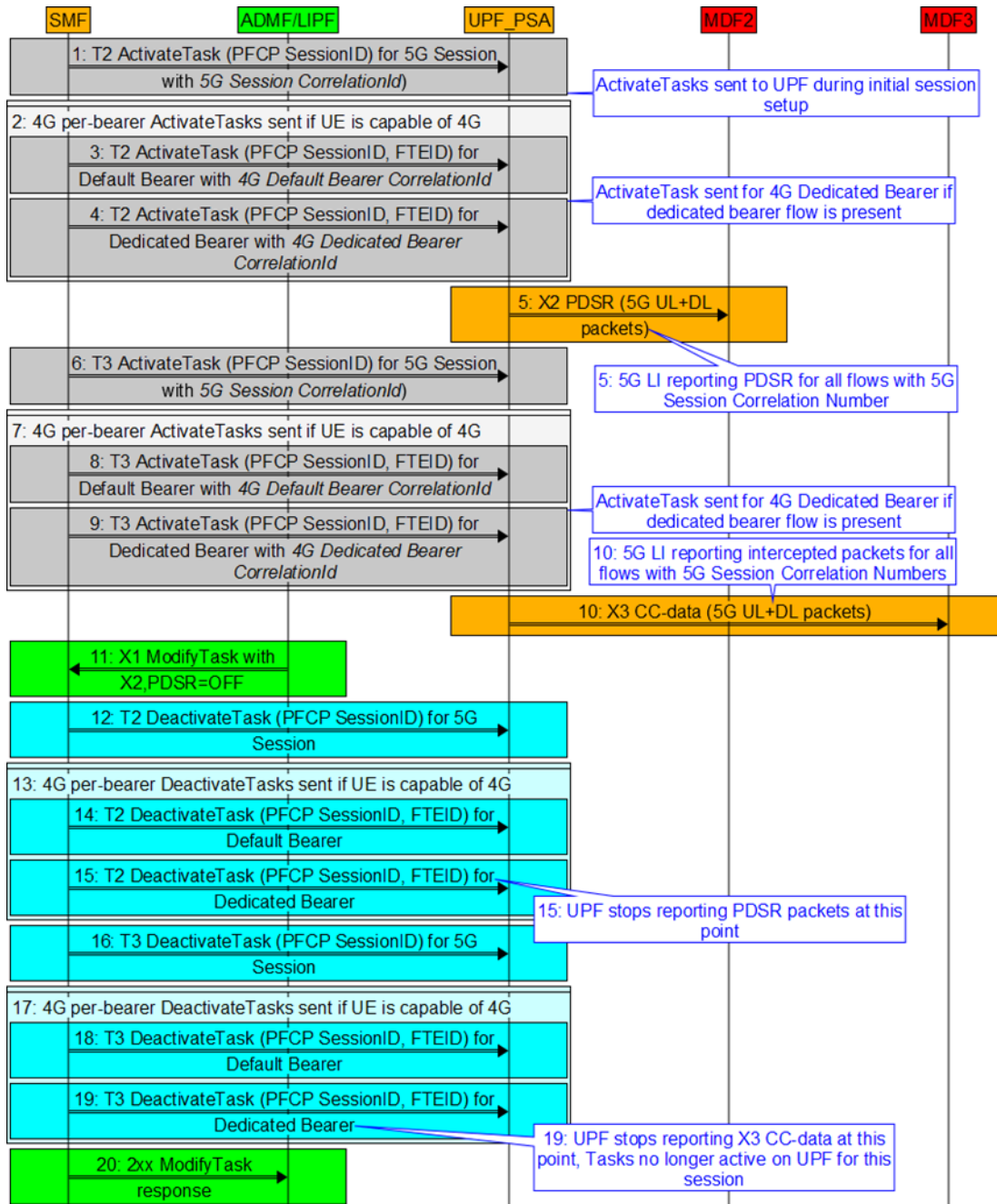
1. SMF sends T3 ActivateTask to UPF with 5G Session Correlation Number.
2. If UE is capable of 4G, SMF sends T3 ActivateTask for 4G bearers to UPF.
3. SMF sends T3 ActivateTask for 4G default bearer with 4G Default Bearer Correlation Number.
4. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T3 ActivateTask to UPF for 4G dedicated bearer with 4G Dedicated Bearer Correlation Number.
5. UPF reports X3 CC-data for all 5G Uplink and Downlink packets.

ModifyTask

6. LIPF sends ModifyTask with DeliveryType=X2Only,PDSR=On.
7. SMF sends T3 DeactivateTask with 5G Session Correlation Number to UPF. At this point, UPF is no longer intercepting X3 CC-data for the 5G session.
8. If UE is capable of 4G, SMF sends T3 DeactivateTask for 4G bearers to UPF.
9. SMF sends T3 DeactivateTask with 4G Default Bearer Correlation Number to UPF.
10. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T3 DeactivateTask to UPF with 4G Dedicated Bearer Correlation Number. At this point, task are no longer active on the UPF for this session. UPF stops reporting intercepted packets for this session.
11. SMF responds to LIPF with 200 OK for ModifyTask.
12. SMF sends T2 ActivateTask to UPF with 5G Session Correlation Number.
13. If UE is capable of 4G, SMF sends T2 ActivateTask for 4G bearers to UPF.
14. SMF sends T2 ActivateTask for 4G default bearer with 4G Default Bearer Correlation Number.
15. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T2 ActivateTask to UPF for 4G dedicated bearer with 4G Dedicated Bearer Correlation Number.
16. UPF reports X2 PDSR events for all 5G Uplink and Downlink packets on the session.

X1 ModifyTask: Previous (DeliveryType=X2andX3,PDSR=ON), New (DeliveryType=X2,PDSR=OFF)

Figure 14. X1 ModifyTask: Previous (DeliveryType=X2andX3,PDSR=ON), New (DeliveryType=X2,PDSR=OFF)



<http://ms-c-generator.sourceforge.net/v0.3.13>

This call flow starts with the initial state where an ActivateTask is X2andX3 with PDSR enabled. The task is then modified to X2Only with PDSR disabled.

Note: When MDF is capable of generating PDSR events from X3 CC-data, it is not expected that the LIPF would provision the SMF for both PDSR and X3 CC-data as shown in this call flow. Instead, when warrants require both X2 PDSR and X3 CC-data, the LIPF provisions the SMF for X3 CC-data, and then the MDF constructs PDSR events from the collected X3 CC-data.

Initial Setup - not shown on the diagram

1. LIPF sends ActivateTask with DeliveryType=X2andX3, PDSR=On.
2. ActivateTask establishes and intercepts the session.

Initial T2 and T3 ActivateTask Signaling

1. SMF sends T2 ActivateTask to UPF with 5G Session Correlation Number.
2. If UE is capable of 4G, SMF sends T2 ActivateTask for 4G bearers to UPF.
3. SMF sends T2 ActivateTask for 4G default bearer with 4G Default Bearer Correlation Number.
4. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T2 ActivateTask to UPF for 4G dedicated bearer with 4G Dedicated Bearer Correlation Number.
5. UPF reports X2 PDSR events for all 5G Uplink and Downlink packets.
6. SMF sends T3 ActivateTask to UPF with 5G Session Correlation Number.
7. If UE is capable of 4G, SMF sends T3 ActivateTask for 4G bearers to UPF.
8. SMF sends T3 ActivateTask for 4G default bearer with 4G Default Bearer Correlation Number.
9. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T3 ActivateTask to UPF for 4G dedicated bearer with 4G Dedicated Bearer Correlation Number.
10. UPF reports X3 CC-data for all 5G Uplink and Downlink packets.

ModifyTask

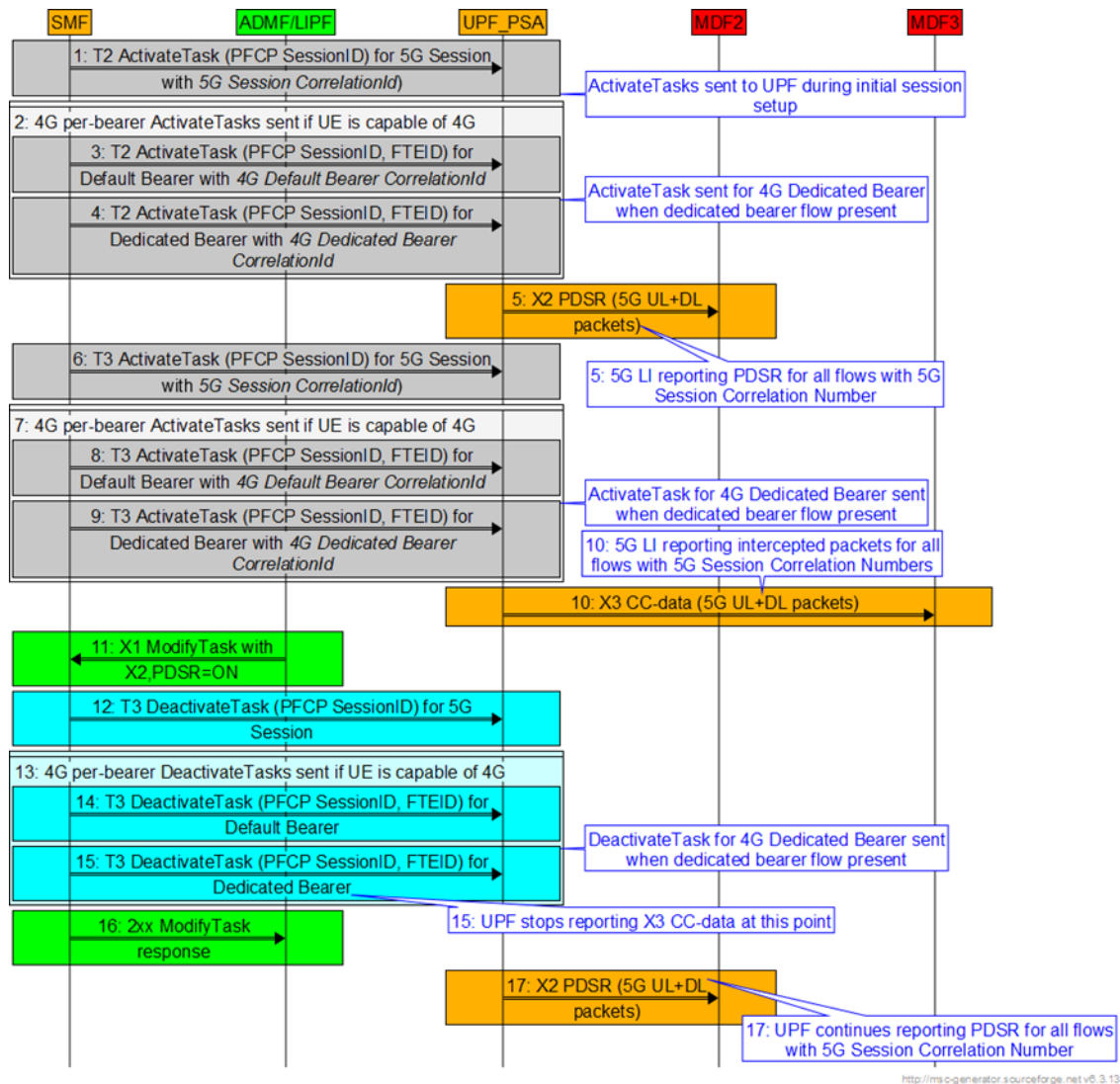
11. LIPF sends ModifyTask with DeliveryType=X2Only,PDSR=Off.
12. SMF sends T2 DeactivateTask with 5G Session Correlation Number to UPF. At this point, UPF is no longer intercepting X2 PDSR events for the 5G session.
13. If UE is capable of 4G, SMF sends T2 DeactivateTask for 4G bearers to UPF.
14. SMF sends T2 DeactivateTask with 4G Default Bearer Correlation Number to UPF.
15. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T2 DeactivateTask to UPF with 4G Dedicated Bearer Correlation Number.

SMF

16. SMF sends T3 DeactivateTask with 5G Session Correlation Number to UPF. At this point, UPF is no longer intercepting X3 CC-data for the 5G session.
17. If UE is capable of 4G, SMF sends T3 DeactivateTask for 4G bearers to UPF.
18. SMF sends T3 DeactivateTask with 4G Default Bearer Correlation Number to UPF.
19. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T3 DeactivateTask to UPF with 4G Dedicated Bearer Correlation Number.
20. SMF responds to LIPF with 200 OK for the ModifyTask request.

X1 ModifyTask: Previous (DeliveryType=X2andX3,PDSR=ON), New (DeliveryType=X2,PDSR=ON)

Figure 15. X1 ModifyTask: Previous (DeliveryType=X2andX3,PDSR=ON), New (DeliveryType=X2,PDSR=ON)



This call flow starts with the initial state where an ActivateTask is X2andX3 with PDSR enabled. The task is then modified to X2Only with PDSR enabled.

Note: When MDF is capable of generating PDSR events from X3 CC-data, it is not expected that the LIPF would provision the SMF for both PDSR and X3 CC-data as shown in this call flow. Instead, when warrants require both X2 PDSR and X3 CC-data, the LIPF provisions the SMF for X3 CC-data, and then the MDF constructs PDSR events from the collected X3 CC-data.

Initial Setup - not shown on the diagram

1. LIPF sends ActivateTask with DeliveryType=X2andX3, PDSR=On.
2. ActivateTask establishes and intercepts the session.

Initial T2 and T3 ActivateTask Signaling

1. SMF sends T2 ActivateTask to UPF with 5G Session Correlation Number.
2. If UE is capable of 4G, SMF sends T2 ActivateTask for 4G bearers to UPF.
3. SMF sends T2 ActivateTask for 4G default bearer with 4G Default Bearer Correlation Number.
4. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T2 ActivateTask to UPF for 4G dedicated bearer with 4G Dedicated Bearer Correlation Number.
5. UPF reports X2 PDSR events for all 5G Uplink and Downlink packets.
6. SMF sends T3 ActivateTask to UPF with 5G Session Correlation Number.
7. If UE is capable of 4G, SMF sends T3 ActivateTask for 4G bearers to UPF.
8. SMF sends T3 ActivateTask for 4G default bearer with 4G Default Bearer Correlation Number.
9. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T3 ActivateTask to UPF for 4G dedicated bearer with 4G Dedicated Bearer Correlation Number.
10. UPF reports X3 CC-data for all 5G Uplink and Downlink packets.

ModifyTask

11. LIPF sends ModifyTask with DeliveryType=X2Only,PDSR=On.
12. SMF sends T3 DeactivateTask with 5G Session Correlation Number to UPF. At this point, UPF is no longer sending X3 CC-data events for the 5G session. UPF is still reporting X2 PDSR events for all Uplink and Downlink packets on the 5G session.
13. If UE is capable of 4G, SMF sends T3 DeactivateTask for 4G bearers to UPF.
14. SMF sends T3 DeactivateTask with 4G Default Bearer Correlation Number to UPF.
15. If flow exists in the session that corresponds to a 4G dedicated bearer, SMF sends T3 DeactivateTask to UPF with 4G Dedicated Bearer Correlation Number.
16. SMF responds to LIPF with 200 OK for the ModifyTask request.
17. UPF continues to report X2 PDSR events for the 5G session.

SMF-LI 4G PGW Message Flows with Lawful Intercept

This section describes Message Flows for Lawful Intercept 4G sessions handled by the SMF/PGW-C product. The following Lawful Intercept X2 IRI events are defined for 4G EPS sessions in the 3GPP TS 33.108 specification:

- Bearer Activation
- Start of Interception with Active Bearer
- Bearer Modification
- Bearer Deactivation
- UE Requested Bearer Resource Modification (Not yet supported by CNA SMF/PGW-C product)

Chart of 4G LI X2 IRI events and 4G LI Message Flows (Success Cases)

The chart below indicates which message flow includes usage of the LI X2 events of interest:

Table 15. Message Flows Including LI X2 events of interest

4G IRI Event	Bearer Activation	Bearer Modification	Bearer Deactivation	Start of Interception with Active Bearer	UE Requested Bearer Resource Modification
UE/Access side initiated call flow:	E-UTRAN Initial Attach with Lawful Intercept		UE Initiated Detach with Lawful Intercept	PGW Start of Interception with Active Bearer	Unsupported
UE/Access side initiated call flow:		HSS Initiated Subscribed QoS Modification with Lawful Intercept	MME Initiated Dedicated Bearer Deactivation with Lawful Intercept		
NW Initiated call flow:	Dedicated Bearer Activation with Lawful Intercept	PGW Initiated Bearer Modification Bearer QoS Update with Lawful Intercept	NW Initiated Detach with Lawful Intercept		Unsupported
NW Initiated call flow:			NW Initiated Dedicated Bearer Deactivation with Lawful Intercept		Unsupported

4G LI Message Flows (Success cases)

- [Dedicated Bearer Activation with Lawful Intercept](#)
- [E-UTRAN Initial Attach with Lawful Intercept](#)
- [HSS Initiated Subscribed QoS Modification with Lawful Intercept](#)
- [NW Initiated Detach with Lawful Intercept](#)
- [PGW Initiated Bearer Modification Bearer QoS Update with Lawful Intercept](#)
- [PGW Start of Interception with Active Bearer](#)
- [UE Initiated Detach with Lawful Intercept](#)
- [MME Initiated Dedicated Bearer Deactivation with Lawful Intercept](#)
- [NW Initiated Dedicated Bearer Deactivation with Lawful Intercept](#)
- [4G Handovers with SGW Change with Lawful Intercept](#)

Chart of 4G LI X2 IRI events and 4G LI Message Flows (Failure Cases)

The chart below indicates which message flow includes usage of the LI X2 events of interest:

Table 16. 4G LI X2 IRI events and 4G LI Message Flows (Failure Cases)

4G IRI Event	Bearer Activation	Bearer Modification	Bearer Deactivation	Start of Interception with Active Bearer	UE Requested Resource Modification
UE Initiated call flow:	E-UTRAN Initial Attach (Failure) with Lawful Intercept	HSS Initiated Subscriber QoS Modification (Failure) with Lawful Intercept	Not Applicable (Failure of Bearer Deactivation event is not reported to MDF)	Not Applicable	Unsupported
NW Initiated call flow:	Dedicated Bearer Activation (Failure) with Lawful Intercept	PGW Initiated Bearer Modification (Failure) with Lawful Intercept	Not Applicable (Failure of Bearer Deactivation event is not reported to MDF)	Not Applicable	Unsupported

4G LI Message Flows (Failure cases)

- [Dedicated Bearer Activation \(Failure\) with Lawful Intercept](#)
- [E-UTRAN Initial Attach \(Failure\) with Lawful Intercept](#)

- [HSS Initiated Subscriber QoS Modification \(Failure\) with Lawful Intercept](#)
- [PGW Initiated Bearer Modification \(Failure\) with Lawful Intercept](#)

Standard Specification Reference

- 3GPP TS 33.108 V16.1.0 (2020-07), Section 10 Evolved Packet System (EPS)
- ETSI TS 103.221-2 v1.3.1 (2020-08)

4G Handovers with SGW Change with Lawful Intercept

LI Tasks are activated on the UPF during initial session setup

1. SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Default Bearer FTEID) for Default Bearer with 4G Default Bearer Correlation Number to UPF.
2. If a dedicated bearer is present, SMF/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Dedicated Bearer FTEID) for with 4G Dedicated Bearer Correlation Number to UPF.
3. If UE is 4G/5G capable, SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID) with 5G Session Correlation Number to UPF-PSA during initial session setup.
4. X3 CC-data containing intercepted packets for each bearer is reported to MDF3 with per- bearer correlation numbers.
5. Handover is triggered by receipt of Modify Bearer Request. SMF/PGW-C can receive Modify bearer Request from SGW. In case there is a UE location change due to which SGW change happens (inter SGW handover), then SMF/PGW-C receives Modify Bearer Request with new Location, new Sender Control FTEID, new SGW-U TEID for all the Bearers. Also if changes then new Serving PLMN ID.
6. MME sends a Modify Bearer Request to the new SGW.
7. SGW forwards Modify Bearer Request to SMF/PGW-C. SMF/PGW-C analyses Modify Bearer Request content to identify if it is a new SGW based on the Sender Control FTEID. If the received Sender Control FTEID is different compared to the previously stored one then SMF/PGW-C consider this as SGW Handover. In this case, SMF/PGW-C expects new S5/S8-U SGW F-TEID in the BearContexts for all the existing bearers.
8. SMF/PGW-C informs PCF about the new Location, ANGW Address, Serving Network Id (if PCF subscribed for these events).
9. PCF sends PCF Update Response to SMF/PGW-C to acknowledge.
10. SMF/PGW-C sends Sx Session Modification Request to UPF to program the UPF with new Remote TEIDs for all bearers. Includes a trigger so that UPF can trigger the End Marker towards the old SGW.

11. UPF acknowledges by sending Sx Session Modification Response to SMF/PGW-C.
12. SMF/PGW-C sends Modify Bearer Response to SGW.
13. SGW forwards Modify Bearer Response to MME.
14. SMF/PGW-C sends IRI CONTINUE: Bearer Modification (LI_X2 message) to MDF2 to inform MDF2 of the new SGW. UPF continues to report all intercepted UE traffic to the MDF3 through X3 CC-data with appropriate 4G Bearer Correlation Numbers.

Dedicated Bearer Activation (Failure) with Lawful Intercept

The PCF performs SM Policy Association Modification procedure to notify SMF/PGW-C about the modification of policies. This is triggered by a policy decision or upon AF requests. This procedure is used to create dedicated bearers.

1. The receipt of an SM Policy Association Modification message from the PCF starts the first transaction. If the new rules to install belongs to a new QoS flows, then SMF/PGW-C starts create bearer procedure to setup dedicated bearers.
2. SMF/PGW-C sends 200 OK to acknowledge the request.
3. SMF/PGW-C sends Sx Session Modification request message to UPF to program new dedicated bearer information.
4. Resource failure on UPF causes dedicated bearer creation to fail. UPF responds with Sx Session Modification Response Failure.
5. SMF/PGW-C sends IRI REPORT: Bearer Activation Unsuccessful with Failed Bearer Activation Reason to MDF2 (LI_X2 interface).

Dedicated Bearer Activation with Lawful Intercept

The PCF performs a PCF initiated SM Policy Association Modification procedure to notify SMF/PGW-C about the modification of policies. This can be triggered by a policy decision or upon AF requests. The procedure is used to create dedicated bearers.

1. The receipt of an SM Policy Association Modification message from the PCF starts the first transaction. If the new rules to install belongs to a new QoS flows, then SMF/PGW-C starts create bearer procedure to setup dedicated bearers.
2. The Sx Session Modification request message is sent to UPF to program new dedicated bearer information. UPF acknowledges with SX Session Modification response message.
3. SMF/PGW-C sends Create Bearer Request message with a list of new bearer context to SGW.
4. SGW sends Create Bearer Request to MME.

5. MME acknowledges bearer activation by sending Create Bearer Response to SGW.
6. SGW acknowledges bearer activation by sending Create Bearer Response message to SMF/PGW-C.
7. SMF/PGW-C sends Sx Session Modification request message to UPF to program remote tunnel ID information for the dedicated bearer.
8. SMF/PGW-C sends X1 ActivateTask with Dedicated Bearer Correlation Number to UPF (LI_T2/LI_T3 interface) to instruct UPF to start intercepting all traffic on the new bearer.
9. UPF starts intercepting Uplink traffic from UE on the Dedicated Bearer. UPF reports intercepted traffic in X3 CC-data message to MDF3 (LI_X3 interface).
10. SMF/PGW-C sends IRI BEGIN: Bearer Activation with Dedicate Bearer Correlation Number to MDF2 (LI_X2 interface).

Note: As shown, it is possible that X3 CC-data for the Dedicated Bearer can start arriving at MDF3 (Step 9) before the Correlation Number has been reported to MDF2 (Step 10).
Mediation Functions must be prepared to handle this condition.

E-UTRAN Initial Attach (Failure) with Lawful Intercept

1. MME sends Create Session Request message to SGW is also forwarded to SMF/PGW-C.
2. Error occurs during session setup (resource allocation failure, authentication failure, etc).
3. SMF/PGW-C sends IRI REPORT: Bearer Activation Unsuccessful to MDF2 (LI_X2 interface).
4. SMF/PGW-C sends Create Session Response Failure to SGW.
5. SGW sends Create Session Response Failure to MME.

E-UTRAN Initial Attach with Lawful Intercept

1. MME sends Create Session Request message to SGW is also forwarded to SMF/PGW-C.
2. The Sx Session Establishment PFCP exchange with UPF creates session resources on UPF.
3. SMF/PGW-C sends LI X1 ActivateTask with 4G Default Bearer Correlation Number to UPF (LI_T2/LI_T3 interface) to instruct UPF to start interception of default bearer data.

4. For 4G/5G capable UE, SMF/PGW-C sends LI X1 ActivateTask with 5G Session Correlation Number to UPF (LI_T2/LI_T3 interface) to put 5G Task in place in preparation for possible handover to 5G.
5. UPF starts intercepting session data (Downlink traffic from UE). UPF reports intercepted data in X3 CC-data messages to MDF3.
6. SMF/PGW-C sends X2 IRI BEGIN: Bearer Activation with 4G Default Bearer Correlation Number (X2 message) to MDF2.
7. SMF/PGW-C sends Create Session Response to SGW.
8. SGW sends Create Session Response to MME.
9. Once Uplink traffic is flowing, UPF intercepts all session data (Uplink and Downlink traffic). UPF reports intercepted data in X3 CC-data messages to MDF3.
10. MME sends Modify Bearer Request to SGW.
11. SGW responds with Modify Bearer Response.

Note: As shown, it is possible that X3 CC-data can start arriving at MDF3 (Step 5) before the Correlation Number has been reported to MDF2 (Step 6). Mediation Functions must be prepared to handle this condition.

HSS Initiated Subscribed QoS Modification with Lawful Intercept

The HSS initiated Subscribed QoS Modification procedure for a UE in GTP-based S5/S8 interface. HSS initiates **Insert Subscriber Data** to MME.

1. UPF is programmed to intercept traffic on session. UPF intercepts all traffic to/from UE. UPF reports intercepted traffic in X3 CC-data events to MDF3 (LI_X3 interface).
2. HSS initiates **Insert Subscriber Data** to MME. If UE is reachable, MME initiates Modify Bearer Command message towards SGW.
3. SGW forwards Modify Bearer Command to SMF/PGW-C.
4. SMF/PGW-C sends Sx Session Modification request message (PFCP) to the UPF.
5. UPF acknowledges with Sx Session Modification response message (PFCP).
6. SMF/PGW-C sends Update Bearer Request to SGW.
7. SGW forwards Update Bearer Request to MME.

8. MME acknowledges with Update Bearer Response to SGW.
9. SGW forwards Update Bearer Response to SMF/PGW-C.
10. SMF/PGW-C informs MDF2 of Bearer Modification by sending IRI CONTINUE: Bearer Modification event to MDF2 (LI_X2 interface).

HSS Initiated Subscriber QoS Modification (Failure) with Lawful Intercept

The HSS initiated Subscribed QoS Modification procedure for a UE in GTP-based S5/S8 interface. HSS initiates **Insert Subscriber Data** to MME.

1. HSS initiates **Insert Subscriber Data** to MME. If UE is reachable, MME initiates Modify Bearer Command message towards SGW.
2. SGW forwards Modify Bearer Command to SMF/PGW-C.
3. SMF/PGW-C sends Sx Session Modification request message (PFCP) to the UPF.
4. UPF acknowledges with Sx Session Modification response message (PFCP).
5. SMF/PGW-C sends Update Bearer Request to SGW.
6. SGW forwards Update Bearer Request to MME.
7. Update Bearer Request fails on MME/UE. MME sends Update Bearer Response Failure to SGW.
8. SGW forwards Update Bearer Response Failure to SMF/PGW-C.
9. SMF/PGW-C informs MDF2 of Bearer Modification Failure by sending IRI CONTINUE: Bearer Modify event with Failed Bearer Modification Reason to MDF2 (LI_X2 interface).

MME Initiated Dedicated Bearer Deactivation with Lawful Intercept

MME initiated deactivation of Dedicated Bearer with Lawful Intercept deactivates only dedicated bearers. Default Bearer is unaffected by this procedure. RAN side can release dedicated bearers for various local reasons and informs the same to MME.

1. X3 CC-data containing intercepted packets for the Dedicated Bearer is reported to MDF3 with the Dedicated Bearer Correlation Number.
2. MME initiates Delete Bearer Command towards SMF/PGW-C through SGW. In the message, it can include ULI, UE TimeZone, RANNAS Cause, etc.
3. SGW forwards the same towards SMF/PGW-C in GTPC message.

4. Upon receiving Delete Bearer Command, SMF/PGW-C identifies the UE session and bearer. SMF/PGW-C informs PCF about the PCC rules deletion.
5. PCF sends PCC Update Response to SMF/PGW-C.
6. SMF/PGW-C sends Delete Bearer Request to SGW.
7. SGW forwards Delete Bearer Request to MME.
8. MME acknowledges by sending Delete Bearer Response to SGW.
9. SGW forwards Delete Bearer Response to SFM/PGW-C.
10. SMF/PGW-C sends IRI END: Bearer Deactivation with Dedicated Bearer Correlation Number (LI_X2 message) to MDF2 to indicate Dedicated Bearer is removed.
11. SMF/PGW-C informs UPF about the deletion of bearer by sending Sx Session Modification Request PFCP message to update PDR, FAR, QER about bearer being removed.
12. UPF acknowledges by sending Sx Session Modification Response to SMF/PGW-C.
13. SMF/PGW-C sends X1 DeactivateTask (LI_T2/LI_T3 message) to UPF to deactivate the LI Task associated with the Dedicated Bearer being removed.
14. No further packet flow occurs for the dedicated bearer hence no further X3 CC-data with the Dedicated Bearer Correlation Number is reported to MDF3.

NW Initiated Dedicated Bearer Deactivation with Lawful Intercept

PCF initiates SM Policy Association Modification request to notify SMF/PGW-C about the modification of policies. The procedure is used to create/modify/delete dedicated bearers. The call flow described here is for dedicated bearer deletion.

1. X3 CC-data containing intercepted packets for the Dedicated Bearer is reported to MDF3 with the Dedicated Bearer Correlation Number.
2. PCF sends SM Policy Association Modification request to SFM/PGW-C to initiate the Bearer Deactivation procedure.
3. SMF/PGW-C acknowledges by sending PCF Response message to PCF.
4. SMF/PGW-C sends Sx Session Modification Request (PFCP message with updated PDR, FAR, QER related to the bearer being removed) to UPF.
5. UPF acknowledges by sending Sx Session Modification Response to SMF/PGW-C.
6. SMF/PGW-C sends Delete Bearer Request to SGW.
7. SGW forwards Delete Bearer Request to MME.
8. MME acknowledges by sending Delete Bearer Response to SGW.

9. SGW forwards Delete Bearer Response to SMF/PGW-C.
10. SMF/PGW-C sends IRI END: Bearer Deactivation with Dedicated Bearer Correlation Number (LI_X2 message) to MDF2 to indicate Dedicated Bearer is removed.
11. SMF/PGW-C sends X1 DeactivateTask (LI_T2/LI_T3 message) to UPF to remove the X1 Task for the Dedicated Bearer being removed.
12. No further packet flow occurs for the dedicated bearer hence no further X3 CC-data with the Dedicated Bearer Correlation Number is reported to MDF3.
13. SMF/PGW-C sends PCF Update report message to PCF to notify PCF of the deletion of bearer.
14. PCF acknowledges by sending PCF Update Response.

NW Initiated Detach with Lawful Intercept

The Network initiated PDN Session Detach procedure allows SMF/PGW-C to request release of a PDN session. It is triggered when PCF initiates deletion of SM Policy Association (SM Policy Association termination request) or by Management intervention using a CLI command.

1. The receipt of an SM Policy Association Terminate Request from the PCF triggers session terminate procedure.
2. SMF/PGW-C releases the PFCP session by sending Sx Session Release Request message towards the UPF.
3. SMF/PGW-C sends LI X1 DeactivateTask with 4G Default Bearer Correlation Number to the UPF (LI_T2/LI_T3 interface) to instruct UPF to stop intercepting traffic on the default bearer.
4. SMF/PGW-C sends LI X1 DeactivateTask with 4G Dedicated Bearer Correlation Number to the UPF (LI_T2/LI_T3 interface) to instruct UPF to stop intercepting traffic on the dedicated bearer.
5. SMF/PGW-C sends LI X1 DeactivateTask with 5G Session Correlation Number to the UPF (LI_T2/LI_T3 interface). When UE is 4G/5G capable, this 5G Task is installed on UPF during initial session setup.
6. SMF/PGW-C sends Delete Bearer Request (EBI : Default Bearer) message to the Serving GW to delete the session.
7. SGW sends Delete Bearer Request (EBI: Default Bearer) to MME to delete the session.
8. MME acknowledges session deletion by sending Delete Bearer Response to SGW.
9. SGW forwards Delete Bearer Response to SMF/PGW-C.
10. SMF/PGW-C sends X2 IRI END: Bearer Deactivation with 4G Default Bearer Correlation Number to MDF2 (LI_X2 interface).

11. SMF/PGW-C sends X2 IRI END: Bearer Deactivation with 4G Dedicated Bearer Correlation Number to MDF2 (LI_X2 interface).

12. If Dynamic PCC is applied to this session, the SMF/PGW-C invokes SM Policy Association Termination procedure towards the PCF to delete the PDN session.

PGW Initiated Bearer Modification (Failure) with Lawful Intercept

PCF initiates SM Policy Association Modification request to notify SMF/PGW-C about the modification of policies. This procedure is used to create/modify/delete dedicated bearers. This call flow is related to dedicated bearer modification.

1. PCF sends SM Policy Association Modification request to SMF/PGW-C to initiate the Modify.
2. SMF/PGW-C acknowledges modification request with 200 OK.
3. SMF/PGW-C sends Update Bearer Request with list of updated bearers to SGW.
4. SGW forwards Update Bearer Request to MME.
5. Update Bearer Request fails at MME.UE. MME sends Update Bearer Response Failure to SGW.
6. SGW forwards Update Bearer Response Failure to SMF/PGW-C.
7. SMF/PGW-C sends IRI CONTINUE: Bearer Modify with Failed Bearer Modification Reason to MDF2 (LI_X2 interface).
8. SMF/PGW-C sends PCF Update Request Failure to notify PCF of the modify failure (if PCF has subscriber for notification).
9. PCF acknowledges with PCF Update Response sent to SMF/PGW-C.

PGW Initiated Bearer Modification Bearer QoS Update with Lawful Intercept

PCF initiates SM Policy Association Modification request to notify SMF/PGW-C about the modification of policies. The procedure is used to create/modify/delete dedicated bearers. This call flow is related to dedicated bearer modification.

1. During session setup, SMF programs UPF to intercept the session. UPF is intercepting UE traffic and reporting intercepted traffic through X3 CC-data events to MDF3 (LI_X3 interface).
2. PCF sends SM Policy Association Modification request to SMF/PGW-C to initiate the Modify.
3. SMF/PGW-C acknowledges modification request with 200 OK.
4. SMF/PGW-C sends Update Bearer Request with list of updated bearers to SGW.

5. SGW forwards Update Bearer Request to MME.
6. MME acknowledges modification by sending Update Bearer Response to SGW.
7. SGW forwards Update Bearer Response to SMF/PGW-C.
8. SMF/PGW-C sends IRI CONTINUE: Bearer Modification to MDF2 (LI_X2 interface).
9. SMF/PGW-C sends Sx Session Modification request (PFCP message) to the UPF with updated/deleted PDR/FAR/QER and URR query.
10. UPF acknowledges modification with Sx Session Modification response (PFCP message).
11. SMF/PGW-C sends PCF Update Request to notify PCF of the modification (if PCF has subscriber for notification).
12. PCF acknowledges with PCF Update Response sent to SMF/PGW-C.

PGW Start of Interception with Active Bearer

1. MME sends Create Session Request message to SGW is also forwarded to SMF/PGW-C.
2. Sx Session Establishment PFCP exchange with UPF creates session resources on UPF.
3. SMF/PGW-C sends Create Session Response to SGW.
4. SGW sends Create Session Response to MME.
5. MME sends Modify Bearer Request (with GTP-U TEID of eNB) to SGW.
6. SGW acknowledges by sending Modify Bearer Response to MME.
7. When required, Dedicated Bearer is added to the session.
8. At this point, session is established, UE sends/receives data. No interception is occurring.
9. ADMF/LIPF sends X1 ActivateTask message to SMF/PGW-C. X1 Task contains Target Identifier of subscriber that established the session.
10. SMF/PGW-C sends X1 ActivateTask (PFCP SessionID, FTEID of Default Bearer) with 4G Default Bearer Correlation Number to UPF (LI_T2/LI_T3 interface) to instruct UPF to intercept all traffic on Default Bearer.
11. SMF/PGW-C sends X1 ActivateTask (PFCP SessionID, FTEID of Dedicated Bearer) with 4G Dedicated Bearer Correlation Number to UPF (LI_T2/LI_T3 interface) to instruct UPF to intercept all traffic on Dedicated Bearer.
12. If UE is 4G/5G capable, SMF/PGW-C sends X1 ActivateTask (PFCP SessionID) with 5G Session Correlation Number to UPF.

13. At this point, UPF intercepts all traffic on both Default and Dedicated Bearers. UPF reports intercepted traffic by sending X3 CC-data messages with the appropriate per-bearer Correlation Number to MDF3 (LI_X3 interface).

14. SMF/PGW-C sends IRI BEGIN: StartOfInterceptionWithActiveBearer with Default Bearer Correlation Number to MDF2 (LI_X2 interface).

15. SMF/PGW-C send IRI BEGIN: StartOfInterceptionWithActiveBearer with Dedicated Bearer Correlation Number to MDF2 (LI_X2 interface).

Note: As shown, it is possible that X3 CC-data can start arriving at MDF3 (Step 13) before the Correlation Numbers have been reported to MDF2 (Steps 14-15). Mediation Functions must be prepared to handle this condition.

SMF-LI 4G PGW Message Flows with Lawful Intercept

This section describes Message Flows for Lawful Intercept 4G sessions handled by the SMF/PGW-C product. The following Lawful Intercept X2 IRI events are defined for 4G EPS sessions in the 3GPP TS 33.108 specification:

- Bearer Activation
- Start of Interception with Active Bearer
- Bearer Modification
- Bearer Deactivation
- UE Requested Bearer Resource Modification (Not yet supported by CNA SMF/PGW-C product)

Chart of 4G LI X2 IRI events and 4G LI Message Flows (Success Cases)

The chart below indicates which message flow includes usage of the LI X2 events of interest:

Table 17. Message Flows Including LI X2 events of interest

4G IRI Event	Bearer Activation	Bearer Modification	Bearer Deactivation	Start of Interception with Active Bearer	UE Requested Bearer Resource Modification
UE/Access side initiated call flow:	E-UTRAN Initial Attach with Lawful Intercept		UE Initiated Detach with Lawful Intercept	PGW Start of Interception with Active Bearer	Unsupported

4G IRI Event	Bearer Activation	Bearer Modification	Bearer Deactivation	Start of Interception with Active Bearer	UE Requested Bearer Resource Modification
UE/Access side initiated call flow:		HSS Initiated Subscribed QoS Modification with Lawful Intercept	MME Initiated Dedicated Bearer Deactivation with Lawful Intercept		
NW Initiated call flow:	Dedicated Bearer Activation with Lawful Intercept	PGW Initiated Bearer Modification Bearer QoS Update with Lawful Intercept	NW Initiated Detach with Lawful Intercept		Unsupported
NW Initiated call flow:			NW Initiated Dedicated Bearer Deactivation with Lawful Intercept		Unsupported

4G LI Message Flows (Success cases)

- [Dedicated Bearer Activation with Lawful Intercept](#)
- [E-UTRAN Initial Attach with Lawful Intercept](#)
- [HSS Initiated Subscribed QoS Modification with Lawful Intercept](#)
- [NW Initiated Detach with Lawful Intercept](#)
- [PGW Initiated Bearer Modification Bearer QoS Update with Lawful Intercept](#)
- [PGW Start of Interception with Active Bearer](#)
- [UE Initiated Detach with Lawful Intercept](#)
- [NW Initiated Dedicated Bearer Deactivation with Lawful Intercept](#)
- [MME Initiated Dedicated Bearer Deactivation with Lawful Intercept](#)
- [4G Handovers with SGW Change with Lawful Intercept](#)

Chart of 4G LI X2 IRI events and 4G LI Message Flows (Failure Cases)

Table 18. 4G LI X2 IRI events and 4G LI Message Flows (Failure Cases)

4G IRI Event	Bearer Activation	Bearer Modification	Bearer Deactivation	Start of Interception with Active Bearer	UE Requested Resource Modification
UE Initiated call flow:	E-UTRAN Initial Attach (Failure) with Lawful Intercept	HSS Initiated Subscriber QoS Modification (Failure) with Lawful Intercept	Not Applicable (Failure of Bearer Deactivation event is not reported to MDF)	Not applicable	Unsupported
NW Initiated call flow:	Dedicated Bearer Activation (Failure) with Lawful Intercept	PGW Initiated Bearer Modification (Failure) with Lawful Intercept	Not Applicable (Failure of Bearer Deactivation event is not reported to MDF)	Note applicable	Unsupported

4G LI Message Flows (Failure cases)

- [Dedicated Bearer Activation \(Failure\) with Lawful Intercept](#)
- [E-UTRAN Initial Attach \(Failure\) with Lawful Intercept](#)
- [HSS Initiated Subscriber QoS Modification \(Failure\) with Lawful Intercept](#)
- [PGW Initiated Bearer Modification \(Failure\) with Lawful Intercept](#)

Standard Specification Reference

- 3GPP TS 33.108 V16.1.0 (2020-07), Section 10 Evolved Packet System (EPS)
- ETSI TS 103.221-2 v1.3.1 (2020-08)

UE Initiated Detach with Lawful Intercept

UE triggers the detach procedure by sending a Detach Request to MME.

1. MME sends Delete Session Request message to SGW to trigger the release of the PDN Session.
2. SGW forwards Delete Session Request message to SMF/PGW-C.
3. SMF/PGW-C releases the PFCP session by sending the Sx Session Release Request message towards the UPF.

4. SMF/PGW-C sends LI X1 DeactivateTask with 4G Default Bearer Correlation Number to UPF (LI_T2/LI_T3 interface).
5. SMF/PGW-C sends LI X1 DeactivateTask with 4G Dedicated Bearer Correlation Number to UPF (LI_T2/LI_T3 interface).
6. When UE is 4G/5G capable, a 5G Task is also activated on UPF. SMF/PGW-C sends LI X1 DeactivateTask with 5G Session Correlation Number to UPF (LI_T2/LI_T3 interface).
7. If Dynamic PCC is applied to this session, the SMF/PGW-C invokes SM Policy Association Termination procedure towards the PCF to delete the PDU session.
8. SMF/PGW-C acknowledges by sending Delete Session Response to SGW.
9. SGW acknowledges by sending Delete Session Response to MME.
10. SMF/PGW-C sends X2 IRI END: Bearer Deactivation with 4G Default Bearer Correlation Number to MDF2 (LI_X2 interface).
11. SMF/PGW-C sends X2 IRI END: Bearer Deactivation with 4G Dedicated Bearer Correlation Number to MDF2 (LI_X2 interface).

SMF-LI 5G SMF Message Flows with Lawful Intercept

product. The following Lawful Intercept X2 IRI events are defined for 5G sessions in the 3GPP TS 33.128 specification:

- SMF PDU Session Establishment
- SMF PDU Session Modification
- SMF PDU Session Release
- SMF Start Of Interception With Established PDU Session
- SMF Unsuccessful Procedure

Chart of 5G LI X2 Events and 5G LI Message Flows (Success Cases)

The chart below indicates which message flow includes usage of the LI X2 events of interest:

Table 19. Flow Usage of LI X2 Events

Procedure	Session Establishment – Success	Session Modification – Success	Session Release - Success	Start of Intercept with Established PDU Session
UE Initiated call flow:	UE Requested PDU Session Establishment with Lawful Intercept	UE Initiated PDU Session Modification with Lawful Intercept	Release SM Context Transaction with Lawful Intercept	
UE Initiated call flow:			UE Requested PDU Session Release with Lawful Intercept	
NW Initiated call flow:		PCF Initiated Session Modification with Lawful Intercept	Network Requested PDU Session Release with Lawful Intercept	SMF Start of Interception with Established PDU Session

5G LI Message Flows (Success Cases)

- [UE Requested PDU Session Establishment with Lawful Intercept](#)
- [UE Initiated PDU Session Modification with Lawful Intercept](#)
- [PCF Initiated Session Modification with Lawful Intercept](#)
- [UE Requested PDU Session Release with Lawful Intercept](#)
- [Release SM Context Transaction with Lawful Intercept](#)
- [Network Requested PDU Session Release with Lawful Intercept](#)
- [SMF Start of Interception with Established PDU Session](#)

Chart of 5G LI X2 Events and 5G LI Message Flows (Failure Cases)

The chart below indicates message flows for failure scenarios that cause SMF to report SMF Unsuccessful Procedure to MDF:

Table 20. Flows for failure scenarios

Procedure	Session Establishment – Failure	Session Modification – Failure	Session Release - Failure
UE Initiated call flow:	UE Requested PDU Session Establishment (Failure) with Lawful Intercept	UE Initiated PDU Session Modification (Failure) with Lawful Intercept	

Procedure	Session Establishment – Failure	Session Modification – Failure	Session Release - Failure
NW Initiated call flow:		PCF Initiated Session Modification (Failure) with Lawful Intercept	

5G LI Message Flows (Failure Cases)

- [UE Requested PDU Session Establishment \(Failure\) with Lawful Intercept](#)
- [UE Initiated PDU Session Modification \(Failure\) with Lawful Intercept](#)
- [PCF Initiated Session Modification \(Failure\) with Lawful Intercept](#)

Standard Specification Reference

- 3GPP TS 33.128 V16.4.0 (2020-09)
- ETSI TS 103.221-2 v1.3.1 (2020-08)

Network Requested PDU Session Release with Lawful Intercept

The Network requested PDU Session Release procedure allows the SMF to request the release of a PDU session. It is triggered when the PCF initiates deletion of SM Policy Association (SM Policy Association termination request) or by Management intervention using a CLI command. When triggered, the procedure is started by the SMF with the transmission of an N1N2 Transfer Request message to the AMF. The procedure consists of 3 transactions.

During previous message exchanges UPF was programmed to intercept all packets from/to UE. UPF sends intercepted packets in X3 CC-data events to MDF3 (LI_X3 interface).

N1N2 Transfer Request with NAS PDU Session Release Command and N2 Resource Release Request

1. The receipt of an SM Policy Association Terminate Request from the PCF starts the first transaction.
2. SMF releases the PFCP session by sending the N4 Session Release Request message towards the UPF.
3. At this point, UPF is no longer intercepting data for the session. No further X3 CC-data is sent to MDF3 for this session (LI_X3 interface).

4. SMF sends an N1N2 Transfer Request message with N2 Resource Release request and N1 PDU Session Release command messages in it.

Update SM Context Transaction with N2 SM Resource Release Response

The second transaction is triggered when AMF sends Update SM Context to SMF. Update SM Context request specifies N2 SM Resource Release Response and the SMF ends the transaction after responding with an Update SM Context response.

Update SM Context Transaction with NAS PDU Session Release Complete

1. The third transaction is triggered when AMF sends Update SM Context message with NAS PDU Session Release Complete message that acknowledges the PDU Session Release Command. The SMF responds to the AMF with an Update SM Context response.
2. SMF sends X2 SMF PDU Session Release event with Session Correlation Number to MDF2 (LI_X2 interface).
3. If Dynamic PCC applied to this session, then the SMF invokes SM Policy Association Termination procedure towards the PCF to delete the PDU session.
4. The SMF then notifies AMF that the SM context for the PDU Session is released.
5. If it is the last PDU Session, the SMF is handling for the UE for the associated DNN and S-NSSAI, SMF unsubscribes from the UDM by means of the Nudm_SDM_Unsubscribe service operation. SMF then invokes the Nudm_UECM_Deregistration service operation towards the UDM.

PCF Initiated Session Modification (Failure) with Lawful Intercept

PCF initiated SM Policy Association Modification

1. PCF sends SM Policy Association Modification message to SMF to start the first transaction.
2. SMF sends 200 OK to PCF to acknowledge SM Policy Association Modification request.
3. SMF sends N1N2 Transfer Request to AMF with N2 PDU Session Modify Request and N1 PDU Session Modify Command messages in it.
4. AMF sends 200 OK to SMF to acknowledge N1N2 Transfer Request.
5. AMF sends Update SM Context Request with N2 PDU Session Modify Response to SMF to initiate second transaction.
6. SMF sends N4 Session Modification Request to UPF to modify the QoS information at the UPF.
7. Error occurs (resource allocation failure) on UPF during session modification.
8. UPF sends N4 Session Modification Response - Failure to SMF.

9. SMF sends SMF Unsuccessful Procedure event to MDF2 (LI_X2 interface).
10. SMF sends Update SM Context Response - Failure to AMF to notify AMF of the failure.
11. SMF notifies PCF of failure of modification request.

PCF Initiated Session Modification with Lawful Intercept

PCF initiated SM Policy Association Modification

1. During previous message exchanges UPF was programmed to intercept all packets from/to UE. UPF sends intercepted packets in X3 CC-data events to MDF3 (LI_X3 interface).
2. PCF sends SM Policy Association Modification message to SMF to start the first transaction.
3. SMF sends 200 OK to PCF to acknowledge SM Policy Association Modification request.
4. SMF sends N1N2 Transfer Request to AMF with N2 PDU Session Modify Request and N1 PDU Session Modify Command messages in it.
5. AMF sends 200 OK to SMF to acknowledge N1N2 Transfer Request.
6. AMF sends Update SM Context Request with N2 PDU Session Modify Response to SMF to initiate second transaction.
7. SMF sends N4 Session Modification Request to UPF to modify the QoS information at the UPF.
8. UPF sends N4 Session Modification Response to SMF.
9. SMF sends SMF PDU Session Modification event to MDF2 (LI_X2 interface).
10. SMF sends Update SM Context Response to AMF to complete second transaction.
11. AMF sends an Update SM Context Request with the N1 PDU Session Modification Complete.
12. SMF sends Update SM Context Response.
13. SMF informs PCF about the status of the rules if requested by PCF.

Release SM Context Transaction with Lawful Intercept

1. From previous message exchanges UPF was programmed to intercept Uplink & Downlink packets from/to UE. UPF reports intercepted packets in X3 CC-data events to MDF3 (LI_X3 interface).
2. AMF sends Release SM Context Request.
3. SMF sends N4 Session Release Request (PFCP message) to UPF.

4. UPF releases session resources and sends N4 Session Release Response (PFCP message) to SMF.
5. At this point, session no longer exists on UPF. No further X3 CC-data events are sent to MDF3 for the session which is releasing.
6. SMF sends Release SM Context Response to AMF to indicate session has been released.
7. SMF sends X2 SMF PDU Session Release event with Session's Correlation Number to MDF2 (LI_X2 interface).
8. SMF performs SM Policy Control Delete procedure to delete the SM Policy Association at the PCF.

SMF Start of Interception with Established PDU Session

1. AMF sends Create SM Context Request to SMF.
2. SMF sends 200 OK to acknowledge Create SM Context Request.
3. SMF performs SM Policy Association Establishment procedure to establish SM Policy Association with the PCF. If dynamic PCC is not enabled or if PCF down, SMF uses local policy configured in the subscriber-policy-profile, see Policy Control Service in UnityCloud User Guide.

Note: In the absence of this information from PCF, SMF receives the addresses from UDM Subscription data or from local configuration as described in the Charging Service section.

4. SMF sends N4 Session Establishment Request to UPF including Packet detection, enforcement, and reporting rules to install on UPF for this PDU Session. If CN Tunnel Information and UE IP address are allocated by SMF, they are provided to UPF.
5. UPF sends N4 Session Establishment Response.
6. SMF notifies AMF of UE IP address, CN Tunnel Info, QoS Rules and Profiles by sending N1N2 Transfer message to AMF.
7. AMF sends Update SM Context Request to SMF.
8. SMF sends N4 Session Modification Request (PFCP message) to UPF.
9. UPF sends N4 Session Modification Response (PFCP message) to SMF.
10. SMF sends Update SM Context Response to AMF.
11. At this point, session is established, UE is transmitting data, no interception is occurring.

12. ADMF/LIPF sends X1 ActivateTask which intercepts existing session.
13. SMF sends X1 ActivateTask to UPF with PFCP SessionId and Session Correlation Number (LI_T2/LI_T3 interface).
14. UPF starts intercepting Uplink & Downlink packets from/to UE. Intercepted packets are sent via X3 CC-PDU events to MDF3 (LI_X3 interface).
15. SMF sends X2 StartOfInterceptionOnEstablishedPduSession event with Session Correlation Number to MDF2 (LI_X2 interface).

Note: X3 CC-data events for the Session Correlation Number may arrive at MDF3 prior to MDF2's receipt of the corresponding X2 StartOfInterceptionOnEstablishedPduSession event.

UE Initiated PDU Session Modification (Failure) with Lawful Intercept

The UE requested PDU Session modification procedure is used by the UE to modify one or several of the QoS parameters exchanged between the UE and the network. The procedure may be used to create a new QoS rule, delete an existing QoS rule or to modify an existing QoS rule in order to add or remove packet filters. The procedure is triggered when the UE transmits a N1 PDU Session Modification Request message.

1. The receipt of an Update SM Context message containing the N1 PDU Session Modify Request message starts the transaction.
2. Error (such as resource allocation failure) occurs during session modification.
3. If the PCF has subscribed for event triggers and SMF receives trigger information in the Update SM Context, SMF reports the events to PCF.
4. SMF responds to AMF with an Update SM Context response - Failure.
5. SMF sends X2 SMF Unsuccessful Procedure event to MDF2 (LI_X2 interface).

UE Initiated PDU Session Modification with Lawful Intercept

The UE requested PDU Session modification procedure is used by the UE to modify one or several of the QoS parameters exchanged between the UE and the network. The procedure is used to create a new QoS rule, delete an existing QoS rule or to modify an existing QoS rule to add or remove packet

filters. The procedure is triggered when the UE transmits a N1 PDU Session Modification Request message. The procedure consists of 3 transactions.

In previous message exchanges UPF was programmed to intercept all traffic for session. UPF sends each intercepted packet in X3 CC-PDU events to MDF3 (LI_X3 interface).

Update SM Context Transaction with N1 PDU Session Modify Request Message

1. The receipt of an Update SM Context message containing the N1 PDU Session Modify Request message starts the first transaction.
2. If the PCF has subscribed for event triggers and SMF receives trigger information in the Update SM Context, SMF reports the events to PCF.
3. SMF then responds with an Update SM Context response message including N2 Session Modify request and N1 PDU Session Modification command messages in it.

Update SM Context Transaction with N2 PDU Session Modify Response

The second transaction is triggered by the receipt of an Update SM Context request with N2 PDU Session Modify Response and the SMF ends the transaction after responding with an Update SM Context response.

Update SM Context Transaction with N1 PDU Session Modification Complete

1. The third transaction is triggered when AMF sends an Update SM Context message with the N1 PDU Session Modification Complete message that acknowledges the PDU Session Modification Command.
2. Upon receipt of the request, SMF instance initiates an N4 Session Modification procedure with the UPF to modify the QoS information at the UPF.
3. SMF sends X2 SMF PDU Session Modification event to MDF2 (LI_X2 interface).
4. SMF sends Update SM Context Response to AMF to end the transaction.
5. SMF may inform the PCF about the status of the rules if requested by PCF.

UE Requested PDU Session Establishment (Failure) with Lawful Intercept

1. AMF sends Create SM Context Request to SMF.
2. SMF sends 200 OK to acknowledge Create SM Context Request.

3. SMF performs SM Policy Association Establishment procedure to establish SM Policy Association with the PCF. If dynamic PCC is not enabled or if PCF down, SMF uses local policy configured in the subscriber-policy-profile, see Policy Control Service in UnityCloud User Guide.

Note: In the absence of this information from PCF, SMF receives the addresses from UDM Subscription data or from local configuration as described in the Charging Service section.

4. SMF sends N4 Session Establishment Request to UPF including Packet detection, enforcement and reporting rules to be installed on UPF for this PDU Session. If CN Tunnel Information and UE IP address are allocated by SMF, they are provided to UPF.
5. Error occurs on UPF during session setup (e.g. resource allocation failure, etc).
6. UPF sends N4 Session Establishment Response - Failure to report the error to SMF.
7. SMF sends X2 SMF Unsuccessful Procedure event to MDF2.

UE Requested PDU Session Establishment with Lawful Intercept

1. AMF sends Create SM Context Request to SMF.
2. SMF sends 200 OK to acknowledge Create SM Context Request.
3. MSF performs SM Policy Association Establishment procedure to establish SM Policy Association with the PCF. If dynamic PCC is not enabled or if PCF down, SMF uses local policy configured in the subscriber-policy-profile, see Policy Control Service in UnityCloud User Guide.

Note: In the absence of this information from PCF, SMF receives the addresses from UDM Subscription data or from local configuration as described in the Charging Service section.

4. SMF sends N4 Session Establishment Request to UPF including Packet detection, enforcement, and reporting rules to be installed on UPF for this PDU Session. If CN Tunnel Information and UE IP address are allocated by SMF, they are provided to UPF.
5. UPF sends N4 Session Establishment Response.
6. SMF sends LI X1 ActivateTask for PFCP Session ID to UPF with Session Correlation Number (LI_T2/LI_T3 interface).

7. UPF starts intercepting Uplink packets from UE. Intercepted packets are sent via X3 CC- PDU events to MDF3 (LI_X3 interface).
8. SMF notifies AMF of UE IP address, CN Tunnel Info, QoS Rules and Profiles by sending N1N2 Transfer message to AMF.
9. AMF sends Update SM Context Request to SMF.
10. SMF sends N4 Session Modification Request (PCFP message) to UPF.
11. SMF sends LI X2 PDU Session Establishment event with Session Correlation Number to MDF2.
12. UPF responds to N4 Session Modification request.
13. UPF is now intercepting Uplink & Downlink packets from/to UE. Intercepted packets are sent through X3 CC-PDU events to MDF3 (LI_X3 interface).
14. SMF responds to AMF with Update SM Context Response.

UE Requested PDU Session Release with Lawful Intercept

The UE requested PDU Session Release procedure allows the UE to request the release of a PDU session. It is triggered when the UE transmits a NAS PDU Session Release Request message.

This message is relayed to the SMF through N2 and the AMF. The procedure consists of 3 transactions that are triggered by receipt of Update SM Context messages.

During previous message exchanges UPF was programmed to intercept all packets from/to UE. UPF sends intercepted packets in X3 CC-data events to MDF3 (LI_X3 interface).

Update SM Context Transaction with NAS PDU Session Release Request Message

1. The receipt of an Update SM Context message containing the N1 PDU Session Release Request message starts the first transaction.
2. SMF-CC releases the PCFP session by sending the N4 Session Release Request message towards the UPF.
3. At this point, UPF is no longer intercepting data for the session. No further X3 CC-data is sent to MDF3 for this session (LI_X3 interface).
4. SMF then responds to AMF with Update SM Context response message including N2 SM Resource Release request and N1 PDU Session Release command messages in it.

Update SM Context Transaction with N2 SM Resource Release Response

The second transaction is triggered when AMF sends to SMF the Update SM Context request with N2 SM Resource Release Response and the SMF ends the transaction after responding with an Update SM Context response.

Update SM Context Transaction with NAS PDU Session Release Complete

1. The third transaction is triggered when AMF sends to SMF the Update SM Context message with NAS PDU Session Release Complete message acknowledging PDU Session Release Command.
2. SMF responds to AMF with Update SM Context response.
3. If Dynamic PCC is applied to this session the SMF invokes SM Policy Association Termination procedure towards the PCF to delete the PDU session.
4. SMF notifies AMF that the SM context for the PDU Session is released.
5. SMF notifies MDF2 that session is released by sending X2 SMF PDU Session Release with Session Correlation Number to MDF2 (LI_X2 interface).
6. If it is last PDU Session the SMF is handling for the UE for the associated DNN and S- NSSAI, smfcc unsubscribes from the UDM by means of the Nudm_SDM_Unsubscribe service operation. smfcc then invokes the Nudm_UECM_Deregistration service operation towards the UDM.

SMF-LI 5G Handover Message Flows with Lawful Intercept

This section documents Message Flows for Lawful Intercept 5G Handover scenarios handled by the SMF product.

- [Xn based Inter NG-RAN Handover with Lawful Intercept](#)
- [N2 based Inter NG-RAN Handover with Lawful Intercept](#)
- [Xn based Inter NG-RAN Handover-Insertion of Intermediate UPF with Lawful Intercept](#)
- [Xn based Inter NG-RAN Handover-Reallocation of Intermediate UPF with Lawful Intercept](#)

N2 based Inter NG-RAN Handover with Lawful Intercept

The N2 based handover procedure hands over a UE from a source NG-RAN node to a target NG-RAN node using the N2 reference point. This is triggered by the source NG-RAN, for example, if there is no Xn connectivity to the target NG-RAN.

Lawful Intercept Tasks are activated on the UPF-PSA as part of the initial session setup

1. SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID) with 5G Session Correlation Number to UPF-PSA during initial session setup.
2. If UE is 4G/5G capable, SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Default Bearer FTEID) for Default Bearer with 4G Default Bearer Correlation Number to UPF-PSA.
3. If UE is 4G/5G capable and a dedicated bearer is present, SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Dedicated Bearer FTEID) for with 4G Dedicated Bearer Correlation Number to UPF-PSA.
4. UPF-PSA is intercepting UE traffic and reporting X3 CC-data to MDF with the 5G Session Correlation Number.

The handover procedure is started on the SMF upon receipt of an Update SM Context message with N2 Handover Required message. This procedure consists of three transactions.

Update SM Context transaction with N2 Handover Required Message

5. The receipt of an Update SM Context message containing the N2 Handover Required message and hoState set to PREPARING starts the first transaction.
6. If an I-UPF has been selected, the SMF instance initiates an N4 Session Establishment procedure with the I-UPF including or requesting for the Tunnel IDs for N3 and N9 interfaces. The N9 tunnel information for the original IP Anchor UPF is also provided. The Uplink and Downlink PDRs for all Qos Flows are installed on the I-UPF with only catch-all packet filters. The QERs are installed with only QFIs set.
7. SMF responds with an Update SM Context response message including N2 PDU Session Resource Setup Request message with N3 Tunnel ID information of the UPF in it.

Update SM Context transaction with N2 Handover Request Ack

8. The second transaction is triggered by the receipt of an Update SM Context request with N2 Handover Request Ack and hoState set to PREPARED. SMF receives the Target NG-RAN N3 Tunnel IDs as well as data forwarding Tunnel information.
9. SMF ends the transaction after responding with an Update SM Context response with hoState PREPARED and an N2 Handover Command message including the forwarding Tunnel Info to be sent to the source NG-RAN.

Update SM Context transaction with hoState COMPLETED

10. The third transaction is triggered when AMF sends an Update SM Context message with hoState set to COMPLETED.

11. UPF-PSA continues to report intercepted UE traffic (X3 CC-data with 5G Session Correlation Number) to MDF3.
12. SMF instance initiates an N4 Session Modification procedure with the new (Target) I-UPF to set the target NG-RAN Tunnel information at the new (Target) I-UPF.
13. SMF then initiates an N4 Session Modification procedure with the original UPF to update it with the N9 tunnel information of the I-UPF.
14. SMF ends the transaction after responding with an Update SM Context response with hoState COMPLETED.
15. SMF sends X2 SMF PDU Session Modification with 5G Session Correlation Number to MDF2 to notify MDF2 of the handover procedure success.
16. UPF-PSA continues to report all intercepted UE traffic (X3 CC-data with 5G Session Correlation Number) to MDF3. Note: UPF-PSA has been intercepting all UE traffic before, during, and after the handover.
17. SMF starts a timer to release resources on the Source I-UPF. When the timer expires, SMF releases the PFCP session at the Source I-UPF.

3GPP Specification Reference

TS 23.502, Section 4.9.1.3

SMF-LI 5G Handover Message Flows with Lawful Intercept

This section documents Message Flows for Lawful Intercept 5G Handover scenarios handled by the SMF product.

- [Xn based Inter NG-RAN Handover with Lawful Intercept](#)
- [N2 based Inter NG-RAN Handover with Lawful Intercept](#)
- [Xn based Inter NG-RAN Handover-Insertion of Intermediate UPF with Lawful Intercept](#)
- [Xn based Inter NG-RAN Handover-Reallocation of Intermediate UPF with Lawful Intercept](#)

Xn based Inter NG-RAN Handover with Lawful Intercept

This procedure hands over a UE from a source NG-RAN to target NG-RAN using Xn when the AMF is unchanged and the SMF decides to keep the existing UPF, which terminates the N3 interface in the 5GC. The presence of IP connectivity between the Source UPF and Target NG-RAN is assumed.

1. X1 ActivateTask for 5G Session is sent to UPF during the initial session setup.

2. If UE is 4G/5G capable, X1 ActivateTasks for 4G Default and Dedicated Bearers are sent to UPF during the initial session setup.
3. UE is intercepting 5G session data and reporting X3 CC-data to MDF3.
4. The handover procedure consists of one transaction that is triggered by receipt of an Update SM Context message with a Path Switch Request from AMF.
5. If PCF is subscribed for event triggers (for example RAT Change) and SMF receives trigger information in the Update SM Context, SMF reports the events to PCF.
6. Upon receipt of the request, the SMF instance initiates an N4 Session Modification procedure with the UPF to modify the N3 F-TEID information to point to the target NG-RAN.
7. SMF sends an Update SM Context Response to AMF with Path Switch Request Ack.
8. SMF sends X2 SMF PDU Session Modification with 5G Session Correlation Number to MDF2 to notify MDF2 of successful handover.

3GPP Specification Reference

TS 23.502, Section 4.9.1.2.2

Xn based Inter NG-RAN Handover-Insertion of Intermediate UPF with Lawful Intercept

This procedure hands over a UE from a Source NG-RAN to a Target NG-RAN using Xn when the AMF is unchanged and the SMF decides that insertion of a new additional intermediate UPF is needed.

1. SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID) with 5G Session Correlation Number to UPF-PSA during initial session setup.
2. If UE is 4G/5G capable, SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Default Bearer FTEID) for Default Bearer with 4G Default Bearer Correlation Number to UPF-PSA.
3. If UE is 4G/5G capable and a dedicated bearer is present, SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Dedicated Bearer FTEID) with 4G Dedicated Bearer Correlation Number to UPF-PSA.
4. UPF-PSA is intercepting all UE traffic for the session and reporting it through X3 CC-data with 5G Session Correlation Number to MDF3.
5. The procedure consists of one transaction that is triggered by receipt of an Update SM Context message with a Path Switch Request from AMF.
6. If PCF is subscribed for event triggers (example RAT Change) and SMF receives trigger information in the Update SM Context, SMF reports the events to PCF.

7. If an I-UPF has been selected, SMF initiates an N4 Session Establishment procedure with the I-UPF including the Tunnel IDs for N3 and N9 interfaces as well as the N9 tunnel information for the original IP Anchor UPF. The Uplink and Downlink PDRs for all Qos Flows are installed on the I-UPF with only catch-all packet filters. The QERs are installed with only QFIs set.
8. SMF then initiates an N4 Session Modification procedure with the original UPF-PSA to update it with the N9 tunnel information of the I-UPF.
9. SMF sends an Update SM Context Response to AMF with Path Switch Request Ack.
10. SMF reports X2 SMF PDU Session Modification with 5G Session Correlation Number to MDF2 to inform MDF2 of the handover.
11. UPF-PSA continues to intercept all UE traffic for the session and report it through X3 CC- data with 5G Session Correlation Number to MDF3.

3GPP Specification Reference

TS 23.502, Section 4.9.1.2.3

Xn based Inter NG-RAN Handover-Reallocation of Intermediate UPF with Lawful Intercept

This procedure is used to hand over a UE from a Source NG-RAN to a Target NG-RAN using Xn when the AMF is unchanged and the SMF decides that the intermediate UPF (I-UPF) is to be changed.

1. SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID) with 5G Session Correlation Number to UPF-PSA during initial session setup.
2. If UE is 4G/5G capable, SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Default Bearer FTEID) for Default Bearer with 4G Default Bearer Correlation Number to UPF-PSA.
3. If UE is 4G/5G capable and a dedicated bearer is present, SMF sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, Dedicated Bearer FTEID) with 4G Dedicated Bearer Correlation Number to UPF-PSA.
4. UPF-PSA is intercepting all UE traffic for the session and reporting it through X3 CC-data with 5G Session Correlation Number to MDF3.
5. The procedure consists of one transaction that is triggered by receipt of an Update SM Context message with a Path Switch Request from AMF.
6. If PCF is subscribed for event triggers (example RAT Change) and SMF receives trigger information in the Update SM Context, SMF reports the events to PCF.
7. If a new I-UPF has been selected, the SMF instance initiates an N4 Session Establishment procedure with the Target I-UPF including the Tunnel IDs for N3 and N9 interfaces as well as the N9

tunnel information for the original IP Anchor UPF. The Uplink and Downlink PDRs for all Qos Flows are installed on the I-UPF with only catch-all packet filters. The QERs are installed with only QFIs set.

8. SMF initiates an N4 Session Modification procedure with the IP Anchor UPF to update it with the N9 tunnel information of the Target I-UPF.
9. SMF reports X2 SMF PDU Session Modification with 5G Session Correlation Number to MDF2.
10. SMF starts a timer to release the PFCP session on the Source I-UPF.
11. SMF sends an Update SM Context Response to AMF with Path Switch Request Ack.
12. When the timer started in step 11 times-out, SMF releases the PFCP session at the Source I-UPF.

3GPP Specification Reference

TS 23.502, Section 4.9.1.2.4

SMF-LI X2 Events for 4G EPC Sessions

Bearer Activation (Unsuccessful) REPORT Record

See 3GPP TS 33.108 V16.1.0 (2020-07) Table 10.5.1.1.3: Bearer Activation (unsuccessful) REPORT Record.

Table 21. Bearer Activation (Unsuccessful) REPORT Record

Parameter	M/C/O	Description/Conditions	Supported?	Notes
observed MSISDN observed IMSI observed ME Id	C	Provide at least one and others when available	Yes	SMF/PGW-C reports all available identifiers when present in the GTP signaling for session/bearer activation
PDN address allocation	C	Provides the PDN type and PDN address(es) used by the network.	Yes	
Event type	C	Provide EPS Bearer Activation event type.	Yes	
Event date Event time	M	Provide the date and time the event is detected.	Yes	

SMF

Parameter	M/C/O	Description/Conditions	Supported?	Notes
access point name	C	Provide to identify the packet data network to which the attempt to connect was made; this information may be provided by the UE; the parameter is applicable only for default bearer activation.	Yes	
RAT type	C	Provide the Radio Access Type used by the target.	Yes	
Initiator	C	Provide to indicate whether the EPS bearer activation is network-initiated, target-initiated, or not available.	Yes	
Network identifier	M	Shall be provided.	Yes	
logicalFunctionInformation	O	Used to distinguish between multiple logical functions operating in a single physical network element.	No	
Lawful intercept identifier	M	Shall be provided.	Yes	Per 3GPP TS 33.128 section 6.3.2.2 this field will be populated with the string "LIIDNotPresent" . The relevant XID will be populated in the X2 PDU Header field XID as defined in ETSI TS 103 221-2
Location information	C	Provide, when authorized, to identify location information for the target's UE, including the primary cell ID from each of the Master Cell (Pcell) and Secondary Cell (PSCell) groups of the target, if available.	Yes	Will be provided when available
Time of Location	C	Date/Time of Location (if target location provided).	Yes	Will be provided when available

SMF

Parameter	M/C/O	Description/Conditions	Supported?	Notes
Failed bearer activation reason	C	Provide information about the reason for failed bearer activation attempts of the target.	Yes	
EPS bearer QOS	C	Provide to identify the QOS parameters. The parameter carries the requested EPS bearer QOS.	Yes	
Bearer activation type	C	Provides information on default or dedicated bearer failed activation	Yes	
APN-AMBR	C	The Aggregate Maximum Bit Rate foreseen for the APN. The parameter carries the subscribed APN-AMBR.	Yes	
Protocol configuration options	C	Provide information about the protocol configuration options requested by the UE	Yes	
Procedure transaction identifier	C	Used to associate the EPS bearer activation attempt to other messages triggering the procedure.	Yes	
Linked EPS bearer identity	C	Provides, in case of failed dedicated bearer activation attempt, the EPS bearer id of the associated default bearer; not applicable in case of default bearer activation attempt.	Yes	
Traffic Flow Template TFT	C	The TFT associated to the dedicated bearer activation attempt; not applicable in case of default bearer activation attempt	Yes	
Handover indication	C	Provide information that the procedure is triggered as part of a handover	Yes	
UE Local IP Address	C	The UE local IP address reported over GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported

Parameter	M/C/O	Description/Conditions	Supported?	Notes
UE UDP Port	C	The UE UDP Port number provided in case of GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported
WLAN location information	C	The TWAN identifier provided in case of GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported
WLAN location timestamp	C	The TWAN identifier timestamp provided in case of GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported

Bearer Activation (Successful) and Start of Interception with Active Bearer BEGIN Record

See 3GPP TS 33.108 V16.1.0 (2020-07) Table 10.5.1.2.1: Bearer Activation (successful) and Start of Interception with active bearer BEGIN Record.

Table 22. Bearer Activation (Successful) and Start of Interception with Active Bearer BEGIN Record

Parameter	M/C/O	Description/Conditions	Supported?	Notes
observed MSISDN observed IMSI observed ME Id	C	Provide at least one and others when available	Yes	SMF/PGW-C reports all available identifiers when present in the GTP signaling for session/bearer activation
event type	C	Provide, depending on the reported event: Bearer activation or: Start of interception with active bearer event type.	Yes	When X1 Task is present prior to bearer activation, the Bearer Activation event will be reported during bearer activation. When X1 Task is not present at time of bearer activation, the Start of interception with active bearer will be reported when the X1 Task is added for a session/bearer that is already present.
event date event time	M	Provide the date and time the event is detected.	Yes	
access point name	C	Provided to identify the packet data network to	Yes	

SMF

Parameter	M/C/O	Description/Conditions	Supported?	Notes
		which the connection is made (valid only for default bearer activation).		
PDN address allocation	C	Provides the PDN type and PDN address(es) used by the network.	Yes	
initiator	C	Provide to indicate whether the EPS bearer activation is network-initiated, target-initiated, or not available.	Yes	Note: In Start of Interception with active bearer event, initiator will be set to "not-Available"
network identifier	M	Shall be provided	Yes	
logicalFunctionInformation	O	Used to distinguish between multiple logical functions operating in a single physical network element.	No	
lawful intercept identifier	M	Shall be provided.	Yes	Per 3GPP TS 33.128 section 6.3.2.2 this field will be populated with the string "LIIDNotPresent" . The relevant XID will be populated in the X2 PDU Header field XID as defined in ETSI TS 103 221-2
location information	C	Provide, when authorized, to identify location information for the target's UE.	Yes	
Time of Location	C	Date/Time of Location (if target location provided).	Yes	
EPS bearer QOS	C	Provide to identify the QOS parameters. It carries the EPS bearer QOS associated to the established bearer.	Yes	
Bearer activation type	C	Provides information on default or dedicated bearer activation.	Yes	
APN-AMBR	C	The Aggregate Maximum Bit Rate foreseen for the APN. The parameter carries the APN-AMBR used for the established bearer	Yes	

SMF

Parameter	M/C/O	Description/Conditions	Supported?	Notes
Protocol configuration options	C	Provide information about the protocol configuration options requested by the UE.	Yes	
Procedure transaction identifier	C	Used to associate the EPS bearer activation to other messages triggering the procedure.	Yes	
EPS bearer id	C	Provides the EPS bearer id allocated by the network.	Yes	
Linked EPS bearer identity	C	Provides, in case of dedicated bearer activation, the EPS bearer id of the associated default bearer; not applicable in case of default bearer activation.	Yes	
Traffic Flow Template(s) TFT	C	The TFT associated to the dedicated bearer activation; not applicable in case of default bearer activation.	Yes	
Handover indication	C	Provide information that the procedure is triggered as part of a handover.	Yes	
RAT type	C	The Radio Access Type used by the target subscriber (only applicable to default bearer activation).	Yes	
Correlation number	M	Shall be provided to uniquely identify the EPS bearer delivered to the LEMF and to correlate IRI records with CC.	Yes	
UE Local IP Address	C	The UE local IP address reported over GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported
UE UDP Port	C	The UE UDP Port provided in case of GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported
WLAN location information	C	The TWAN identifier provided in case of GTP	No	Non-3GPP LI not yet supported

Parameter	M/C/O	Description/Conditions	Supported?	Notes
		based S2b interface TS 29.274 [46].		
WLAN location timestamp	C	The TWAN identifier timestamp provided in case of GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported

Bearer Modification CONTINUE Record

See 3GPP TS 33.108 V16.1.0 (2020-07) Table 10.5.1.3.1: Bearer Modification CONTINUE Record.

Table 23. Bearer Modification CONTINUE Record

Parameter	M/C/O	Description/Conditions	Supported?	Notes
observed MSISDN observed IMSI observed ME Id	C	Provide at least one and others when available.	Yes	SMF/PGW-C reports all available identifiers when present in the GTP signaling for session/bearer activation
event type	C	Provide Bearer modification event type.	Yes	
event date event time	M	Provide the date and time the event is detected.	Yes	
initiator	C	Provide to indicate whether the EPS bearer modification is network-initiated, target-initiated, or not available.	Yes	
network identifier	M	Shall be provided.	Yes	
logicalFunctionInformation	O	Used to distinguish between multiple logical functions operating in a single physical network element.	No	
lawful intercept identifier	M	Shall be provided.	Yes	Per 3GPP TS 33.128 section 6.3.2.2 this field will be populated with the string "LIIDNotPresent". The relevant XID will be populated in the X2 PDU Header field XID as defined in ETSI TS 103 221-2
location information	C	Provide, when authorized, to identify location	Yes	

SMF

Parameter	M/C/O	Description/Conditions	Supported?	Notes
		information for the target's UE.		
Time of Location	C	Date/Time of Location (if target location provided).	Yes	
EPS bearer QOS	C	Provide to identify the QOS parameters.	Yes	
APN-AMBR	C	The Aggregate Maximum Bit Rate for the APN.	Yes	
Procedure transaction identifier	C	Used to associate the EPS bearer modification to other messages triggering the procedure.	Yes	
EPS bearer id	C	Provides the EPS bearer id allocated by the network.	Yes	
Traffic Flow Template(s) TFT	C	The TFT associated to the EPS bearer modification.	Yes	
RAT type	C	The Radio Access Type used by the target.	Yes	
APN-AMBR	C	The Aggregate Maximum Bit Rate foreseen for the APN.	Yes	
Handover indication	C	Provide information that the procedure is triggered as part of a handover.	Yes	
Correlation number	M	Shall be provided to uniquely identify the EPS bearer delivered to the LEMF and to correlate IRI records with CC.	Yes	
Failed bearer modification reason	C	Provide information about the reason for failed bearer modification	Yes	
UE Local IP Address	C	The UE local IP address reported over GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported
UE UDP Port	C	The UE UDP Port provided in case of GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported
WLAN location information	C	The TWAN identifier provided in case of GTP	No	Non-3GPP LI not yet supported

Parameter	M/C/O	Description/Conditions	Supported?	Notes
		based S2b interface TS 29.274 [46].		
WLAN location timestamp	C	The TWAN identifier timestamp provided in case of GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported
Secondary RAT Usage Indication	O	The Secondary RAT Usage Indication (see TS 37.340 [99]).	No	

Bearer Deactivation END Record

See 3GPP TS 33.108 V16.1.0 (2020-07) Table 10.5.1.4.1: Bearer Deactivation END Record.

Table 24. Bearer Deactivation END Record

Parameter	M/C/O	Description/Conditions	Supported?	Notes
observed MSISDN observed IMSI observed ME Id	C	Provide at least one and others when available.	Yes	SMF/PGW-C reports all available identifiers when present in the GTP signaling for session/bearer activation
event type	C	Provide Bearer modification event type.	Yes	
event date event time	M	Provide the date and time the event is detected.	Yes	
initiator	C	Provide to indicate whether the EPS deactivation is network-initiated, target-initiated, or not available.	Yes	
network identifier	M	Shall be provided.	Yes	
logicalFunctionInformation	O	Used to distinguish between multiple logical functions operating in a single physical network element.	No	
correlation number	M	Shall be provided to uniquely identify the PDP context delivered to the LEM and to correlate IRI records with CC.	Yes	

SMF

Parameter	M/C/O	Description/Conditions	Supported?	Notes
lawful intercept identifier	M	Shall be provided.	Yes	Per 3GPP TS 33.128 section 6.3.2.2 this field will be populated with the string " LIIDNotPresent ". The relevant XID will be populated in the X2 PDU Header field XID as defined in ETSI TS 103 221-2
location information	C	Provide, when authorized, to identify location information for the target's MS.	Yes	
Time of Location	C	Date/Time of Location (if target location provided).	Yes	
Bearer deactivation type	C	Provides information on default or dedicated bearer deactivation.	Yes	
Bearer deactivation cause	C	Provide to indicate reason for deactivation.	Yes	
EPS bearer id	O	Provides the identity of the deactivated bearer.	No	
Procedure Transaction Identifier	C	Used to associate the EPS bearer deactivation to other messages triggering the procedure.	Yes	
ULI Timestamp	O	Indicates the time when the User Location Information was acquired.	Yes	
UE Local IP Address	C	The UE local IP address reported over GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported
UE UDP Port	C	The UE UDP Port provided in case of GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported
WLAN location information	C	The TWAN identifier provided in case of GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported
WLAN location timestamp	C	The TWAN identifier timestamp provided in case of GTP based S2b interface TS 29.274 [46].	No	Non-3GPP LI not yet supported

Parameter	M/C/O	Description/Conditions	Supported?	Notes
Secondary RAT Usage Information	O	Secondary RAT Usage Information (see TS 37.340 [99]).	No	

SMF-LI X2 Events for 5G Sessions

SMF PDUSessionEstablishment

See 3GPP TS 33.128 v16.4.0 (2020-09) Table 6.2.3-1: Payload for SMF PDUSessionEstablishment record.

Table 25. SMF PDUSessionEstablishment

Field Name	Description	M/C/O	Supported?	Notes
sUPI	SUPI associated with the PDU session (e.g. as provided by the AMF in the associated Nsmf_PDU_Session_CreateSMContext service operation). Shall be present except for PEI-only unauthenticated emergency sessions (see NOTE).	C	Yes	
sUPIUnauthenticated	Shall be present if a SUPI is present in the message and set to “true” if the SUPI has not been authenticated, or “false” if it has been authenticated.	C	Yes	
pEI	PEI associated with the PDU session if available (see NOTE).	C	Yes	
gPSI	GPSI associated with the PDU session if available (see NOTE).	C	Yes	
pDUSessionID	PDU Session ID See clause 9.4 of TS 24.501 [13].	M	Yes	
gTPTunnelID	Contains the F-TEID identifying the GTP tunnel used to encapsulate the traffic, as defined in TS 29.244 [15] clause 8.2.3. Non-GTP encapsulation is for further study.	M	Yes	

SMF

Field Name	Description	M/C/O	Supported?	Notes
pDUSessionType	Identifies selected PDU session type, see TS 24.501 [13] clause 9.11.4.11.	M	Yes	
sNSSAI	Slice identifiers associated with the PDU session, if available. See TS 23.003 [19] clause 28.4.2 and TS 23.501 [2] clause 5.12.2.2.	C	Yes	
uEEndpoint	UE endpoint address(es) if available.	C	Yes	
non3GPPAccessEndpoint	UE's local IP address used to reach the N3IWF, if available. IP addresses are given as 4 octets (for IPv4) or 16 octets (for IPv6) with the most significant octet first (network byte order).	C	No	In future SMF will add Lawful Intercept support for Non-3GPP sessions
location	Location information provided by the AMF, if available. Encoded as a userLocation parameter (location>location Info>userLocation), see Annex A.	C	Yes	Provided by SMF when available (i.e. when AMF provided to SMF)
dNN	Data Network Name associated with the target traffic, as defined in TS 23.003[19] clause 9A and described in TS 23.501 [2] clause 4.3.2.2.	M	Yes	
aMFID	Identifier of the AMF associated with the target UE, as defined in TS 23.003 [19] clause 2.10.1 when available.	C	Yes	
hSMFURI	URI of the Nsmf_PDUSession service of the selected H-SMF, if available. See TS 29.502 [16] clause 6.1.6.2.2.	C	Yes	
requestType	Type of request as described in TS 24.501 [13] clause 9.11.3.47 if available. In the case where the network does not support Multi Access (MA) PDU sessions, but receives a MA PDU session request, a request type of "Initial request" shall be reported.	C	Yes	

SMF

Field Name	Description	M/C/O	Supported?	Notes
accessType	Access type associated with the session (i.e. 3GPP or non-3GPP access) if provided by the AMF (see TS 24.501 [13] clause 9.11.2.1A).	C	Yes	Currently only 3GPP access is supported
rATType	RAT Type associated with the access if provided by the AMF as part of session establishment (see TS 23.502 [4] clause 4.3.2). Values given as per TS 29.571 [17] clause 5.4.3.2.	C	Yes	
sMPDUDNRequest	Contents of the SM PDU DN Request container, if available, as described in TS 24.501 [13] clause 9.11.4.15.	C	Yes	

Note: At least one of the SUPI, PEI or GPSI fields shall be present.

SMF PDUSessionModification record

See 3GPP TS 33.128 v16.4.0 (2020-09) Table 6.2.3-2: Payload for SMF PDUSessionModification record.

Table 26. SMF PDUSessionModification record

Field Name	Description	M/C/O	Supported?	Notes
sUPI	SUPI associated with the PDU session (e.g. as provided by the AMF in the associated Nsmf_PDU_Session_CreateSMContext service operation). Shall be present except for PEI-only unauthenticated emergency sessions.	C	Yes	
sUPIUnauthenticated	Shall be present if a SUPI is present in the message, and set to "true" if the SUPI was not authenticated, or "false" if it has been authenticated.	C	Yes	
pEI	PEI associated with the PDU session if available.	C	Yes	

SMF

Field Name	Description	M/C/O	Supported?	Notes
gPSI	GPSI associated with the PDU session if available.	C	Yes	
sNSSAI	Slice identifier associated with the PDU session, if available. See TS 23.003 [19] clause 28.4.2 and TS 23.501 [2] clause 5.12.2.2.	C	Yes	
non3GPPAccessEndpoint	UE's local IP address used to reach the N3IWF, if available. IP addresses are given as 4 octets (for IPv4) or 16 octets (for IPv6) with the most significant octet first (network byte order).	C	No	Support for Non-3GPP LI will be added in future release
location	Location information provided by the AMF, if available. Encoded as a userLocation parameter (location>locationInfo>userLocation), see Annex A.	C	Yes	Provided by SMF when available (i.e. when AMF provided to SMF)
accessType	Access type associated with the session (i.e. 3GPP or non-3GPP access) if provided by the AMF (see TS 24.501 [13] clause 9.11.2.1A).	C	Yes	
rATType	RAT type associated with the access, if available. Values given as per TS 29.571 [17] clause 5.4.3.2.	C	Yes	

SMF PDUSessionRelease record

See 3GPP TS 33.128 v16.4.0 (2020-09) Table 6.2.3-3: Payload for SMF PDUSessionRelease record.

Table 27. SMF PDUSessionRelease record

Field Name	Description	M/C/O	Supported?	Notes
sUPI	SUPI associated with the PDU session.	M	Yes	
pEI	PEI associated with the PDU session if available.	C	Yes	
gPSI	GPSI associated with the PDU session if available.	C	Yes	

SMF

Field Name	Description	M/C/O	Supported?	Notes
pDUSessionID	PDU Session ID as assigned by the AMF.	M	Yes	
timeOfFirstPacket	Time of first packet for the PDU session.	C	Yes	
timeOfLastPacket	Time of last packet for the PDU session.	C	Yes	
uplinkVolume	Number of uplink octets for the PDU session.	C	Yes	
downlinkVolume	Number of downlink octets for the PDU session.	C	Yes	
location	Location information, if available. Encoded as a userLocation parameter (location>locationInfo>userLocation), see Annex A.	C	Yes	Provided by SMF when available (i.e. when AMF provided to SMF)

SMFStartOfInterceptionWithEstablishedPDUSession

See 3GPP TS 33.128 v16.4.0 (2020-09) Table 6.2.3-4: Payload for SMFStartOfInterceptionWithEstablishedPDUSession record.

Table 28. SMFStartOfInterceptionWithEstablishedPDUSession

Field Name	Description	M/C/O	Supported?	Notes
sUPI	SUPI associated with the PDU session (e.g. as provided by the AMF in the associated Nsmf_PDU_Session_CreateSMContext service operation). Shall be present except for PEI-only unauthenticated emergency sessions.	C	Yes	
sUPIUnauthenticated	Shall be present if a SUPI is present in the message and set to "true" if the SUPI has not been authenticated, or "false" if it has been authenticated.	C	Yes	
pEI	PEI associated with the PDU session if available.	C	Yes	

SMF

Field Name	Description	M/C/O	Supported?	Notes
gPSI	GPSI associated with the PDU session if available.	C	Yes	
pDUSessionID	PDU Session ID as assigned by the AMF, as defined in TS 24.007 [14] clause 11.2.3.1b.	M	Yes	
gTPTunnelID	Contains the F-TEID identifying the tunnel used to encapsulate the traffic, as defined in TS 29.244 [15] clause 8.2.3. Non-GTP encapsulation is for further study.	M	Yes	
pDUSessionType	Identifies selected PDU session type, see TS 24.501 [13] clause 9.11.4.11.	M	Yes	
sNSSAI	Slice identifier associated with the PDU session, if available. See TS 23.003 [19] clause 28.4.2 and TS 23.501 [2] clause 5.12.2.2.	C	Yes	
uEEndpoint	UE endpoint address(es) if available. IP addresses are given as 4 octets (for IPv4) or 16 octets (for IPv6) with the most significant octet first (network byte order). MAC addresses are given as 6 octets with the most significant octet first.	C	Yes	
non3GPPAccessEndpoint	UE's local IP address used to reach the N3IWF, if available. IP addresses are given as 4 octets (for IPv4) or 16 octets (for IPv6) with the most significant octet first (network byte order).	C	No	Support for Non-3GPP Lawful Intercept will be added in future release
location	Location information provided by the AMF at session establishment, if available. Encoded as a userLocation parameter (location>locationInfo>userLocation), see Annex A.	C	Yes	Provided by SMF when available (i.e. when AMF provided to SMF)
dNN	Data Network Name associated with the target traffic, as defined in TS 23.003 [19] clause 9A and	M	Yes	

SMF

Field Name	Description	M/C/O	Supported?	Notes
	described in TS 23.501 [2] clause 4.3.2.2.			
aMFID	Identifier of the AMF associated with the target UE, as defined in TS 23.003 [19] clause 2.10.1, if available.	C	Yes	
hSMFURI	URI of the Nsmf_PDUSession service of the selected H-SMF, if available. See TS 29.502 [16] clause 6.1.6.2.2.	C	Yes	
requestType	Type of request as described in TS 24.501 [13] clause 9.11.3.47 if available.	C	Yes	
accessType	Access type associated with the session (i.e. 3GPP or non-3GPP access) if provided by the AMF (see TS 24.501 [13] clause 9.11.2.1A).	C	Yes	
rATType	RAT type associated with the access if provided by the AMF as part of session establishment (see TS 23.502 [4] clause 4.3.2). Values given as per TS 29.571 [17] clause 5.4.3.2.	C	Yes	
sMPDUDNRequest	Contents of the SM PDU DN request container, if available, as described in TS 24.501 [13] clause 9.11.4.15.	C	Yes	

SMFUnsuccessfulProcedure

See 3GPP TS 33.128 v16.4.0 (2020-09) Table 6.2.3-5: Payload for SMFUnsuccessfulProcedure record.

Table 29. SMFUnsuccessfulProcedure

Field Name	Description	M/C/O	Supported?	Notes
failedProcedureType	Specifies the procedure which failed or is aborted at the SMF.	M	Yes	
failureCause	Provides the value of the 5GSM cause, see TS 24.501 [13], clause 9.11.4.2. In case the	M	Yes	

SMF

Field Name	Description	M/C/O	Supported?	Notes
	procedure is aborted due to a 5GSM STATUS message, the 5GSM cause is the one included in the 5GSM status message.			
requestedSlice	Slice requested for the procedure, if available, given as a NSSAI (a list of S-NSSAI values as described in TS 24.501 [13] clause 9.11.3.37).	C	Yes	
initiator	Specifies whether the network (SMF) or the UE is initiating the rejection or indicating the failure.	M	Yes	
sUPI	SUPI associated with the procedure, if available (see NOTE).	C	Yes	
sUPIUnauthenticated	Shall be present if a SUPI is present in the message and set to "true" if the SUPI has not been authenticated, or "false" if it has been authenticated.	C	Yes	
pEI	PEI used in the procedure, if available (see NOTE).	C	Yes	
gPSI	GPSI used in the procedure, if available (see NOTE).	C	Yes	
pDUSessionID	PDU Session ID See clause 9.4 of TS 24.501 [13], if available.	C	Yes	
uEEndpoint	UE endpoint address(es) if available.	C	Yes	
non3GPPAccessEndpoint	UE's local IP address used to reach the N3IWF, if available.	C	No	Support for Non-3GPP Lawful Intercept will be added in future release
location	Location information provided by the AMF, if available. Encoded as a userLocation parameter (location>locationInfo>userLocation), see Annex A.	C	Yes	Provided by SMF when available (i.e. when AMF provided to SMF)
dNN	Data Network Name associated with the target traffic, as defined in TS 23.003 [19] clause 9A and described in TS 23.501 [2] clause 4.3.2.2, if available.	C	Yes	

SMF

Field Name	Description	M/C/O	Supported?	Notes
aMFID	Identifier of the AMF associated with the target UE, as defined in TS 23.003 [19] clause 2.10.1 when available.	C	Yes	
hSMFURI	URI of the Nsmf_PDUSession service of the selected H-SMF, if available. See TS 29.502 [16] clause 6.1.6.2.2.	C	Yes	
requestType	Type of request as described in TS 24.501 [13] clause 9.11.3.47 if available.	C	Yes	
accessType	Access type associated with the session (i.e. 3GPP or non-3GPP access) if provided by the AMF (see TS 24.501 [13] clause 9.11.2.1A).	C	Yes	
rATType	RAT Type associated with the access if provided by the AMF as part of session establishment (see TS 23.502 [4] clause 4.3.2). Values given as per TS 29.571 [17] clause 5.4.3.2.	C	Yes	
sMPDUDNRequest	Contents of the SM PDU DN Request container, if available, as described in TS 24.501 [13] clause 9.11.4.15.	C	Yes	

Note: At least one identity shall be provided, the others shall be provided if available.

SMF-LI Traffic Steering

This feature adds Lawful Intercept support for all Multiple UPF call flows supported by the Traffic Steering feature. This feature is only applicable to 5G PDU Sessions (since Traffic Steering is only applicable to 5G Sessions).

Traffic Steering on 5G sessions includes:

- ULCL insertion/update/removal
- Local Anchor (LA) UPF insertion/update/removal

UPF Type Definitions

The following list describes the UPF type definitions.

- PSA-IPA = PDU Session Anchor - IP Anchor
 - UPF that supports N6 interface
 - IP anchor point of the IP address/prefix allocated to the UE
- PSA-LA = PDU Session Anchor - Local Anchor
 - UPF that allows selective routing of traffic to Local DN (Data Network)
- ULCL = Uplink Classifier
 - UPF that diverts some traffic toward PSA-LA UPF and all other traffic to PSA-IPA UPF
- I-UPF = Intermediate UPF (aka N3-UPF)
 - UPF that supports both N3 and N9 interfaces
 - Inserted to provide connectivity to PSA-IPA when UE moves out of service area of the PSA-IPA

Call Flows

This section contains call flow graphics for Traffic Steering.

Figure 16. Traffic Steering during PDU Session Establishment - IPA UPF and ULCL with LA Role Call Flow

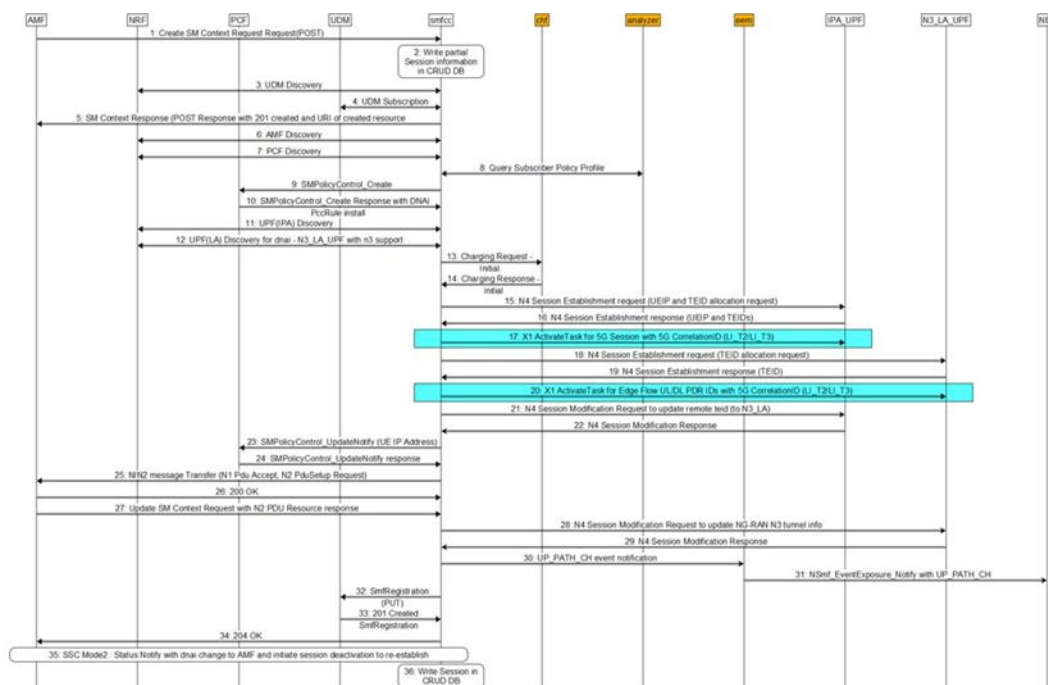


Figure 17. Traffic Steering during PDU Session Establishment - IPA ULCL and Stand Alone LA UPFs

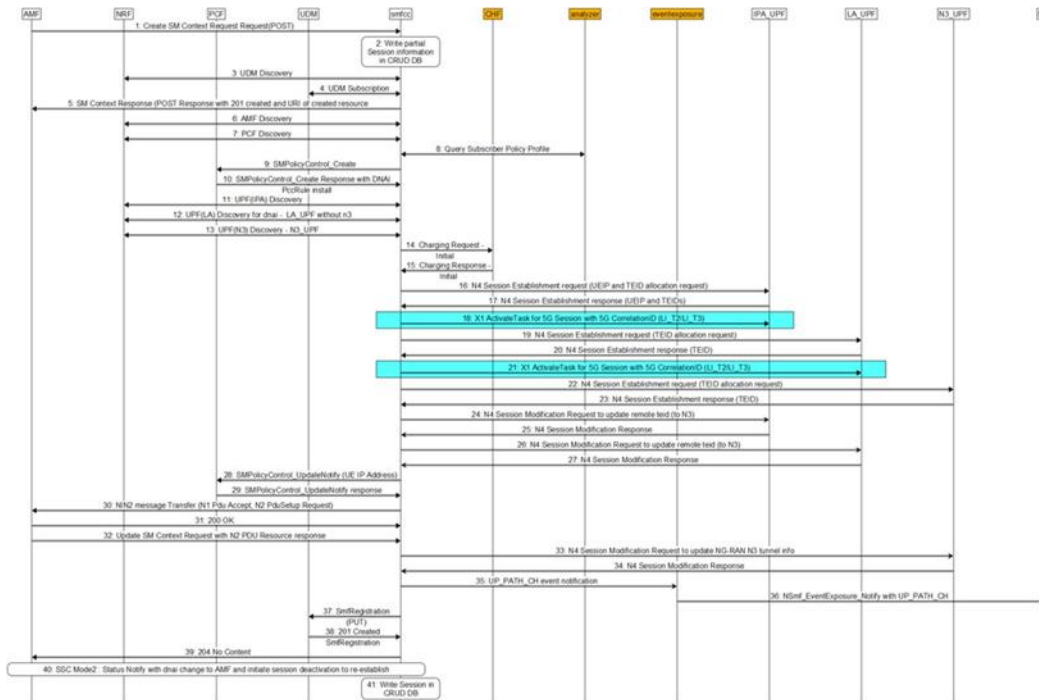


Figure 18. Traffic Steering during Mid Session DNAI Rule Install - ULCL with LA Role Insertion

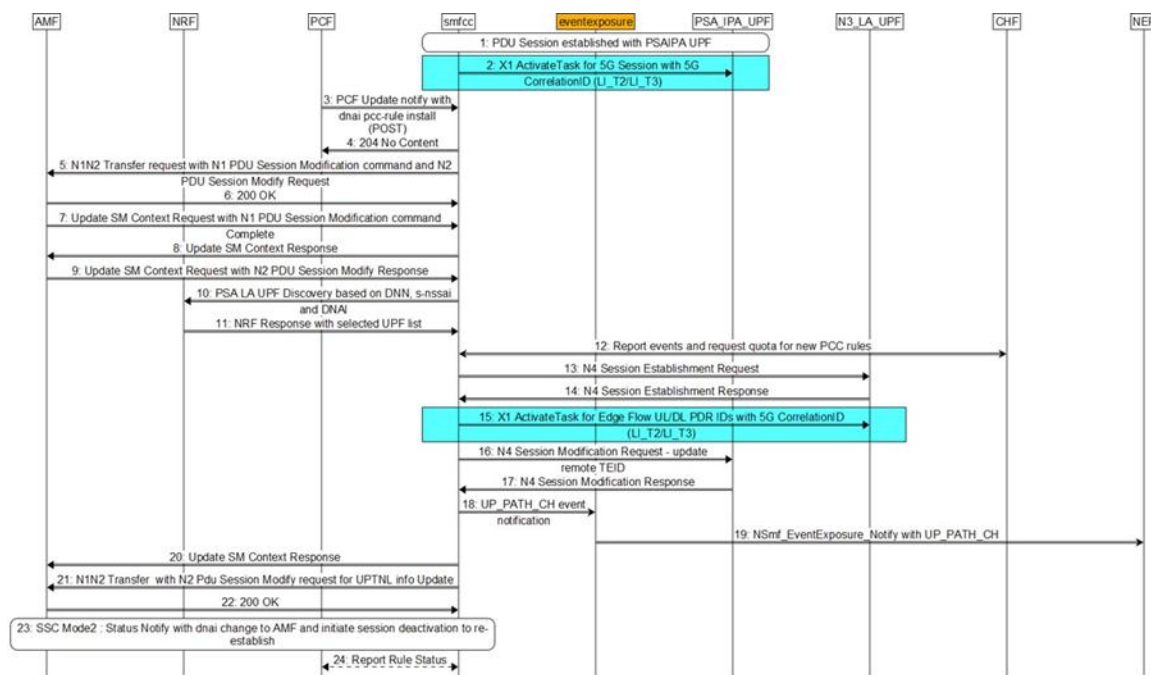


Figure 19. Selective Traffic Routing during Mid Session DNAI Rule Install - ULCL and LA UPF Insertion

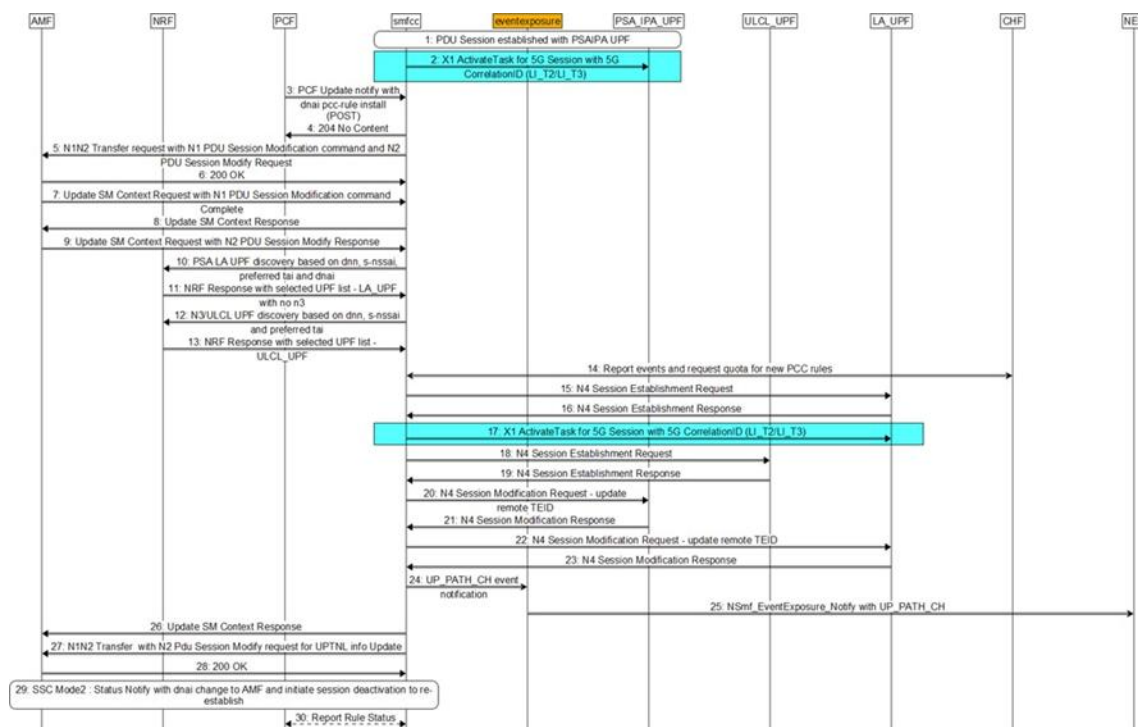


Figure 20. Selective Traffic Routing during Mid Session DNAI Rule Install after IUPF Insertion

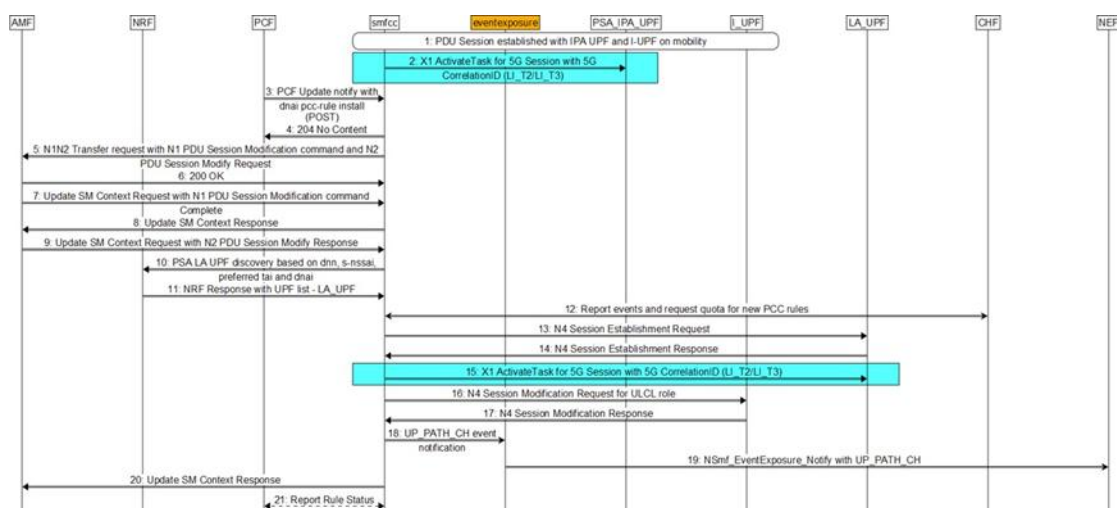


Figure 21. Selective Traffic Routing during Mid Session DNAI Rule Install with Combined I-UPF/ULCL/LA-UPF

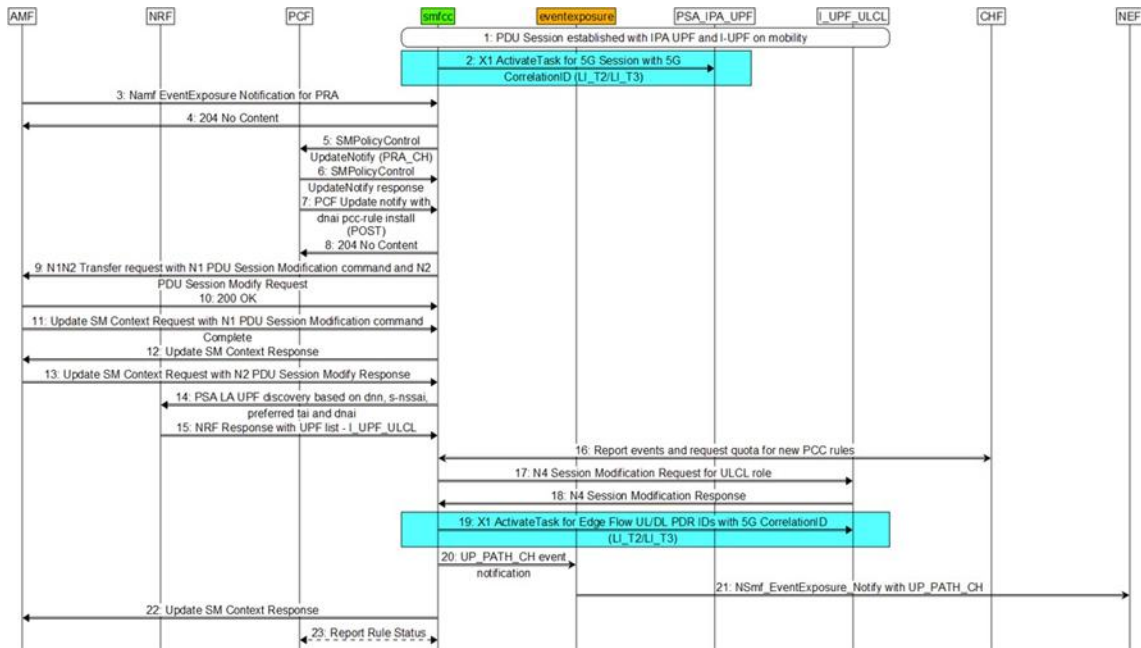
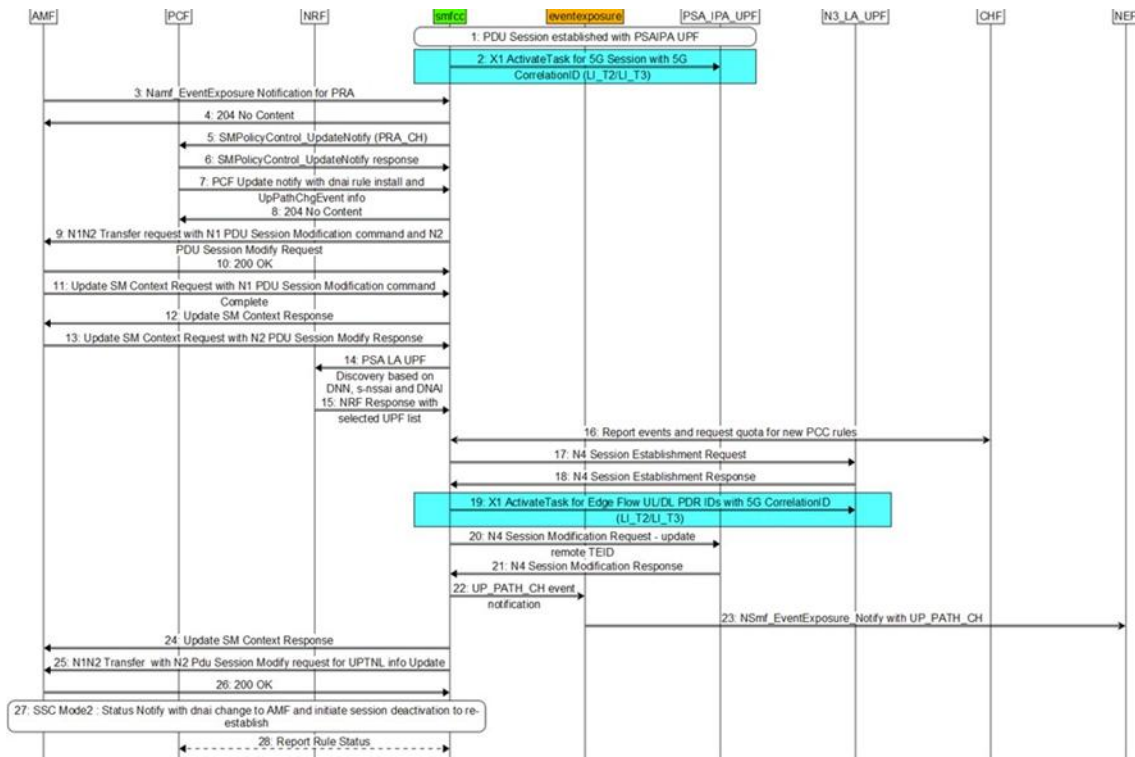


Figure 22. Selective Traffic Routing based on PRA Reporting - DNAI Rule Install

Error Handling

The SMF deactivates the session if any of the session setup/modification UPF programming fails (N4 Create/modify/delete) during traffic steering activation or deactivation. The SMF does not deactivate the session if any LI related UPF programming fails during traffic steering activation or deactivation.

SME Set Support

This feature adds Lawful Intercept support to the PGW-C/SMF Set functionality, where multiple SMF instances can form a SET to provide distribution, redundancy and scalability together.

When a new SMF in the Set takes over ownership of a session, if that session is targeted for Lawful Intercept, the interception will continue:

- The new SMF reports LI-X2 events for the session
- The UPF continues to send X2 PDSR and/or X3 CC-data to the MDF as applicable based on the LI-X1 Task
- When the X1 trigger is received in any of SMFs in the set, it only processes the records that belong to itself, or it could takeover if the current owner is currently not alive

Note: The ADMF/LIPF is unaware of the SMF Set concept. The ADM/LIPF provisions the exact same LI-X1 Tasks on all the SMFs in the set.

Statistics

Existing stat `li_x2_callcontrol_iri_request_attempts` is used to track the event sent from `smfcc` to LI-X2 in SMF. New stat `li_x1_task_local_cleanup` is introduced to track local cleanup of tasks when LI-X1 in UPF decides to cleanup the task when it receives T2T3Activate for a task with same correlationID.

SMF Outbound Roaming Support for 5G SA and LTE

This feature provides support for outbound roamers at the SMF (the SMF acts as the H-SMF).

5G session functionality

- SMF PDUSessionEstablishment record - The SMF in the HPLMN (i.e. H-SMF) sends the N16: `Nsmf_PDU_Session_Create` response message with `n1SmInfoToUe` IE containing the PDU SESSION ESTABLISHMENT ACCEPT
- SMF PDUSessionModification record - For a home-routed roaming scenario, the SMF in the HPLMN (i.e. H-SMF) receives the N16: `Nsmf_PDU_Session_Update` response message with `n1SmInfoFromUe` IE containing the PDU SESSION MODIFICATION COMMAND COMPLETE (see TS 29.502 [16]). This applies to the following three cases:
 - UE initiated PDU session modification
 - Network (VPLMN) initiated PDU session modification
 - Network (HPLMN) initiated PDU session modification
- SMF PDUSessionRelease record - For a home-routed roaming scenario, the SMF in the HPLMN (i.e. H-SMF) receives the N16: `Nsmf_PDU_Session_Update` response message with `n1SmInfoFromUe` IE containing the PDU SESSION RELEASE COMMAND COMPLETE (see TS 29.502 [16]) from the V-SMF. This applies to the following three cases:
 - UE initiated PDU session release
 - Network (VPLMN) initiated PDU session release
 - Network (HPLMN) initiated PDU session release

- SMFStartOfInterceptionWithEstablishedPDUSession record - In a home-routed roaming scenario, the IRI-POI in the H-SMF shall generate the xIRI containing the SMFStartOfInterceptionWithEstablishedPDUSession record when it detects that a new interception for a UE is activated (i.e. provisioned by the LIPF) for the following case:
 - The H-SMF had not sent a Nsmf_PDU_Session_Update Request (n1SmInfoToUe: PDU SESSION RELEASE COMMAND) to the V-SMF for a PDU session and H-SMF had previously sent a Nsmf_PDU_Session_Create response (n1SmInfoToUE: PDU SESSION ESTABLISHMENT ACCEPT) to the V-SMF for that PDU session.
- SMFUnsuccessfulProcedure record
 - SMF sends a PDU SESSION ESTABLISHMENT REJECT message to the target UE.
 - SMF sends a PDU SESSION MODIFICATION REJECT message to the target UE.
 - SMF sends a PDU SESSION RELEASE REJECT message to the target UE.
 - SMF receives a PDU SESSION MODIFICATION COMMAND REJECT message from the target UE.
- CC-TF triggers such as T2T3Activate, T2T3Deactivate to be supported.

4G session functionality

- BEGIN record - Bearer Activation, Start Of Interception with active bearer
- CONTINUE record - Bearer Modification
- END record - Bearer Deactivation
- REPORT record - Unsuccessful Bearer Activation
- CC-TF triggers such as T2T3Activate, T2T3Deactivate to be supported.

Handover functionality

- EPS to 5G Idle mode HO using N26 interface - Session Modification X2 IRI to be generated for the SMF in the HPLMN (i.e. H-SMF) sends the N16: Nsmf_PDU_Session_Create response message with n1SmInfoToUe IE containing the PDU SESSION ESTABLISHMENT ACCEPT (see TS 29.502 [16]) while it had received a N16 Nsmf_PDU_Session_Create request message with an existing PDU Session Id
- 5G to EPS HO using N26 interface - Bearer Modification X2 IRI to be generated

Figure 23. Session Establishment



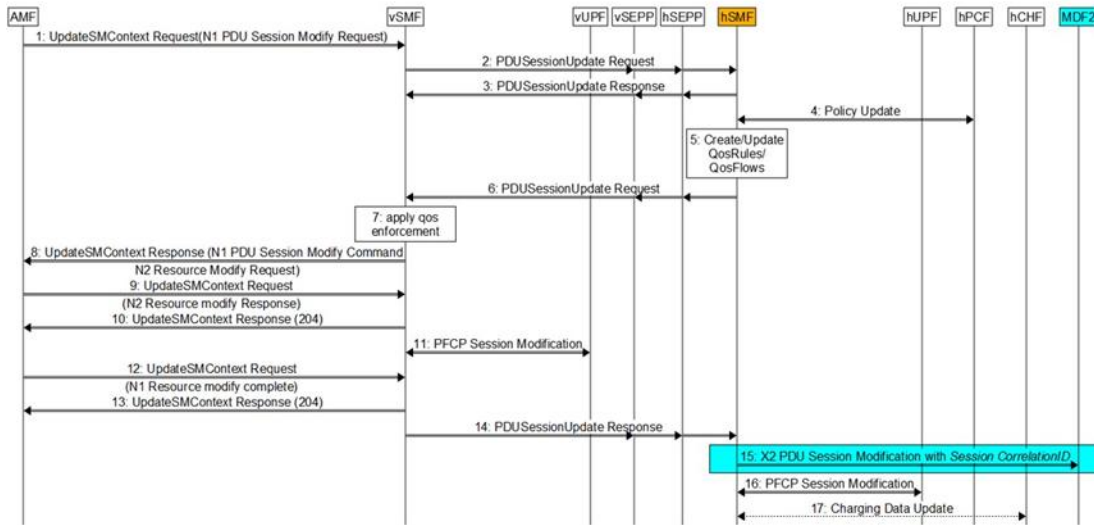
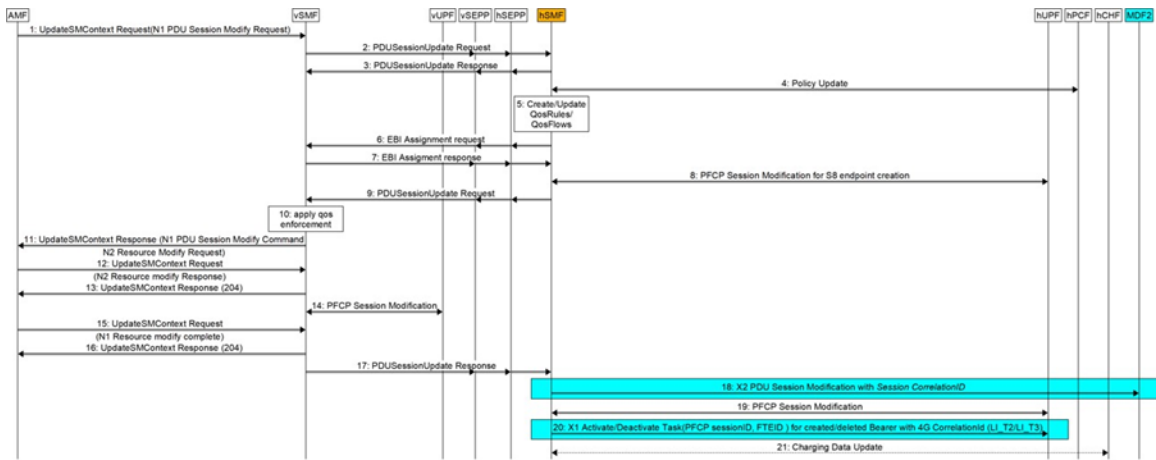
Figure 24. UE INIT Session Modification**UE INIT Session Modification****Figure 25. UE initiated PDU session modification with EBI Assignment success**

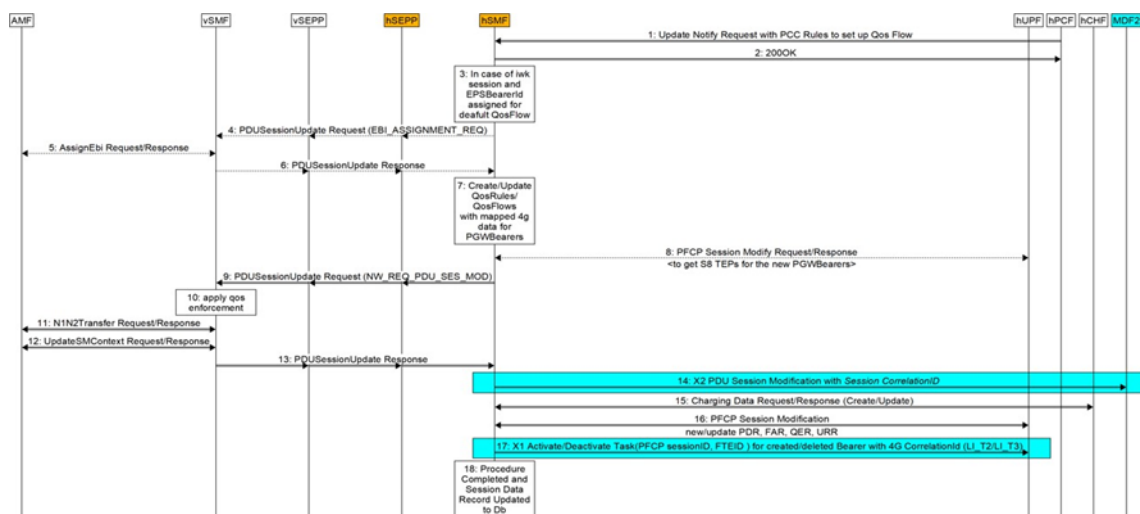
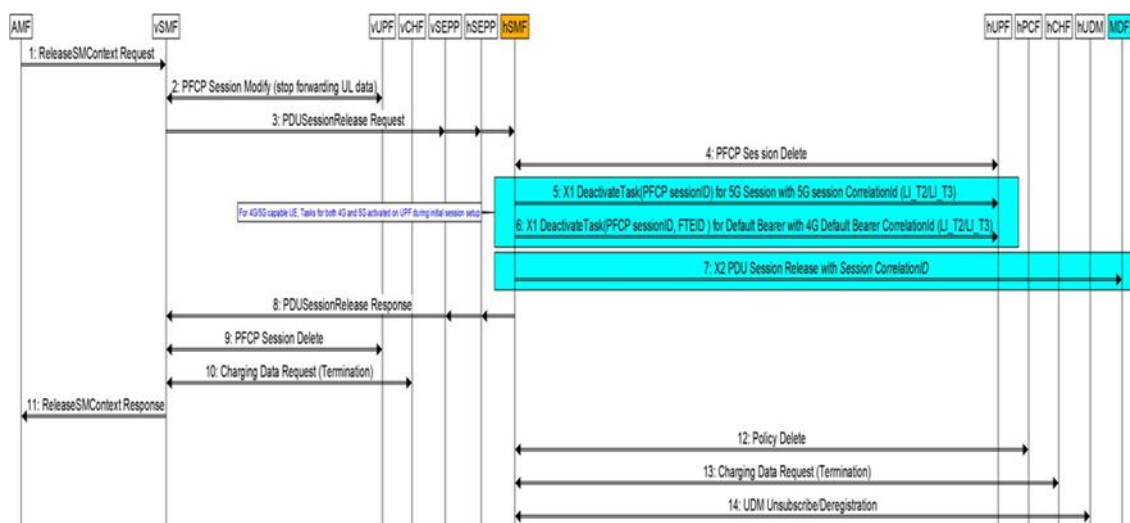
Figure 26. Nw-Init Session Modification**Figure 27. AMF initiated PDU Session Release**

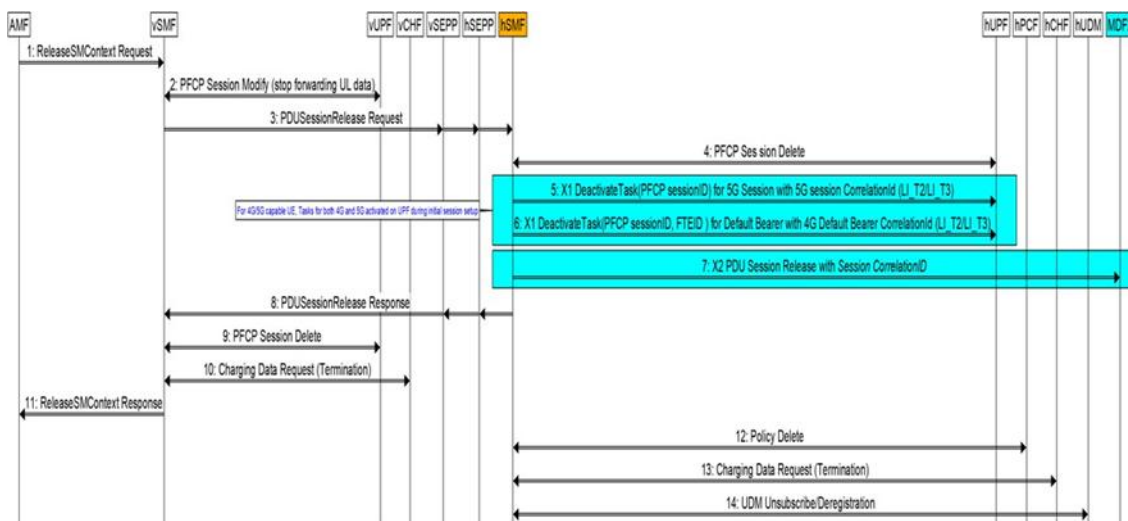
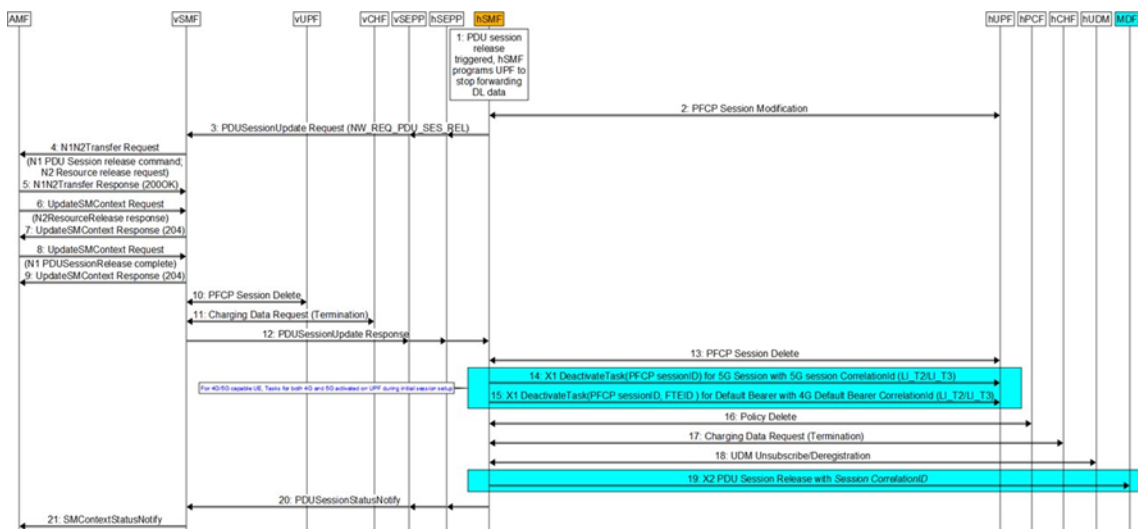
Figure 28. vSMF initiated PDU Session Release**Figure 29. hSMF initiated PDU session Release – outbound roamer**

Figure 30. UE initiated PDU Session Release – outbound roamer

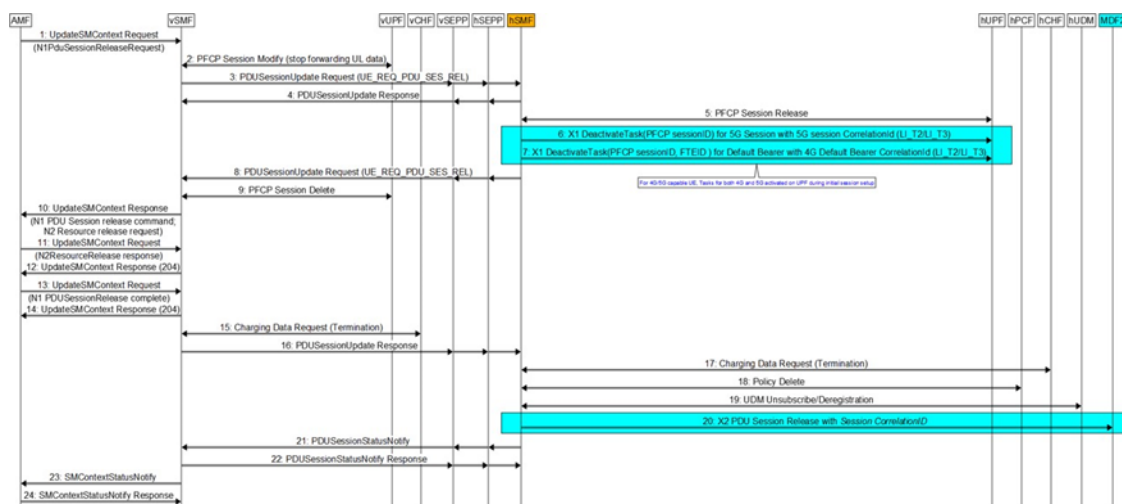


Figure 31. Collision at vSMF (hSMF handling of 403/409)

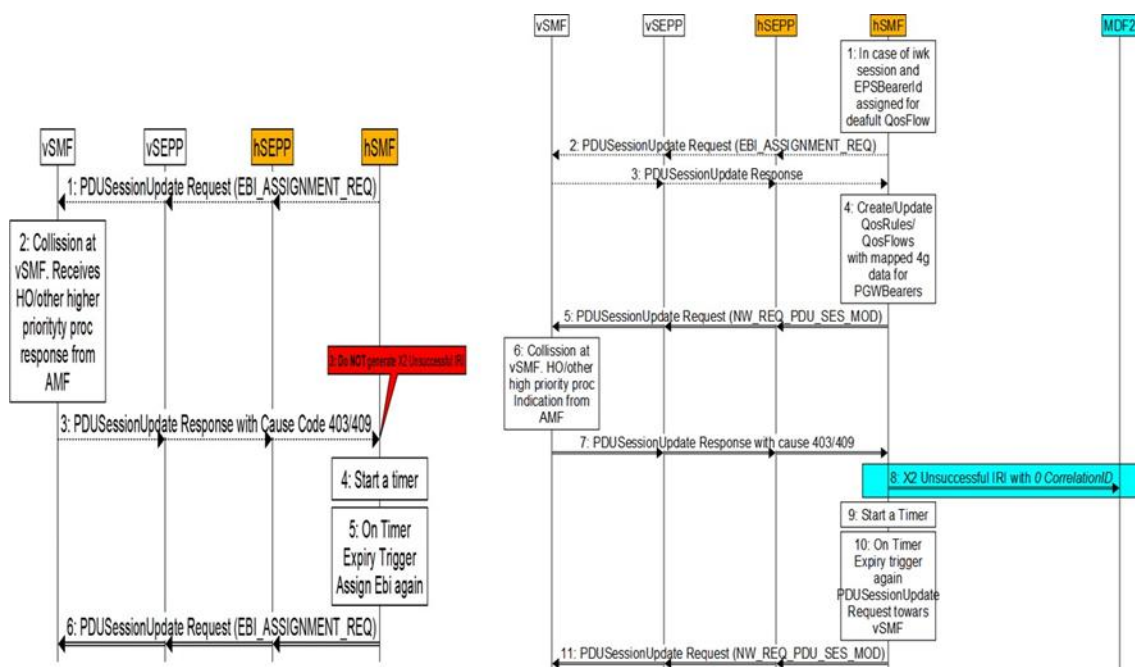


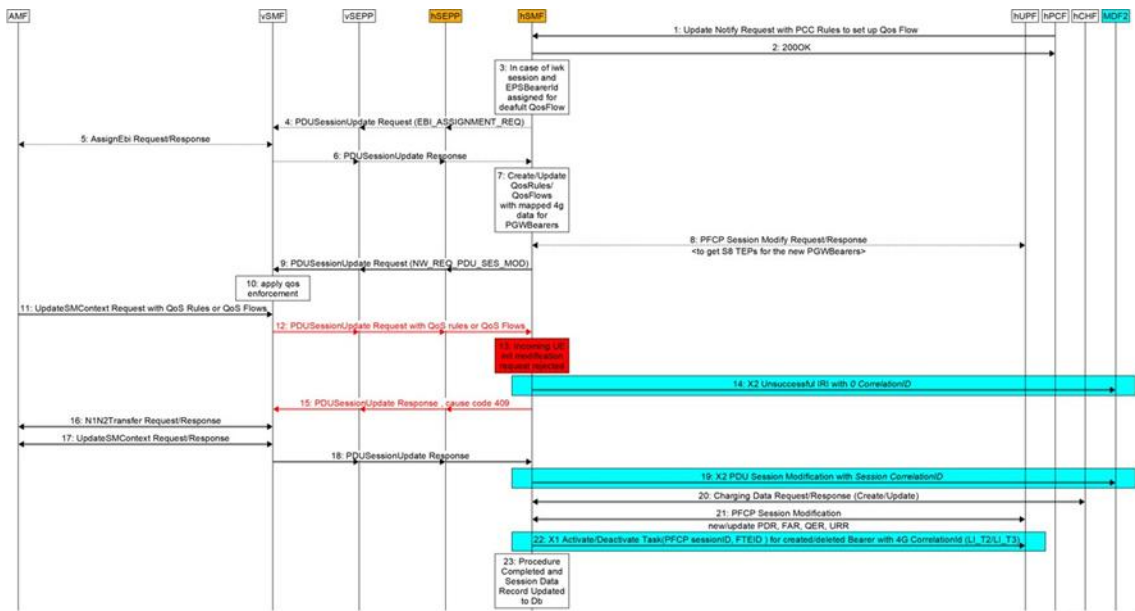
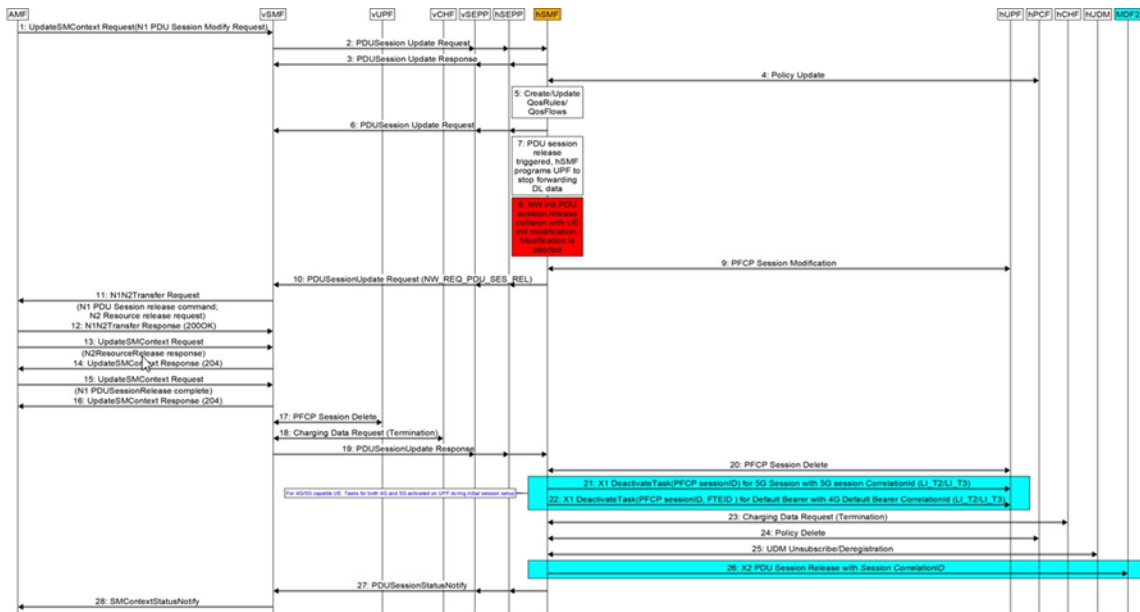
Figure 32. Collision at hSMF – Network Init Modify with UE Init Modify**Figure 33. Collisions at hSMF – UE Init Session Modify with NW Init release**

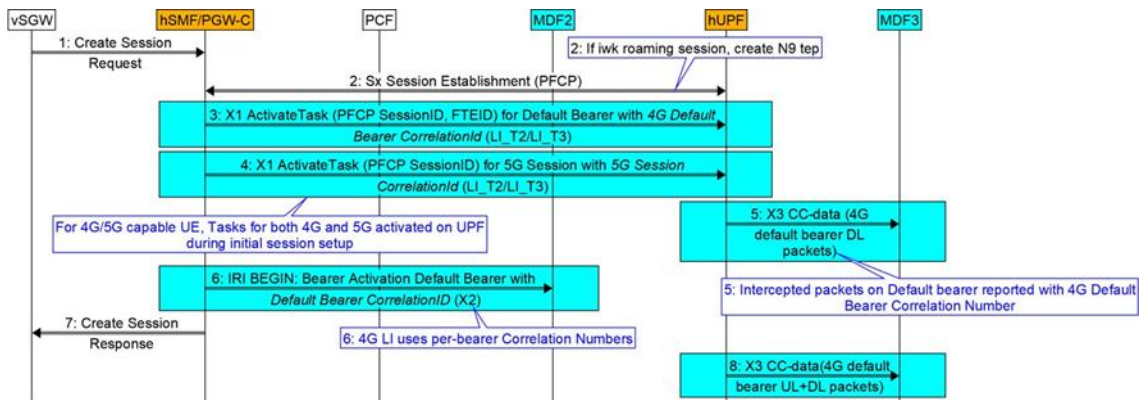
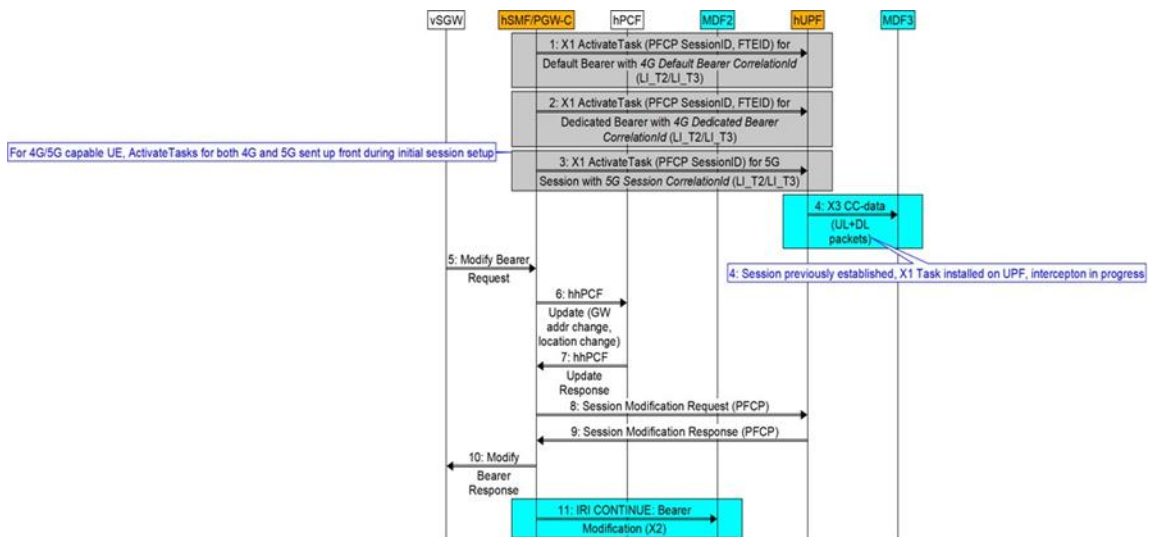
Figure 34. 4G initial attach**Figure 35. 4G Session modification**

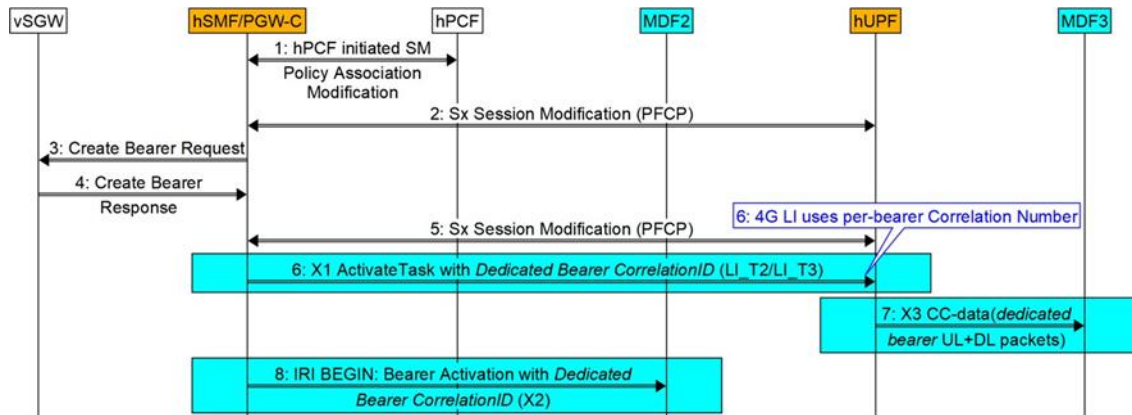
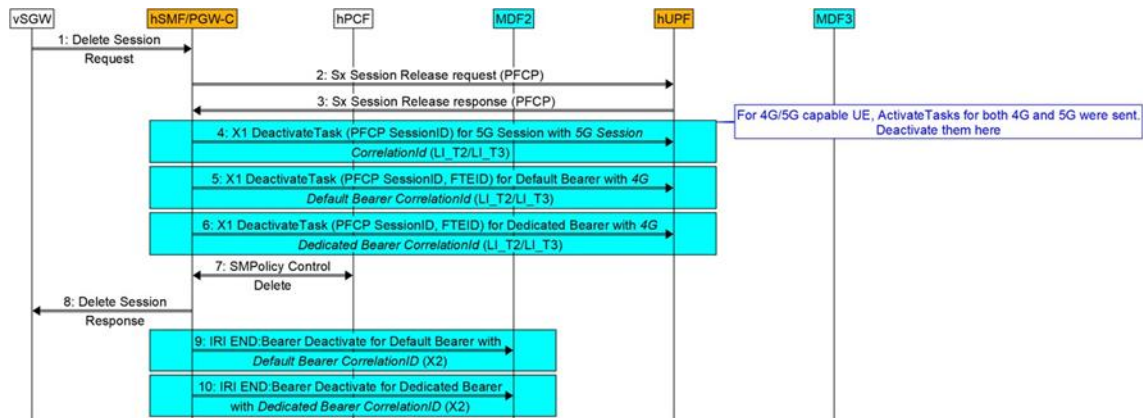
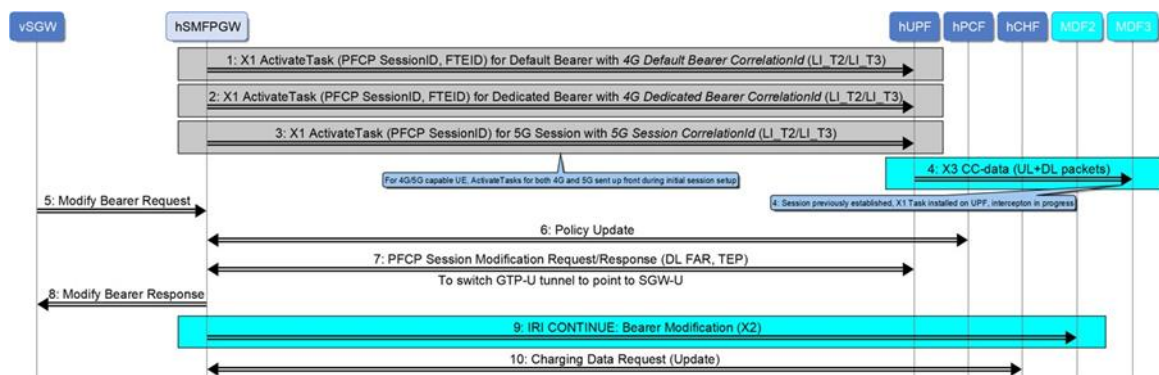
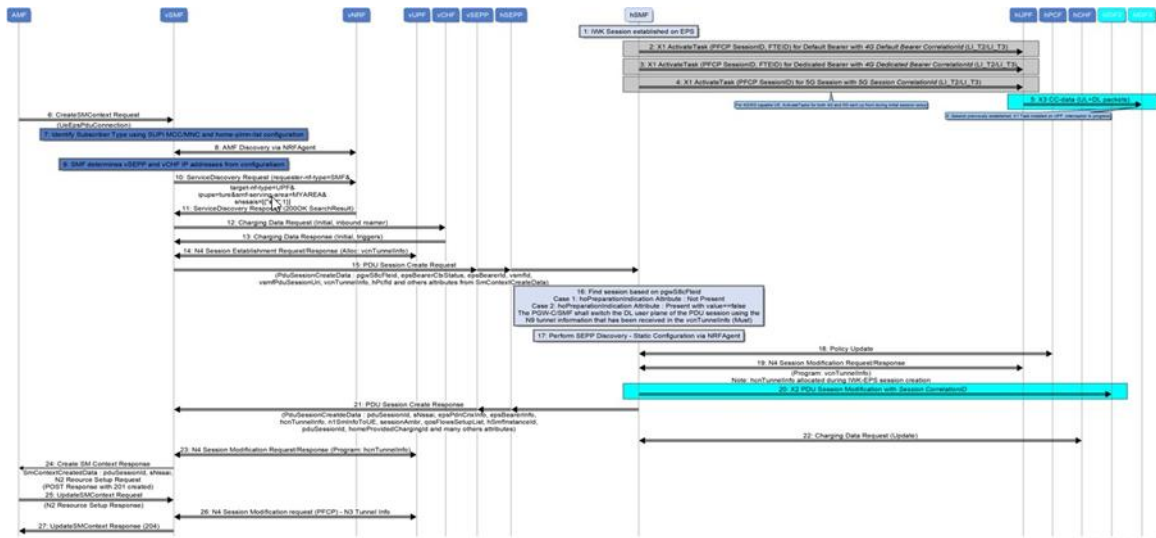
Figure 36. hSMF init session modification**Figure 37. 4G Session release****Figure 38. 5G to EPS handover using N26 interface**

Figure 39. EPS to 5GS Idle Mode Mobility using N26 interface



Spec reference

- 3GPP TS 33.106 – Lawful intercept requirements
- 3GPP TS 33.107 – LI architecture and functions
- 3GPP TS 33.108 – Protocol and procedures for LI
- 3GPP TS 33.126 V16 – Lawful intercept requirements
- 3GPP TS 33.127 V16 – LI architecture and functions
- 3GPP TS 33.128 V16 – Protocol and procedures for LI
- ETSI TS 103 221-1 – LI Internal Network Interfaces Part1: X1
- ETSI TS 103 221-2 – LI Internal Network Interfaces Part2: X2/X3

The following section provides UPF LI sessions reporting.

- X2 PDSR for 5G sessions is reported using ETSI TS 103.221-2 X2 PDU header encapsulating 3GPP TS 33.128 ASN.1 encoded X2 PDSR events (ETSI TS 103.221-2 payload format 2)
- X2 PDSR for 4G bearers is reported using ETSI TS 103.221-2 X2 PDU header encapsulating 3GPP TS 33.108 ASN.1 encoded X2 PDSR events (ETSI TS 103.221-2 payload format 3)
- X3 CC-data for 5G and 4G sessions is reported using ETSI TS 103.221-2 X3 PDU header encapsulating the intercepted GTP-U message (ETSI TS 103.221-2 payload format 12)

A single session could report both 5G and 4G events, if the session undergoes handover between 5G and 4G Radio Access Networks:

- 5G LI events will be reported with the 5G session correlation number when the session is on 5G RAN
- 4G LI events for each bearer will be reported with the corresponding 4G bearer correlation number when the session is on 4G RAN

This section lists of topics related to UPF Lawful Intercept:

- [LI Support for LiApp Push of X1 Task Statistics to LI-X1](#)
- [Operational CLIs](#)
- [UPF](#)
- [UPF-LI – Support KeepAlives for X2 and X3](#)

5G LI Services

The 5G LI services provide standard-compliant lawful intercept functionality as specified in 3GPP TS 33.128.

- SMF X1 signaling supports LI_T2 and LI_T3 interfaces along with the following functions:
 - Create and Remove XID
 - Create and Remove DID
 - Get XID details
 - Get DID details
- Support for X3 CC (full packet) interception only:

UPF

- X3 PDU format described in 3GPP TS 33.128 section 6.2.3.6
- Uses GTP-U payload format defined in ETSI TS 103 221-2 clause 5.4.13
- Session ID is the only supported target type.

Note: Two new attributes are added to the li-profile, named **x2-tcp-connections-per-destination** and **x3-tcp-connections-per-destination**. They set the number of TCP connections that are established by every PPE pod instance to a given MDF for X2 and X3 traffic, respectively.

Note: LI_X3 interface to MDF is currently plain text TCP.
LI X2 supports the conditional attributes Network Function ID (NPID) and Interception Point ID (IPID) in X2 PDU towards MDF. Network Function ID is the IP address of the UPF, and Interception Point ID (IPID) is the unique part of the name of the PPE pod.

Configuration

LI Profile is used to specify the name of the port-range profile and a global IPv4 loopback and/or IPv6 loopback to be used for LI on the upf:

show running-config upf service-construct

```
upf service-construct li-profile ipv4-loopback-name FOO
upf service-construct li-profile port-profile PORTPROF-1
```

show running-config interface-manager port-profile

```
interface-manager
port-profile PORTPROF-1 start-port-range 10000
end-port-range 30000
ports-per-pod 50
!
```

show running-config interface-manager network-context

```
interface-manager
```

```

network-context N6

loopback-ipv4-interface FOO

loopback-start-address 1.2.3.4

loopback-end-address 1.2.3.4

admin-state    enabled

loopback-scope global

!

show ppe vl li resource-info ppeId <pod-id>

{
  "connect-bind-failure":0,
  "connect-disconnected":0,
  "connect-event-notifier-failure":0,
  "connect-no-local-port":0,
  "connect-reject-or-timeout":0,
  "connect-socket-allocation-failure":0,
  "connect-socket-not-opened":0,
  "connect-socket-option-failure":0,
  "connections-succeeded":2,
  "destinations-configured":1,
  "destinations-connected":1,
  "destinations-connecting":0,
  "destinations-disconnected":0,
  "loopbackIPv4Addr":"0.0.0.0",
  "loopbackIPv6Addr":"0:0:0:0::0",
  "partial-sends":0,
  "partial-sends-completed":0,
  "partial-sends-repeated":0
}

```

```

"port-range-end":0,

"port-range-start":0

"truncated-pdus-discarded":0

"unknown-pdus-discarded":0,

"x3-pdus-destination-not-connected":0

"x3-pdus-destination-not-found":0,

"x3-pdus-received":0

"x3-pdus-send-error":0,

"x3-pdus-sent":0,

"x3-pdus-socket-buffer-full":0

}

```

Features

- [LI Support for LiApp Push of X1 Task Statistics to LI-X1](#)
- [Support DSCP Marking on SBA and LI interfaces for UPF](#)
- [Operational CLIs](#)
- [UPF-LI – Support KeepAlives for X2 and X3](#)

Spec Reference

LI_T2 and LI_T3 interfaces are specified in 3GPP TS 33.128.

Related Topics

For more information on Lawful Intercept microservice, refer the following sections:

- [LI-X1](#)
- [LI-X2](#)

LI Support for LiApp Push of X1 Task Statistics to LI-X1

UPF populates X1 Task statistics for currently active Tasks for intercepted sessions on the UPF as described in the ETSI 103.221-1 specification.

These statistics are available to the LIPF through X1 requests such as **GetTaskDetailsRequest**. The **TaskStatus** structure defined in ETSI 103.221-1 is populated with correct statistics for **AmountOfX3Data**, **TimeOfLastIntercept**, and other fields as applicable to the Task.

Note: Support for error handling of LI Statistics Response from X1 on UPF is not available for this release.

UPF populates the local-NEID configured in the li-x1 local-profile as **NFID** in X2/X3 PDU Headers.

```
liadmin@xyz-10-163-66-96-08-25-16-14 06:03:18# show running- config li-x1
local-profile
```

```
li-x1 local-profile local-ne-identifier neID li-x1 local-profile operator-
identifier att li-x1 local-profile time-p1 60
```

```
li-x1 local-profile time-p2 0
```

```
li-x1 local-profile time-p1-x2x3 60 li-x1 local-profile time-p2-x2x3 180
```

Note: Interception Point ID is not supported for this release.

Operational CLIs

LI microservice supports for Operational CLIs - LI-X1 and LI-X2.

For LI-X1 Operational CLIs, see [Operational CLI](#).

For LI-X2 Operational CLIs, [Operational CLI](#).

This feature supports the following:

- [Keep Alive from LIPF - CPF or UPF](#)

- [UPF-LI – Support KeepAlives for X2 and X3](#)
- [KeepAlives](#)
- [KeepAlives](#)
- [KeepAlive-P2](#)

UPF Operational CLIs

LI-PPE side peer-status and keep-alives CLIs are as follows:

```
show li-ppe keep-alive x2 {command} (command could be verbose, summary, did)
```

```
show li-ppe keep-alive x3 {command} (command could be verbose, summary, did)
```

```
show li-ppe peer-status x2 {command} (command could be verbose, summary, did)
```

```
show li-ppe peer-status x3 {command} (command could be verbose, summary, did)
```

Ingress Keep-Alive Status

Following command is used to query the incoming keep-alive request status on LI-X1/LI-T2/LI- T3 interfaces.

Table 30. show-ingress-keep-alive Parameters

Parameters	Description
LocalNEID	Configured Local NE identifier.
PeerADMFID	ADMF identifier of the peer.
TIME_P2	Configured time at the NF. If no message is received from ADMF within the configured time, it leads to DeactivateAllTasks on NF.
KeepAliveReqRcvdTime	Timestamp when last keep alive/any other message received from the ADMF.

```
# show li-x1 destination-details did 21ce308c-ee32-42bd- a27b-287bcb5b3469

{

"destinationDetails": { "DIDPeerEntryValue": { "listOfDIDPeers": {
```

```

"value": [ { "admfIdentifier": "li-admin" } ]

}, "version": "1.0.0"

},

"admfIdPair": [ {

"admfId": "li-admin", "obfuscatedAdmfId":

"ff597fae551330757930b93ed782aa482d287c70788b2f685afaec5481e d34",

"taskCountValue": 1 } ],

"destinationEntryValue": { "destinationDetails": {

"n1_dId": "21ce308c-ee32-42bd-a27b-287bcb5b3469", "n1_deliveryAddress": {

"deliveryAddressType": "ipAddressAndPort",

"n1_ipAddressAndPort": {"etsi103280_address": { "etsi103280_IPv4Address":

"10.163.232.133" },

"etsi103280_port": { "etsi103280_TCPPort": 7117 } } }, "n1_deliveryType":

"X2Only" }, "destinationProvisioningSource": "provSourceLocalConfig",

"switchBackTimer": 60, "time_p1": 60, "time_p2": 180,

"version": "1.0.0" },

"destinationStatus": "

{ \"claimant\": \"unassigned\", \"version\": \"1.0.0\", \"secretKeyVersion\": 1, \"obfuscateddid\": \"9c1fc4717e0e9a84991d8b392db6338c93cf2c842a961f52a6fa13ebce4c37\", \"data\":

{ \"destinationstatus\": \"eActiveAndWorking\" }, \"obfuscatedadmfid\":

[ \"ff597fae551330757930b93ed782aa482d287c70788b2f685afaec5481ed34\" ] }\",

"obfuscatedDid":

"9c1fc4717e0e9a84991d8b392db6338c93cf2c842a961f52a6fa13ebce4 c37",

"taskCountValue": 1 } }

# show li-x1 keep-alive trigger-function

Possible completions:

t2 t3

```

show li-x1 keep-alive trigger-function t2

Possible completions:

all show li-x1 keep-alive trigger-function t2 status all

neid optional peer neid

| Output modifiers

<cr>

show li-x1 keep-alive ingress-x1 status

Possible completions:

admfid optional admfid

all show li-x1 keep-alive ingress-x1 status all

show upf v1 provisioned

Possible completions:

allowed-nssai show commands that may be run to show confd state

allowed-visited-plmn-list show commands that may be run to show confd state

dnn-list show commands that may be run to show confd state

home-plmn-list show commands that may be run to show confd state

interface show commands that may be run to show confd state

ip-pool show commands that may be run to show confd state

plmn show commands that may be run to show confd state

smf-serving-area show commands that may be run to show confd state

Note: LI status reporting in PPE pod is not supported.

Support DSCP Marking on SBA and LI interfaces for UPF

UPF supports configuration of DSCP marking to all IP traffic on LI-X1 and LI-X2 interfaces.

For more information on LI-X1 DSCP marking, see [DSCP Marking](#).

For more information on LI-X2 DSCP marking, see [DSCP Marking](#).

Configuration

Execute the following command to display DSCP Marking:

```
liadmin@xxx-10-163-65-1-01-14-04-19 06:46:12(config)# li-x1 dscp-marking ?
```

Possible completions:

```
lipf      LIPF Interface DSCP marking.
```

```
lipf-mgmt      LIPF Management Interface DSCP marking. t2      LI_T2
Interface DSCP marking.
```

```
t3 LI_T3 Interface DSCP marking. □ This is missing in the document.
```

```
x2 LI_X2 Interface DSCP marking.
```

```
x3 LI_X3 Interface DSCP marking.
```

```
liadmin@xxx-10-163-65-1-01-14-04-19 06:46:12(config)# li-x1 dscp-marking
```

UPF-LI – Support KeepAlives for X2 and X3

Support for KeepAlive mechanism is as specified in ETSI TS 103 221-2 Section 6.2.4 on X2 and X3 interfaces between UPF-LI and MDF.

- Send KeepAlive PDU to every primary and secondary endpoint for each destination MDF, once every TIME_P1 seconds.
- Select a different TCP connection to the endpoint for every KeepAlive interval.
- Match the KeepAlive Acknowledgement PDU on any TCP connection to the endpoint.
- If a KeepAlive Acknowledgement PDU is not received within TIME_P2 seconds after sending the KeepAlive PDU, disconnect all TCP connections to that endpoint and attempt to reconnect.
 - If the endpoint is a primary endpoint for the destination, switch to the secondary endpoint (if available) and it is then connected.

- If the endpoint is a secondary endpoint for the destination, switch to the primary endpoint if it is connected.
- TIME_P1 and TIME_P2 are configured as part of the “li-x1 local-profile” liadmin CLI configuration.
- TIME_P1 is configured as “li-x1 local-profile time-p1-x2x3” (default 60 seconds).
- TIME_P2 is configured as “li-x1 local-profile time-p2-x2x3” (default 180 seconds)

Note: If TIME_P1 is set to zero, KeepAlive PDUs are not sent.

If TIME_P2 is set to zero, TCP connections to the MDF is not disconnected, if KeepAlive Acknowledgement PDUs are not received.

Statistics

Statistics displayed as part of “show ppe v1 li resource-info” CLI command output:

- keepalives-sent – number of KeepAlive PDUs sent
- keepalive-acks-received – number of KeepAlive Acknowledgement PDUs received which match the KeepAlive PDUs that were sent
- keepalive-timeouts – number of TIME_P2 timeouts resulting in endpoint disconnections
- mdf-pdus-received – number of total PDUs received from MDFs, including KeepAlive Acknowledgement PDUs
- mdf-bytes-receives – total number of bytes received from MDFs

Enable or Disable CipherSuite Option

UPF LI can turn on/off TLS and selects TLS CipherSuite through liadmin configuration manager.

UPF TLS Configuration

- liadmin / liadmin (i.e., LI-X1 in UPF)
- ```
show running-config li-x1 | include x3-over

li-x1 x2-x3-over-tls-egress enabled

li-x1 x2-x3-over-tls-egress-cipher DEFAULT
```

- li-x1 x2-x3-over-tls-egress
  - TLS over TCP enabled/disabled on egress interface towards MDF Peers (LI\_X2 or LI\_X3)
- li-x1 x2-x3-over-tls-egress-cipher
  - TLS Cipher for the egress interface towards MDF Peers (LI\_X2 or LI\_X3)
  - Valid values are: DEFAULT, AES128-SHA, NULL-MD5. You can provide multiple ciphers by concatenating with: (Example: AES128- SHA:ECDH-ECDSA-RC4-SHA)

**Note:** Most important values are “**DEFAULT**” and “**NULL-MD5**”. **DEFAULT** will be used for most of the time and NULL-MD5 can be used in lab environment for debugging purpose.

When NULL-MD5 is used, the TLS connection is using TLS protocol (i.e., TLS handshake and other options) and can see the plaintext in captured packets. Ensure the server side (i.e., F5 loadbalancer) is configured to support NULL-MD5 as well.

The other ciphers are unlikely to be used.

## LI for Multiple UPFs

This feature adds Lawful Intercept support for all Multiple UPF call flows supported by the Traffic Steering feature.

Traffic Steering on 5G sessions includes:

- Uplink Classifier (ULCL) insertion/update/removal
- Local Anchor (LA) UPF insertion/update/removal

**Note:** Most important values are “**DEFAULT**” and “**NULL-MD5**”. **DEFAULT** will be used for most of the time and NULL-MD5 can be used in lab environment for debugging purpose.

When NULL-MD5 is used, the TLS connection is using TLS protocol (i.e., TLS handshake and other options) and can see the plaintext in captured packets. Ensure the server side (i.e., F5 loadbalancer) is configured to support NULL-MD5 as well.

The other ciphers are unlikely to be used.

LI call flows for Traffic Steering involves:

- Traffic Steering during PDU Session Establishment – IP Anchor (IPA) UPF and ULCL with LA role
- Traffic Steering during PDU Session Establishment – IPA ULCL and stand alone LA UPFs
- Traffic Steering – mid session dnai rule install – ULCL with LA role insertion
- Selective Traffic Routing – mid session dnai rule install – (ULCL and LA UPF insertion)
- Selective Traffic Routing – mid session dnai rule install after IUPF insertion
- Selective Traffic Routing – mid session rule install with combined I-UPF/ULCL/LA-UPF
- Selective Traffic Routing based on PRA reporting – DNAI rule install

## Failure Handling

- SMF deactivates the session if any of the session setup/modification UPF programming fails (N4 create/modify/delete) during traffic steering activation or deactivation.
- SMF will not deactivate the session if any LI related UPF programming fails during traffic steering activation or deactivation.

# GGSN

---

The Gateway GPRS Support Node (GGSN) supports LI events in GGSN-C call flows. IRI reporting for packet domain at the GGSN is also supported. The UnityCloud GGSN-C supports the following deployment models:

- Standalone GGSN
- Combined GGSN/PGW
- Combined GGSN/PGW/SMF

**Note:** *LI is supported in all the deployment models.*

The following GGSN 3GPP GTP procedures are supported:

- PDP Context Activation
- PDP Context Modification
- PDP Context Deactivation
- Network Requested PDP Context Modification
- Network Requested PDP Context Deactivation
- Inter RAT Handover between E-UTRAN and UTRAN/GERAN
- Error Indication

The following list of GGSN targets are supported by the LI-X1 microservice for the GGSNC:

- MSISDN
- E. 164 number for Non-Local ID
- IMSI
- IMEI

## GGSN IRI Events

Presented here are the Message Flows for Lawful Intercept 3G sessions handled by the GGSN/PGW-C product. The following Lawful Intercept X2 IRI events are defined for 3G EPS sessions in the 3GPP TS 33.108 specification:

- PDP Context Activation
- Start of Interception with PDP context active
- PDP Context Modification
- PDP Context Deactivation

## Chart of 3G LI X2 IRI events and 3G LI Message Flows (Success Cases)

The chart below indicates which message flow includes usage of the LI X2 events of interest:

**Table 31. Message Flows Including LI X2 events of interest**

| 3G IRI Event                        | PDP Context Activation      | PDP Context Modification                               | PDP Context Deactivation              | Start of Interception with Active PDP Context      |
|-------------------------------------|-----------------------------|--------------------------------------------------------|---------------------------------------|----------------------------------------------------|
| UE Access side- initiated call flow | UTRA Initial Attach with LI | HSS Initiated Qos modification with LI                 | UE Initiated Detach with LI           | GGSN Start of Interception with Active PDP Context |
| NW Initiated call flow              |                             | NW initiated PDP Context Modification, 3G-4G handovers | NW initiated PDP Context Deactivation |                                                    |

## 3G LI Message Flows (Success cases)

- PDP Context Activation (Primary PDP Context Only) (IRI Begin Record)
- PDP Context Modification (IRI Continue Record)
- PDP Context Deactivation (IRI End Record)
- Network Requested PDP Context Modification (IRI Continue Record)
- Network Requested PDP Context Deactivation (IRI End Record)
- Inter RAT Handover between E-UTRAN and UTRAN/GERAN (IRI Continue Record)

### 3G LI Message Flows (Failure cases)

- PDP Context Activation (Failure) with Lawful Intercept (IRI Report Record)
- UTRAN Initial Attach (Failure) with Lawful Intercept (IRI Report Record)

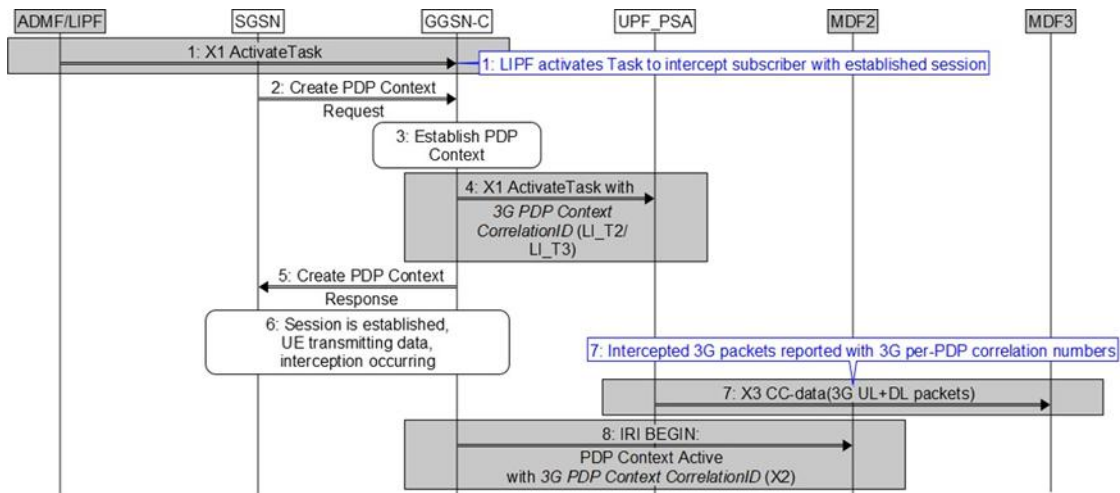
### Standard Specification Reference

- 3GPP TS 33.108 V16.1.0 (2020-07), Section 6 Packet data domain.
- ETSI TS 103.221-2 v1.3.1 (2020-08)

### UTRAN Initial Attach with Lawful Intercept

1. ADMF/LIPF sends X1 ActivateTask message to GGSN/PGW-C. X1 Task contains Target Identifier of subscriber that established the session.
2. RNC sends Activate/Create PDP Context Request message to SGSN and is also forwarded to GGSN/PGW-C.
3. The Sx Session Establishment PFCP exchange with UPF creates session resources on UPF.
4. GGSN/PGW-C sends LI X1 ActivateTask with 3G PDP Context Correlation Number to UPF (LI\_T2/LI\_T3 interface) to instruct UPF to start interception of primary pdp context.
5. For 3G/4G capable UE, GGSN/PGW-C same LI X1 ActivateTask with same correlation number will be used in case of 4G handover with modify bearer request.
6. UPF starts intercepting session data (Downlink traffic from UE). UPF reports intercepted data in X3 CC-data messages to MDF3.
7. GGSN/PGW-C sends X2 IRI BEGIN: PDP Context Activation with 3G Primary PDP Correlation Number (X2 message) to MDF2.
8. GGSN/PGW-C sends Create PDP Context Response to SGSN.
9. SGSN sends Activate PDP Context Response to RNC.
10. Once Uplink traffic is flowing, UPF intercepts all session data (Uplink and Downlink traffic). UPF reports intercepted data in X3 CC-data messages to MDF3.

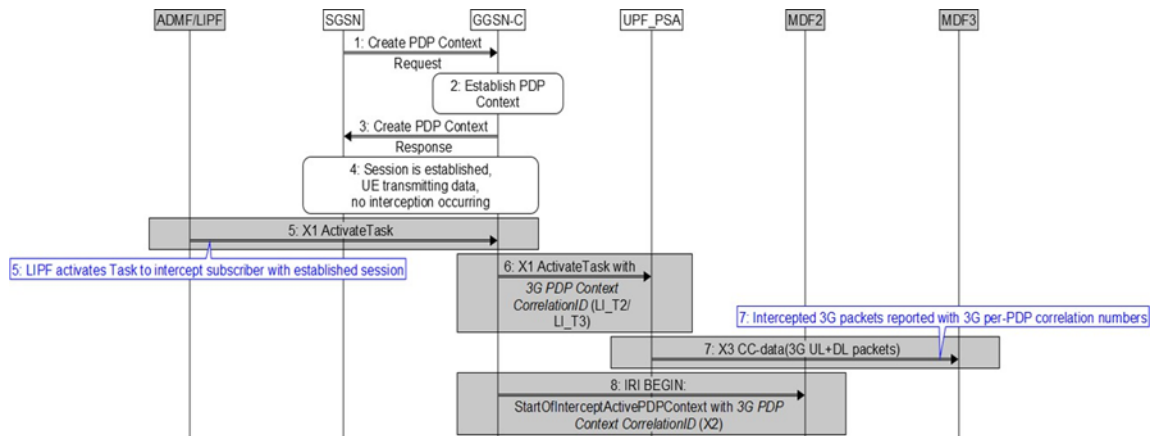
**Figure 40. PDP Context Activation**



## GGSN Start of Interception with Active PDP Context

1. RNC sends Activate/Create PDP Context Request message to SGSN and is also forwarded to GGSN/PGW-C.
2. The Sx Session Establishment PFCP exchange with UPF creates session resources on UPF.
3. At this point, session is established, UE sends/receives data. No interception is occurring.
4. ADMF/LIPF sends X1 ActivateTask message to GGSN/PGW-C. X1 Task contains Target Identifier of subscriber that established the session.
5. GGSN/PGW-C sends X1 ActivateTask (PFCP SessionID, FTEID of PDP Context) with 3G PDP Context Correlation Number to UPF (LI\_T2/LI\_T3 interface) to instruct UPF to intercept all traffic on PDP context.
6. At this point, UPF intercepts all traffic in default PDP context. UPF reports intercepted traffic by sending X3 CC-data messages with the appropriate per-pdp Correlation Number to MDF3 (LI\_X3 interface).
7. GGSN/PGW-C sends IRI BEGIN: StartOfInterceptionWithActivePDPContext with PDP Context Correlation Number to MDF2 (LI\_X2 interface).

**Figure 41. Start of interception with PDP Context Active**

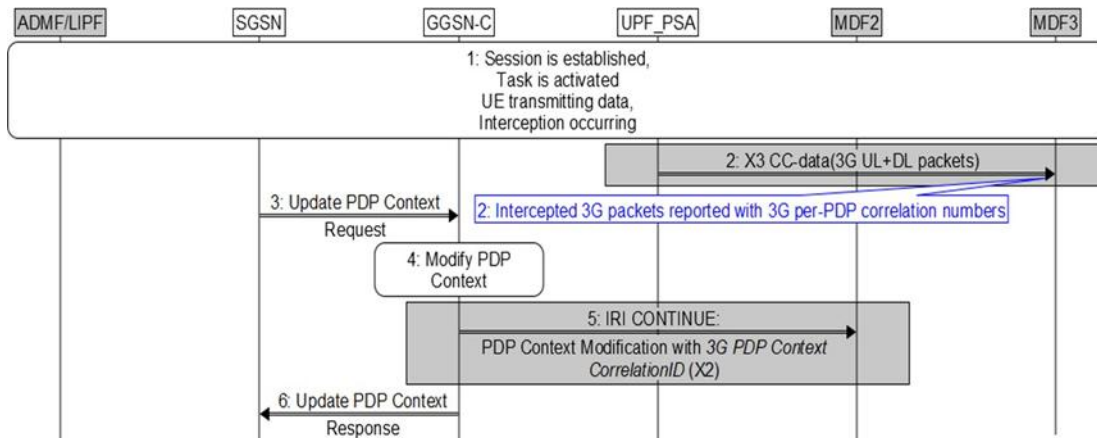


## HSS/NW Initiated Subscribed QoS Modification with Lawful Intercept

The HSS/NW initiated Subscribed QoS Modification procedure for a UE in GTP-based gn-gp- ggsn interface. HSS initiates Insert Subscriber Data to MME.

1. UPF is programmed to intercept traffic on session. UPF intercepts all traffic to/from UE. UPF reports intercepted traffic in X3 CC-data events to MDF3 (LI\_X3 interface).
2. HSS initiates Insert Subscriber Data to SGSN. If UE is reachable, SGSN initiates Update PDP Context message towards GGSN/PGW-C.
3. GGSN/PGW-C sends Sx Session Modification request message (PFCP) to the UPF.
4. UPF acknowledges with Sx Session Modification response message (PFCP).
5. GGSN/PGW-C sends Update PDP Context Response to SGSN.
6. GGSN/PGW-C informs MDF2 of PDP Context Modification by sending IRI CONTINUE: PDP Context Modification event to MDF2 (LI\_X2 interface).

**Figure 42. PDP Context Modification**

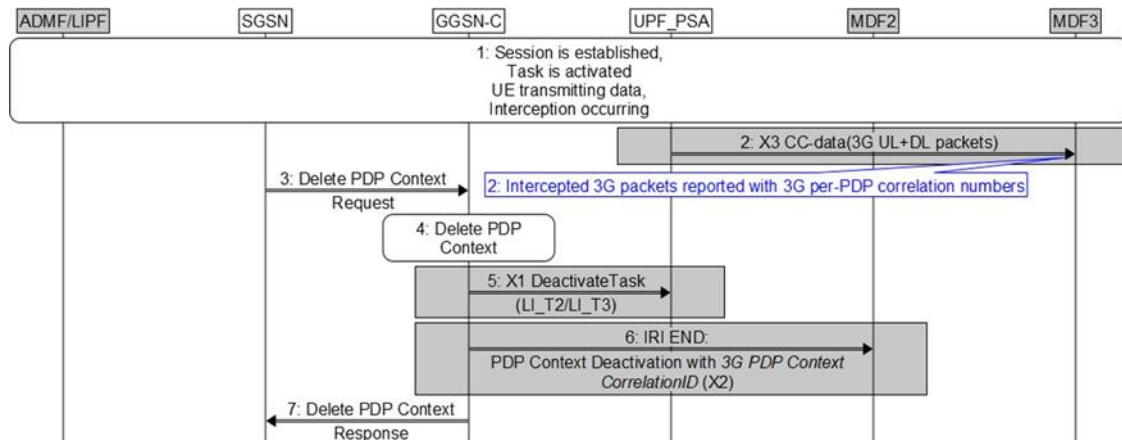


## NW/UE Initiated Detach with Lawful Intercept

The Network initiated PDN Session Detach procedure allows GGSN/PGW-C to request release of a PDP session. It is triggered when PCF initiates deletion of SM Policy Association (SM Policy Association termination request) or by Management intervention using a CLI command.

1. The receipt of an SM Policy Association Terminate Request from the PCF triggers session terminate procedure.
2. GGSN/PGW-C releases the PFCP session by sending Sx Session Release Request message towards the UPF.
3. GGSN/PGW-C sends LI X1 DeactivateTask with 3G PDP Context Correlation Number to the UPF (LI\_T2/LI\_T3 interface) to instruct UPF to stop intercepting traffic on the pdp context.
4. GGSN/PGW-C sends LI X1 DeactivateTask with 3G PDP Context Correlation Number to the UPF (LI\_T2/LI\_T3 interface) to instruct UPF to stop intercepting traffic on the pdp context.
5. GSN/PGW-C sends Delete PDP Context Request (EBI : NSAPI ID) message to the Serving GSN to delete the session.
6. SGSN sends Delete PDP Context (EBI: NSAPI ID) to RNC to delete the session.
7. GGSN/PGW-C sends X2 IRI END: PDP Context Deactivation with 3G PDP Context Correlation Number to MDF2 (LI\_X2 interface).
8. If Dynamic PCC is applied to this session, the GGSN/PGW-C invokes SM Policy Association Termination procedure towards the PCF to delete the PDP session.

**Figure 43. PDP Context Deactivation**

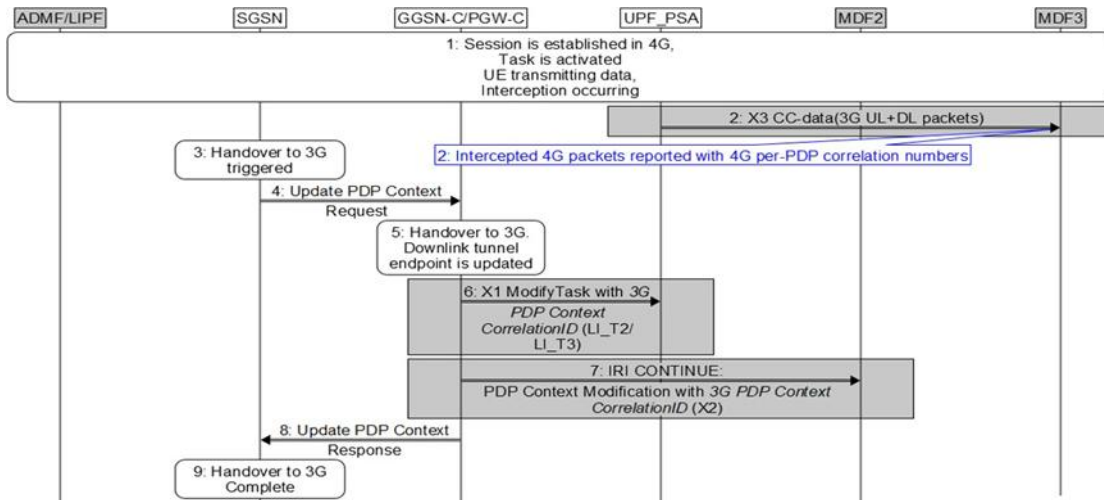


## Inter RAT Handover between E-UTRAN and UTRAN/GERAN

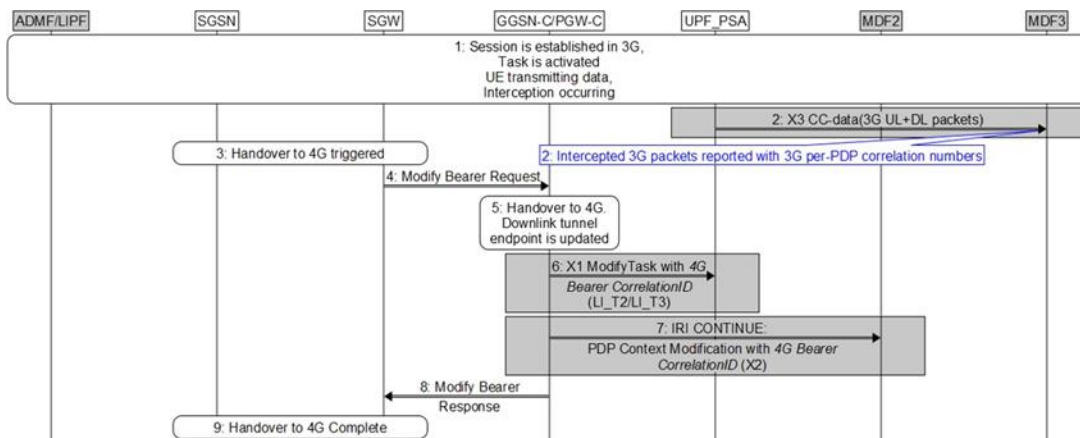
LI Tasks are activated on the UPF during initial session setup (3g - 4g, 4g-3g) handovers.

1. GGSN/PGW-C sends X1 ActivateTask (TargetIdentifier: PFCP SessionID, PDP FTEID) for Primary PDP with 3G PDP Context Correlation Number to UPF.
2. X3 CC-data containing intercepted packets for each PDP context is reported to MDF3 with PDP correlation number.
3. Handover is triggered by receipt of Modify Bearer Request/update PDP context. GGSN/PGW-C can receive Modify bearer Request from SGW in case of 3g to 4g handoff. In case there is a UE location change due to which 4g to 3g handoff happens, then GGSN/PGW-C receives update PDP context with new Location, new Sender Control FTEID, new SGW/SGSN-U TEID for all the PDP Contexts. Also, if changes then new Serving PLMN ID.
4. GGSN/PGW-C informs PCF about the new Location, ANGW Address, Serving Network Id (if PCF subscribed for these events).
5. GGSN/PGW-C sends Sx Session Modification Request to UPF to program the UPF with new Remote TEIDs for PDP contexts.
6. UPF acknowledges by sending Sx Session Modification Response to GGSN/PGW-C.
7. GGSN/PGW-C sends Modify Bearer/Update PDP Context Response to SGW/SGSN.
8. GGSN/PGW-C sends IRI CONTINUE: Bearer Modification/PDP Context Modification (LI\_ X2 message) to MDF2 to inform MDF2 of the new SGW. UPF continues to report all intercepted UE traffic to the MDF3 through X3 CC-data with appropriate 4G Bearer Correlation Numbers.

**Figure 44. 4G to 3G Handoff**



**Figure 45. 3G to 4G Handoff**



## UTRAN Initial Attach (Failure) with Lawful Intercept

1. SGSN sends Activate PDP Context message to GGSN/PGW-C.
2. Error occurs during session setup (resource allocation failure, authentication failure, etc).
3. GGSN/PGW-C sends IRI REPORT: PDP Context Activation Unsuccessful to MDF2 (LI\_X2 interface).

GGSN/PGW-C sends Activate PDP Context Response Failure to SGSN.

## SMF LI X2 Events for 3G GGSN Sessions

### PDP Context Activation (Unsuccessful) REPORT Record

See 3GPP TS 33.108 V16.1.0 (2020-07) Table 6.5: PDP Context Activation (unsuccessful) REPORT Record.

**Table 32. PDP Context Activation (Unsuccessful) Report Record**

| Parameter                                          | M/C/O | Description / Condition                                                                                                                                                                                                                                                                                                                                                                        | Supported? | Notes                                                                                                             |
|----------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------|
| Observed MSISDN<br>Observed IMSI<br>Observed ME Id | C     | Provide at least one and others when available                                                                                                                                                                                                                                                                                                                                                 | Yes        | GGSN/PGW-C reports all available identifiers when present in the GTP signaling for session/pdp context activation |
| Observed PDP Address                               |       | When a:<br>- static address requested by the target's MS in association with a target-initiated PDP context activation request is unsuccessful; or<br>- address offered by the network in association with a network-initiated PDP context activation request and the target's MS rejects the network-initiated PDP context activation,<br>The address requested or offered shall be reported. | Yes        |                                                                                                                   |
| IP Assignment                                      | C     | When an observed PDP address is reported, shall provide to indicate observed PDP address is statically or dynamically assigned.                                                                                                                                                                                                                                                                | Yes        |                                                                                                                   |
| Event type                                         | C     | Shall provide PDP Context Activation event type.                                                                                                                                                                                                                                                                                                                                               | Yes        |                                                                                                                   |
| Event date<br>Event time                           | M     | Provide the date and time the event is detected.                                                                                                                                                                                                                                                                                                                                               | Yes        |                                                                                                                   |
| Access Point Name                                  | C     | If available (i.e., provided by the UE) shall identify either the:<br>- packet data network to which the target requested to be connected when the target's                                                                                                                                                                                                                                    | Yes        |                                                                                                                   |

## GGSN

| Parameter                        | M/C/O | Description / Condition                                                                                                                                                                                                                                                                        | Supported? | Notes                                                                                                                                                 |
|----------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  |       | mobile station is unsuccessful at performing a PDP context activation procedure (MS to Network); or<br>- access point of the packet data network that requested to be connected to the MS when the target's mobile station rejects a network-initiated PDP context activation (Network to MS). |            |                                                                                                                                                       |
| PDP Type                         | C     | When an observed PDP address is reported, provide to describe the PDP type of the observed PDP address. The PDP Type defines the end user protocol to be used between the external packet data network and the MS.                                                                             | Yes        |                                                                                                                                                       |
| Initiator                        | M     | Shall be provided to indicate whether the PDP context activation is network- initiated, target-initiated, or not available.                                                                                                                                                                    | Yes        |                                                                                                                                                       |
| Network identifier               | M     | Shall be provided.                                                                                                                                                                                                                                                                             | Yes        |                                                                                                                                                       |
| Lawful intercept identifier      | M     | Shall be provided.                                                                                                                                                                                                                                                                             | Yes        | This field will be populated with "LIIDNotPresent". The relevant XID will be populated in the X2 PDU Header field XID as defined in ETSI TS 103 221-2 |
| Location Information             | C     | Provide, when authorized, to identify location information for the target's MS.                                                                                                                                                                                                                | Yes        | Will be provided when available                                                                                                                       |
| Time of Location                 | C     | Date/Time of Location. (if target location provided).                                                                                                                                                                                                                                          | Yes        |                                                                                                                                                       |
| Failed Context Activation reason | M     | Information about the reason for failed context activation attempts of the target shall be provided.                                                                                                                                                                                           | Yes        |                                                                                                                                                       |
| UMTS QOS                         | C     | Provide to identify the QOS parameters.                                                                                                                                                                                                                                                        | Yes        |                                                                                                                                                       |

## PDP Context Activation (Unsuccessful) BEGIN Record

See 3GPP TS 33.108 V16.1.0 (2020-07) Table 6.10: PDP Context Activation (successful) BEGIN Record.

**Table 33. PDP Context Activation (Unsuccessful) BEGIN Record**

| Parameter                                          | M/C/O | Description / Condition                                                                                                                                                                                                                                                                                                                                                                        | Supported? | Notes                                                                                                             |
|----------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------|
| Observed MSISDN<br>Observed IMSI<br>Observed ME Id | C     | Provide at least one and others when available                                                                                                                                                                                                                                                                                                                                                 | Yes        | GGSN/PGW-C reports all available identifiers when present in the GTP signaling for session/pdp context activation |
| Observed PDP Address                               |       | When a:<br>- static address requested by the target's MS in association with a target-initiated PDP context activation request is unsuccessful; or<br>- address offered by the network in association with a network-initiated PDP context activation request and the target's MS rejects the network-initiated PDP context activation,<br>The address requested or offered shall be reported. | Yes        |                                                                                                                   |
| IP Assignment                                      | C     | When an observed PDP address is reported, shall provide to indicate observed PDP address is statically or dynamically assigned.                                                                                                                                                                                                                                                                | Yes        |                                                                                                                   |
| Event type                                         | C     | Shall provide PDP Context Activation event type.                                                                                                                                                                                                                                                                                                                                               | Yes        |                                                                                                                   |
| Event date<br>Event time                           | M     | Provide the date and time the event is detected.                                                                                                                                                                                                                                                                                                                                               | Yes        |                                                                                                                   |
| Access Point Name                                  | C     | If available (i.e., provided by the UE) shall identify either the:<br>- packet data network to which the target requested to be connected when the target's mobile station is unsuccessful at performing a PDP context                                                                                                                                                                         | Yes        |                                                                                                                   |

# GGSN

| Parameter                        | M/C/O | Description / Condition                                                                                                                                                                                                             | Supported? | Notes                                                                                                                                                 |
|----------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  |       | activation procedure (MS to Network); or<br>- access point of the packet data network that requested to be connected to the MS when the target's mobile station rejects a network-initiated PDP context activation (Network to MS). |            |                                                                                                                                                       |
| PDP Type                         | C     | When an observed PDP address is reported, provide to describe the PDP type of the observed PDP address. The PDP Type defines the end user protocol to be used between the external packet data network and the MS.                  | Yes        |                                                                                                                                                       |
| Initiator                        | M     | Shall be provided to indicate whether the PDP context activation is network- initiated, target-initiated, or not available.                                                                                                         | Yes        |                                                                                                                                                       |
| Network identifier               | M     | Shall be provided.                                                                                                                                                                                                                  | Yes        |                                                                                                                                                       |
| Lawful intercept identifier      | M     | Shall be provided.                                                                                                                                                                                                                  | Yes        | This field will be populated with "LIIDNotPresent". The relevant XID will be populated in the X2 PDU Header field XID as defined in ETSI TS 103 221-2 |
| Location Information             | C     | Provide, when authorized, to identify location information for the target's MS.                                                                                                                                                     | Yes        | Will be provided when available                                                                                                                       |
| Time of Location                 | C     | Date/Time of Location. (if target location provided).                                                                                                                                                                               | Yes        |                                                                                                                                                       |
| Failed Context Activation reason | M     | Information about the reason for failed context activation attempts of the target shall be provided.                                                                                                                                | Yes        |                                                                                                                                                       |
| UMTS QOS                         | C     | Provide to identify the QOS parameters.                                                                                                                                                                                             | Yes        |                                                                                                                                                       |
| NSAPI                            | O     | Provided additional information.                                                                                                                                                                                                    | Yes        |                                                                                                                                                       |

## Start Of Interception (with PDP Context Active) BEGIN Record

See 3GPP TS 33.108 V16.1.0 (2020-07) Table 6.11: Start Of Interception BEGIN Record.

**Table 34. PDP Context Activation (Unsuccessful) BEGIN Record**

| Parameter                                          | M/C/O | Description / Condition                                                                                                                                                                                                                                                                                                                                                                        | Supported? | Notes                                                                                                             |
|----------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------|
| Observed MSISDN<br>Observed IMSI<br>Observed ME Id | C     | Provide at least one and others when available                                                                                                                                                                                                                                                                                                                                                 | Yes        | GGSN/PGW-C reports all available identifiers when present in the GTP signaling for session/pdp context activation |
| Observed PDP Address                               |       | When a:<br>- static address requested by the target's MS in association with a target-initiated PDP context activation request is unsuccessful; or<br>- address offered by the network in association with a network-initiated PDP context activation request and the target's MS rejects the network-initiated PDP context activation,<br>The address requested or offered shall be reported. | Yes        |                                                                                                                   |
| IP Assignment                                      | C     | When an observed PDP address is reported, shall provide to indicate observed PDP address is statically or dynamically assigned.                                                                                                                                                                                                                                                                | Yes        |                                                                                                                   |
| Event type                                         | C     | Shall provide PDP Context Activation event type.                                                                                                                                                                                                                                                                                                                                               | Yes        |                                                                                                                   |
| Event date<br>Event time                           | M     | Provide the date and time the event is detected.                                                                                                                                                                                                                                                                                                                                               | Yes        |                                                                                                                   |
| Access Point Name                                  | C     | If available (i.e., provided by the UE) shall identify either the:<br>- packet data network to which the target requested to be connected when the target's mobile station is unsuccessful at performing a PDP context activation procedure (MS to Network); or                                                                                                                                | Yes        |                                                                                                                   |

# GGSN

| Parameter                        | M/C/O | Description / Condition                                                                                                                                                                                            | Supported? | Notes                                                                                                                                                 |
|----------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  |       | - access point of the packet data network that requested to be connected to the MS when the target's mobile station rejects a network-initiated PDP context activation (Network to MS).                            |            |                                                                                                                                                       |
| PDP Type                         | C     | When an observed PDP address is reported, provide to describe the PDP type of the observed PDP address. The PDP Type defines the end user protocol to be used between the external packet data network and the MS. | Yes        |                                                                                                                                                       |
| Initiator                        | M     | Shall be provided to indicate whether the PDP context activation is network- initiated, target-initiated, or not available.                                                                                        | Yes        |                                                                                                                                                       |
| Network identifier               | M     | Shall be provided.                                                                                                                                                                                                 | Yes        |                                                                                                                                                       |
| Correlation Number               | M     | Shall provide to uniquely identify the PDP context delivered to the LEMF and to correlate IRI records with CC.                                                                                                     | Yes        |                                                                                                                                                       |
| Lawful intercept identifier      | M     | Shall be provided.                                                                                                                                                                                                 | Yes        | This field will be populated with "LIIDNotPresent". The relevant XID will be populated in the X2 PDU Header field XID as defined in ETSI TS 103 221-2 |
| Location Information             | C     | Provide, when authorized, to identify location information for the target's MS.                                                                                                                                    | Yes        | Will be provided when available                                                                                                                       |
| Time of Location                 | C     | Date/Time of Location. (if target location provided).                                                                                                                                                              | Yes        |                                                                                                                                                       |
| Failed Context Activation reason | M     | Information about the reason for failed context activation attempts of the target shall be provided.                                                                                                               | Yes        |                                                                                                                                                       |
| UMTS QOS                         | C     | Provide to identify the QOS parameters.                                                                                                                                                                            | Yes        |                                                                                                                                                       |
| NSAPI                            | O     | Provided additional information.                                                                                                                                                                                   | Yes        |                                                                                                                                                       |

## PDP Context Modification CONTINUE Record

See 3GPP TS 33.108 V16.1.0 (2020-07) Table 6.12: PDP Context Modification CONTINUE Record.

**Table 35. Context Modification CONTINUE Record**

| Parameter                                          | M/C/O | Description / Condition                                                                                                                                                                                                                                                                                                                                                                                                                                    | Supported? | Notes                                                                                                             |
|----------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------|
| Observed MSISDN<br>Observed IMSI<br>Observed ME Id | C     | Provide at least one and others when available                                                                                                                                                                                                                                                                                                                                                                                                             | Yes        | GGSN/PGW-C reports all available identifiers when present in the GTP signaling for session/pdp context activation |
| Observed PDP Address                               |       | When a:<br>- static address requested by the target's MS in association with a target-initiated PDP context activation request is unsuccessful; or<br>- address offered by the network in association with a network-initiated PDP context activation request and the target's MS rejects the network-initiated PDP context activation,<br>The address requested or offered shall be reported.                                                             | Yes        |                                                                                                                   |
| Event type                                         | C     | Shall provide PDP Context Activation event type.                                                                                                                                                                                                                                                                                                                                                                                                           | Yes        |                                                                                                                   |
| Event date<br>Event time                           | M     | Provide the date and time the event is detected.                                                                                                                                                                                                                                                                                                                                                                                                           | Yes        |                                                                                                                   |
| Access Point Name                                  | C     | If available (i.e., provided by the UE) shall identify either the:<br>- packet data network to which the target requested to be connected when the target's mobile station is unsuccessful at performing a PDP context activation procedure (MS to Network); or<br>- access point of the packet data network that requested to be connected to the MS when the target's mobile station rejects a network-initiated PDP context activation (Network to MS). | Yes        |                                                                                                                   |

| Parameter                   | M/C/O | Description / Condition                                                                                                                                                                                            | Supported? | Notes                                                                                                                                                 |
|-----------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| PDP Type                    | C     | When an observed PDP address is reported, provide to describe the PDP type of the observed PDP address. The PDP Type defines the end user protocol to be used between the external packet data network and the MS. | Yes        |                                                                                                                                                       |
| Initiator                   | M     | Shall be provided to indicate whether the PDP context activation is network- initiated, target-initiated, or not available.                                                                                        | Yes        |                                                                                                                                                       |
| Network identifier          | M     | Shall be provided.                                                                                                                                                                                                 | Yes        |                                                                                                                                                       |
| Correlation Number          | M     | Shall provide to uniquely identify the PDP context delivered to the LEMF and to correlate IRI records with CC.                                                                                                     | Yes        |                                                                                                                                                       |
| Lawful intercept identifier | M     | Shall be provided.                                                                                                                                                                                                 | Yes        | This field will be populated with "LIIDNotPresent". The relevant XID will be populated in the X2 PDU Header field XID as defined in ETSI TS 103 221-2 |
| Location Information        | C     | Provide, when authorized, to identify location information for the target's MS.                                                                                                                                    | Yes        | Will be provided when available                                                                                                                       |
| Time of Location            | C     | Date/Time of Location. (if target location provided).                                                                                                                                                              | Yes        |                                                                                                                                                       |
| UMTS QOS                    | C     | Provide to identify the QOS parameters.                                                                                                                                                                            | Yes        |                                                                                                                                                       |
| NSAPI                       | O     | Provided additional information.                                                                                                                                                                                   | Yes        |                                                                                                                                                       |

## PDP Content Deactivation END Record

See 3GPP TS 33.108 V16.1.0 (2020-07) Table 6.14: PDP Context Deactivation END Record.

**Table 36. PDP Context Deactivation**

| Parameter                                          | M/C/O | Description / Condition                                                                                                                                                                                                                                                                                                                                                                                                                                    | Supported? | Notes                                                                                                             |
|----------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------|
| Observed MSISDN<br>Observed IMSI<br>Observed ME Id | C     | Provide at least one and others when available                                                                                                                                                                                                                                                                                                                                                                                                             | Yes        | GGSN/PGW-C reports all available identifiers when present in the GTP signaling for session/pdp context activation |
| Observed PDP Address                               |       | When a:<br>- static address requested by the target's MS in association with a target-initiated PDP context activation request is unsuccessful; or<br>- address offered by the network in association with a network-initiated PDP context activation request and the target's MS rejects the network-initiated PDP context activation,<br>The address requested or offered shall be reported.                                                             | Yes        |                                                                                                                   |
| Event type                                         | C     | Shall provide PDP Context Activation event type.                                                                                                                                                                                                                                                                                                                                                                                                           | Yes        |                                                                                                                   |
| Event date<br>Event time                           | M     | Provide the date and time the event is detected.                                                                                                                                                                                                                                                                                                                                                                                                           | Yes        |                                                                                                                   |
| Access Point Name                                  | C     | If available (i.e., provided by the UE) shall identify either the:<br>- packet data network to which the target requested to be connected when the target's mobile station is unsuccessful at performing a PDP context activation procedure (MS to Network); or<br>- access point of the packet data network that requested to be connected to the MS when the target's mobile station rejects a network-initiated PDP context activation (Network to MS). | Yes        |                                                                                                                   |
| PDP Type                                           | C     | When an observed PDP address is reported, provide to describe the PDP type of the observed PDP address. The PDP Type                                                                                                                                                                                                                                                                                                                                       | Yes        |                                                                                                                   |

| Parameter                   | M/C/O | Description / Condition                                                                                                     | Supported? | Notes                                                                                                                                                 |
|-----------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |       | defines the end user protocol to be used between the external packet data network and the MS.                               |            |                                                                                                                                                       |
| Initiator                   | M     | Shall be provided to indicate whether the PDP context activation is network- initiated, target-initiated, or not available. | Yes        |                                                                                                                                                       |
| Network identifier          | M     | Shall be provided.                                                                                                          | Yes        |                                                                                                                                                       |
| Correlation Number          | M     | Shall provide to uniquely identify the PDP context delivered to the LEMF and to correlate IRI records with CC.              | Yes        |                                                                                                                                                       |
| Lawful intercept identifier | M     | Shall be provided.                                                                                                          | Yes        | This field will be populated with "LIIDNotPresent". The relevant XID will be populated in the X2 PDU Header field XID as defined in ETSI TS 103 221-2 |
| Location Information        | C     | Provide, when authorized, to identify location information for the target's MS.                                             | Yes        | Will be provided when available                                                                                                                       |
| Time of Location            | C     | Date/Time of Location. (if target location provided).                                                                       | Yes        |                                                                                                                                                       |
| UMTS QOS                    | C     | Provide to identify the QOS parameters.                                                                                     | Yes        |                                                                                                                                                       |
| NSAPI                       | O     | Provided additional information.                                                                                            | Yes        |                                                                                                                                                       |

## Statistics

LiX2 statistics are generated for the following LI events, PDPCContextActivation, PDPCContextModification, startOfInterceptionWithPDPCContextActive and PDPCContextDeactivation. They are:

- li\_x2\_callcontrol\_iri\_request attempts
- li\_x2\_callcontrol\_iri\_request success
- li\_x2\_callcontrol\_iri\_request failures
- li\_x2\_callcontrol\_iri\_request\_punted\_rcvd

- li\_x2\_callcontrol\_iri\_request\_punted\_sent
- li\_x2\_callcontrol\_msg\_request\_attempts
- li\_x2\_callcontrol\_msg\_request\_success
- li\_x2\_callcontrol\_msg\_request\_failures
- li\_x2\_callcontrol\_msg\_request\_punted\_rcvd
- li\_x2\_callcontrol\_msg\_request\_punted\_sent
- li\_x2\_callcontrol\_iri\_targetid\_request\_attempts
- li\_x2\_callcontrol\_iri\_targetid\_request\_success
- li\_x2\_callcontrol\_iri\_targetid\_request\_failures
- li\_x2\_callcontrol\_iri\_targetid\_request\_punted\_rcvd
- li\_x2\_callcontrol\_iri\_targetid\_request\_punted\_sent
- li\_x1\_x1msg\_ingress\_request\_attempts
- li\_x1\_x1msg\_ingress\_request\_success
- li\_x1\_x1msg\_ingress\_request\_failures
- li\_x1\_x1msg\_ingress\_request\_punted\_rcvd
- li\_x1\_x1msg\_ingress\_request\_punted\_sent
- li\_x1\_x1msg\_egress\_request\_attempts
- li\_x1\_x1msg\_egress\_request\_success
- li\_x1\_x1msg\_egress\_request\_failures
- li\_x1\_error\_response\_sent
- li\_x1\_error\_response\_rcvd
- li\_x1\_create\_destination\_request\_attempts
- li\_x1\_create\_destination\_request\_success
- li\_x1\_error\_response\_create\_destination\_request
- li\_x1\_create\_destination\_request\_punted\_rcvd
- li\_x1\_x1msg\_ingress\_request\_attempts
- li\_x1\_get\_all\_details\_request\_attempts

- li\_x1\_deactivate\_task\_request\_attempts
- li\_x1\_keep\_alive\_request\_attempts
- li\_x1\_ingress\_ping\_request\_attempts
- li\_x1\_report\_task\_issue\_request\_attempts
- li\_x1\_report\_destination\_issue\_request\_attempts
- li\_x1\_report\_ne\_issue\_request\_attempts
- li\_x1\_keep\_alive\_p2\_timer\_expired
- li\_x1\_local\_cleanup\_deactivate\_all\_tasks
- li\_x1\_create\_destination\_request\_punted\_sent
- li\_x1\_modify\_destination\_request\_punted\_sent
- li\_x1\_remove\_destination\_request\_punted\_sent
- li\_x1\_get\_destination\_details\_request\_punted\_sent
- li\_x1\_remove\_all\_destinations\_request\_punted\_sent
- li\_x1\_activate\_task\_request\_punted\_sent
- li\_x1\_modify\_task\_request\_punted\_sent
- li\_x1\_get\_ne\_status\_request\_attempts
- li\_x1\_x1msg\_ingress\_request\_attempts
- li\_x1\_total\_punt\_request\_sent
- li\_x1\_tf\_ingress\_x1msg\_request\_attempts
- li\_x1\_tf\_egress\_x1msg\_request\_attempts
- li\_x1\_tf\_create\_destination\_request\_attempts
- li\_x1\_tf\_modify\_destination\_request\_attempts
- li\_x1\_tf\_remove\_destination\_request\_attempts
- li\_x1\_tf\_get\_destination\_details\_request\_attempts
- li\_x1\_tf\_remove\_all\_destinations\_request\_attempts
- li\_x1\_tf\_activate\_task\_request\_attempts
- li\_x1\_tf\_modify\_task\_request\_attempts

- li\_x1\_tf\_deactivate\_task\_request\_attempts
- li\_x1\_tf\_deactivate\_all\_tasks\_request\_attempts
- li\_x1\_tf\_get\_task\_details\_request\_attempts
- li\_x1\_tf\_list\_all\_details\_request\_attempts
- li\_x1\_tf\_get\_all\_details\_request\_attempts
- li\_x1\_tf\_keep\_alive\_request\_attempts
- li\_x1\_tf\_egress\_ping\_request\_attempts
- li\_x1\_tf\_ingress\_ping\_request\_attempts
- li\_x1\_tf\_report\_task\_issue\_request\_attempts
- li\_x1\_tf\_report\_destination\_issue\_request\_attempts
- li\_x1\_tf\_report\_ne\_issue\_request\_attempts
- li\_x1\_tf\_get\_ne\_status\_request\_attempts
- li\_x1\_callcontrol\_msg\_request\_attempts
- li\_x1\_callcontrol\_activate\_task\_request\_attempts
- li\_x1\_callcontrol\_modify\_task\_request\_attempts
- li\_x1\_callcontrol\_deactivate\_task\_request\_attempts
- li\_x1\_consumer\_report\_msg\_request\_attempts
- li\_x1\_consumer\_report\_task\_stat\_request\_attempts
- li\_x1\_notify\_msg\_request\_attempts
- li\_x1\_activate\_task\_notify\_request\_attempts
- li\_x1\_modify\_task\_notify\_request\_success
- li\_x1\_deactivate\_task\_notify\_request\_attempts
- li\_x1\_timer\_start\_request\_attempts
- li\_x1\_timer\_stop\_request\_attempts
- li\_x1\_istioconfigif\_peer\_register\_request\_attempts
- li\_x1\_istioconfigif\_peer\_update\_request\_attempts
- li\_x1\_procedure\_duration

- li\_x1\_activate\_task\_setup\_duration
- li\_x1\_modify\_task\_setup\_duration
- li\_x1\_remove\_all\_destinations\_duration

## Protocol Interfaces

LI\_T2, LI\_T3 and LI\_X1 interfaces.

## Spec reference

- 3GPP TS 33.106 – Lawful intercept requirements
- 3GPP TS 33.107 – LI architecture and functions
- 3GPP TS 33.108 – Protocol and procedures for LI
- ETSI TS 103 221-1 – LI Internal Network Interfaces Part1: X1
- ETSI TS 103 221-2 – LI Internal Network Interfaces Part2: X2/X

# Certificate Management in Istio LI Ingress and LI Egress Gateways with HashiCorp Vault enabled

---

Istio LI Ingress and Istio LI Egress Gateway provides mTLS encryption through the TLS certificates provisioned to these gateways in the form of K8s secrets. TLS certificates in the form of secrets are provisioned by cert-manager which interacts with Hashicorp Vault for fetching the certificates.

Changes are applicable for both SMF and UPF.

**Note:** For HashiCorp Vault Configuration with cert-manager and for li cluster issuer config Refer: *UnityCloud\_Deployment\_Guide\_UC5.2.0* document -section “Hashicorp Vault Configuration with cert-manager”.

## TLS Certificates Provisioning through Cert-Manager/Hashicorp Vault

Istio LI Ingress and LI Egress Gateways helm chart via *values.yaml* provides the interface for configuring the certificates provisioned through cert-manager/HashiCorp Vault mode. The following is example configuration:

1. certManager changes under section ingressGwTLSLi

```
This controls use of TLS in GWs
 ingressGwTLSLi:
 enabled: true
 mode: MUTUAL

The below section covers the TLS certificates provisioning for
Istio Ingress Gateway
 certManager:
 enabled: true
 certDuration: 2160h
 renewBefore: 648h
 organization: affirmednetworks
 issuerName: affirmed-clusterissuer-li
```

```

 issuerType: "vault" <===== This must be set
to vault for the cert-manager to fetch the certificates from HashiCorp
Vault

```

## 2. certManager changes under section egressGwTLSLi

```

egressGwTLSLi:
 enabled: true
 mode: MUTUAL
The below section covers the TLS certificates provisioning for
Istio Egress Gateway
 certManager:
 enabled: true
 certDuration: 2160h
 renewBefore: 648h
 organization: affirmednetworks
 issuerName: affirmed-clusterissuer-li
 issuerType: "vault" <===== This must
be set to vault for the cert-manager to fetch the certificates from
HashiCorp Vault

```

When Istio LI Ingress and LI Egress Gateways are configured during the deployment like the preceding, users can verify if the LI certificates are provisioned correctly using the following command:

```

kubectl get secrets -n fed-smf

```

| NAME                              | TYPE              | DATA | AGE |
|-----------------------------------|-------------------|------|-----|
| smfli-com-li-ingress-cert-manager | kubernetes.io/tls | 3    | 11d |

Note: Data = 3 indicates certificates pulled successfully from hashicorp vault.

```

kc get certificates -n fed-smf

```

| NAME                             | READY | SECRET                            | AGE |
|----------------------------------|-------|-----------------------------------|-----|
| smf-com-li-egress-certificate    | True  | egress-li-cert-manager            | 25d |
| smfli-com-li-ingress-certificate | True  | smfli-com-li-ingress-cert-manager | 25d |

Note: Ingress certificate name depends on virtualservice host name like in above case "smli" and it may change.

In the preceding output, Certificates provisioning is considered to be successful only when READY status is set to true. With the READY status set to true, egress-li-cert-manager and smfli-com-li-ingress-cert-manage are the secrets created with the TLS LI certificates present inside it for Istio LI Egress Gateway and Istio LI Ingress Gateway respectively.

The following is an example secret that is created by cert-manager:

```
apiVersion: v1
data:
 ca.crt: <...>
 tls.crt: <...>
 tls.key: <...>
kind: Secret
metadata:
 annotations:
 cert-manager.io/alt-names: smf11i.com
 cert-manager.io/certificate-name: smf11i-com-li-egress-certificate
 cert-manager.io/common-name: smf11i.com
 cert-manager.io/ip-sans: ""
 cert-manager.io/issuer-group: ""
 cert-manager.io/issuer-kind: ClusterIssuer
 cert-manager.io/issuer-name: affirmed-clusterissuer-li
 cert-manager.io/uri-sans: ""
 creationTimestamp: "2024-11-08T06:46:51Z"
 labels:
 controller.cert-manager.io/fao: "true"
 name: egress-li-cert-manager
 namespace: fed-smf
 resourceVersion: "36589827"
 uid: af912fd8-97c2-4e3d-a7d7-3b7b9cef869c
type: kubernetes.io/tls
```

## User Provided certificates from Hashicorp vault.

If user provided certificates are used, they need to be stored in Hashicorp vault and below helm variables need to be configured.

```
global:
 secretNames:
 egressgwli:
 mountCSI: true
 userProvided:
```

```
 tlscrtPath: secrets/data/tls-crt-egressli <===== Should match
with tls cert path configured in the hashicorp vault
 cacertPath: secrets/data/ca-crt <===== Should match with ca cert
path configured in the hashicorp vault
 tlskeyPath: secrets/data/tls-key-egressli <===== Should match
with tls key path configured in the hashicorp vault
```

```
ingressgwli:
 mountCSI: true
 userProvided:
 tlscrtPath: secrets/data/tls-crt-ingressli <===== Should match
with tls cert path configured in the hashicorp vault
 cacertPath: secrets/data/ca-crt <===== Should match with ca cert
path configured in the hashicorp vault
 tlskeyPath: secrets/data/tls-key-ingressli <===== Should match
with tls key path configured in the hashicorp vault
```

**Note:** Only one of the preceding needs enabled for certificates, not both.

# LI Encryption Key Management with HashiCorp Vault enabled

---

The functionality of this feature is:

- Prior to this feature the keys to decrypt LI data records were stored in clear text inside Helm Chart
- With this feature SMF/UPF uses the infra support to retrieve and uses the keys stored in Hashicorp Vault
- SMF and UPF support storing two versions of the key
- On key rotation LI Applications update the records stored in DB and etcd with the new key
- Keys created in HashiCorp vault should match with Helm configured names as highlighted in the example

**Note:** For HashiCorp Vault Configuration Refer: *UnityCloud\_Deployment\_Guide\_UC5.2.0* document.

## Creating secrets in HashiCorp Vault:

```
vault kv put secrets/li-key-prev value="testKey"

vault kv put secrets/li-key-current value="NewtestKey"
```

**Note:** First time moving from helm key to hashicorp vault keys configure the current and previous keys in hashicorp with the current configured helm key.

```
global:
 encryptDecryptKey:
 key: CHANGEME
```

## Helm Chart Configuration

```
global:
```

```

secretNames:
 likey:
 mountCSI: true
 prevSecretName: secrets/data/li-key-prev <===== Should
match with prev secret name configured in the hashicorp vault
 secretName: secrets/data/li-key-current <===== Should
match with current name configured in the hashicorp vault

```

*Note: After "secrets/" path add "/data" in the helm configuration as shown in the preceding.*

## Steps to modify LI Encryption Keys

- To change the existing encryption key first update previous key with current key in hashicorp vault.
- Now update the current key with the new key.

### Example:

1. Get the current key from the vault with command "vault kv get secrets/**li-key-current**"

```

vault kv get secrets/li-key-current

```

```

===== Secret Path =====

```

```

secrets/data/li-key-current

```

```

.....

```

```

===== Data =====

```

```

Key Value

```

```

--- -----

```

```

value testKey

```

2. Update the previous key in the vault with the current key value received from the above step

```

vault kv put secrets/li-key-prev value="testKey"

```

3. Update the current key in the vault with new desired key.

## Key Management

```
vault kv put secrets/li-key-current value=" NewtestKey"
```

4. After changing the current key, Li Applications will begin refreshing the db entries with new key. And also this will not have impact on the current ongoing LI procedures.
5. It is recommended to wait for 120 mins before performing another key rotation.

**Note:** *If there is need to rotate key 120 mins before the previous key rotation, it is suggested to clean up the tasks and destinations and reprovision them after key rotation.*

# ISSU Support

---

## SMF

ISSU for SMF allows you to upgrade or rollback the SMF component without affecting the existing LI Tasks. ISSU for SMF also uses helm hooks to perform pre and post checks for the upgrade and rollback, such as verifying the pod readiness, the session state, and the service availability.

SMF supports pre and post upgrade/rollback hooks to validate and track the status of ISSU.

**Note:** Please refer *UnityCloud In-Service Software Upgrade Guide* for more information.

Following the upgrade, this table shows the new LI monitoring KPIs that are validated.

**Table 37. LI monitoring KPIs added in 5.2.0**

| Monitoring KPIs                                       |
|-------------------------------------------------------|
| li_x1_activate_task_request_acceptable_failures       |
| li_x1_modify_task_request_acceptable_failures         |
| li_x1_deactivate_task_request_acceptable_failures     |
| li_x1_create_destination_request_acceptable_failures  |
| li_x1_modify_destination_request_acceptable_failures  |
| li_x1_remove_destination_request_acceptable_failures  |
| li_x1_tf_activate_task_request_acceptable_failures    |
| li_x2_x2_iri_msg_dispatch_request_acceptable_failures |